Course Notes for Linear Algebra

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Chapter 1

Week 1

1.1 Introduction

1.1.1 Themes of Linear Algebra

Most students entering this course will have experience with high school mathematics and first year calculus (and maybe a smattering of other experienes, such as statistics). Linear algebra is not only a new topic in mathematics; compared to these previous experiences, it involves a substantially new way of thinking. In this introduction, I want to briefly lay out the main themes and ideas that I'll be working through for the entire term. I encourage you to keep these ideas in the back of your head and notice how the material of the course builds up these theme.

- *Geometry and Algebra.* Much of modern mathematics is based on the cartesian coordinate system. The major strength of this system is that is builds a connection between geometry and algebra. Shapes are connected to their equations, so that you want determine the properties of shapes by working with those equations. Likewise, equations can be interpreted as shapes, so that you can use geometry to visualize what is happening in algebra. In calculus, this happened through graphs: a graph was a geometric visualization of the behaviour of a function. Linear algebra continues to build on this connection, though the specific types of geometry and equations are different from calculus. All of the equations and algebraic manipulation in the course can be visualized geometrically and, likewise, all of the geometry in the course can be accessed algebraically. Though it is called an 'algebra' course, this is as much a geometry course as anything.

- *Flatness.* If linear algebra is a geometry course, what kind of geometry does it do? It does flat geometry: straight lines, planes, flat surfaces, and higher dimensional analogues. Flat objects are inherently more accessible and understandable than more complicated curved objects, so they are a natural start to geometry. Much of more complicated geometry is built on the linear algebra foundation of flat geometry. It's a natural place to start.

- *Linearity.* In geometry, the word 'linearity' refers to lines, which fits with the previous point about flat objects. However, the word also has another important meaning in algebra: an operation or structure is **linear** if it interacts well with addition, subtraction and multiplication by constants.

That's a bit vague for now, but I'll be developing this idea in detail over the course. From the algebriac side, this course is about understanding how these linear operations behave and what they can and cannot do. And, of course, the geometric linear and the algebraic linear do match up with each other.

- *Symmetry.* In ordinary usage, a symmetry is a change of a shape that preserves it, like a reflection. The mathematical notion is similar, but not precisely the same. If I have a transformation (such as a reflection), its symmetries are the things its preserves. I've just flipped the definition here: in conventional usage, the shape has a symmetry, but in mathematical usage, the transformation (the reflection) has the symmetry. Mathematically, whenever I define some kind of transformation, I want to ask for its symmetries: for what is preserves. If can preserve shapes, and I will talk about that in detail. But it can also preserve other mathematical structures, particularly algebraic structures. to study symmetry is to ask, for a mathematical operation: what is preserved? Asking this quesiton can build a deep understanding of the transformation. I'll keep coming back to this question of what is perserved throughout the course.

- *Definition, Rigor and Proof.* Mathematics is a formal language, with strict rules about logic, definition and arguments. Your mathematical education to date has included this formalization, but this course represents a substantial step up in commitment to that formalization. All the definitions in the course will be given in mathematically formal language. I will try to provide proofs for many of the statements. I'll also ask you to try to write proofs in your exercises and assignments. I'll try to convince you that writing proofs, along side doing calculations or solving problems, is 'doing mathematics'.

1.1.2 Mathematical Thinking and Writing

The themes listed above are the content goals I have for the course. However, in addition to the content of linear algebra itself, I want to focus on the more general skills of mathematical thinking and writing. I'm going to present material more formally and expect a higher standard of formal writing from you. A large part of learning mathematics is being apprenticed into the way that mathematics is written and, evern more so, the way that mathematics is thought. This is a much trickier task than just learning how to do problems: it's learning a style, not an algorithm. It's learning a way of thinking, not just an individual fact. As challening as it is, working on thinking and writing is a major goal of this course. I hope you buy into this challenge, that you consciously work on your mathematical writing and thinking.

Of the undergraduate mathematics courses, linear algebra is the most natural place to start talking about mathematical proofs and to work on formalized mathematical writing. The material lends itself to a number of accessible proof examples and have a level of abstraction that suits proof questions. The linear algebra course also serves as a prerequisite for most of the other mathematic courses where proof becomes quite important, including the senior calculus courses, discrete mathematics, abstract algebra, cryptography and theory of computing.

1.2 Euclidean Space

1.2.1 Definitions

Euclidean or cartesian space is the environment for this course. The cartesian plane is familiar to you, but I'm going to give formal definitions regardless. This is part of the process of learning the formal language of mathematics: of taking previously known mathematical ideas and redefining them in more formal, logical ways. We start that process with geometric space and the vectors therein.

Definition 1.2.1 In linear algebra and many other brances of mathematics, ordinary numbers (integers, rational numbers or real numbers) are called **scalars**. $\Diamond$

Definition 1.2.2 A **vector** is a finite ordered list of scalars. Vectors can be written either as columns or rows. If a vector is a list of the numbers 4, 15, π and $-e$ (in that order), I write the vector in one of two ways:

$$\begin{pmatrix} 4 \\ 15 \\ \pi \\ -e \end{pmatrix} \qquad \text{or} \qquad (4, 15, \pi, -e)$$

$\Diamond$

In this course, I will exclusively use column vectors. In many other places, row vectors are common. I find it best to choose one of the two for the whole course instead of switching back and forth. For reason that I'll talk about in future weeks, column vectors are a good choice for this course.

Definition 1.2.3 Let n be a positive integer. **Real Euclidean Space** or **Cartesian Space**, written $\mathbb{R}^n$, is the set of all vectors of length n with real number entries. I write an arbitrary vector with coefficients x_i as follows.

$$\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

Since there are n coefficients in its vectors, the space $\mathbb{R}^n$ has **dimension** n. $\Diamond$

Two dimensional space, $\mathbb{R}^2$, and possibly three dimensional space, $\mathbb{R}^3$, should be familiar. Notice, though, that my new formal definitions work in any dimension. This is the power of mathematical abstraction and formalization. Even though we live in three dimensions and can't visualize higher dimension, there are many reasons (including very practical reasons) to consider higher dimensions. Therefore, my definitions are flexible enough to consider four dimension space, or six dimension vectors, or whatever else I need.

Definition 1.2.4 The scalars x_i in a vector are called the **entries**, **coordinates** or **components** of that vector. Specifically, x_1 is the first coordinate, x_2 is the second coordinate, and so on. For $\mathbb{R}^2$, $\mathbb{R}^3$ and $\mathbb{R}^4$, I use the letters

w, x, y, z instead of x_i.

$$\begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 \qquad \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 \qquad \begin{pmatrix} w \\ x \\ y \\ z \end{pmatrix} \in \mathbb{R}^4$$

$\diamondsuit$

Definition 1.2.5 In any $\mathbb{R}^n$, the **origin** is the unique point given by a vector of zeros. It is also called the **zero vector**. It is considered the centre point of Euclidean space.

$$\begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

$\diamondsuit$

This is geometry, so I want to actully see it. Euclidean space is usually visualized by drawing axes, one in each independent perpendicular direction. In this visualization, the vector $\begin{pmatrix} a \\ b \end{pmatrix}$ corresponds to the unique point I get moving a units in the direction of the x axis and b units in the direction of the y axis. Figure 1.2.6 shows the location of several points in $\mathbb{R}^2$.

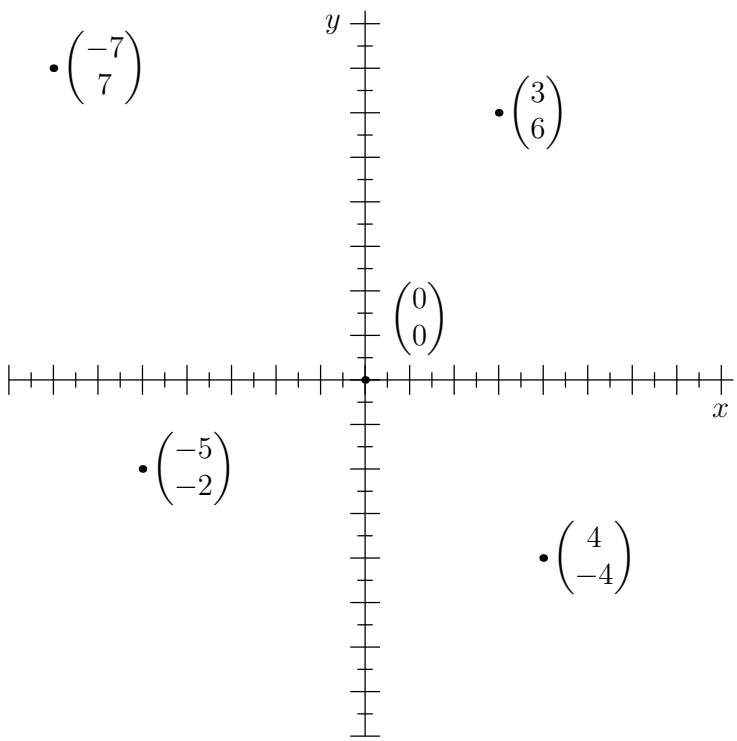

Figure 1.2.6 Points in $\mathbb{R}^2$

As with $\mathbb{R}^2$, the point $\begin{pmatrix} a \\ b \\ c \end{pmatrix} \in \mathbb{R}^3$ is the unique point I find by moving a units in the x direction, b units in the y direction and c units in the z direction. When visualizing $\mathbb{R}^2$, the convention is to draw the x axis horizontally, with a

positive direction to the right, and the y axis vertically, with a positive direction upwards. For $\mathbb{R}^3$, the x and y axes form a flat plane, and the z axis extends vertically from that plane. Figure 1.2.7 shows some points in $\mathbb{R}^3$

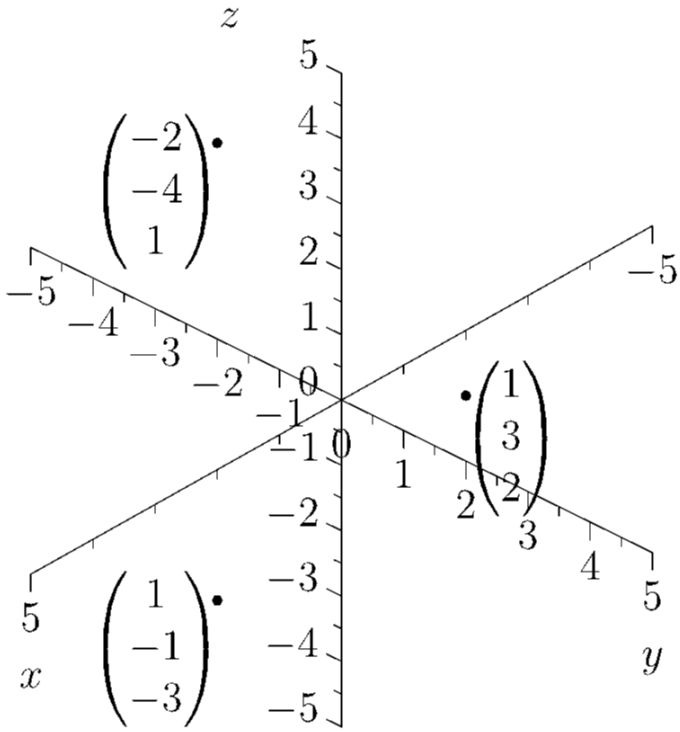

Figure 1.2.7 Points in $\mathbb{R}^3$

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Definition 1.2.8 Note that I needed to choose directions for the axes in both $\mathbb{R}^2$ and $\mathbb{R}^3$. In a visualization of Euclidean space, there are no predetermined directions for axes. I have to make a choice. This choice of axis directions in the visualization of $\mathbb{R}^n$ is called **orientation**. $\Diamond$

While I can visualize $\mathbb{R}^2$ and $\mathbb{R}^3$ relatively easily and efficiently, I can't visualize any higher $\mathbb{R}^n$. However, this doesn't prevent me from working in higher dimensions. I need to rely on the algebraic descriptions of vectors instead of the drawings and visualizations of $\mathbb{R}^2$ and $\mathbb{R}^3$. This is where the algebra/geometry connection becomes truly remarkable: if I build the algebra to describe two and three dimensional geometry, I can use that algebra to describe four, five and higher dimensional geometry as well. Even though I can't see these spaces, I can understand them and work with them. It's amazing.

In the visualizations of $\mathbb{R}^2$ and $\mathbb{R}^3$, I see the different axes as fundamentally different perpendicular directions. I can think of $\mathbb{R}^2$ as the space with two independent directions and $\mathbb{R}^3$ as the space with three independent directions. Similarly, $\mathbb{R}^4$ is the space with four perpendicular, independent directions, even though it is impossible to visualize such a thing. Likewise, $\mathbb{R}^n$ is the space with n independent directions.

1.2.2 Points or Directions?

I can think of an element of $\mathbb{R}^2$, say $\begin{pmatrix}1\\4\end{pmatrix}$, as both the point located at $\begin{pmatrix}1\\4\end{pmatrix}$ and the vector drawn from the origin to the point $\begin{pmatrix}1\\4\end{pmatrix}$, as shown in Figure 1.2.9. Though these two ideas are distinct, I will frequently switch between them, often without explicitly saying so. Part of becoming proficient in vector geometry is becoming accustomed to the switch between the perspectives of points and directions.

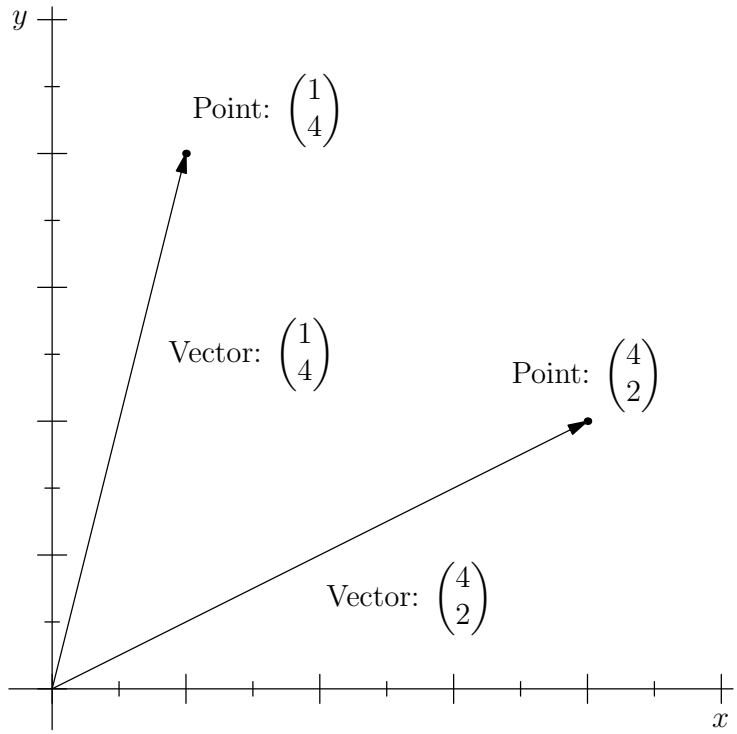

Figure 1.2.9 Vectors as Points and Directions

1.2.3 Local Direction Vectors

I've already spoken about the distinction between elements of $\mathbb{R}^n$ as points and vectors. There is another important subtlety that shows up all throughout vector geometry. In addition to thinking of vectors as directions starting at the origin, I can think of them as directions starting anywhere in $\mathbb{R}^n$. I call these **local direction vectors**.

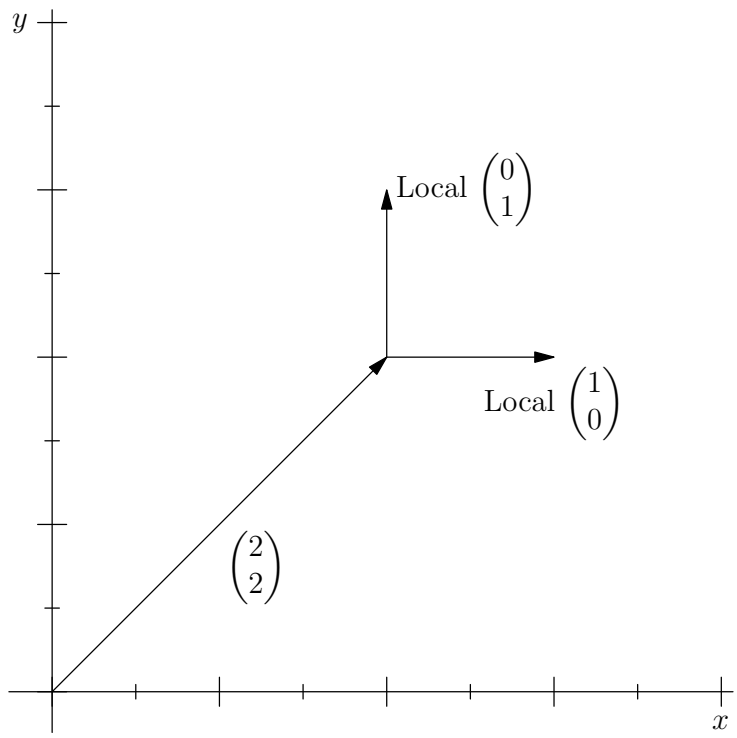

Figure 1.2.10 Local Direction Vectors

Figure 1.2.10 shows local direction vectors starting at the point $\begin{pmatrix}2\\2\end{pmatrix} \in \mathbb{R}^2$. The two local vectors $\begin{pmatrix}1\\0\end{pmatrix}$ and $\begin{pmatrix}0\\1\end{pmatrix}$ are relative to the point $\begin{pmatrix}2\\2\end{pmatrix} \in \mathbb{R}^2$ as if that were their origin.

Using vectors to define local directions is a useful tool. A standard example is a camera in a three-dimensional virtual environment. First, I need to know the location of the camera, which is an ordinary vector starting from the origin. Second, I need to know what direction the camera is pointing, which is a local direction vector which treats the camera location as the current origin.

One of the most difficult things about learning vector geometry is becoming accustomed to local direction vectors. I won't always carefully distinguish between vectors at the origin and local direction vectors; often, the difference is implied, and it is up to the reader/student to figure out how the vectors are being used.

1.3 Vector Arithmetic

1.3.1 Linear Operations

In this section, I go through the basic arithmetic of vectors. In some ways, it mimics the arithmetic of scalars; in other ways, scalar operations no longer work and are replaced with other operations.

Definition 1.3.1 The **sum** of two vectors u and v in $\mathbb{R}^n$ is the sum taken component wise.

$$u + v = \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{pmatrix} + \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix} = \begin{pmatrix} u_1 + v_1 \\ u_2 + v_2 \\ \vdots \\ u_n + v_n \end{pmatrix}$$

$\Diamond$

The sum is visualized by placing the start of the second vector at the end of the first, as in Figure 1.3.2. Note that I can only add two vectors in the same dimension. I can't add a vector in $\mathbb{R}^2$ to a vector in $\mathbb{R}^3$.

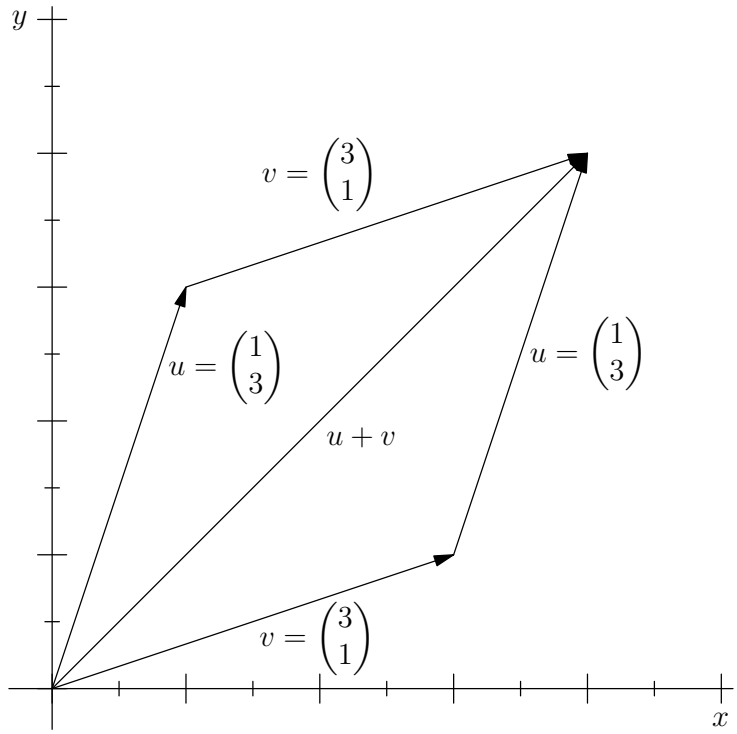

Figure 1.3.2 Visualizing Vector Addition

Definition 1.3.3 If u is a vector in $\mathbb{R}^n$ and $a \in \mathbb{R}$ is a real number, then the **scalar multiplication** of u and a is multiplication by a in each component of u. By convention, scalar multiplication is written with the scalar on the left of the vector.

$$au = a \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{pmatrix} = \begin{pmatrix} au_1 \\ au_2 \\ \vdots \\ au_n \end{pmatrix}$$

$\Diamond$

Though there will be other 'multiplications' to come, it is mostly true that I can't multiply vectors together in any way reminiscent of numbers. Instead, I can only multiply by scalars. Scalar multiplication is visualized by scaling the vector by the value of the scalar. (Hence the term 'scalar'!) If the scalar is negative, the direction is also reversed, as in Figure 1.3.4.

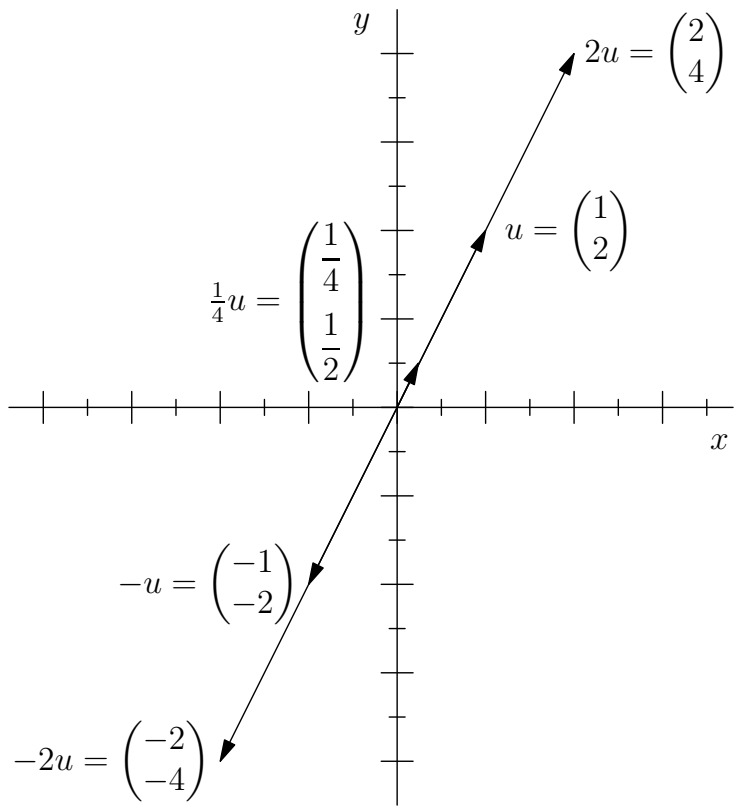

Figure 1.3.4 Visualizing Scalar Multiplication

Scalar multiplication also lets me define the difference between vectors.

Definition 1.3.5 The difference between two vectors u and v is the vector $u + (-1)v$, defined using addition and scalar multiplication. This works out to be componentwise subtraction.

$$u - v = u + (-1)v = \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{pmatrix} + (-1) \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix} = \begin{pmatrix} u_1 - v_1 \\ u_2 - v_2 \\ \vdots \\ u_n - v_n \end{pmatrix}$$

$\Diamond$

Definition 1.3.6 With respect to some set of scalars (such as $\mathbb{R}$), whenever I find a mathematical structure that has the two properties of addition and scalar multiplication, I call the structure **linear**. $\mathbb{R}^n$ is a linear space because vectors allow for addition and scalar multiplication. As I said in Section 1.1, this algebraic sense of linear will be a major theme in the course. This is the first instance of that theme. $\Diamond$

Definition 1.3.7 The **length** of a vector u in $\mathbb{R}^n$ is written $|u|$ and is given by a generalized form of the Pythagorean rule for right triangles.

$$|u| = \sqrt{u_1^2 + u_2^2 + \ldots + u_n^2}$$

This length is also called the **norm** of the vector. A vector of length one is called a **unit vector**. $\Diamond$

If I think of vectors as directions from the origin towards a point, this

definition of length gives exactly what I expect: the physical length of that arrow in $\mathbb{R}^2$ and $\mathbb{R}^3$. Past $\mathbb{R}^3$, I don't have a natural notion of length. This definition serves as a reasonable generalization to $\mathbb{R}^4$ and higher dimensions, which I can't visualize. Note also that $|u| = 0$ only if u is the zero vector. All other vectors have a positive length.

Often the square root is annoying, and I find it convenient to work with the square of the length.

$$|u|^2 = u_1^2 + u_2^2 + \ldots + u_n^2$$

The notions of length and difference allow me to define the distance between two vectors.

Definition 1.3.8 The **distance between two vectors** u and v in $\mathbb{R}^n$ is the length of their difference: $|u - v|$. $\Diamond$

You can check from the definition that $|u-v| = |v-u|$, so distance doesn't depend on which comes first. If $|\cdot|$ were absolute value in $\mathbb{R}$, this definition would match the notion of distance between numbers on the number line. Difference and length are visualized in Figure 1.3.9.

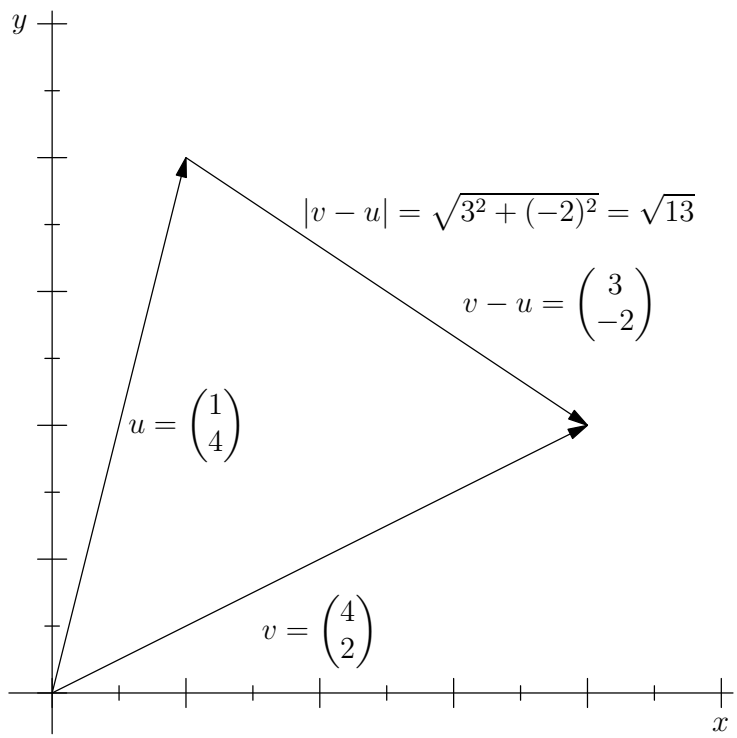

Figure 1.3.9 Visualizing Distance Between Vectors

Proposition 1.3.10 *Having defined all these operations (addition, scalar multiplication, subtraction, length, distance), I want to investigate their properties. For now, I'll just give one example of the kind of properties I am interested in. The following statement is called the* ***triangle inequality****.*

$$|u + v| \leq |u| + |v| \qquad \textit{Triangle Inequality}$$

The proof of the triangle inequality is left to an exercise later in the course.

1.4 Higher Dimensions

1.4.1 New Environments

In the definitions in Section 1.2, I defined vectors and Euclidean space in any dimension, not just the familiar two and three dimensions. If I want to work in seven dimensions, I simply need to write vectors with seven components. This is a strange and new thing. Conventional geometry deals with things I can draw and see, which are things that necessarily exist in two or three dimensions. Higher dimensions are something new and bizzare, but also fascinating. (Higher dimensions are not just a mathematical fascination: doing higher dimensional geometry is also extremely practical. I'll try to provide justification for this practicality before the course is finished.)

In this section, I want to talk through some constructions that might help to build some intuition and familiarity with higher dimensions. Working in higher dimensions is a new kind of challenge, so buidling some intuition can be extremely valuable. I do need to use indirect (mostly algebriac)methods, since I can't visualize higher dimensional spaces.

In some contexts, particularly in physics using Einstein's general relativility, the fourth dimension is often treated as time. That's not the approach in this course. I am interested in treating all these higher dimensions as environments for geometry. I want to imagine objects, paths, intersections, and many other geometric ideas in these higher dimensions. In the applications of linear algebra, higher dimensions can be given a huge variety of interpretation, including treating one dimension as time. Those applications can come afterwards, after the theory is well developed. In linear algebra, I treat all dimensions geometrically.

I'll start building higher dimensional intituion by talking about how some familiar objects generalize into these higher dimensions.

1.4.2 Spheres

In the plane, all points that are exactly one unit from the origin form the unit circle. In three dimensional space, all points that are exactly one unit from the origin form the unit sphere. This is the start of a pattern. Following this logic, I can construct 'spheres' in higher dimensions. In any $\mathbb{R}^n$, all objects which are exactly one unit from the origin form the sphere in that dimension.

What do I know about these higher-dimensional spheres? First, I can look at their equations. In $\mathbb{R}^2$, the circle has the equation $x^2 + y^2 = 1$. In $\mathbb{R}^3$ the sphere has equation $x^2 + y^2 + z^2 = 1$. The sphere in $\mathbb{R}^4$ has equation $w^2 + x^2 + y^2 + z^2 = 1$. The sphere in $\mathbb{R}^n$ has equation $x_1^2 + x_2^2 + \ldots + x_n^2 = 1$.

These equations are useful for studying the algebra of the object. I can determine which points are on any given sphere using these equations. The points

$$\begin{pmatrix}1\\0\\0\\0\end{pmatrix} \qquad \begin{pmatrix}\frac{1}{\sqrt{3}}\\ \frac{-1}{\sqrt{3}}\\ 0\\ \frac{-1}{\sqrt{3}}\end{pmatrix} \qquad \begin{pmatrix}\frac{-2}{\sqrt{12}}\\ \frac{-1}{\sqrt{12}}\\ \frac{2}{\sqrt{12}}\\ \frac{\sqrt{3}}{\sqrt{12}}\end{pmatrix} \qquad \begin{pmatrix}0\\0\\ \frac{7}{\sqrt{74}}\\ \frac{5}{\sqrt{74}}\end{pmatrix}$$

are all points on the 3-sphere in $\mathbb{R}^4$, since they all satisfy the equation of that sphere. However, these equations aren't that useful for getting an intuitive sense of what these spheres look like.

Getting this intuitive sense is fundamentally difficult. It is worth repeating that the major challenge of higher dimensions geometry is the fact that I can't see it, since I am so used to seeing the geometry I am working with. However, there are still some indirect geometric ways of thinking that can help me understand higher-dimensional objects.

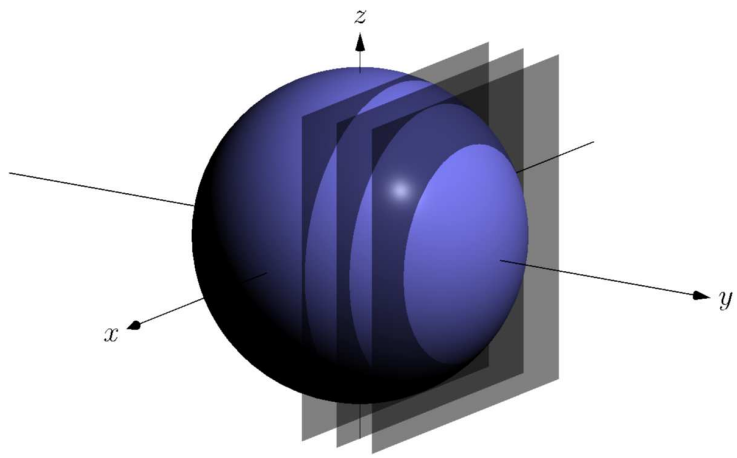

Figure 1.4.1 Slicing A Sphere

If I take the normal sphere in $\mathbb{R}^3$ and slice it vertically, each cross-section is a circle, as illustrated in Figure 1.4.1.. The cross-section at the widest point is a unit circle, and as I move along the y axis, the cross-sections are smaller and smaller circles. I can think of the sphere as an object whose slices are all circles (though it is not the unique object with that property). I could also think of the sphere as an infinite stack of infinitely thin circles, starting with small circles, growing to the the unit circles at the middle, and then shrinking again as I move further across the sphere.

This notion of slicing works in higher dimensions. The sphere in $\mathbb{R}^4$ has slices, which are usual three-dimensional spheres. That's a bizarre idea, because trying to think in four dimensions is a bizarre task, and one that defies visualization. I can't see an infinite stack of spheres all right next to each other. But this slicing into spheres is a useful way of thinking about the sphere in $\mathbb{R}^4$. The sphere in four dimensions is an object whose slices are spheres in three dimensions; it can be thought of as infinitely many spheres stacked up. This is still strange: the stacking happens in a new direction, so that all points on adjacent spheres are right next to each other, just like the circles stack up to make the ordinary sphere are right next to each other. But maybe this helps us think a bit about what kind of object the sphere in $\mathbb{R}^4$ is. This pattern could continue: the five dimensional sphere has slices which are four dimensional spheres, and so on.

1.4.3 Cubes

After the sphere, I want to consider square objects. To understand the higher-dimensional cubes, I want to talk about how to build each cube from lower-dimensional pieces. I'll start at the very beginning, in $\mathbb{R}^1$, which is just the number line.

The 'cube' in $\mathbb{R}$ is the interval $[0, 1]$. It can be defined by the inequality $0 \leq x \leq 1$ for $x \in \mathbb{R}$.

In $\mathbb{R}^2$, the 'cube' is just the ordinary square. It is all vectors $\begin{pmatrix} x \\ y \end{pmatrix}$ such that $0 \leq x \leq 1$ and $0 \leq y \leq 1$. The square can be formed by taking two intervals

(1-dimensional cubes) and connecting the matching vertices. This will become the way in which I build higher-dimensional cubes: each cube is formed of two cubes from the previous dimension with all the matching vertices connected.

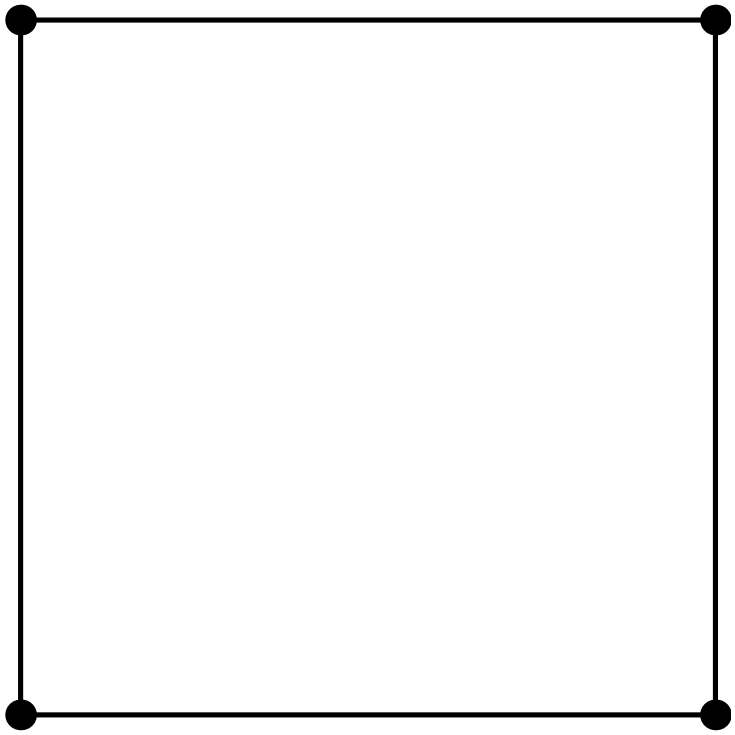

Figure 1.4.2 The Square

Now I'll look at $\mathbb{R}^3$ and follow the construction process. If I take two squares and connect the matching vertices, I do indeed get the ordinary cube. One of the two squares is the front of the cube, and the other is the back (or top and bottom, left and right, depending on how you orient the pieces). To make this a bit more clear, in Figure 1.4.3, I've drawn the two original squares in black and the four connecting lines in red. I could also describe the cube (including its interior) as all points where the coordinates satisfy $x \in [0, 1]$, $y \in [0, 1]$ and $z \in [0, 1]$.

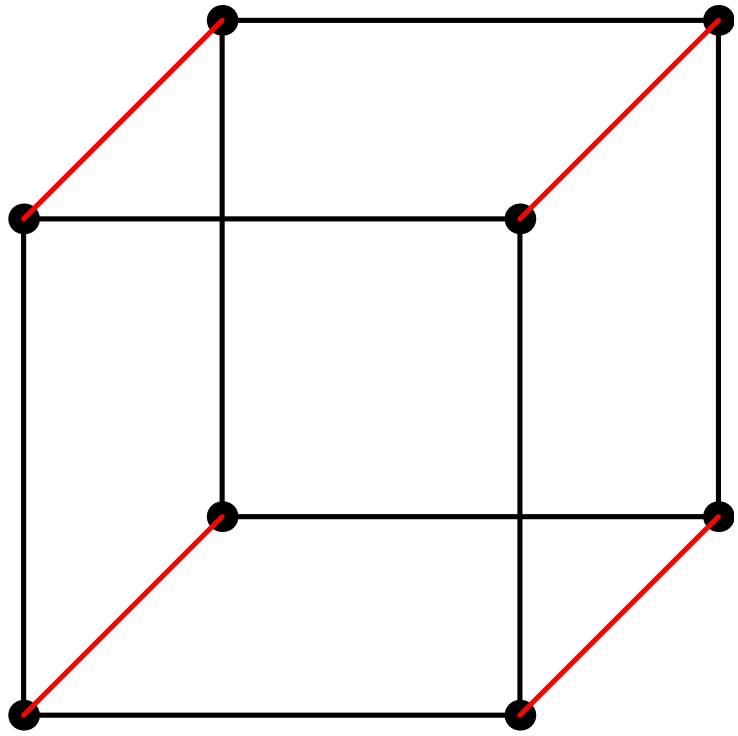

Figure 1.4.3 The Cube

Then I can simply keep extending. In $\mathbb{R}^4$, I will take two copies of the 3D cube and connect all the matching vertices. In Figure 1.4.4, I've shown the connecting lines again in red. This diagram is a poor picture, being a 2D representation of a 4D object, but it does give some idea of the construction of the shape. The 4D cube is often called the tesseract. I could again describe it as all vectors in $\mathbb{R}^4$ such that $w \in [0,1]$, $x \in [0,1]$, $y \in [0,1]$ and $z \in [0,1]$.

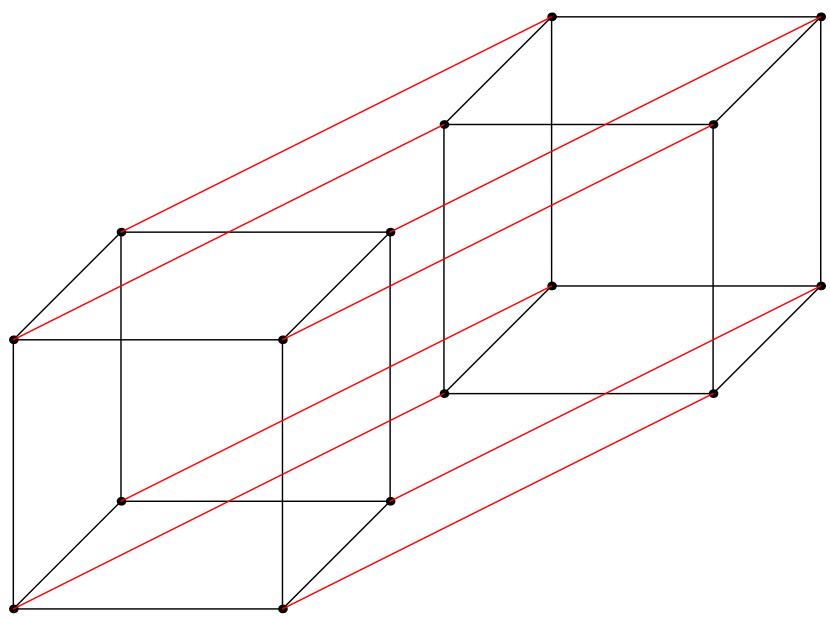

Figure 1.4.4 The Tesseract

There is a cube in each $\mathbb{R}^n$, consisting of all vectors where all components are in the interval $[0,1]$. Each can be constructed by joining two copies of a

lower-dimensional $(n-1)$-cube with edges between matching vertices. I haven't tried to draw diagrams for any of these higher-dimensional cubes, since the diagrams quickly become very difficult to parse. But, hopefully, the diagram of the tesseract gives a sense of how to think about this un-seeable four dimensional object.

The cube is an example of a solid object with flat sides and sharp corners. Such objects are called polytopes (polygons in two dimensions). The study of polytopes in higher dimensions is itself a rich branch of mathematics with a long history.

1.4.4 Cross-Polytopes

In addition to the cubes, I want to introduce another kind of higher-dimensional polytope, called the cross-polytope. It generalizes the idea of a diamond shape in two and three dimensions. It also lets me show another way of trying to get a sense of what higher-dimensional objects look like.

In $\mathbb{R}^2$, the cross-polytope is the diamond with vertices $\begin{pmatrix}1\\0\end{pmatrix}$, $\begin{pmatrix}-1\\0\end{pmatrix}$, $\begin{pmatrix}0\\1\end{pmatrix}$, and $\begin{pmatrix}0\\-1\end{pmatrix}$. In each dimension, cross-polytopes are built by adding two vertices at ± 1 in the new axis direction. Then, edges are added connecting the two new vertices to each existing vertex (but the two new vertices are not connected to each other).

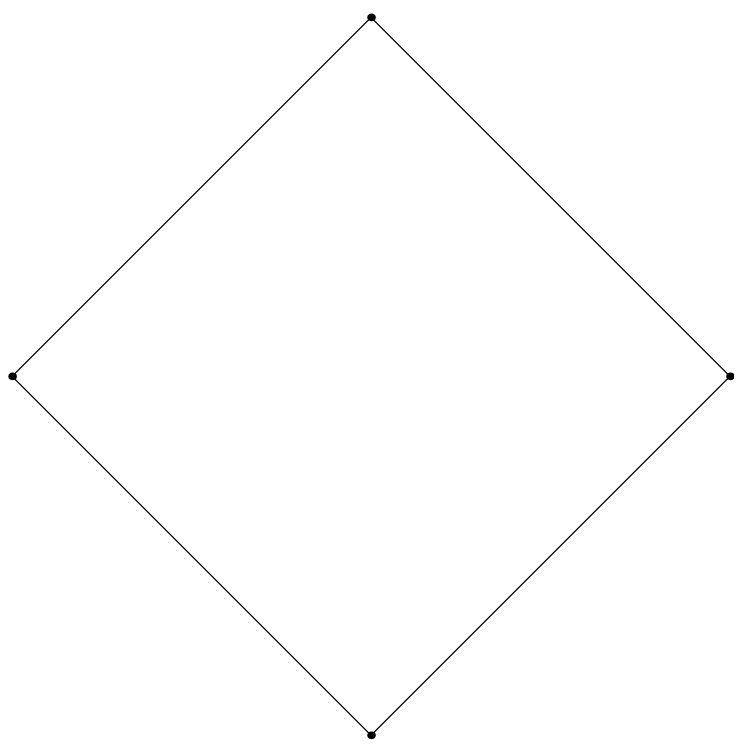

Figure 1.4.5 The 2-Cross-Polytope (Diamond)

In $\mathbb{R}^3$, if I take the diamond, add two points on either side and connect the new points to all four original vertices of the diamond, I am essentially building a square pyramid on either side of the diamond. The result is called

the octahedron and has vertices $\begin{pmatrix}1\\0\\0\end{pmatrix}$, $\begin{pmatrix}-1\\0\\0\end{pmatrix}$, $\begin{pmatrix}0\\1\\0\end{pmatrix}$, $\begin{pmatrix}0\\-1\\0\end{pmatrix}$, $\begin{pmatrix}0\\0\\1\end{pmatrix}$, and $\begin{pmatrix}0\\0\\-1\end{pmatrix}$. In Figure 1.4.6, I've tried to draw the octahedron in perspective, to give a sense of it as a 3D object.

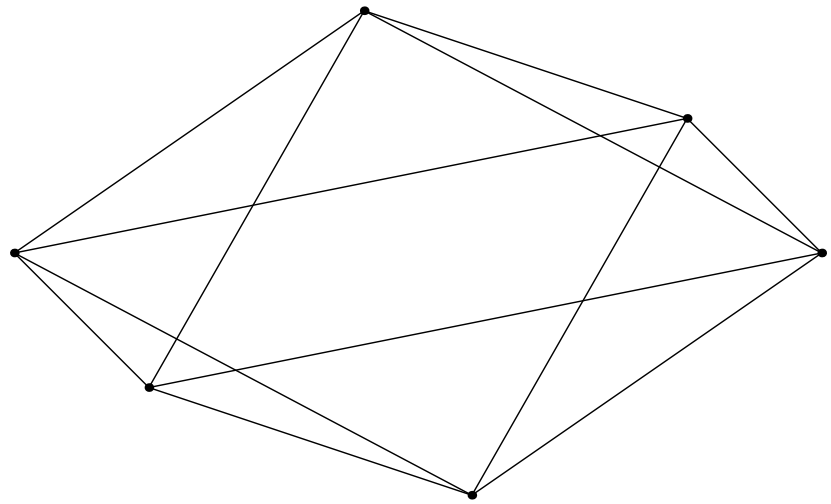

Figure 1.4.6 The 3-Cross-Polytope (Octahedron)

I can continue to build cross-polytopes in higher dimensions using this process: adding vertices on either side and connecting the new vertices to all previous vertices (but not to each other). If I choose the new vertices as points in the new axis direction, then all the vertices of the n-cross-polytope are vertices with ± 1 in one component and zero in all other components. Each vertex is connected to all other vertices except its opposite. I haven't tried to draw diagrams of any of these higher cross-polytopes.

One way of understanding higher-dimensional objects (since I can't directly see them) is to just draw their lower-dimensional components. For polyhedra, I can do this by only drawing their vertices and edges (and not trying to represent the higher-dimensional structures that include those vertices and edges). I'm going to do this for the cross-polytopes to demonstrate the idea. The diamond (2-cross-polytope) has four vertices with all connections except for opposite vertices.

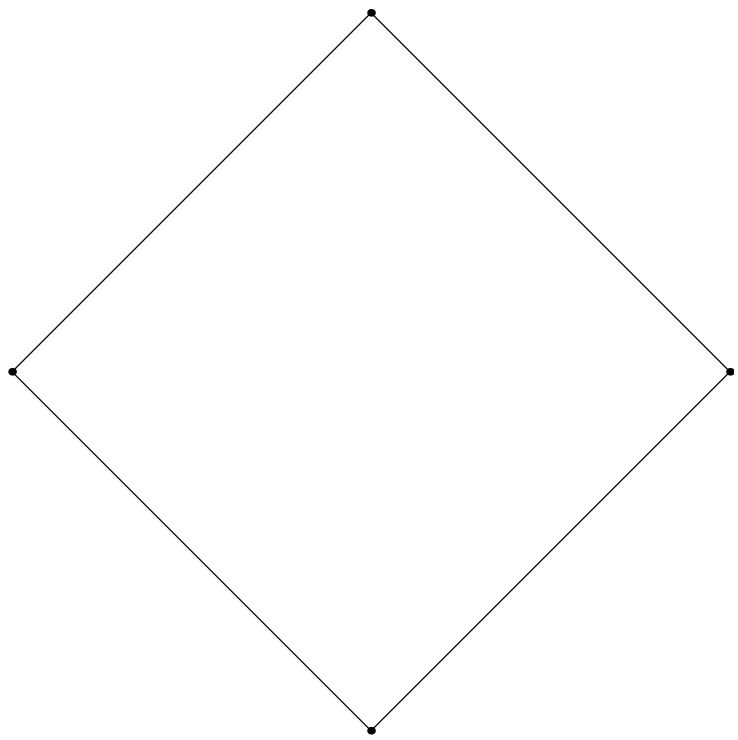

Figure 1.4.7 The 2-Cross-Polytope (Diamond) Vertex Diagram

The octahedron is the 3-cross polytope. I tried to draw a 3D image in perspective in Figure 1.4.6. However, if I don't try to get a 3D perspective and just draw the vertices and edges, I get the vertex diagram instead of the entire object in perspective. I've shown this vertex diagram in Figure 1.4.8. It looks like a flattened version of the octahedron, which is precisely what I want it to do. Like the diamond, all vertices except for those directly opposite are connected.

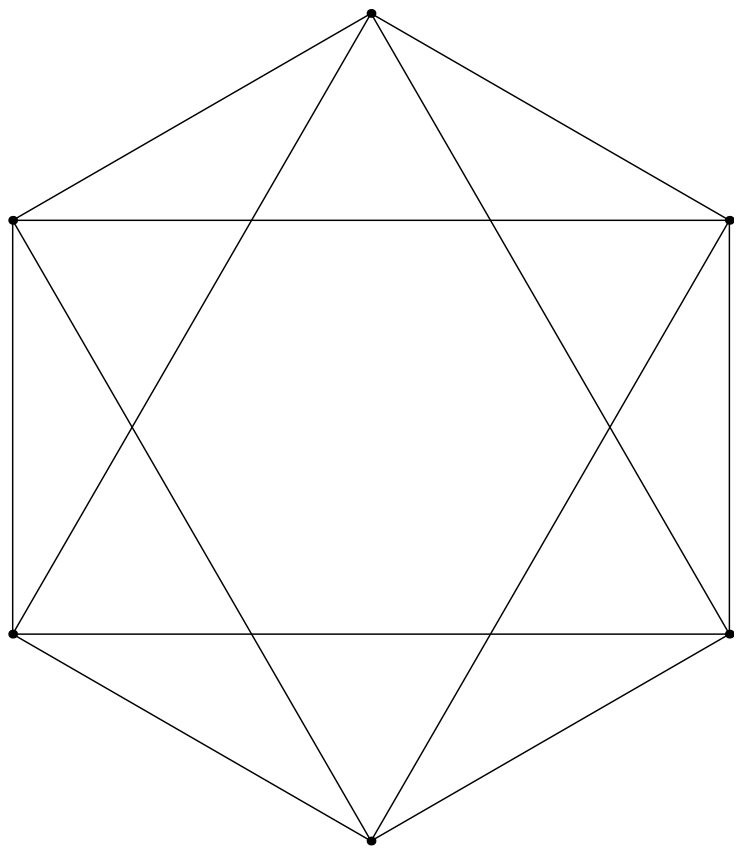

Figure 1.4.8 The 3-Cross-Polytope (Octahedron) Vertex Diagram

The construction of the higher cross-polytopes, where I add two vertices and connect them to all existing vertices, consistently builds a set of vertices where each vertex is attached, by an edge, to all other vertices except for the one directly opposite it. If I collapse the vertices and edges onto a 2D diagram, that means I draw all the lines between our vertices except those directly opposite. The 4-cross polytope has 8 vertices connected in this way and its vertex diagram is shown in Figure 1.4.9.

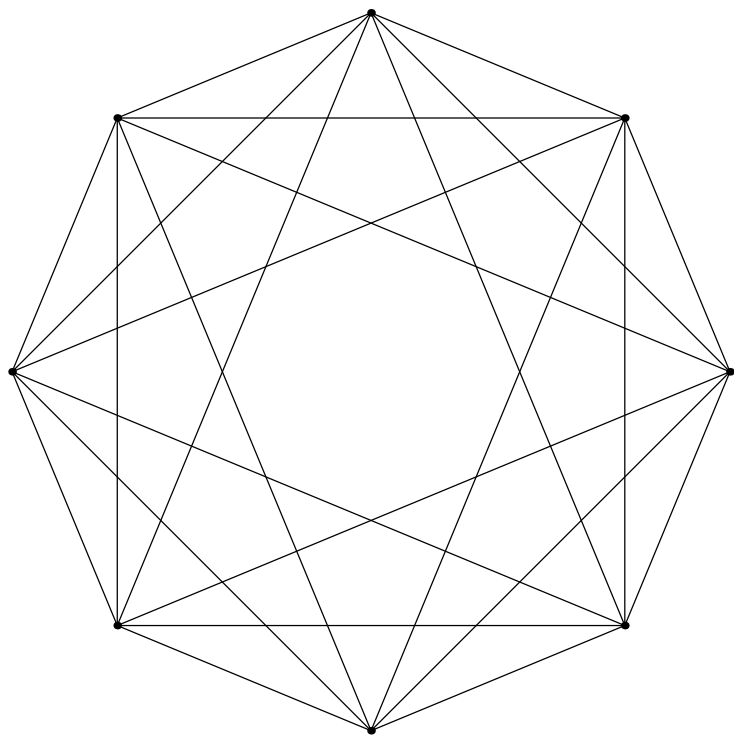

Figure 1.4.9 The 4-Cross-Polytope Vertex Diagram

Now I just keep going. In Figure 1.4.10, I show the 5-cross-polytope, whichs 10 vertices connected this way.

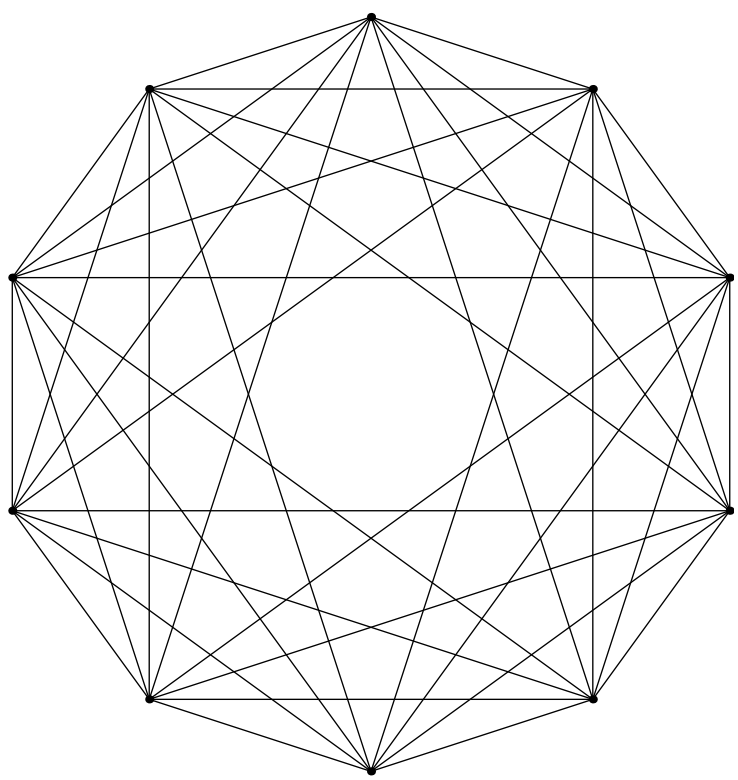

Figure 1.4.10 The 5-Cross-Polytope Vertex Diagram

I can go as far as I want in this way. Each time I go up a dimension, I add

two vertices. The 8-cross-polytope (in $\mathbb{R}^8$) has 16 vertices connected this way, as shown in Figure 1.4.11.

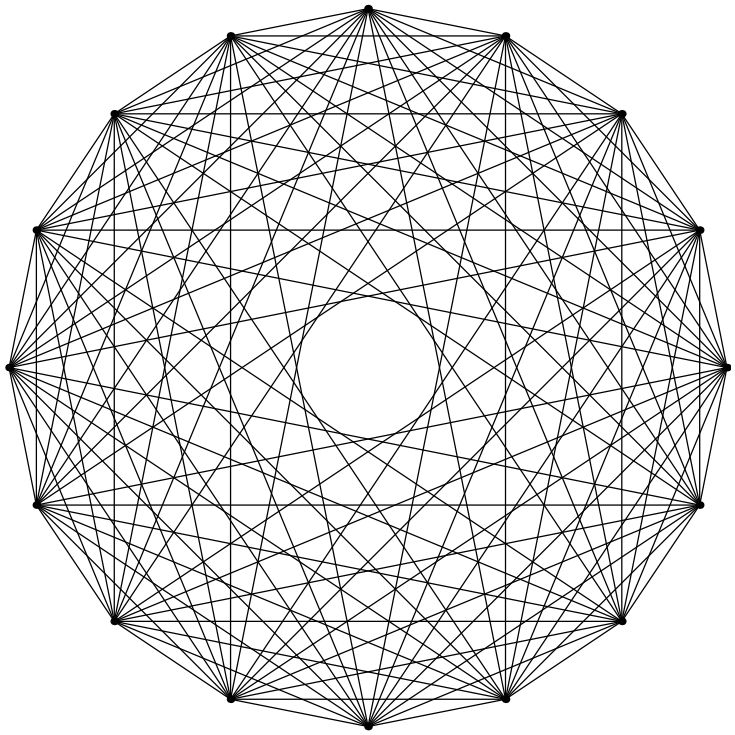

Figure 1.4.11 The 8-Cross-Polytope Vertex Diagram

To show how wild these vertex diagrams get, consider Figure 1.4.12, the vertex diagram of the 16-cross-polytope (in $\mathbb{R}^{16}$), which has 32 vertices connected this way.

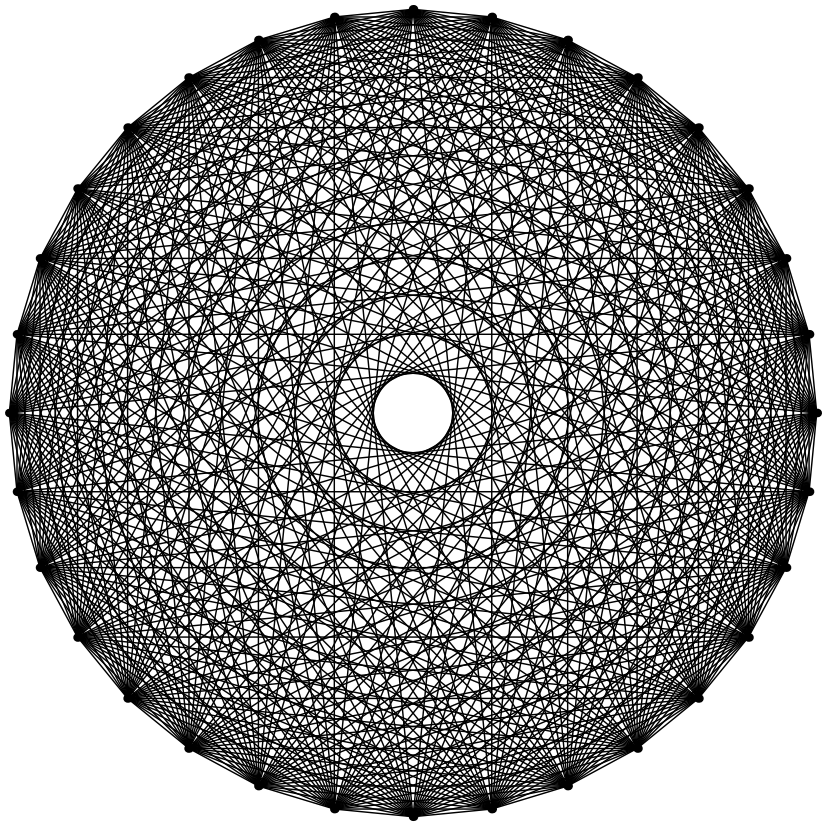

Figure 1.4.12 The 16-Cross-Polytope Vertex Diagram

These images are obviously not full pictures of these higher dimensional objects, but just some kind of 2D shadows which give me some information and intuition for the objects. Everything in higher dimensions must be done indirectly, but working indirectly, I can produce some useful and interesting visualizations that let me start to understand the objects. Hopefully it is a little bit interesting to have some mental pictures of a 16 dimensional diamond-shaped prism, even if that mental picture is just of all the corners and edges collapsed down only a flat surface.

1.5 Writing Proofs

1.5.1 What Is A Proof?

As I said in Section 1.1, a major goal of this course is the introduction of more formalized mathematical language, particular for definitions and proofs. In this section, I want to start that process. I'm going to talk about several general ways to structure a proof, so that you can recognize them in the course and so that you can start to construct them yourself.

So what is a mathematical proof? A simple definition is hard, but my best attempt is very general: a proof is a formal, logical, mathematical argument that succeeds in convincing an informed reader. Let me talk through this definition.

- First, a proof is an argument. It is persuasive: it has a proposition, and it wants to convince its reader that the proposition is true. In this, it shares something with all the other persuasive writing throughout university, including scientific studies, aesthetic critiques, and persuasive essays.

- Second, a proof is logical. Though mathematical discovery can happen in a variety of ways, the presentation of mathematical ideas happen through logic. A proof will be a series of logical statements and implications. If successful, the chain of logic will lead from the assumptions of the proof to the desired conclusion.

- Third, a proof is formal. Mathematics has formal notations, definitions, and rules for manipulating expressions. A proof follows these formalities. It can't treat mathematical ideas and definitions loosely or casually; it has to follow the rules and strict conventions of mathematics.

- Finally, it succeeds in convincing the reader. At the end of a proof, the question should always be: are you convinced? Proofs are accepted by the mathematical community when a sufficient number of qualified readers are convinced by the mathematical formality and logical sequence. In this course, when I present a proof, I will try to repeatedly ask if you are convinced by the proof. If you are not, either the proof is flawed, or your understanding of the argument is weak and you need to go back and study the assumptions, notations, and structure more carefully.

1.5.2 How to Write a Proof

I desperately wish there were something simple and straightforward I could say here about how to write a proof. I wish I could give you a reasonable three-step process that would always produce good proofs. Unfortunately, no such thing exists. Writing proofs is the most difficult skill in mathematics. Writing a good proof is comparable to writing a good essay. I learn by reading good examples, practicing my own, and being patient with the long process.

In this course, I'm not going to give you any short, easy method for writing proofs (since, as I just said, no such method exists). Instead, I'm going to try to start the process of exposing you to proofs. In many ways, learning to write proofs is done by example — by reading lots of proofs and seeing how they work. I'll show you a bunch of short proofs in this course and ask you to try to emulate them in the activities and assignments.

That said, I will give some pointers and ideas on how to start.

- Remember that a proof is a chain of logical arguments. You should see this structure in the proofs you write. Look out for any sudden jumps or gaps in the logic.

- Use your starting assumptions. If you are proving something about a prime number, but never use the fact that it is prime, something very strange is going on. Either you are making a mistake, or the assumption that the number is prime is unnecessary.

- Don't assume what you are trying to prove. The statement that you want to prove should show up at the very end of your proof. You can't start with the statement you want to prove — logical implications don't work that way. A proof should be a logical chain that starts with the assumptions of the proof and end with the desired conclusion.

- There is a huge difference between the rough work that goes into sketching a proof and the final proof. In the rough work, I will often start with the conclusion and make various attempts to see how I might get there. I try things, play around with ideas, do test cases. But when I write up a proof, I take all this rough work and try to put it into a formal logical chain with the desired statement coming at the end, as a conclusion.

- A good proof includes good writing as well as good symbolic mathematics. Write clear, complete sentences. Treat proof-writing with the same level of care as you would any academic writing.

- There are a number of special proof techniques that are worth some attention. I'll spend the next section on two of these techniques. Not all proofs need a named technique, but these two are common enough to point out at the start.

- I can't prove by example. Proof is not inductive reasoning, like that which is used in scientific experiments. A proof should prove all possible instances of the proposition, without exception. A bunch of examples are interesting and can provide hints, but they never constitue a proof. (That said, working with examples is often a great idea when working on a proof. Example gives insight and often show the way to the general arguent. They just can't be used as the persuasive evidence in the proof.)

- Similarly, I can't prove by diagram, drawing or illustration. Drawing pictures of what is going on is often a great way to build intuition, to understand the situation. It can often guild you in the right direction for constructing a proof. But a visualization itself does not constitute a complete logical argument. It doesn't meet the standard that mathematics has set for a proof.

1.5.3 Proof by Contradiction

There are many common structures and techniques used with proofs. I'm going to share with you just one of the most common techniques: proof by contradiction. This technique relies on the assumption that any (well-formed) statement in mathematics is either true or false. To prove a statement is true, I can equivalently show that it is impossible for the statement to be false.

To prove by contradiction, I assume the statement I want to prove is false. Then, starting from that assumption, I make logical conclusions. These logical conclusions lead me to an impossible situation — a contradiction. This contradiction can take many forms: a contradiction of some statement I know is true, an impossibility of a statement being both true and false, or some other situation which cannot make sense. Since I end up with this contradiction, my original assumption must have been wrong. Since I assumed the statement was false, the statement must therefore be true.

This is perhaps easier to see in an example. I'll do two examples, both of which are classic proof examples. (If you've study anything about proofs elsewhere, there is a good chance you've seen these examples.) First, I'll prove that there are infinitely many prime numbers.

- Step 1: Assume the statement is false. The statement says there are infinitely many prime numbers. If that is false, there must be only finitely many prime numbers. So my assumption is that there are finitely many prime numbers.

- Step 2: Work from the assumption to build a contradiction. The assumption says that there are finitely many prime numbers. Since there are only finitely many, I can index them: $p_1, p_2, p_3, \ldots p_k$. This is a finite list of some length k. Then consider a very large number formed by multiplying these all together and adding 1.

$$m = p_1 p_2 \ldots p_k + 1$$

What are the prime factors of this number m? I know that all numbers break down uniquely into prime factors, so it must have prime factors. However, every prime number p_i in my list (which, by assumption, is all prime numbers) is a factor of $m-1$. A prime number can't be the factor of two consecutive numbers, so none of the prime numbers are factors of m.

- Step 3: State the contradiction. The number m must have prime factors, but cannot have prime factors, since none of the primes p_i are factors. This is a contradiction: the number must have prime factors but has none.

- Step 4: Conclude that the original statement was false, which finishes the proof. The assumption that there were only finitely many prime numbers led to a contradiction. Therefore, by contradiction, I have proved that there are infinitely many prime numbers.

This proof is *non-constructive*. It tells me that there are infinitely many prime numbers, but it doesn't tell me anything about them: what they are, how to find them, how they are distributed on the number line, etc. This is often a major weakness of proofs by contradiction: they may not give concrete information about the objects in the proposition.

Another classic proof example is the proof that $\sqrt{2}$ is irrational. I will also prove this by contradiction.

- Step 1: Assume the statement is false. The statement says that $\sqrt{2}$ is irrational. If that is false, then $\sqrt{2}$ must be rational. Therefore, my assumption is that $\sqrt{2}$ is rational.

- Step 2: Work from the assumption to build a contradiction. If $\sqrt{2}$ is rational, then there are integers a and b such that $\sqrt{2} = \frac{a}{b}$. Since any fraction can be written in the simplest terms, I can assume that a and b have no common factors. Then I can do some arithmetic with this equation.

$$\begin{aligned} \sqrt{2} &= \frac{a}{b} \\ 2 &= \frac{a^2}{b^2} \\ 2b^2 &= a^2 \end{aligned}$$

2 divides the left side of the equation, so it must divide the right side as well, meaning that a^2 must be even. But the squares of odd numbers are odd, and the squares of even numbers are even. Therefore, a must be even. I can write $a = 2c$ for some integer c. Then I'll replace a by $2c$ in the equation and do a bit more arithmetic.

$$\begin{aligned} 2b^2 &= a^2 \\ 2b^2 &= (2c)^2 \\ 2b^2 &= 4c^2 \\ b^2 &= 2c^2 \end{aligned}$$

Now 2 divides the right side of the equation, so it must divide the left as well. That means that b^2 is even, so b must be even.

- Step 3: State the contradiction. I assumed that a and b have no common factors since I wrote the fraction in the simplest terms. Now I have proved that both numbers are even, so 2 is a factor of both. This is a contradiction: a and b have no common factors, but 2 is a common factor.
- Step 4: Conclude that the original statement was false, which finishes the proof. The assumption that $\sqrt{2}$ was rational produced to a contradiction, so I have proved that $\sqrt{2}$ must be irrational.

Proof by contradiction, as I mentioned at the start, relies on the assumption that any (well-formed) statement in mathematics is either true or false. That seems like a reasonable assumption in a logical system, and proof by contradiction is an important tool in mainstream mathematics. However, some versions of mathematics have rejected this assumption. There are some versions of mathematics that do not allow proof by contradiction. (These are called intuitionistic or constructivist mathematics).

1.6 Proofs - Vector Arithmetic

1.6.1 Properties of Vector Arithmetic

In Section 1.3, I introduced three structures on $\mathbb{R}^n$: vector addition, scalar multiplication, and vector length. The abstract algebraic question that mathematicians ask at this point is: what are the properties of these structures? How does vector addition resemble normal addition (which is it named after), and how does it differ? How does scalar multiplication resemble normal multiplication (which is it named after), and how does it differ? How does the length differ from the absolute value (which shares the same notation)?

These are abstract questions, but they are also practically useful. If I am in a situation where I need to use linear algebra (of which there are many), I need to know how to manipulate the objects. I have a whole collection of a habits with addition and multiplication, but I can't assume that all those habits immediately transfer to these new structures. If the new structures resemble the old, I have to prove it to be sure. If they don't, I have to adjust my habits. Both situation shappen in this course. In this section, I'll mostly be proving that the properties of ordinary addition and multiplication carry over to these new structures. Later in the course, I'll have new structures which have properties that are different from what I expect for conventional arithmetic.

In this section, I'm going to address three properties of scalar multiplication and vector addition. Some properties of the vector length operation are left for the activities.

1.6.2 Commutativity

Addition and multiplication of numbers are commutative binary operations. This means that I can interchange the order: $4 + 5$ is the same as $5 + 4$, and 7×2 is the same as 2×7. I've defined a new addition and a new multiplication, but I can't just assume that these new structures are commutative. I have to prove it.

Proposition 1.6.1 *Vector addition is commutative. That is, for any two vectors $u, v \in \mathbb{R}^n$,*

$$u + v = v + u.$$

Proof. When approaching a proof, it is often very useful to return to the definition. Let me first work in $\mathbb{R}^2$ to illustrate; I'll generalize to $\mathbb{R}^n$ afterwards. I have two vectors $u, v \in \mathbb{R}^n$. I write them as components.

$$u = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} \qquad v = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$$

Then I use the definition of vector addition. I'll start with the left side of the equation and try to work towards the right.

$$\begin{aligned} u + v &= \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} + \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} \\ &= \begin{pmatrix} u_1 + v_1 \\ u_2 + v_2 \end{pmatrix} \end{aligned}$$

That's the definition. Now each addition in the components (u_1+v_1 and u_2+v_2) is the addition of ordinary numbers, so I can reverse the order of this addition.

$$= \begin{pmatrix} v_1 + u_1 \\ v_2 + u_2 \end{pmatrix}$$

The result is precisely the definition of the right side.

$$= v + u$$

I started on the left side of the desired equation and ended up with the right side, thus proving the identity. This was for $\mathbb{R}^2$, so I still need to do the general case for $\mathbb{R}^n$. Luckily for me, the steps are exactly the same, just for a vector with n components. Here are the steps in the general case: first, I use the definition of $u+v$, then I switch the order of the ordinary addition of numbers in each component, then I use the definition of $v + u$.

$$\begin{aligned} u + v &= \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{pmatrix} + \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix} \\ &= \begin{pmatrix} u_1 + v_2 \\ u_2 + v_2 \\ \vdots \\ u_n + v_n \end{pmatrix} \\ &= \begin{pmatrix} v_1 + u_2 \\ v_2 + u_2 \\ \vdots \\ v_n + u_n \end{pmatrix} \\ &= v + u \end{aligned}$$

That is the proof for the general case, so I am done. ■

I want to point out a couple of things about this proof. First, it is general. I didn't try to prove by choosing specific vectors and checking. I can't prove by example. That said, often working through some examples is extremely valuable. When starting a proof, I'll often try some examples to get a sense of what is going on. But, in the end, the argument needs to be general — it

must cover all possible cases in the statement of the proposition.

Second, I never started with $u + v = v + u$. I started with the left side and worked my way to the right (though I could have just as easily started from the right and worked my way to the left). I should never assume what I want to prove, so I should never write $u + v = v + u$ until the end of the proof when it has been established. If I'm trying to figure out how to do a proof, I'll often work backwards: I'll start with the statement I want to get to, manipulate it and see how the piece works. But when I write up the proof, I'm never starting with my goal, only finishing with it.

This proof establishes the commutativity of vector addition. What about scalar multiplication? For real numbers, I know $ab = ba$. However, scalar multiplication is something quite different. If a is a scalar and u is a vector, then au produces another vector. In defining scalar multiplication, I wrote the scalar on the left and the vector on the right. Right now, the expression ua doesn't even mean anything. I could define it to mean the same thing so that I could do scalar multiplication on either side, but the mathematical notation convention is to always write the scalar on the left and the vector on the right. In this way, scalar multiplication isn't commutative because of the choice of notation.

1.6.3 Associativity

Another property of ordinary addition and multiplication is associativity. This property states that I can bracket a three-fold addition or multiplication either way, and the result is the same. For addition, this is

$$a + (b + c) = (a + b) + c.$$

For multiplication, this is similarly

$$a(bc) = (ab)c.$$

To be more explicit, consider the sum $6 + 4 + 9$. I can add this in two ways. If I start with the left sum, I get $10 + 9 = 19$. If I start with the right sum, I get $6 + 13 = 19$. Both ways produce 19; associativity says that this works for all sums of numbers. Associativity is a property that I very rarely think about. It is naturally built into my number sense, and when doing arithmetic, I use it implicitly and subconsciously all the time. However, it is a property I have to worry about for new operations. There are (strangely enough!) non-associative operations in mathematics.

Happily, both vector addition and scalar multiplication are associative.

Proposition 1.6.2 *If $u, v, w \in \mathbb{R}^n$ are vectors, then*

$$(u + v) + w = u + (v + w)$$

The proof is nearly identical to the proof of commutativity. I write the vectors with components and then use the associativity of ordinary addition in each component.

For scalar multiplication, since there are vectors and scalars involved, I have to be careful how I even state associativity.

Proposition 1.6.3 *If $u \in \mathbb{R}^n$ and $a, b, \in \mathbb{R}$ then*

$$a(bu) = (ab)u$$

This is the only statement of associativity that makes sense. I can't multiply vectors, but I can do successive multiplication by different scalars. The

property here says that I can do two scalar multiplications with the vector u, or I can first multiply the scalars normally then do one scalar multiplication. The proof is also like the proof for commutativity: I write everything in components and use the associativity of ordinary multiplication.

1.6.4 Distribution

For real numbers, multiplication distributes over addition.

$$a(b+c) = ab + ac$$

For vector addition and scalar multiplication, distribution still works. There are two versions of the distributive law here.

Proposition 1.6.4 *If $u, \in \mathbb{R}^n$ and $a, b \in \mathbb{R}$, then*

$$(a+b)u = au + bu.$$

Proposition 1.6.5 *If $u, v \in \mathbb{R}^n$ and $a \in \mathbb{R}$, then*

$$a(u+v) = au + av$$

In the first version, the addition is happening with the scalars. In the second, the addition is happening with the vectors. Since these are different kinds of addition, two different statements are required. The proofs are again like the commutativity proofs: write everything in components and use the distributive rules for ordinary numbers in each component.

1.7 Week 1 Activity

1.7.1 Higher-dimensional Space

Activity 1.7.1 In 2-dimensional space, an infinitely long line divides the space into two pieces: one cannot pass from one side of the line to the other without crossing the line. In 3-dimensional space, however, an infinitely long line does not do this; I can go around the line, and there isn't a division of space into two halves.

In 3-dimensional space, an infinitely extended plane also divides space into two pieces, and it is impossible to pass from one side to the other without going across the plane. In 4-dimensional space, does an infinitely extended plane still divide it into two pieces, or can I go around it, like I go around the line in 3-dimensional space?

Solution. This is a tricky thing to try to visualize (being that 4-dimensional space is not visualizable). However, I can argue by analogy. How do I get around the line in 3-dimensions? I move to the side, then move past the line, instead of running into it. I can do this because I have two degrees of freedom in directions other than the direction of the line.

The same thing works for a plane in 4-dimensional space. While in 3-dimensional space, I only have one dimension of movement apart from the plane. To get to the other side, moving in this one degree of freedom, I have to cross the plane. In 4-dimensional space, I have two other degrees of freedom. Like getting around the line in 3-dimensional space, I can move to the side, then move past the plane to reach "the other side." There really isn't another side, much like a line in 3-dimensional space doesn't really create two sides of space.

Activity 1.7.2 In 2-dimensional space, a square or rectangle is enough to trap or contain something that lives in the plane. In 3-dimensional space, a cube (or other box) is enough to trap or contain something. What do I need to contain something in 4-dimensional space? In 5-dimensional space?

Solution. To contain something in 4-dimensional space, I need a four dimensional box. Since I have four independent and free directions of movement, I have to close of movement in all four of these directions. This box will have a front/back, a top/bottom, a left/right, and then an additional pair of sides which block in the fourth direction of movement. In five dimensional space, I need a five-dimensional box: it will have a front/back, top/bottom, left/right, and two other independent boundary pairs.

Activity 1.7.3 In 2-dimensional space, the slices of a rectangle are lines. In 3-dimensional space, the slices of a box are rectangles. What are the slices of a 4-dimensional box? A 5-dimensional box?

Solution. By a "slice," it seems I am referring to a piece of the object of one smaller dimension. Therefore, the slices of this four-dimensional box must be three-dimensional boxes. In this way, I can think of a four-dimensional box as an infinite stack of three-dimensional boxes, all stacked in some new direction of movement. (The same way I can think of a cube as an infinite stack of squares moving in a direction perpendicular to the squares themselves). Likewise, the slices of a five-dimensional box are themselves four-dimensional boxes.

1.7.2 Vector Arithmetic

Activity 1.7.4 Consider these vectors.

$$u = \begin{pmatrix} 3 \\ 0 \end{pmatrix} \qquad v = \begin{pmatrix} -2 \\ -1 \end{pmatrix} \qquad w = \begin{pmatrix} -3 \\ 2 \end{pmatrix}$$

Calculate and draw the following vectors.

$$\begin{aligned} &-u \\ &-v \\ &-w \\ &u+v \\ &u+w \\ &v+w \\ &u-v \\ &v-u \\ &u-w \\ &w-u \\ &v-w \\ &w-v \end{aligned}$$

In addition, calcluate these lengths.

$$\begin{aligned} &|u| \\ &|v| \\ &|w| \\ &|u-v| \\ &|u-w| \end{aligned}$$

$$|v - w|$$

Solution. I'll do the arithmetic one by one. Then, at the end, all the vectors will be drawn in Figure 1.7.1.

$$\begin{aligned}
-u &= -\begin{pmatrix}3\\0\end{pmatrix} = \begin{pmatrix}-3\\0\end{pmatrix}\\
-v &= -\begin{pmatrix}-2\\-1\end{pmatrix} = \begin{pmatrix}2\\1\end{pmatrix}\\
-w &= -\begin{pmatrix}-3\\2\end{pmatrix} = \begin{pmatrix}3\\-2\end{pmatrix}\\
u+v &= \begin{pmatrix}3\\0\end{pmatrix} + \begin{pmatrix}-2\\-1\end{pmatrix} = \begin{pmatrix}1\\-1\end{pmatrix}\\
u+w &= \begin{pmatrix}3\\0\end{pmatrix} + \begin{pmatrix}-3\\2\end{pmatrix} = \begin{pmatrix}0\\2\end{pmatrix}\\
v+w &= \begin{pmatrix}-2\\-1\end{pmatrix} + \begin{pmatrix}-3\\2\end{pmatrix} = \begin{pmatrix}-5\\1\end{pmatrix}\\
u-v &= \begin{pmatrix}3\\0\end{pmatrix} - \begin{pmatrix}-2\\-1\end{pmatrix} = \begin{pmatrix}5\\1\end{pmatrix}\\
v-u &= \begin{pmatrix}-2\\-1\end{pmatrix} - \begin{pmatrix}3\\0\end{pmatrix} = \begin{pmatrix}-5\\-1\end{pmatrix}\\
u-w &= \begin{pmatrix}3\\0\end{pmatrix} - \begin{pmatrix}-3\\2\end{pmatrix} = \begin{pmatrix}6\\-2\end{pmatrix}\\
w-u &= \begin{pmatrix}-3\\2\end{pmatrix} - \begin{pmatrix}3\\0\end{pmatrix} = \begin{pmatrix}-6\\2\end{pmatrix}\\
v-w &= \begin{pmatrix}-2\\-1\end{pmatrix} - \begin{pmatrix}-3\\2\end{pmatrix} = \begin{pmatrix}1\\-3\end{pmatrix}\\
w-v &= \begin{pmatrix}-3\\2\end{pmatrix} - \begin{pmatrix}-2\\-1\end{pmatrix} = \begin{pmatrix}-1\\3\end{pmatrix}
\end{aligned}$$

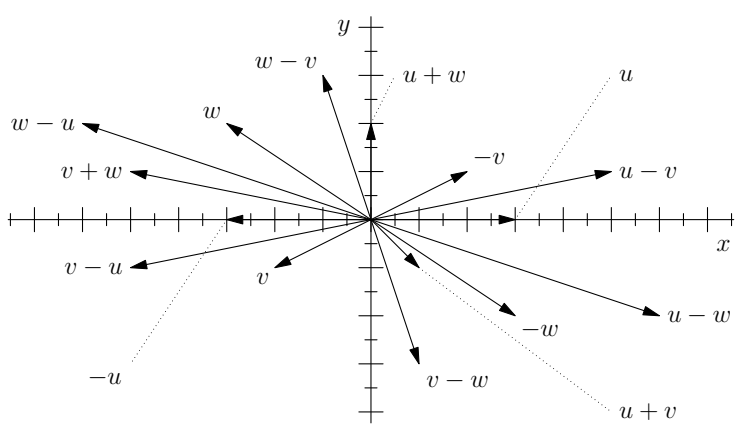

Figure 1.7.1 Vector Arithmetic

In addition, calcluate these lengths.

$$\begin{aligned}
|u| &= \sqrt{3^2 + 0^2} = \sqrt{9} = 3\\
|v| &= \sqrt{(-2)^2 + (-1)^2} = \sqrt{5}
\end{aligned}$$

$$|w| = \sqrt{(-3)^2 + 2^2} = \sqrt{13}$$
$$|u - v| = \sqrt{5^2 + 1^2} = \sqrt{26}$$
$$|u - w| = \sqrt{6^2 + (-2)^2} = \sqrt{40} = 2\sqrt{10}$$
$$|v - w| = \sqrt{1^2 + (-3)^2} = \sqrt{10}$$

Activity 1.7.5 Consider these vectors.

$$u = \begin{pmatrix} 6 \\ 0 \\ -1 \\ -1 \\ 1 \end{pmatrix} \qquad v = \begin{pmatrix} 3 \\ -3 \\ 1 \\ -3 \\ 1 \end{pmatrix} \qquad w = \begin{pmatrix} 0 \\ -1 \\ 0 \\ 3 \\ 5 \end{pmatrix}$$

Calculate the following vectors and length. (Since this is in $\mathbb{R}^5$, drawing these is no longer an option.)

$$-u$$
$$-v$$
$$-w$$
$$u + v$$
$$u + w$$
$$v + w$$
$$u - v$$
$$v - u$$
$$u - w$$
$$w - u$$
$$v - w$$
$$w - v$$
$$|u|$$
$$|v|$$
$$|w|$$
$$|u - v|$$
$$|u - w|$$
$$|v - w|$$

Solution. I'll do the calculation one by one.

$$-u = -\begin{pmatrix} 6 \\ 0 \\ -1 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} -6 \\ 0 \\ 1 \\ 1 \\ -1 \end{pmatrix}$$

$$-v = -\begin{pmatrix} 3 \\ -3 \\ 1 \\ -3 \\ 1 \end{pmatrix} \begin{pmatrix} -3 \\ 3 \\ -1 \\ 3 \\ -1 \end{pmatrix}$$

$$-w = -\begin{pmatrix} 0 \\ -1 \\ 0 \\ 3 \\ 5 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ -3 \\ -5 \end{pmatrix}$$

$$u + v = \begin{pmatrix} 6 \\ 0 \\ -1 \\ -1 \\ 1 \end{pmatrix} + \begin{pmatrix} 3 \\ -3 \\ 1 \\ -3 \\ 1 \end{pmatrix} = \begin{pmatrix} 9 \\ -3 \\ 0 \\ -4 \\ 2 \end{pmatrix}$$

$$u + w = \begin{pmatrix} 6 \\ 0 \\ -1 \\ -1 \\ 1 \end{pmatrix} + \begin{pmatrix} 0 \\ -1 \\ 0 \\ 3 \\ 5 \end{pmatrix} = \begin{pmatrix} 6 \\ -1 \\ -1 \\ 2 \\ 6 \end{pmatrix}$$

$$v + w = \begin{pmatrix} 3 \\ -3 \\ 1 \\ -3 \\ 1 \end{pmatrix} + \begin{pmatrix} 0 \\ -1 \\ 0 \\ 3 \\ 5 \end{pmatrix} = \begin{pmatrix} 3 \\ -4 \\ 1 \\ 0 \\ 6 \end{pmatrix}$$

$$u - v = \begin{pmatrix} 6 \\ 0 \\ -1 \\ -1 \\ 1 \end{pmatrix} - \begin{pmatrix} 3 \\ -3 \\ 1 \\ -3 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \\ -2 \\ 2 \\ 0 \end{pmatrix}$$

$$v - u = \begin{pmatrix} 3 \\ -3 \\ 1 \\ -3 \\ 1 \end{pmatrix} - \begin{pmatrix} 6 \\ 0 \\ -1 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} -3 \\ -3 \\ 2 \\ -2 \\ 0 \end{pmatrix}$$

$$u - w = \begin{pmatrix} 6 \\ 0 \\ -1 \\ -1 \\ 1 \end{pmatrix} - \begin{pmatrix} 0 \\ -1 \\ 0 \\ 3 \\ 5 \end{pmatrix} = \begin{pmatrix} 6 \\ 1 \\ -1 \\ -4 \\ -4 \end{pmatrix}$$

$$w - u = \begin{pmatrix} 0 \\ -1 \\ 0 \\ 3 \\ 5 \end{pmatrix} - \begin{pmatrix} 6 \\ 0 \\ -1 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} -6 \\ -1 \\ 1 \\ 4 \\ 4 \end{pmatrix}$$

$$v - w = \begin{pmatrix} 3 \\ -3 \\ 1 \\ -3 \\ 1 \end{pmatrix} - \begin{pmatrix} 0 \\ -1 \\ 0 \\ 3 \\ 5 \end{pmatrix} = \begin{pmatrix} 3 \\ -2 \\ 1 \\ -6 \\ -4 \end{pmatrix}$$

$$w - v = \begin{pmatrix} 0 \\ -1 \\ 0 \\ 3 \\ 5 \end{pmatrix} - \begin{pmatrix} 3 \\ -3 \\ 1 \\ -3 \\ 1 \end{pmatrix} = \begin{pmatrix} -3 \\ 2 \\ -1 \\ 6 \\ 4 \end{pmatrix}$$

$$|u| = \sqrt{6^2 + 0^2 + (-1)^2 + (-1)^2 + 1^2} = \sqrt{39}$$
$$|v| = \sqrt{3^2 + (-3)^2 + 1^2 + (-3)^2 + 1^2} = \sqrt{29}$$
$$|w| = \sqrt{0^2 + (-1)^2 + 0^2 + 3^2 + 5^2} = \sqrt{35}$$
$$|u - v| = \sqrt{3^2 + 3^2 + (-2)^2 + 2^2 + 0^2} = \sqrt{26}$$
$$|u - w| = \sqrt{6^2 + 1^2 + (-1)^2 + (-4)^2 + (-4)^2} = \sqrt{70}$$
$$|v - w| = \sqrt{3^2 + (-2)^2 + 1^2 + (-6)^2 + (-4)^2} = \sqrt{66}$$

1.7.3 Proof Questions

Activity 1.7.6 If $a \in \mathbb{R}$ and $u \in \mathbb{R}^3$, prove that $|au| = |a||u|$. (Use the definition of scalar multiplication and the definition of length of a vector. Start with the left side and try to turn it into the right or vice-versa. Do not reduce both equations to equivalent expressions -- that's not a valid proof technique. You may try some examples out if you wish, but your final solution should be general. Finally, note that the notation here is a bit subtle: $|au|$ and $|u|$ are the lengths of vectors, defined by Pythagorean calculations, but $|a|$ is just the absolute value of a real number.)

Solution. I will start with the left side and try to manipulate it into the right side. I'll write u in components and use the definition of the scalar multiplication to start.

$$|au| = \left| a \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} \right| = \left| \begin{pmatrix} au_1 \\ au_2 \\ au_3 \end{pmatrix} \right|$$

This is the definition of scalar multiplication: multiply the scalar by each component of the vector. Now I use the square root definition of the length of a vector.

$$\left| \begin{pmatrix} au_1 \\ au_2 \\ au_3 \end{pmatrix} \right| = \sqrt{(au_1)^2 + (au_2)^2 + (au_3)^2}$$

Now I need to do some algebra with the square root. I can distribute the exponent, using the exponent rule $(xy)^a = x^a y^a$.

$$\sqrt{(au_1)^2 + (au_2)^2 + (au_3)^2} = \sqrt{a^2u_1^2 + a^2u_2^2 + a_2u_3^2}$$

Now I can factor out the a^2. Once it is factored out, I can also remove it from the square root.

$$\sqrt{a^2u_1^2 + a^2u_2^2 + a_2u_3^2} = \sqrt{a^2(u_1^2 + u_2^2 + u_3^2)} = |a|\sqrt{u_1^2 + u_2^2 + u_3^2}$$

When I take a^2 out of the square root, since the square root also returns a positive number, I don't get a back. If a was negative to start, then I will have turned a from negative to positive. Therefore, I get the absolute value

of a back. Finally, I can recognize the remaining term as the definition of the length of a vector.

$$|a|\sqrt{u_1^2 + u_2^2 + u_3^2} = |a||u|$$

Now I have started with the left side and produced the right. This proves that $|au| = |a||u|$, as desired.

Activity 1.7.7 If u and v are vectors in $\mathbb{R}^3$, prove that $|u - v| = |v - u|$. Then adapt your argument to work for vectors in $\mathbb{R}^n$.

Solution. I will write u and v in components, take their difference and then apply the length operation. I will start with the left side and try to produce the right.

$$|u - v| = \left| \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} - \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} \right|$$

The difference is done componentwise.

$$\left| \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} - \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} \right| = \left| \begin{pmatrix} u_1 - v_1 \\ u_2 - v_2 \\ u_3 - v_3 \end{pmatrix} \right|$$

Then I use the square root definition of the length.

$$\left| \begin{pmatrix} u_1 - v_1 \\ u_2 - v_2 \\ u_3 - v_3 \end{pmatrix} \right| = \sqrt{(u_1 - v_1)^2 + (u_2 - v_2)^2 + (u_3 - v_3)^2}$$

Now, each of the three terms is square. Since they are square, any positive or negative will become positive since squares are always positive. Therefore, I could multiply by (-1) inside the square root to no effect. This will interchange the order of the difference in each term.

$$\sqrt{(u_1 - v_1)^2 + (u_2 - v_2)^2 + (u_3 - v_3)^2} = \sqrt{(v_1 - u_1)^2 + (v_2 - u_2)^2 + (v_3 - u_3)^2}$$

Then I repeat the steps from before in reverse. This is the definition of the length of a vector with the terms in the squares as components.

$$\sqrt{(v_1 - u_1)^2 + (v_2 - u_2)^2 + (v_3 - u_3)^2} = \left| \begin{pmatrix} v_1 - u_1 \\ v_2 - u_2 \\ v_3 - u_3 \end{pmatrix} \right|$$

Since each term is a difference, I can write this as the difference of two vectors.

$$\left| \begin{pmatrix} v_1 - u_1 \\ v_2 - u_2 \\ v_3 - u_3 \end{pmatrix} \right| = \left| \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} - \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} \right|$$

Finally, these are just the component of the vectors u and v.

$$= \left| \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} - \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} \right| = |v - u|$$

I have started from the left and produced the right in a string of equality, so I have proven the desired result.

If I wanted to adapt this to a more general proof for $\mathbb{R}^n$, the steps of the proof would be identical, but I would have vectors with n components instead of just three components. I would use ellipses to indicate the list of components. The proof would start like this.

$$|u - v| = \left| \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{pmatrix} - \begin{pmatrix} v_1 \\ v_2 \\ \\ \vdots \\ v_n \end{pmatrix} \right|$$

The difference is done componentwise.

$$\left| \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{pmatrix} - \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix} \right| = \left| \begin{pmatrix} u_1 - v_1 \\ u_2 - v_2 \\ \vdots \\ u_n - v_n \end{pmatrix} \right|$$

I won't repeat all the steps, but they would proceed exactly in the same sequence as before, resulting in $|v - u|$ eventually at the end.

Activity 1.7.8 Prove that the zero vector is the *only* vector in $\mathbb{R}^n$ of length zero. (To prove that it is the only vector with this property, assume that $u \in \mathbb{R}^n$ is a vector with length zero and prove that u must be the zero vector. This will show that the zero vector is the only vector with the desired property).

Solution. Let u be an arbitrary vector in $\mathbb{R}^n$ of length 0. I will start by writing u in components and use the definition of length. Note I need ellipses to show all the components of u, since I am working in an arbitrary dimension n.

$$|u| = \left| \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{pmatrix} \right|$$

Then I use the definition of length. By assumption, I set this length equal to zero.

$$|u| = \sqrt{(u_1)^2 + (u_2)^2 + \ldots + (u_n)^2} = 0$$

I can square both sides of the equation.

$$\left(\sqrt{(u_1)^2 + (u_2)^2 + \ldots + (u_n)^2} \right)^2 = 0^2$$

The square root and squaring cancel each other out.

$$(u_1)^2 + (u_2)^2 + \ldots + (u_n)^2 = 0$$

On the left, I have the sum of a bunch of non-negative numbers (since the square of any real number must be zero or postive). On the right I have zero. The only way that a bunch of non-negative numbers add up to zero is if they are all zero to begin with. Therefore, I see that for each index i, the component u_i must be zero. But that means that

$$\begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

This is precisely the definition of the zero vector: the vector where all components are zero. Therefore, u must be the zero vector. To conclude, I assumed that u had length zero and thus concluded that it must be the zero vector. This proved that the zero vector is the only vector with length zero.

1.7.4 Conceptual Review Questions

- How is it reasonable to talk about higher-dimensional geometric spaces?
- What is cartesian or euclidean space? What are coordinates?
- What is a vector?
- What is a scalar?
- How is vector arithmetic similar to and different from number arithmetic.
- What is the length of a vector?

Chapter 2

Week 2

2.1 Dot Product

2.1.1 Definition

Earlier, I said that I couldn't multiply two vectors together. That's mostly true: there is no general product of two vectors uv, which is still a vector, as least none that has any useful or reasonable geometric meaning. However, there are other kinds of 'multiplication' which combine two vectors. The operation defined in this lecture multiplies two vectors, but the result is a scalar.

Definition 2.1.1 The **dot product** or **inner product** or **scalar product** of two vectors u and v is given by the following formula.

$$u \cdot v = \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{pmatrix} \cdot \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix} = u_1 v_1 + u_2 v_2 + \ldots + u_n v_n$$

$\Diamond$

I can think of the dot product as a scalar measure of the similarity of direction between the two vectors. If the two vectors point in a similar direction, their dot product is large, but if they point in very different directions, their dot product is small. This is a bit strange, since there already is a measure, at least in $\mathbb{R}^2$, of this difference: the angle between two vectors. Thankfully, the two measures of difference agree, and the dot product can be expressed in terms of angles.

Definition 2.1.2 The **angle** θ between two non-zero vectors u and v in $\mathbb{R}^n$ is given by the follow equation.

$$\cos\theta = \frac{u \cdot v}{|u||v|}$$

$\Diamond$

This definition agrees with the angles in $\mathbb{R}^2$ and $\mathbb{R}^3$, which I can visualize. However, this serves as a new *definition* for angles between vectors in all $\mathbb{R}^n$ when $n \geq 4$. Since I can't visualize those spaces, I don't have a way of drawing angles and calculating them with conventional trigonometry. This definition allows me to extend angles in a completely algebraic way. Notes that this angle $\theta \in [0, \pi]$, since I can always choose the smallest possible angle between two

vectors. This means that the inverse cosine is unique for this calculation.

Definition 2.1.3 Two vectors u and v in $\mathbb{R}^n$ are called **orthogonal** or **perpendicular** or **normal** if $u \cdot v = 0$. $\diamondsuit$

2.1.2 Properties of the Dot Product

There are many pairs of orthogonal vectors. Thinking of the dot product as a multiplication, I have uncovered a serious difference between the dot product and the conventional multiplication of numbers. If $a, b \in \mathbb{R}$, then $ab = 0$ implies that one of a or b must be zero. For vectors, it can be true that $\begin{pmatrix} 1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \end{pmatrix} = 0$ even though neither factor in the product is the zero vector. Here is a definition to keep track of this new property.

Definition 2.1.4 Assume A is a set with addition and some kind of multiplication. Also assume that $0 \in A$. If $u \neq 0$ and $v \neq 0$ but $uv = 0$, then u and v are called **zero divisors**. $\diamondsuit$

An important property of ordinary numbers, as I just noted, is that there are no zero divisors. Other algebraic structures, such as vectors with the dot product, may have many zero divisors.

Now that I have defined a new operation, it is useful to see how it interacts with previously defined structures. I'll take up the proofs of these in Section 2.4 and in the activity for this week.

Proposition 2.1.5 *Let u, v, w be vectors in $\mathbb{R}^n$ and let a be a scalar in $\mathbb{R}$. The following four identities are true for all vectors and scalars.*

$$\begin{aligned} u \cdot v &= v \cdot u && \textit{Commutative} \\ u \cdot u &= |u|^2 \\ u \cdot (v + w) &= u \cdot v + u \cdot w && \textit{Distributive} \\ u \cdot (av) &= (au) \cdot v = a(u \cdot v) \end{aligned}$$

In $\mathbb{R}^2$, norms and dot products allow me to recreate some well-known geometric constructions. For example, now that I have defined lengths and angles, I can state the cosine law in terms of vectors. The visualization of the vector relationships of the cosine law is shown in Figure 2.1.6.

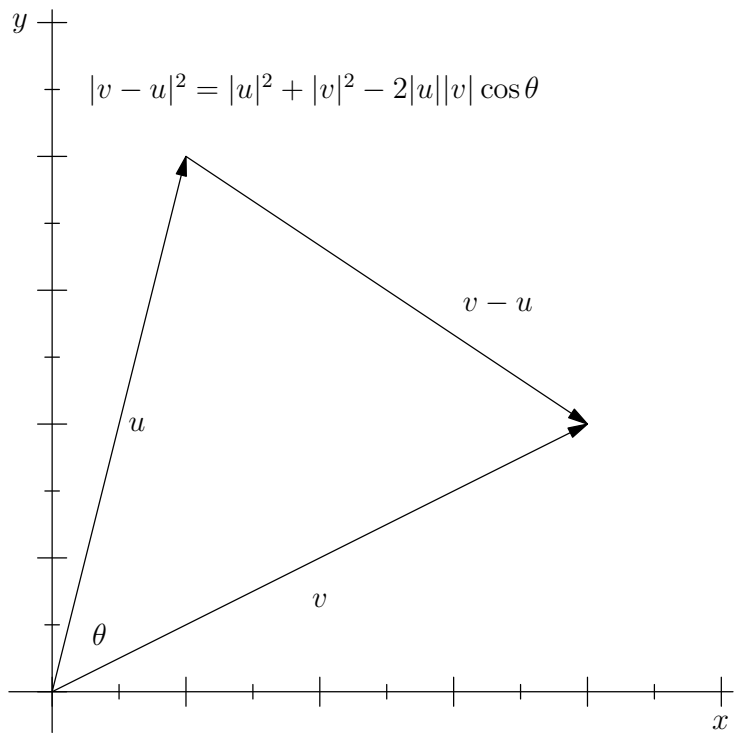

Figure 2.1.6 The Cosine Law

Proposition 2.1.7 The Cosine Law. *Let u and v be vectors in $\mathbb{R}^n$.*

$$|u-v|^2 = |u|^2 + |v|^2 - 2|u||v|\cos\theta = |u|^2 + |v|^2 - 2u \cdot v$$

Proof. To prove the cosine law using vectors, I'm going to interpret the left side, $|u-v|^2$, as $(u-v)\cdot(u-v)$, using the second identity in Proposition 2.1.5. Then I can go through some algebraic manipulation with the dot product. First, I'll distribute the dot product.

$$|u-v|^2 = (u-v)\cdot(u-v) = u \cdot u - v \cdot u - u \cdot v + v \cdot v$$

Then I can interchange the order of the second term, since the dot product is commutative.

$$|u-v|^2 = u \cdot u - u \cdot v - u \cdot v + v \cdot v$$

Then I can group the middle terms.

$$|u-v|^2 = u \cdot u - 2u \cdot v + v \cdot v$$

Then I can interpret the first and last terms as length squared.

$$|u-v|^2 = |u|^2 - 2u \cdot v + |v|^2$$

Then I can reorder the terms.

$$|u-v|^2 = |u|^2 + |v|^2 - 2u \cdot v$$

Finally, using the definition of angle, I can replace the dot product with $|u||v|\cos\theta$, where θ is the angle between the two vectors.

$$|u-v|^2 = |u|^2 + |v|^2 - 2|u||v|\cos\theta$$

This is the cosine law. ■

2.2 Cross Product

2.2.1 Definition of the Cross Product

The dot product is an operation that can be performed on any two vectors in $\mathbb{R}^n$ for any $n \geq 1$. There are no other conventional products that work in all dimensions. However, there is a special product that works in three dimensions.

Definition 2.2.1 Let $u = \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix}$ and $v = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$ be two vectors in $\mathbb{R}^3$. The **cross product** of u and v is written $u \times v$ and defined by the following formula.

$$u \times v = \begin{pmatrix} u_2v_3 - u_3v_2 \\ u_3v_1 - u_1v_3 \\ u_1v_2 - u_2v_1 \end{pmatrix}$$

$\Diamond$

The cross product differs from the dot product in several important ways. First, it produces a new vector in $\mathbb{R}^3$, not a scalar. For this reason, when working in $\mathbb{R}^3$, the dot product is often referred to as the **scalar product** and the cross product as the **vector product**. Second, the dot product measures, in some sense, the similarity of two vectors. The cross product measures, in some sense, the difference between two vectors. The cross product has a greater magnitude if the vectors are closer to being perpendicular. If θ is the angle between u and v, the dot product was expressed in terms of $\cos\theta$. This measures similarity, since $\cos 0 = 1$. There is a similar identity for the cross product.

$$|u \times v| = |u||v| \sin\theta$$

This identity tells me that the cross product measures difference in direction, since $\sin 0 = 0$. In particular, this tells me that $|u \times u| = 0$, implying that $u \times u = 0$ (the zero vector is the only vector which has zero length). This is another new and strange property: in this particular multiplication, everything squares to zero. The cross product is obviously very different from the multiplication of scalars, where $a^2 = 0$ cannot happen unless $a = 0$.

Also consider the relationship between u and $u \times v$ as calculated through the dot product.

$$\begin{aligned} u \cdot (u \times v) &= \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} \cdot \begin{pmatrix} u_2v_3 - u_3v_2 \\ u_3v_1 - u_1v_3 \\ u_1v_2 - u_2v_1 \end{pmatrix} \\ &= u_1u_2v_3 - u_1u_3v_2 + u_2u_3v_1 - u_2u_1v_3 + u_3u_1v_2 - u_3u_2v_1 = 0 \end{aligned}$$

A similar calculation shows that $v \cdot (u \times v) = 0$. Since a dot product of two vectors is zero if and only if the vectors are perpendicular, the vector $v \times u$ is perpendicular to both u and v. This turns out to be a very useful property of the cross product.

Finally, a calculation from the definition shows that $u \times v = -(v \times u)$. So far, multiplication of scalars and the dot product of vectors have not depended on order. The cross product is one of many products in mathematics which depends on order. If I change the order of the cross product, I introduce a negative sign.

Definition 2.2.2 Products that do not depend on the order of the factors, such as multiplication of scalars and the dot product of vectors, are called **commutative products**. Products where changing the order of the factors introduces a negative sign are called **anti-commutative products**. The cross product is an anti-commutative product. Other products that have neither of these properties are called **non-commutative products**. ◊

2.2.2 Angular Motion

An important application of the cross product is found in describing rotational motion. Linear mechanics describes the motion of an object through space, but rotational mechanics describes the rotation of an object independent of its movement through space. A force on an object can cause both kinds of movement, obviously. The following table summarizes the parallel questions of linear motion and rotational motion in $\mathbb{R}^3$.

Linear Motion	Rotational Motion
Moving in a straight line	Continual spinning
Direction of motion	Axis of spin
Force	Torque
Momentum	Angular Momentum
Mass (resistance to motion)	Moment of Inertia (resistance to spin)
Velocity	Frequency (Angular Velocity)
Acceleration	Angular Acceleration

How should I describe torque? If there is a linear force applied to an object which can only rotate around an axis, and if the linear force is applied at a distance r from the axis, I can think of the force F and the distance r as vectors. The torque is then $\tau = r \times F$. Notice that $|\tau| = |r||F| \sin\theta$, indicating that linear force perpendicular to the radius gives the greatest angular acceleration. That makes sense. If F and r share a direction, then I am pushing directly along the axis, and no rotation can occur.

The use of cross products in rotational dynamics is ubiquitous. In fluid dynamics, local rotational motion of the fluid result in turbulence, vortices and similar effects. Tornadoes and hurricanes are particularly extreme examples of vortices. All the descriptions of the force and motion of these vortices involve cross products in the vectors describing the fluid.

2.3 Sets and Structures

2.3.1 Sets

Before I talk about properties of dot/cross products and proof, I want to step back to take a much more abstract look at what I am accomplishing. In academic mathematics, algebra can be defined as the study of sets with (algebraic) structures. I want to use this section to explain that idea.

I should start with sets, very briefly. Set theory is a whole branch of mathematics itself, but I'm not going into the details. It is enough for to know that a set is a collection of things. In many places in mathematics, sets are collections of numbers. In other courses, you may have seen some of these: the set of natural numbers ($\mathbb{N}$); the set of integers ($\mathbb{Z}$); the set of rational numbers ($\mathbb{Q}$); the set of real numbers ($\mathbb{R}$); or intervals on the number line (such as $(1, 4]$). However, sets can contain anything I want. In this course, I'll have sets of vectors and sets of matrices.

2.3.2 Sets with Structure

So what do I mean when I say 'sets with structure'? What is a structure? A structure is a mathematical construction on a set — something that I can do, mathematically, with the elements of the set. This is best illustrated by some very familiar examples.

- Addition is a structure on many number sets.
- Subtraction is a structure on many number sets. Subtraction is not a structure on $\mathbb{N}$, however. 4 and 7 are natural numbers, but $4 - 7 = -3$ is no longer a natural number.
- Multiplication is a structure on many number sets.
- Division by non-zero elements is a structure on many number sets. Note that zero must be excluded, since for sets that have zero, division by zero is undefined. Division is not a structure on $\mathbb{N}$; like with subtraction, even though 4 and 7 are natural numbers, $\frac{4}{7}$ is not a natural number.
- The four previous examples are all types of *binary relations*. They are structures that take two elements from the set and combine them together to produce something new from the set.
- The dot product is a new binary operation on $\mathbb{R}^n$, and a strange one, since its output is a different kind of object. It takes two vectors and produces a scalar.
- The cross product is a binary operation on vectors in $\mathbb{R}^3$.
- There are also structures that are not binary operations. Less than and greater than are structures on many number sets.
- Absolute value is also a structure on many number sets.

This list is just the start of many, many kinds of algebraic structures that I can consider on sets. When I think about linear algebra abstractly, I am building new structures on sets of vectors and analyzing their properties. The dot product and the cross product are the new structures I've introduced in Week 2; there will be more to come.

I'm also interested in the properties of structures. In Section 1.6 I introduced the properties of commutativity, associativity and distribution. In Section 2.2 I talked about how the cross product is anti-commutative. Other properties are possible as well. When I have structure, I will try to investigate their properties.

In the activity for this week, I'll ask you to think about the abstract idea of sets with structure.

2.4 Proofs - Dot and Cross Products

2.4.1 Prooving the Dot Product Properties

In Proposition 2.1.5, I listed four properties of the dot product. In this section, I'll demonstrate how to prove some of these properties.

Proposition 2.4.1 *The dot product is commutative. That is, if $u, v \in \mathbb{R}^n$,*

$$u \cdot v = v \cdot u.$$

Proof. I first write the vector in coordinates. This is the full generality of $\mathbb{R}^n$, so I write vectors with n coordinates using ellipses

$$u = \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ \vdots \\ u_n \end{pmatrix} \qquad v = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \\ \vdots \\ v_n \end{pmatrix}$$

Then I'll start with the left side and see if I can make a chain of equality that results in the right side. First I use the definition of the dot product.

$$u \cdot v = u_1 v_1 + u_2 v_2 + u_3 v_3 + \ldots + u_n v_n$$

Then each of the products $u_i v_i$ are products of ordinary real numbers. Multiplicaton of real numbers is commutative, so I can interchange the order of each of these products.

$$u \cdot v = v_1 u_1 + v_2 u_2 + v_3 u_3 + \ldots + v_n u_n$$

Then I use the definition of the dot product again. The right side of the previous equation is, by definition, the dot product $v \cdot u$.

$$u \cdot v = v \cdot u$$

That is the desired equality, so I have finished the proof. Notice that the desired equation only shows up at the end of the proof. I never start with what I want to prove. ■

Proposition 2.4.2 *If $u \in \mathbb{R}^n$, then*

$$u \cdot u = |u|^2$$

Proof. I write u in its coordinates.

$$u = \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ \ldots \\ u_n \end{pmatrix}$$

Then I'll start with the left side and use the definition of the dot product.

$$u \cdot u = u_1 u_1 + u_2 u_2 + u_3 u_3 + \ldots u_n u_n$$

I can write these products on the left as squares.

$$u \cdot u = u_1^2 + u_2^2 + u_3^2 + \ldots + u_n^2$$

Since the right side is a sum of squares, it is a positive number (or possibly zero if all the u_i are zero). As a positive number or zero, I can square root it and then square it without effect: $\alpha = (\sqrt{\alpha})^2$ for any non-negative α.

$$u \cdot u = \left(\sqrt{u_1^2 + u_2^2 + u_3^2 + \ldots + u_n^2} \right)^2$$

Then the expression inside the brackets is, by definition, the length of the vector u.

$$u \cdot u = (|u|)^2 = |u|^2$$

This is the desired equation, so the proof is completed. ■

Proposition 2.4.3 *The dot product distributes over vector addition. That is, if $u, v, w \in \mathbb{R}^n$,*

$$u \cdot (v + w) = u \cdot v + u \cdot w$$

Proof. I'll write the three vectors in components.

$$u = \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ \dots \\ u_n \end{pmatrix} \qquad v = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \\ \dots \\ v_n \end{pmatrix} \qquad w = \begin{pmatrix} w_1 \\ w_2 \\ w_3 \\ \dots \\ w_n \end{pmatrix}$$

Then I'll start with the left side. I'll use the definition of vector addition and the dot product.

$$\begin{aligned} u \cdot (v + w) & \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ \dots \\ u_n \end{pmatrix} \cdot \begin{pmatrix} v_1 + w_1 \\ v_2 + w_2 \\ v_3 + w_3 \\ \dots \\ v_n + w_n \end{pmatrix} \\ & = u_1(v_1 + w_1) + u_w(v_2 + w_2) + u_3(v_3 + w_3) + \ldots + u_n(v_n + w_n) \end{aligned}$$

Each of these products can be distributed, since this is all arithmetic of ordinary real numbers.

$$\begin{aligned} u \cdot (v + w) = (u_1 v_1 + u_1 w_1) + (u_2 v_2 + u_2 w_2) \\ + (u_3 v_3 + u_3 w_3) + \ldots + (u_n v_n + u_n w_n) \end{aligned}$$

The right side is the sum of many products. I can rearrange this sum as I please, so I'll put all the terms with v_i first, then all those with w_i afterwards.

$$\begin{aligned} u \cdot (v + w) = (u_1 v_1 + u_2 v_2 + u_3 v_3 + \ldots + u_n v_n) \\ + (u_1 w_1 + u_2 w_2 + u_3 w_3 + \ldots + u_n w_n) \end{aligned}$$

Then I can use the definition of the dot product again. On the right side, the term in the first bracket is, by definition, $u \cdot v$. The term in the second bracket is, again by definition, $u \cdot w$.

$$u \cdot (v + w) = u \cdot v + u \cdot w$$

This is the desired equation, so the proof is finished. ■

Proposition 2.4.4 *If $u, v \in \mathbb{R}^n$ and $a \in \mathbb{R}$, then*

$$u \cdot (av) = (au) \cdot v = a(u \cdot v)$$

Proof. I'll write the two vectors in components.

$$u = \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ \dots \\ u_n \end{pmatrix} \qquad v = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \\ \dots \\ v_n \end{pmatrix}$$

Then I want to consider this sum:

$$au_1 v_1 + au_2 v_2 + au_3 v_3 + \ldots au_n v_n.$$

I'm going to organize this sum in three different ways, which will correspond to the three desired versions in the equation in the proposition. Since I will only be arranging brackets for addition and multiplication of numbers in acceptable ways, all three of these organizations are just calculating the same sum, so I can conclude that all three are equal.
First, I'll factor the a out of the entire sum.

$$a(u_1v_2 + u_2v_2 + u_3v_3 + \dots u_nv_n)$$

What is in brackets is the definition of the dot product, so this is $a(u \cdot v)$. Next I'll bracket the a with the u_i components.

$$(au_1)v_2 + (au_2)v_2 + (au_3)v_3 + \dots (au_n)v_n$$

This is, by definition, also a dot product.

$$\begin{pmatrix} au_1 \\ au_2 \\ au_3 \\ \vdots \\ au_n \end{pmatrix} \cdot \begin{pmatrix} v_1 \\ v_2 \\ v_3 \\ \vdots \\ v_n \end{pmatrix}$$

The first vector in this dot product is, by definition, the scalar multiple of u by a. Therefore, this organization of the sum gives $(au) \cdot v$. Finally, I'll change the order of the multiplications and group a with the v_i components.

$$u_1(av_2) + u_2(av_2) + u_3(av_3) + \dots u_n(av_n)$$

This is, by definition, also a dot product.

$$\begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ \vdots \\ u_n \end{pmatrix} \cdot \begin{pmatrix} av_1 \\ av_2 \\ av_3 \\ \vdots \\ av_n \end{pmatrix}$$

The second vector in this dot product is, by definition, the scalar multiple of v by a. Therefore, this organization of the sum gives $u \cdot (av)$. Since all three organizations are valid ways of calculating the sum, I conclude that

$$u \cdot (av) = (au) \cdot v = a(u \cdot v),$$

which concludes the proof. ■

2.5 Week 2 Activity

2.5.1 Dot Products

Thoughout this activity, (WA) means that I used Wolfram Alpha to calculate an approximate value.

Activity 2.5.1 Calculate the dot product.

$$\begin{pmatrix} 4 \\ -2 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ -1 \end{pmatrix}$$

Solution. The dot product is calculated by multiplying matching components and adding these products together.

$$\begin{pmatrix} 4 \\ -2 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ -1 \end{pmatrix} = (4)(0) + (-2)(-1) = 2$$

Activity 2.5.2 Calculate the dot product.

$$\begin{pmatrix} 2 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ -7 \end{pmatrix}$$

Solution. The dot product is calculated by multiplying matching components and adding these products together.

$$\begin{pmatrix} 2 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ -7 \end{pmatrix} = (2)(-2) + (3)(-7) = -25$$

Activity 2.5.3 Calculate the dot product.

$$\begin{pmatrix} 0 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 19 \\ -29 \end{pmatrix}$$

Solution. The dot product is calculated by multiplying matching components and adding these products together.

$$\begin{pmatrix} 0 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 19 \\ -29 \end{pmatrix} = (0)(19) + (0)(-29) = 0$$

Activity 2.5.4 Calculate the dot product.

$$\begin{pmatrix} \frac{1}{4} \\ \frac{-4}{7} \end{pmatrix} \cdot \begin{pmatrix} \frac{-2}{7} \\ \frac{-4}{5} \end{pmatrix}$$

Solution. The dot product is calculated by multiplying matching components and adding these products together.

$$\begin{pmatrix} \frac{1}{4} \\ \frac{-4}{7} \end{pmatrix} \cdot \begin{pmatrix} \frac{-2}{7} \\ \frac{-4}{5} \end{pmatrix} = \frac{1}{4}\frac{-2}{7} + \frac{-4}{7}\frac{-4}{5} = \frac{-1}{14} + \frac{16}{35} = \frac{-5}{70} + \frac{32}{70} = \frac{27}{70}$$

Activity 2.5.5 Calculate the dot product.

$$\begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ -1 \\ 9 \end{pmatrix}$$

Solution. The dot product is calculated by multiplying matching components and adding these products together.

$$\begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ -1 \\ 9 \end{pmatrix} = (-2)(0) + (1)(-1) + (1)(9) = 8$$

Activity 2.5.6 Calculate the dot product.

$$\begin{pmatrix} 4 \\ -2 \\ -6 \end{pmatrix} \cdot \begin{pmatrix} -6 \\ -7 \\ -2 \end{pmatrix}$$

Solution. The dot product is calculated by multiplying matching components and adding these products together.

$$\begin{pmatrix} 4 \\ -2 \\ -6 \end{pmatrix} \cdot \begin{pmatrix} -6 \\ -7 \\ -2 \end{pmatrix} = (4)(-6) + (-2)(-7) + (-6)(-2) = 2$$

Activity 2.5.7 Calculate the dot product.

$$\begin{pmatrix} \sqrt{2} \\ 4 \\ \sqrt[3]{2} \end{pmatrix} \cdot \begin{pmatrix} \sqrt{7} \\ \pi^2 \\ 0 \end{pmatrix}$$

Solution. The dot product is calculated by multiplying matching components and adding these products together.

$$\begin{pmatrix} \sqrt{2} \\ 4 \\ \sqrt[3]{2} \end{pmatrix} \cdot \begin{pmatrix} \sqrt{7} \\ \pi^2 \\ 0 \end{pmatrix} = (\sqrt{2})(\sqrt{7}) + (4)(\pi^2) + (\sqrt[3]{2})(0) = \sqrt{14} + 4\pi^2$$

There isn't really a simpler way to write this number. This form is just fine. If I wanted an approximate value, I could ask a computer (WA): $\sqrt{14} + 4\pi^2 \doteq 217.78$. (When you write approximate values, please use this equals with a dot to indicate that this is approximately equal to, not exactly equal to.)

Activity 2.5.8 Calculate the dot product.

$$\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} \sqrt{\pi} \\ \frac{1}{e^2+6} \\ \sqrt{1+\sqrt{8}} \end{pmatrix}$$

Solution. The dot product is calculated by multiplying matching components and adding these products together.

$$\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} \sqrt{\pi} \ \frac{1}{e^2+6} \\ \sqrt{1+\sqrt{8}} \end{pmatrix} = 0$$

No matter how complicated the other vector is, anything multiplied by zero is still zero.

Activity 2.5.9 Calculate the dot product.

$$\begin{pmatrix} -2 \\ -3 \\ 1 \\ 4 \\ -2 \end{pmatrix} \cdot \begin{pmatrix} -6 \\ -5 \\ 2 \\ -2 \\ 3 \end{pmatrix}$$

Solution. The dot product is calculated by multiplying matching components and adding these products together.

$$\begin{pmatrix} -2 \\ -3 \\ 1 \\ 4 \\ -2 \end{pmatrix} \cdot \begin{pmatrix} -6 \\ -5 \\ 2 \\ -2 \\ 3 \end{pmatrix} = (-2)(-6) + (-3)(-5) + (1)(2) + (4)(-2) + (-2)(3)$$

$$= 12 + 15 + 2 - 8 - 6 = 15$$

Activity 2.5.10 Calculate the dot product.

$$\begin{pmatrix} 6 \\ -7 \\ 1 \\ 0 \\ -2 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 4 \\ -3 \\ -3 \\ -4 \end{pmatrix}$$

Solution. The dot product is calculated by multiplying matching components and adding these products together.

$$\begin{pmatrix} 6 \\ -7 \\ 1 \\ 0 \\ -2 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 4 \\ -3 \\ -3 \\ -4 \end{pmatrix} = (6)(-2) + (-7)(4) + (1)(-3) + (0)(-3) + (-2)(-4)$$

$$= -12 - 28 - 3 + 0 + 8 = -35$$

Activity 2.5.11 Calculate the dot product.

$$\begin{pmatrix} 5 \\ -3 \\ -7 \\ 10 \\ 1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ -2 \\ 0 \\ -4 \\ -4 \\ 0 \end{pmatrix}$$

Solution. The dot product is calculated by multiplying matching components and adding these products together.

$$\begin{pmatrix} 5 \\ -3 \\ -7 \\ 10 \\ 1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ -2 \\ 0 \\ -4 \\ -4 \\ 0 \end{pmatrix} = (5)(0) + (-3)(-2) + (-7)(0) + (10)(-4) + (1)(-4) + (0)(0)$$

$$= 0 + 6 + 0 - 40 - 4 + 0 = -38$$

2.5.2 Angles Between Vectors

Activity 2.5.12 Find the angle between these two vectors.

$$u = \begin{pmatrix} -2 \\ -5 \end{pmatrix} \text{ and } v = \begin{pmatrix} 3 \\ -4 \end{pmatrix}$$

Solution. The cosine of the angle between two vectors is the dot product divided by the product of the lengths. I'll calculate the three pieces of this and put them into the expression for the cosine of the angle.

$$\cos\theta = \frac{u \cdot v}{|u||v|}$$

$$|u| = \sqrt{(-2)^2 + (-5)^2} = \sqrt{29}$$

$$\begin{aligned}
|v| &= \sqrt{3^2 + (-4)^2} = \sqrt{25} = 5 \\
u \cdot v &= \begin{pmatrix} -2 \\ -5 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -4 \end{pmatrix} = (-2)(3) + (-5)(-4) = 14 \\
\cos\theta &= \frac{14}{5\sqrt{29}}
\end{aligned}$$

Then I can ask a computer (WA) to give me the inverse cosine to get an approximation for the angle. (All angles are given in radians. The inverse cosine function will give an angle between 0 and π, which is good since the angle between two vector should be bounded by π radians.)

$$\theta = \arccos \frac{14}{5\sqrt{29}} \doteq 1.024$$

Activity 2.5.13 Find the angle between these two vectors.

$$u = \begin{pmatrix} 1 \\ 3 \end{pmatrix} \text{ and } v = \begin{pmatrix} 1 \\ 7 \end{pmatrix}$$

Solution. The cosine of the angle between two vectors is the dot product divided by the product of the lengths. I'll calculate the three pieces of this and put them into the expression for the cosine of the angle.

$$\begin{aligned}
\cos\theta &= \frac{u \cdot v}{|u||v|} \\
|u| &= \sqrt{1^2 + 3^2} = \sqrt{10} \\
|v| &= \sqrt{1^2 + 7^2} = \sqrt{50} \\
u \cdot v &= \begin{pmatrix} 1 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 7 \end{pmatrix} = (1)(1) + (3)(7) = 22 \\
\cos\theta &= \frac{22}{\sqrt{500}} = \frac{22}{10\sqrt{5}} = \frac{11}{5\sqrt{5}}
\end{aligned}$$

Then I can ask a computer (WA) to give me the inverse cosine to get an approximation for the angle.

$$\theta = \arccos \frac{11}{5\sqrt{5}} \doteq 0.180$$

Activity 2.5.14 Find the angle between these two vectors.

$$u = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \text{ and } v = \begin{pmatrix} 1 \\ -2 \\ -2 \end{pmatrix}$$

Solution. The cosine of the angle between two vectors is the dot product divided by the product of the lengths. I'll calculate the three pieces of this and put them into the expression for the cosine of the angle.

$$\begin{aligned}
\cos\theta &= \frac{u \cdot v}{|u||v|} \\
|u| &= \sqrt{0^2 + 1^2 + 2^2} = \sqrt{5} \\
|v| &= \sqrt{1^2 + (-2)^2 + (-2)^2} = \sqrt{9} = 3
\end{aligned}$$

$$u \cdot v = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -2 \\ -2 \end{pmatrix} = (0)(1) + (1)(-2) + (2)(-2) = -6$$

$$\cos\theta = \frac{-6}{3\sqrt{5}}$$

Then I can ask a computer (WA) to give me the inverse cosine to get an approximation for the angle.

$$\theta = \arccos \frac{-6}{3\sqrt{5}} \doteq 2.678$$

Activity 2.5.15 Find the angle between these two vectors.

$$u = \begin{pmatrix} -4 \\ 1 \\ 1 \end{pmatrix} \text{ and } v = \begin{pmatrix} 2 \\ 4 \\ 4 \end{pmatrix}$$

Solution. The cosine of the angle between two vectors is the dot product divided by the product of the lengths. I'll calculate the three pieces of this and put them into the expression for the cosine of the angle.

$$\begin{aligned} \cos\theta &= \frac{u \cdot v}{|u||v|} \\ |u| &= \sqrt{(-4)^2 + 1^2 + 1^2} = \sqrt{18} \\ |v| &= \sqrt{2^2 + 4^2 + 4^2} = \sqrt{36} = 6 \\ u \cdot v &= \begin{pmatrix} -4 \\ 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 4 \\ 4 \end{pmatrix} = (-4)(2) + (1)(4) + (1)(4) = 0 \\ \cos\theta &= 0 \end{aligned}$$

Cosine is zero when angles are perpendicular, so the angle bewteen these two vectors must be $\frac{\pi}{2}$ radians.

2.5.3 Cross Product

Activity 2.5.16 Calculate this cross product.

$$\begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix} \times \begin{pmatrix} -4 \\ 2 \\ 0 \end{pmatrix}$$

Solution. The cross product is calculated by a complicated pattern of multiplication and addition with the components as presented in the notes and the videos.

$$\begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix} \times \begin{pmatrix} -4 \\ 2 \\ 0 \end{pmatrix} = \begin{pmatrix} (0)(0) - (2)(-2) \\ (-2)(-4) - (0)(1) \\ (1)(2) - (0)(-4) \end{pmatrix} = \begin{pmatrix} 4 \\ 8 \\ 2 \end{pmatrix}$$

Activity 2.5.17 Calculate this cross product.

$$\begin{pmatrix} 2 \\ -6 \\ -3 \end{pmatrix} \times \begin{pmatrix} -1 \\ -2 \\ -1 \end{pmatrix}$$

Solution. The cross product is calculated by a complicated pattern of multiplication and addition with the components as presented in the notes and the videos.

$$\begin{pmatrix} 2 \\ -6 \\ -3 \end{pmatrix} \times \begin{pmatrix} -1 \\ -2 \\ -1 \end{pmatrix} = \begin{pmatrix} (-6)(-1) - (-3)(-2) \\ (-3)(-1) - (2)(-1) \\ (2)(-2) - (-6)(-1) \end{pmatrix} = \begin{pmatrix} 0 \\ 5 \\ -10 \end{pmatrix}$$

Activity 2.5.18 Calculate this cross product.

$$\begin{pmatrix} 0 \\ 3 \\ 8 \end{pmatrix} \times \begin{pmatrix} 1 \\ 6 \\ -5 \end{pmatrix}$$

Solution. The cross product is calculated by a complicated pattern of multiplication and addition with the components as presented in the notes and the videos.

$$\begin{pmatrix} 0 \\ 3 \\ 8 \end{pmatrix} \times \begin{pmatrix} 1 \\ 6 \\ -5 \end{pmatrix} = \begin{pmatrix} (3)(-5) - (8)(6) \\ (8)(1) - (0)(-5) \\ (0)(6) - (3)(1) \end{pmatrix} = \begin{pmatrix} -63 \\ 8 \\ -3 \end{pmatrix}$$

Activity 2.5.19 Calculate this cross product.

$$\begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} \times \begin{pmatrix} -4 \\ -2 \\ 0 \end{pmatrix}$$

Solution. The cross product is calculated by a complicated pattern of multiplication and addition with the components as presented in the notes and the videos.

$$\begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} \times \begin{pmatrix} -4 \\ -2 \\ 0 \end{pmatrix} = \begin{pmatrix} (1)(0) - (0)(-2) \\ (0)(-4) - (2)(0) \\ (2)(-2) - (1)(-4) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

2.5.4 Sets with Structure

Activity 2.5.20 In Section 2.3, I introduced the abstract idea of sets with structure. In the next two activities, I'm asking you to think through that abstract idea. In the first activity, I've given you an empty table with some number sets and some possible structures. Complete this table by writing "Yes" or "No" in each space. Write "Yes" if the set in question includes the structure (the structure must remain in the set); write "No" if the set does not include the structure.

The set $\mathbb{Z}/12\mathbb{Z}$ is the integers modulo 12. If this set is unfamiliar, you may treat it as the numbers on the clock. So, for example, if I want to add $7 + 9$, I ask what time is 9 hours after 7, which is 4. Therefore $7 + 9 = 4$ in an example of proper arithmetic for this set. In general, whenever I would get a number larger than 11, I subtract multiples of 12 to reduce it down to something between 0 and 11. For another example, $5 \times 4 = 8$ is also proper arithmetic for this set, since $20 - 12 = 8$. Note that 12 is the same thing as 0 in this set; even though on a clock, I would often write 12, in this set, I usually just write 0. So, for example, I would write $5 + 7 = 0$ in this set. Finally, this is just integers modulo 12, so I only consider the whole numbers of the clock — I don't have minutes or any other subdivisions, only hours.

The term 'zero divisor' is probably also unfamiliar. A pair of zero divisors

are numbers a and b, each of them non-zero, but whose product $ab = 0$. This is a strange definition, since it not a property of usual numbers: for integers, rational numbers, real number, two non-zero numbers cannot multiply together to produce zero. However, it does happen in other sets. In the activity, think about how you can find zero divisors in $\mathbb{Z}/12\mathbb{Z}$.

Table 2.5.1 Table for Sets with Structure

Structure	$\mathbb{N}$	$\mathbb{Z}$	$\mathbb{Q}$	$\mathbb{R}$	$\mathbb{Z}/12\mathbb{Z}$
Addition					
Subtraction					
Multiplication					
Division					
Distribution					
Absolute Value					
Order					
Zero Divisors					

Solution. Here is the completed table. I'll comment on some specific entries below.

Table 2.5.2 Table for Sets with Structure

Structure	$\mathbb{N}$	$\mathbb{Z}$	$\mathbb{Q}$	$\mathbb{R}$	$\mathbb{Z}/12\mathbb{Z}$
Addition	Yes	Yes	Yes	Yes	Yes
Subtraction	No	Yes	Yes	Yes	Yes
Multiplication	Yes	Yes	Yes	Yes	Yes
Division by Non-Zero Elements	No	No	Yes	Yes	No
Multiplication Distributes Over Addition	Yes	Yes	Yes	Yes	Yes
Absolute Value	Yes, Trivially	Yes	Yes	Yes	Maybe
Order (Greater Than, Less Than)	Yes	Yes	Yes	Yes	No
Zero Divisors	No	No	No	No	Yes

Hopefully, most of these answers are straightforward. I'll comment on a few, though, which might be less obvious.

- The natural numbers can still be said to have absolute value, but the structure doesn't do anything. The absolute value of a positive number is itself, so the structure is very boring: all numbers are unchanged. This is why I wrote "trivially", to indicate that the structure doesn't do anything. You could argue that this answer should be no; I'd be willing to accept that a trivial absolute value doesn't really count as a structure.

- Division for the integers modulo 12 is tricky. I might think that since I can extend addition, subtraction and multiplication just fine for modulo 12, I could extend division as well. The problem is zero divisors: if I allow division by zero divisors, the arithmetic breaks. This is not immediatly obvious, so I didn't really expect anyone to know this unless you've previously studied modular arithmetic.

- Speaking of zero divisors, take the numbers 3 and 4 in the integers modulo 12. Their product is 12, which is the same as 0 in that system. So, I can reasonably say that $3 \times 4 = 0$ in the integers modulo 12. Since 3 and 4 are not also equivalent to zero themselves, they are zero divisors.

- Whether or not there is an order relation on the integers modulo 12 is also tricky. I can impose an order, saying that $0 < 1 < 2 < 3 < \ldots < 11$.

However, since the numbers loop around, if I keep going from 12 back to 1, this seems intuitively problematic. So, even though I could impose such an order, I think it makes more sense not to. I conclude that the integers modulo 12 is not ordered. If I think about time on a clock, I think I can justify this decision. If it is currently 10 o'clock, then there is a time before that was 5 o'clock as well as a time in the future, which will also be 5 o'clock. Therefore, putting an order seems unreasonable.

Activity 2.5.21 The structures I've included so far have mostly been binary operations (the conventional four of arithmetic) and some other familiar things like order and absolute value. However, structures can be as complicated as I like. Here are some more involved structures. For each structure, give an example of a number set that has the structure. (Some of these are a bit tricky: you are allowed to restrict the conventional number sets to make this work. For example, the set of all positive rational numbers is a number set, as is the set of all natural numbers less than 40.)

1. Decomposition of numbers into prime factors (except for 0 and 1).
2. Reciprocals of non-zero numbers.
3. The ability to do long division with a remainder.
4. You can get the whole set just by starting with 1 and allowing yourself to do any number of additions.
5. All numbers have square roots.
6. The existence of a maximal element (some number in the set which is larger than all other numbers).

Solution. There are a variety of possible answers, but here are my suggestions for some reasonable responses.

1. Decomposition of numbers into prime factors (except for 0 and 1): This is true of $\mathbb{N}$. Positive whole numbers (larger than 1) do decompose into prime factors.
2. Reciprocals of non-zero numbers: $\mathbb{Q}$ or $\mathbb{R}$ would work here, since those number sets have division and reciprocals are just a specific instance of division.
3. The ability to do long division with a remainder: this works in $\mathbb{N}$. Note that the remainder is important here: division is not exact in $\mathbb{N}$, since we don't have fractions. However, I can say that 7 goes into 40 five times, with a remainder of 5 — that still just uses whole numbers.
4. You can get the whole set just by starting with 1 and allowing yourself to do any number of additions: this is true for $\mathbb{N}$. Any positive number can be thought of as adding 1 some number of times. For example: $8 = 1 + 1 + 1 + 1 + 1 + 1 + 1 + 1$. In this way, the number 1 and the addition operation generate all of the natural numbers.
5. All numbers have square roots. This is a bit tricky, since none of the conventional number sets have this property. I have to make a restriction. I certainly want to work with real numbers, since I know that $\sqrt{2}$ is already irrational. I'm not going to find square roots in the natural numbers, the integers, or the rational numbers. But the negative numbers cause a problem: they do not have square roots in the real numbers. I

can exclude these by taking the set of all positive real numbers. This set has the structure of all numbers having square roots.

6. The existence of a maximal element (some number in the set which is larger than all other numbers). Again, no standard number set will have this structure, since they are all unbounded and keep getting larger forever. However, I can make a restriction and take a bounded subset. Consider, for example, all natural numbers less than 40 (as I suggested in the statement of the problem). This set has a maximal element of 39, since everything else in the set is a whole number less than or equal to 39.

2.5.5 Proof Questions

Activity 2.5.22 This is a guided proof to prove the triangle inequality using the dot product. The triangle inequality states that for any vector $u, v \in \mathbb{R}^n$, the length of the sum is bounded by the sum of the lengths of the individual vectors. In symbols:

$$|u + v| \leq |u| + |v|.$$

I'm going to give you several small steps which fit together into a complete proof. I want you to write out the details of each step and make it fit together into a whole argument.

1. Write $|u + v|^2$ as a dot product.
2. Expand that dot product (distribute the pieces).
3. Replace $u \cdot v$ with $|u||v| \cos\theta$.
4. Use the fact that $\cos\theta \leq 1$ to make the equality into an inequality.
5. Manipulate the inequality to show that $|u + v|^2 \leq (|u| + |v|)^2$.
6. Take the square root of both sides (which preserves the inequality and is well defined since both sides are positive).
7. Conclude the triangle inequality.

Solution. I'll follow the guided steps for this proof. First, I'll write $|u + v|^2$ as a dot product. I use the fact that the square of the length of a vector is the same as the dot product of a vector with itself.

$$|u + v|^2 = (u + v) \cdot (u + v)$$

The next step is to distribute the right side. The dot product distributes over vector addition like ordinary multipliction. Since the dot product is commutative, $u \cdot v$ is the same as $v \cdot u$, so I can combine the two term when I distribute the dot product.

$$|u + v|^2 = u \cdot u + 2u \cdot v + v \cdot v$$

The middle term has the dot product which I can replace as instructed in the third step.

$$|u + v|^2 = u \cdot u + 2|u||v| \cos\theta + v \cdot v$$

The cosine term is at most 1, so the maximum for the right side happens when I replace the cosine with 1.

$$|u + v|^2 \leq u \cdot u + 2|u||v| + v \cdot v$$

Then I need to turn the right side into a square term. The first step in that direction is to write the dot products as squares of lengths.

$$|u+v|^2 \leq |u|^2 + 2|u||v| + |v|^2$$

Then the right side is the square of a binomial.

$$|u+v|^2 \leq (|u| + |v|)^2$$

Finally, I take the square root of both sides. The square root of positive numbers preserves inequalities ($a \leq b$ if an only if $\sqrt{a} \leq \sqrt{b}$), so applying the square root is a valid operations for sovling an inequality like this.

$$|u+v| \leq |u| + |v|$$

I have produced the triangle inequality.

Activity 2.5.23 Prove that the cross product is anticommutative. (Take any two vectors $u, v \in \mathbb{R}^3$, write them in their components and prove that $u \times v = -(v \times u)$.)

Solution. I will start with the right side and try to manipulate it into the left side. I'll write the vectors in their components, use the definition of the cross product, and bring in the (-1) using the definition of scalar multiplication.

$$\begin{aligned}
-(v \times u) &= -\left(\begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} \times \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix}\right) \\
&= -\left(\begin{pmatrix} v_2u_3 - v_3u_2 \\ v_3u_1 - v_1u_3 \\ v_1u_2 - v_2u_1 \end{pmatrix}\right) \\
&= \left(\begin{pmatrix} -v_2u_3 + v_3u_2 \\ -v_3u_1 + v_1u_3 \\ -v_1u_2 + v_2u_1 \end{pmatrix}\right)
\end{aligned}$$

Now I just write each component in the other order, interchanging the order of each multiplication, and recognize the definition of the cross product in the result.

$$= \left(\begin{pmatrix} v_3u_2 - v_2u_3 \\ v_1u_3 - v_3u_1 \\ v_2u_1 - v_1u_2 \end{pmatrix}\right) = \left(\begin{pmatrix} u_2v_3 - u_3v_2 \\ u_3v_1 - u_1v_3 \\ u_1v_2 - u_2v_1 \end{pmatrix}\right) = u \times v$$

I started with the right side and made a chain of equality to the left side, thus proving the identity.

2.5.6 Conceptual Review Questions

- What is a dot product? What does it mean?
- What is a cross product? Why is it only in three dimensions?
- What is a binary operation?
- What is a structure on a set (of numbers)?
- Why do different sets (of numbers) have different structures on them?

Chapter 3

Week 3

3.1 Spans and Subspaces

3.1.1 Definitions

This chapter also introduces linear combinations, spans and linear dependence. These are some of the most important and central definitions in linear algebra. Let me start by defining the these three important terms.

Definition 3.1.1 A **linear combination** of a set of vectors $\{v_1, v_2, \ldots, v_k\}$ is a sum of the form $a_1v_1 + a_2v_2 + \ldots a_kv_k$ where the $a_i \in \mathbb{R}$. $\Diamond$

Definition 3.1.2 The **span** of a set of vectors $\{v_1, v_2, \ldots, v_k\}$, written

$$\text{Span}\{v_1, v_2, \ldots, v_k\},$$

is the set of all possible linear combinations of the vectors. $\Diamond$

Spans have an important geometric interpretation, but I'll get to that in the next section.

Definition 3.1.3 A set of vectors $\{v_1, v_2, \ldots, v_k\}$ in $\mathbb{R}^n$ is called **linearly independent** if the equation

$$a_1v_1 + a_2v_2 + a_3v_3 + \ldots + a_kv_k = 0$$

has only the trivial solution: for all i, $a_i = 0$. If a set of vectors isn't linearly independent, it is called **linearly dependent**. $\Diamond$

This may seem like a strange definition, but it algebraically captures the idea of independent directions. A set of vectors is linearly independent if all of them point in fundamentally different directions. We could also say that a set of vectors is linearly independent if no vector is in the span of the other vectors. No vector is a redundant piece of information; if we remove any vectors, the span of the set gets smaller.

In order for a set like this to be linearly independent, I need $k \leq n$. $\mathbb{R}^n$ has only n independent directions, so it is impossible to have more than n linearly independent vectors in $\mathbb{R}^n$.

3.1.2 Linear and Affine Subspaces

Using the language of linear combinations and spans, I can define the major objects that live in Euclidean space.

Definition 3.1.4 A **linear subspace** of $\mathbb{R}^n$ is a non-empty set of vectors L which satisfies the following two properties.

- If $u, v \in L$ then $u + v \in L$. The term for this property is: **closed under vector addition**.
- If $u \in L$ and $a \in \mathbb{R}$ then $au \in L$. The term for this property is: **closed under scalar multiplication**.

$\diamondsuit$

There are two basic operations on $\mathbb{R}^n$: adding vectors and multipling vectdors by scalars. Linear subspaces are just subsets where both operations stay within the subset.

Geometrically, vector addition and scalar multiplication produce flat objects: lines, planes, and their higher-dimensional analogues. Also, since $a = 0$ is possible, it must be true that $0 \in L$. So linear subspaces can be informally defined as 'flat' subsets which include the origin. If I relax the definition slighlty to consider 'flat' subsets which do not contain the origin, I get a related definition.

Definition 3.1.5 An **affine subspace** of $\mathbb{R}^n$ is a non-empty set of vectors A which can be described as a sum $v + u$ where v is a fixed vector and u is any vector in some fixed linear subspace L. With some abuse of notation, this sum of a fixed vector and a linear subspace is often written

$$A = v + L.$$

Affine subspaces are flat spaces that may be offset from the origin. The vector v is called the **offset vector**. Affine spaces include linear spaces since $v = 0$ is always possible, leading to $A = L$. Affine objects are the lines, planes and higher-dimensional flat objects that may or may not pass through the origin. $\diamondsuit$

Notice that I defined both affine and linear subspaces to be non-empty. The empty set $\emptyset$ is *not* a linear or affine subspace. The smallest linear subspace is $\{0\}$: just the origin. The smallest affine subspace is any isolated point.

Now let me connect the two idea presented so far in this section.

Proposition 3.1.6 *Any span is a linear subspace. Conversely, any linear subspace can be written as a span.*

By this proposition, spans provide an algebraic description of the geometric notion of a linear subspace. Since 'any linear combination' can include the trivial linear combination where all the real constants are zero, spans always go through the origin. To use spans to define affine subspaces, I have to add an offset vector.

Definition 3.1.7 An **offset span** is an affine subspace formed by adding a fixed vector u, called the **offset vector**, to the span of some set of vectors. $\diamondsuit$

Let me talk about spans of small numbers of vectors. First, the span of one non-zero vector is the line (through the origin) consisting of all multiples of that vector. Similarly, I expect the span of two vectors to be a plane. However, here I have a problem: there may be redundant information. For example, in $\mathbb{R}^2$, we could consider this span.

$$\text{Span}\left\{\begin{pmatrix}1\\2\end{pmatrix}, \begin{pmatrix}2\\4\end{pmatrix}\right\}$$

I would hope the span of two vectors would be the entire plane, but this is just

the line in the direction $\begin{pmatrix}1\\2\end{pmatrix}$. The vector $\begin{pmatrix}2\\4\end{pmatrix}$, since it is already a multiple of $\begin{pmatrix}1\\2\end{pmatrix}$, is redundant.

The problem is magnified in higher dimensions. If I have the span of a large number of vectors in $\mathbb{R}^n$, it is nearly impossible to tell, at a glance, whether any of the vectors are redundant. I would like to have tools to determine this redundancy. I'm going to define the keys ideas behind this redundancy in the next part of this section, but the calculation tools for checking redundancy have to wait until Section 5.1.

3.1.3 Dimension and Basis

I've already casually assumed some intuition about dimensions for linear subspaces. A line through the origin should have dimensions one. A plane through the origin should have dimensions two. However, I can't rely on intuition for mathematical definition: I have to formalize. This second formally defines dimension and related ideas.

Definition 3.1.8 Let L be a linear subspace of $\mathbb{R}^n$. Then L has **dimension k** if L can be written as the span of k linearly independent vectors. ◊

Definition 3.1.9 Let A be an affine subspace of $\mathbb{R}^n$ and write A as $A = u + L$ for L, a linear subspace, and u an offset vector. Then A has **dimension k** if L has dimension k. ◊

This is the proper, complete definition of dimension for linear and affine spaces. It solves the problem of redundant information (either redundant equations for loci or redundant vectors for spans) by insisting on a linearly independent spanning set. It insists that the vectors defining a span are a minimal set. There is a name for such a minimal set.

Definition 3.1.10 Let L be a linear subspace of $\mathbb{R}^n$. A **basis** for L is a minimal spanning set; that is, a set of linearly independent vectors that span L. ◊

Since a span is the set of all linear combinations, I can think of a basis as a way of writing the vectors of L: every vector in L can be written as a *unique* linear combination of the basis vectors. A basis gives a nice way to account for all the vectors in L.

Linear subspaces have many (infinitely many) different bases. There are some standard choices.

Definition 3.1.11 The **standard basis** of $\mathbb{R}^2$ is composed of the two-unit vectors in the positive x and y directions. I can write any vector as a linear combination of the basis vectors.

$$e_1 = \begin{pmatrix}1\\0\end{pmatrix} \qquad e_2 = \begin{pmatrix}0\\1\end{pmatrix} \qquad \begin{pmatrix}x\\y\end{pmatrix} = xe_1 + ye_2$$

The **standard basis** of $\mathbb{R}^3$ is composed of the three-unit vectors in the positive x, y and z directions. I can again write any vector as a linear combination of the basis vectors.

$$e_1 = \begin{pmatrix}1\\0\\0\end{pmatrix} \qquad e_2 = \begin{pmatrix}0\\1\\0\end{pmatrix} \qquad e_3 = \begin{pmatrix}0\\0\\1\end{pmatrix} \qquad \begin{pmatrix}x\\y\\z\end{pmatrix} = xe_1 + ye_2 + ze_3$$

The **standard basis** of $\mathbb{R}^n$ is composed of vectors $e_1, e_2, \ldots, e_n$ where e_i has a 1 in the ith component and zeroes in all other components. e_i is the unit vector in the positive ith axis direction. ◊

3.2 Loci

3.2.1 Definition of a Locus

Spans were the first way to define linear and affine subspaces. There is another approach to describe these object, which I describe in this section. I'll start with the main definition.

Definition 3.2.1 Consider any set of equations in the variables $x_1, x_2, \ldots, x_n$. The **locus** in $\mathbb{R}^n$ of this set of equations is the set of vectors that satisfy *all* of the equations. The plural of a locus is **loci**. $\Diamond$

In general, the equations can be of any sort. The unit circle in $\mathbb{R}^2$ is most commonly defined as the locus of the equation $x^2 + y^2 = 1$. The graph of a function is the locus of the equation $y = f(x)$. However, linear algebra excludes curved objects. The subset is concerned with linear/affine objects: things that are straight and flat. So, I don't want loci of just any kind of equation: I want loci of linear equations. The idea is likely familiar, but let me formally define it anyway.

Definition 3.2.2 Let a_i and c be real numbers. A **linear equation** in the variables $x_1, x_2, \ldots x_n$ is an equation of the following form.

$$a_1x_1 + a_2x_2 + \ldots + a_nx_n = c$$

$\Diamond$

Proposition 3.1.6 claimed that all linear subspaces were described by spans. Shortly afterward, I showed that all affine subspaces were described by offset spans. In addition to spans, I can also use loci to describe any affine or linear subspace.

Proposition 3.2.3 *Any linear or affine subspace of $\mathbb{R}^n$ can be described as the locus of finitely many linear equations. Likewise, the locus of any number of linear equations is either an affine subspace of $\mathbb{R}^n$ or the empty set.*

It is quite valuable to have these two different descriptions of linear and affine subspaces. Each description (spans or loci) has strengths and weaknesses. Depending on what calculations I want to do, using either a span or a locus description may help a great deal.

There is also a nice contrast in the constructions of spans and loci. Spans are essentially built up by adding more and more vectors, more and more directions. The span of no vectors is the origin. The span of one vector is a line. The span of two linearly independent vectors is a plane. This process continues (if the ambient dimension is large enough) to make object of higher and higher dimensions as more linearly independent vectors are added to the span.In contrast, loci are built top-down, by starting with the ambient space and reducing the number of points by imposing restrictions in the form of linear equations. The locus of no equations is the entire space, since all points trivially satisfy no equations. The locus of one equation should be something of one dimension less that the ambient space, as one restriction is imposed. The locus of two equations should be something of two dimensions less than the ambient space, as two restrictions are imposed. This continues: the dimension of a locus, at least ideally, is the dimensions of the ambient space minus the number of equations or restrictions.

To be more specific, in $\mathbb{R}^2$, adding the equation $x = 3$ restricts to a vertical line passing through the x-axis at $\begin{pmatrix} 3 \\ 0 \end{pmatrix}$. Likewise, the equation $y = 4$ restricts

to a horizontal line passing through the y-axis at $\begin{pmatrix} 0 \\ 4 \end{pmatrix}$. If I consider the locus of *both* equations, there is only one point remaining: $\begin{pmatrix} 3 \\ 4 \end{pmatrix}$ is the only point that satisfies both equations. In this way, each additional equation potentially adds an additional restriction and leads to a smaller linear or affine subspaces.

As with spans, there can be the problem of redundant information. In $\mathbb{R}^3$, the locus of the equation $x + y + z = 1$ is a plane. If I also add the equation $2x + 2y + 2z = 2$, however, I don't get a line. The second equation is a multiple of the first, and any points that satisfy the first equation also satisfy the second. The second equation is redundant and unnecessary. Determining what is redundant (and determining the dimension of a locus) will rely on calculation tools introduced in Section 5.3.

3.2.2 Familiar Loci

I can define some familiar objects which are presented as the locus of one linear equation.

Definition 3.2.4 A **line** in $\mathbb{R}^2$ is the locus of the equation $ax + by = c$ for $a, b, c \in \mathbb{R}$. In general, the line is affine. The line is linear if $c = 0$. $\Diamond$

Definition 3.2.5 A **plane** in $\mathbb{R}^3$ is the locus of the linear equation $ax+by+cz = d$. In general, the plane is affine. The plane is linear if $d = 0$. $\Diamond$

If we think of a plane in $\mathbb{R}^3$ as the locus of one linear equation, the important dimensional fact about a plane is not that it has dimension two but that it has dimension one less than its ambient space $\mathbb{R}^3$.

Definition 3.2.6 A **hyperplane** in $\mathbb{R}^n$ is the locus of one linear equation: $a_1x_1 + a_2x_2 + \ldots + a_nx_n = c$. It has dimension $n - 1$. It is, in general, affine. The hyperplane is linear if $c = 0$. $\Diamond$

3.2.3 Intersection

Definition 3.2.7 If A and B are sets, their intersection $A \cap B$ is the set of all points they have in common. The intersection of affine subspaces is also an affine subspace. If A and B are both linear, the intersection is also linear. $\Diamond$

Example 3.2.8 Loci of more than one equation can be understood as intersections. Consider the locus of two equations, say the example I used from $\mathbb{R}^2$ before: the locus of $x = 3$ and $y = 4$. I defined this directly as a single locus. However, I could just as easily think of this as the intersection of the two lines given by $x = 3$ and $y = 4$ separately. In this way, it is the intersection of two loci. Similarly, all loci are the intersection of the planes or hyperplanes defined by each individual linear equation. $\square$

3.3 Equations of Planes

3.3.1 Dot Products and Loci

Loci are the solutions to one of more linear equations. These linear equation can be re-interpreted and expressed via via dot products. Consider, again, the general linear equation in $\mathbb{R}^n$.

$$a_1u_1 + a_2u_2 + \ldots + a_nu_n = c$$

Think of the variables u_i as the components of a vector $u \in \mathbb{R}^n$. There also are n scalars a_i, which I can likewise treat as components of the vector $a \in \mathbb{R}^n$. The vector u is variable; the vector a is constant. Then I can re-write the linear equation using these two vectors.

$$a_1u_1 + a_2u_2 + \ldots + a_nu_n = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} \cdot \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{pmatrix} = a \cdot u = c$$

In this way, a linear equation specifies that the dot product result of a variable vector u with a fixed vector a must have the result c. In this light, an affine plane in $\mathbb{R}^3$ is given by the equation

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} \cdot \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = c.$$

This plane is precisely all vectors whose dot product with the vector $\begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ is the fixed number c. If $c = 0$, the equation become

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} \cdot \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = 0.$$

A dot product of zero indicates that the two inputs to the dot product are perpendicular. Therefore, a plane through the origin is the set of all vectors which are perpendicular to a fixed vector $\begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$.

Definition 3.3.1 Let P be a plane in $\mathbb{R}^3$ determined by the equation

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} \cdot \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = c.$$

The vector $\begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ is called the **normal to the plane**. Let H be a hyperplane in $\mathbb{R}^n$ determined by the equivalent equation in $\mathbb{R}^n$.

$$u \cdot a = \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{pmatrix} \cdot \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} = c$$

The vector a is called the **normal to the hyperplane**. $\Diamond$

If $c = 0$, the plane or hyperplane is perpendicular to its normal. This notion of orthogonality still works when $c \neq 0$. In this case, the normal is a *local* perpendicular direction from any point on the affine plane. Treating any such point as a local origin, the normal points in a direction perpendicular to all the *local direction* vectors which lie on the plane.

3.3.2 An Algorithm for Equations of Planes

Now I can build a general process for finding the equation of a plane in $\mathbb{R}^3$. Any time I have a point p on the plane and two *local direction vectors* u and v which remain on the plane, I can find a normal to the plane by calculating $u \times v$. Then I can find the equation of the plane by calculating the dot product $p \cdot (u \times v)$ to find the constant c.

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} \cdot (u \times v) = c$$

If I am given three points on a plane (p, q and r), then I can use p as the local origin and construct the local direction vectors as $q - p$ and $r - p$. The normal is $(q-p) \times (r-p)$. In this way, I can construct the equation of a plane given three points or a single point and two local directions.

3.4 Proofs - Linear Subspaces

3.4.1 Proving Some Properties of Linear Subspaces

Throughout the course, I want to keep developing the idea of mathematical proof. I'll do two proofs using the material from this section of the course.

Proposition 3.4.1 *The intersection of any two linear subspaces is another linear subspace.*

Proof. A linear subspace is a subset of $\mathbb{R}^n$ with two properties: closed under vector addition and closed under scalar multiplication. If L_1 and L_2 are subspaces, I have to check that both properties hold for the intersection $L_1 \cap L_2$. If two vectors u and v are in the intersection, then they are contained in both L_1 and L_2. Since they are both in L_1 and L_1 is a subspace (meaning closed under vector addition), the sum $u + v$ is also in L_1. The same is true for L_2. Therefore, the sum $u + v$ is in both subspaces, thus in the intersection. Since I made no assumptions on the vectors, this shows the intersection is closed under vector addition.
A very similar argument works for scalar multiplication. If a vector u is in the intersection (hence in both subspaces) and a is a scalar, then since each subspace is closed under scalar multiplication, the scalar multiple au is in both subspaces, hence in the intersection. This shows the intersection is closed under scalar multiplication. ■

Proposition 3.4.2 *A hyperplane through the origin is a linear subspace.*

Proof. I stated that loci were linear or affine subspaces and that loci through the origins were linear subspaces. However, I never proved this. Here, I'm going to prove it for the special case of a hyperplane.
A hyperplane is the locus of one linear equation.

$$a_1x_1 + a_2x_2 + \ldots a_nx_n = c$$

However, if the hyperplane contains the origin, the constant c must be zero. (I stated this earlier; I could argue this by saying that if all the x_i are set to zero, the left side of the equation is zero, so the right side must also be zero).

$$a_1x_1 + a_2x_2 + \ldots a_nx_n = 0$$

All points on the hyperplane satisfy this equation, and I will use this fact to prove the proposition. To make this easier, I will write the equation in terms

of dot products. If n is the vector with all the a_i (n for normal) and x is the vector with all the x_i, then the equation is

$$n \cdot x = 0.$$

Let u and v be vectors on the hyperplane, then $n \cdot u = 0$ and $n \cdot v = 0$. Then I can consider

$$n \cdot (u + v).$$

The dot product is linear, so it distributes here.

$$= n \cdot u + n \cdot v$$

I just said that both of these dot products are zero, so the equation gives $0 + 0 = 0$. Therefore, $u + v$ also satisfies the equation of the hyperplane, so is contained in the hyperplane.
I do something very similar with the dot product to proof that the hyperplane is closed under scalar multiplication. If a is a scalar, consider the vector au to see if it satisfies the equation of the hyperplane. I start with the dot product with the normal.

$$n \cdot (au)$$

By the properties of the dot product, again, I can pull out the scalar.

$$= a(n \cdot u)$$

Then I get $a(0)$, since u is on the hyperplane. This shows that the scalar multiple is also on the hyperplane. ■

3.5 Week 3 Activity

3.5.1 Linear Combinations

Activity 3.5.1 Write the first vector as a linear combination of the second and third vectors.

$$\begin{pmatrix} 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Solution. I need to write the first vector as a multiple of the second plus a multiple of the third.

$$\begin{pmatrix} 2 \\ 3 \end{pmatrix} = a \begin{pmatrix} 1 \\ -1 \end{pmatrix} + b \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

If I look at the components, I get two equations.

$$\begin{aligned} 2 &= a + b \\ 3 &= -a + b \end{aligned}$$

I have to solve this system. If I isolate a in the first equation, I get $a = 2 - b$. If I replace that in the second equation, I get $3 = -2 + b + b$. I can solve this to get $b = \frac{5}{2}$. Then the expression for a gives $a = 2 - b = 2 - \frac{5}{2} = \frac{-1}{2}$. That lets me write the linear combination.

$$\begin{pmatrix} 2 \\ 3 \end{pmatrix} = \frac{-1}{2} \begin{pmatrix} 1 \\ -1 \end{pmatrix} + \frac{5}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Activity 3.5.2 Write the first vector as a linear combination of the second and third vectors.

$$\begin{pmatrix}-4\\-1\end{pmatrix}, \begin{pmatrix}1\\-1\end{pmatrix}, \begin{pmatrix}1\\1\end{pmatrix}$$

Solution. I need to write the first vector as a multiple of the second plus a multiple of the third.

$$\begin{pmatrix}-4\\-1\end{pmatrix} = a\begin{pmatrix}1\\-1\end{pmatrix} + b\begin{pmatrix}1\\1\end{pmatrix}$$

If I look at the components, I get two equations.

$$\begin{aligned}-4 &= a + b\\ -1 &= -a + b\end{aligned}$$

I have to solve this sytem. If I isolate a in the first equation, I get $a = -4 - b$. If I replace that in the second equation, I get $-1 = 4 + b + b$. I can solve this to get $b = \frac{-5}{2}$. Then the expression for a gives $a = -4 - b = -4 - \frac{5}{2} = \frac{-3}{2}$. That lets me write the linear combination.

$$\begin{pmatrix}-4\\-1\end{pmatrix} = \frac{-3}{2}\begin{pmatrix}1\\-1\end{pmatrix} + \frac{-5}{2}\begin{pmatrix}1\\1\end{pmatrix}$$

Activity 3.5.3 Write the first vector as a linear combination of the second and third vectors.

$$\begin{pmatrix}0\\3\end{pmatrix}, \begin{pmatrix}1\\-1\end{pmatrix}, \begin{pmatrix}1\\1\end{pmatrix}$$

Solution. I need to write the first vector as a multiple of the second plus a multiple of the third.

$$\begin{pmatrix}0\\3\end{pmatrix} = a\begin{pmatrix}1\\-1\end{pmatrix} + b\begin{pmatrix}1\\1\end{pmatrix}$$

If I look at the components, I get two equations.

$$\begin{aligned}0 &= a + b\\ 3 &= -a + b\end{aligned}$$

I have to solve this sytem. If I isolate a in the first equation, I get $a = -b$. If I replace that in the second equation, I get $3 = b + b$. I can solve this to get $b = \frac{3}{2}$. Then the expression for a gives $a = -b = -\frac{3}{2} = \frac{-3}{2}$. That lets me write the linear combination.

$$\begin{pmatrix}0\\3\end{pmatrix} = \frac{-3}{2}\begin{pmatrix}1\\-1\end{pmatrix} + \frac{3}{2}\begin{pmatrix}1\\1\end{pmatrix}$$

Activity 3.5.4 Write the first vector as a linear combination of the second third and fourth vectors.

$$\begin{pmatrix}3\\0\\2\end{pmatrix}, \begin{pmatrix}1\\0\\0\end{pmatrix}, \begin{pmatrix}0\\1\\1\end{pmatrix}, \begin{pmatrix}0\\-1\\1\end{pmatrix}$$

Solution. I need to write the first vector as a multiple of the second plus a

multiple of the third.

$$\begin{pmatrix}3\\0\\2\end{pmatrix} = a\begin{pmatrix}1\\0\\0\end{pmatrix} + b\begin{pmatrix}0\\1\\1\end{pmatrix} + c\begin{pmatrix}0\\-1\\1\end{pmatrix}$$

I could try to solve this by linear system as I did before, but I'll try to work by inspection first, so I can see if I can informally find a solution. Since the second vector is the only vector of the last three that has an x component, I know that $a = 3$ is the only way to get a 3 in the first coordinate. That gives me one constant. Then, looking at the y and z coordinates, we can see that if I simply add the third and fourth vectors togother, the y cancels and the z adds to 2. Therefore, I can conclude that $b = c = 1$

$$\begin{pmatrix}3\\0\\2\end{pmatrix} = 3\begin{pmatrix}1\\0\\0\end{pmatrix} + \begin{pmatrix}0\\1\\1\end{pmatrix} + \begin{pmatrix}0\\-1\\1\end{pmatrix}$$

3.5.2 Linear Independence

Activity 3.5.5 Determine if this set of vectors is linearly independent.

$$\left\{\begin{pmatrix}4\\5\end{pmatrix}, \begin{pmatrix}-6\\-9\end{pmatrix}, \begin{pmatrix}-3\\11\end{pmatrix}\right\}$$

Solution. There are three vectors here, but they are all vectors in $\mathbb{R}^2$. In two-dimensional space, there are only two independent directions of movement, so three vectors can never be linearly independent. I conclude that this is a linearly dependent set.

Activity 3.5.6 Determine if this set of vectors is linearly independent.

$$\left\{\begin{pmatrix}5\\0\\1\end{pmatrix}, \begin{pmatrix}15\\0\\3\end{pmatrix}, \begin{pmatrix}-25\\0\\-5\end{pmatrix}\right\}$$

Solution. The second and third vectors are multiplied (by 3 and (-5), respectively) of the first vector. That means that all three vectors point along the same line, so these are not different directions and cannot be linearly independent.

Activity 3.5.7 Determine if this set of vectors is linearly independent.

$$\left\{\begin{pmatrix}5\\9\\0\end{pmatrix}, \begin{pmatrix}0\\0\\-4\end{pmatrix}, \begin{pmatrix}-9\\5\\0\end{pmatrix}\right\}$$

Solution. The second vector is clearly independent of the other two since it only has a z coordinate and the others have 0 in the z coordinate. It points in the z direction, while the others are restricted to the $x - y$ plane. For the first and third vector, the dot product shows that they are perpendicular, so they point in fundamentally different directions. I conclude that this is a linearly independent set.

Activity 3.5.8 Determine if this set of vectors is linearly independent.

$$\left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right\}$$

Solution. Working by inspection again, I can see that the sum of the first two vectors is the third vector. Any relation like this, where some linear combinations of some of the vectors produce another vector, means that the set is linearly dependent.

3.5.3 Spans and Bases

Activity 3.5.9 Determine a basis for the given span.

$$\text{Span}\left\{ \begin{pmatrix} 4 \\ 1 \end{pmatrix}, \begin{pmatrix} -9 \\ -4 \end{pmatrix}, \begin{pmatrix} 0 \\ 5 \end{pmatrix}, \begin{pmatrix} 6 \\ -2 \end{pmatrix} \right\}$$

Solution. This is in $\mathbb{R}^2$, so the maximum number of linearly independent vectors is 2. These are not all in the same direction, so this span is all of $\mathbb{R}^2$. I can take any two of these as a basis since no vector is a multiple of any other vector. Since the span is $\mathbb{R}^2$, we could also take the standard basis $\{e_1, e_2\}$.

Activity 3.5.10 Determine a basis for the given span.

$$\text{Span}\left\{ \begin{pmatrix} 0 \\ 4 \\ 13 \end{pmatrix}, \begin{pmatrix} 0 \\ 5 \\ -6 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \\ -1 \end{pmatrix} \right\}$$

Solution. None of the vectors have a non-zero x component, so this set cannot span all of $\mathbb{R}^3$. However, I do have different directions in the yz plane, so this span does span all of the yz plane. Therefore, since none of the vectors is a multiple of the others, any two vectors form a basis. I can also take the subset of the standard basis that spans the yz plane: $\{e_2, e_3\}$.

Activity 3.5.11 Determine a basis for the given span.

$$\text{Span}\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ -4 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ -6 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ -9 \end{pmatrix} \right\}$$

Solution. All of the vector point in axis directions in $\mathbb{R}^5$, and all the axis directions are different. I conclude that this is a linearly independent set, so I can take all four vectors as a basis. Since I get the first, second, third and fifth axis directions, I could also take a portion of the standard basis that spans those four directions: $\{e_1, e_2, e_3, e_5\}$.

Activity 3.5.12 Determine a basis for the given span.

$$\text{Span}\left\{ \begin{pmatrix} 0 \\ 6 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ -13 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 5 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 34 \\ 0 \\ 0 \end{pmatrix} \right\}$$

Solution. All four vectors point in the second axis direction, so they are all multiples of the standard basis vector e_2. Therefore, the span is one-dimensional and any individual vector, or for that matter e_2, could be taken as a basis.

3.5.4 Equations of Planes and Hyperplanes

Activity 3.5.13 Write the equation of the plane with normal $\begin{pmatrix} -1 \\ -1 \\ 0 \end{pmatrix}$ if $\begin{pmatrix} 0 \\ 1 \\ 3 \end{pmatrix}$ is a point on the plane.

Solution. The normal determines the coefficients of the equation of the plane. I put those coefficients into the general form, with the constant d undetermined.

$$(-1)x + (-1)y + 0z = d$$

I can calculate the constant using the point. Put the coordinates of the point into the left side of the equation and calculate d.

$$d = (-1)(0) + (-1)(1) + (0)(3) = -1$$

Therefore, the equation of the plane is

$$-x - y = -1.$$

Activity 3.5.14 Write the equation of the plane with normal $\begin{pmatrix} 2 \\ 0 \\ 3 \end{pmatrix}$ if $\begin{pmatrix} -4 \\ -1 \\ 2 \end{pmatrix}$ is a point on the plane.

Solution. The normal determines the coefficients of the equation of the plane. I put those coefficients into the general form, with the constant d undetermined.

$$2x + 0y + 3z = d$$

I can calculate the constant using the point. Put the coordinates of the point into the left side of the equation and calculate d.

$$d = 2(-4) + 0(-1) + 3(2) = -2$$

Therefore, the equation of the plane is

$$2x + 3z = -2.$$

Activity 3.5.15 Write the equation of the plane with local directions $\begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}$ if $\begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}$ is a point on the plane.

Solution. The cross product of the local directions gives the normal to the plane.

$$\begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix} \times \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} (-1)(0) - (1)(1) \\ (1)(-2) - (-1)(0) \\ (-1)(1) - (-1)(-2) \end{pmatrix} = \begin{pmatrix} -1 \\ -2 \\ -1 \end{pmatrix}$$

This is the normal. The normal determines the coefficients of the equation of

the plane. I put those coefficients into the general form, with the constant d undetermined.

$$-x - 2y - z = d$$

I can calculate the constant using the point. Put the coordinates of the point into the left side of the equation and calculate d.

$$d = -(1)(2) - (2)(1) - (1) = -5$$

Therefore, the equation of the plane is

$$-x - 2y - z = -5.$$

Activity 3.5.16 Write the equation of the plane with local directions $\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}$ if $\begin{pmatrix} -1 \\ 0 \\ -3 \end{pmatrix}$ is a point on the plane.

Solution. The cross product of the local directions gives the normal to the plane.

$$\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \times \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} (1)(1) - (1)(1) \\ (1)(-2) - (0)(1) \\ (0)(1) - (1)(-2) \end{pmatrix} = \begin{pmatrix} 0 \\ -2 \\ 2 \end{pmatrix}$$

This is the normal. The normal determines the coefficients of the equation of the plane. I put those coefficients into the general form, with the constant d undetermined.

$$0x - 2y + 2z = d$$

I can calculate the constant using the point. Put the coordinates of the point into the left side of the equation and calculate d.

$$d = 0(-1) - 2(0) + 2(-3) = -6$$

Therefore, the equation of the plane is

$$-2y + 2z = -6.$$

Activity 3.5.17 Write the equation of the plane with points $p = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$, $q = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ and $r = \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}$.

Solution. The local directions are given by the differences of the point: $(q-p)$ and $(r-p)$. I calculate those two local directions.

$$q - p = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}$$

$$r - p = \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} - \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -2 \\ 0 \\ 0 \end{pmatrix}$$

These are the two local directions. The cross product of the local directions

gives the normal to the plane.

$$\begin{pmatrix}1\\-1\\-1\end{pmatrix}\times\begin{pmatrix}-2\\0\\0\end{pmatrix}=\begin{pmatrix}(-1)(0)-(-1)(0)\\(-1)(-2)-(1)(0)\\(1)(0)-(-1)(-2)\end{pmatrix}=\begin{pmatrix}0\\2\\-2\end{pmatrix}$$

This is the normal. The normal determines the coefficients of the equation of the plane. I put those coefficients into the general form, with the constant d undetermined.

$$0x+2y-2z=d$$

I can calculate the constant using the point (I can use any of the three points — all will determine the same constant). Put the coordinates of the point into the left side of the equation and calculate d.

$$d=(0)(0)+2(1)-2(1)=0$$

Therefore, the equation of the plane is

$$2y-2z=0.$$

Activity 3.5.18 Write the equation of the plane with points $p=\begin{pmatrix}-3\\-1\\0\end{pmatrix}$, $q=\begin{pmatrix}-2\\1\\-4\end{pmatrix}$ and $r=\begin{pmatrix}0\\5\\0\end{pmatrix}$.

Solution. The local directions are given by the difference of the points: $(q-p)$ and $(r-p)$. I calculate those two local directions.

$$q-p=\begin{pmatrix}-2\\1\\-4\end{pmatrix}-\begin{pmatrix}-3\\-1\\0\end{pmatrix}=\begin{pmatrix}1\\2\\-4\end{pmatrix}$$

$$r-p=\begin{pmatrix}0\\5\\0\end{pmatrix}-\begin{pmatrix}-3\\-1\\0\end{pmatrix}=\begin{pmatrix}3\\6\\0\end{pmatrix}$$

These are the two local directions. The cross product of the local directions gives the normal to the plane.

$$\begin{pmatrix}1\\2\\-4\end{pmatrix}\times\begin{pmatrix}3\\6\\0\end{pmatrix}=\begin{pmatrix}(2)(0)-(-4)(6)\\(-4)(3)-(1)(0)\\(1)(6)-(2)(3)\end{pmatrix}=\begin{pmatrix}24\\-12\\0\end{pmatrix}$$

This is the normal. The normal determines the coefficients of the equation of the plane. I put those coefficients into the general form, with the constant d undetermined.

$$24x-12y+0z=d$$

I can calculate the constant using the point (I can use any of the three points — all will determine the same constant). Put the coordinates of the point into the left side of the equation and calculate d.

$$d=24(0)-12(5)+0z=-60$$

Therefore, the equation of the plane is

$$24x-12y=-60.$$

Activity 3.5.19 Write the equation of the hyperplane with normal $\begin{pmatrix} -3 \\ 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}$ if $\begin{pmatrix} 4 \\ 0 \\ 1 \\ -1 \\ -2 \end{pmatrix}$ is a point on the hyperplane.

Solution. The normal determines the coefficients of the equation of the hyperplane. I put those coefficients into the general form, with the constant d undetermined.

$$-3x_1 + 1x_2 + 1x_3 + 0x_4 + 0x_5 = d$$

I can calculate the constant using the point. Put the coordinates of the point into the left side of the equation and calculate d.

$$d = -3(4) + 1(0) + 1(1) + 0(-1) + 0(-2) = -11$$

Therefore, the equation of the hyperplane is

$$-3x_1 + 1x_2 + 1x_3 + 0x_4 + 0x_5 = -11.$$

Activity 3.5.20 Write the equation of the hyperplane with normal $\begin{pmatrix} -1 \\ -1 \\ -4 \\ 1 \\ -2 \\ 2 \end{pmatrix}$ if $\begin{pmatrix} 3 \\ 1 \\ -2 \\ -2 \\ -6 \\ 0 \end{pmatrix}$ is a point on the hyperplane.

Solution. The normal determines the coefficients of the equation of the hyperplane. I put those coefficients into the general form, with the constant d undetermined.

$$-1x_1 - 1x_2 - 4x_3 + 1x_4 - 2x_5 + 2x_6 = d$$

I can calculate the constant using the point. Put the coordinates of the point into the left side of the equation and calculate d.

$$d = -1(3) - 1(1) + (-4)(-2) + 1(-2) - 2(-6) + 2(0) = 14$$

Therefore, the equation of the hyperplane is

$$-1x_1 - 1x_2 - 4x_3 + 1x_4 - 2x_5 + 2x_6 = 14.$$

3.5.5 Proof Questions

Activity 3.5.21 Consider a set of three vectors in $\mathbb{R}^3$ such that all three vectors have a 0 x coordinate. Prove that these three vectors cannot be linearly independent.

Solution. There are a number of possible approaches here. First, I can talk about fundamentally different directions. If all three vectors have a 0 x component, then they are all found in the yz. In that plane (like any plane), there are only two fundamentally different directions. Therefore, a set of three vectors cannot have fundamentally different directions, hence cannot be linearly independent.

Alternatively, I could delete the first coordinate of each vector and think of the resulting vectors in $\mathbb{R}^2$. Then I would have three vectors in $\mathbb{R}^2$, so they are linearly independent, and there exists a way to write one of the vectors as a linear combination of the others: $u = av + bw$. Since all I did was deleted a zero component from each vector, this relationship will still work with the original vectors in $\mathbb{R}^3$, proving that they are linearly independent.

Activity 3.5.22 Prove that if a set of non-zero vectors are pairwise orthogonal (the dot product of any two vectors in the set is zero), then the set must be linearly independent.

Solution. Again, there are several possible approaches. First, I can think about fundamentally different directions. Since all the vectors in the set are perpendicular to all the other vectors, there is no way that they can share any kind of common direction. This means that they all need a fundamentally new, perpendicular-to-all-others direction. Therefore, they must be a basis.

More formally, I could prove by contradiction. Assume that the vectors are linearly independent. Let the vectors be written $\{v_i\}$ (each index indicates a different vector, not the components of one vector). Linear dependence means I can find constant a_i such that not all of the a_i are zero and

$$a_1v_1 + a_2v_2 + \ldots + a_nv_n = 0.$$

Now take the dot product of this equation with the vector v_i.

$$v_i \cdot (a_1v_1 + a_2v_2 + \ldots + a_nv_n) = v_i \cdot 0$$

On the right, the dot product of any vector with the zero vector is zero. On the right, the dot product is linear, so I can distribute and pull out constants.

$$a_1(v_i \cdot v_1) + a_2(v_i \cdot v_2) + \ldots + a_n(v_i \cdot v_n)) = v_i \cdot 0$$

But all the vectors are perpendicular, so on the left, everything is zero except for $v_i \cdot v_i$.

$$a_i(v_i \cdot v_i) = 0$$

Then I can write the dot product as a length squared.

$$a_i|v_i| = 0$$

Since the vector v_i is not zero, the constant a_i must be. But I did this arbitrarily for any index i, so this means all of the constant a_i must be zero. This contradicts the assumption that at least one of the constants is non-zero. This contradiction means that the original assumption must be wrong. I conclude that the vectors are linearly independent.

Activity 3.5.23 Prove that any linear subspace has a basis of vector which all have length 1.

Solution. Any linear subspace has a basis of non-zero vectors. For any linear subspace, take any basis whatever. Then, for every vector v in that basis, replace v with $\frac{1}{|v|}v$. This new vector is a scalar multiple of the old, so it points in the same direction. Since the directions have not changed, these new replacement vectors are still a basis: they still all point in fundamentally different directions. But, by dividing by the original lengths, the lengths of these new vectors are all 1. Therefore, I have constructed a basis of vectors of length 1. Since I made no assumption on the linear subspace, I conclude that all linear subspaces have a basis of vectors of length 1.

3.5.6 Conceptual Review Questions

- What is a linear combination?
- What is a span?
- What is linear independence, and how it is checked?
- What is a basis? What does it mean to represent a vector as a linear combination of a basis?
- What does it mean if geometrically a set of vectors is linearly independent?
- What is a normal vector to a plane or hyperplane? Why are planes and hyperplanes determined by normals?
- Why do cross products help calculate normals to planes in three dimensions?

Chapter 4

Week 4

4.1 Matrices

4.1.1 Definition of a Matrix

Definition 4.1.1 A **matrix** is a rectangular array of scalars. If the matrix has m rows and n columns, it is called an $m \times n$ matrix. The scalars in the matrix are called the **entries**, **components** or **coefficients**. $\Diamond$

Definition 4.1.2 The rows of a matrix are called the **row vectors** of the matrix. Likewise, the columns are called **column vectors** of the matrix. $\Diamond$

The array of numbers is written in rows and columns, enclosed in either square or round brackets. Here are two ways of writing a particular 3×2 matrix with integer coefficients.

$$\begin{pmatrix} 5 & 6 \\ -4 & -4 \\ 0 & -3 \end{pmatrix} \qquad \begin{bmatrix} 5 & 6 \\ -4 & -4 \\ 0 & -3 \end{bmatrix}$$

The rows of this matrix are the following three vectors.

$$\begin{pmatrix} 5 \\ 6 \end{pmatrix} \qquad \begin{pmatrix} -4 \\ -4 \end{pmatrix} \qquad \begin{pmatrix} 0 \\ -3 \end{pmatrix}$$

The columns of this matrix are the following two vectors.

$$\begin{pmatrix} 5 \\ -4 \\ 0 \end{pmatrix} \qquad \begin{pmatrix} 6 \\ -4 \\ -3 \end{pmatrix}$$

In these notes, I'll use curved brackets for matrices; however, in many texts and books, square brackets are very common. Both notations are conventional and acceptable.

If I wanted to write a general matrix, I can use a double subscript.

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nm} \end{pmatrix}$$

By convention, in an arbitrary matrix entry a_{ij} the first subscript tells us the row and the second subscript tells us the column. For example, a_{64}

is the entry in the sixth row and the fourth column. In the rare occurrence where matrices with more than 10 rows or columns are used, I can separate the indices by commas: $a_{12,15}$ would be in the twelfth row and the fifteenth column. Sometimes I write $A = a_{ij}$ as short-hand for the entire matrix when the size is understood or undetermined.

Definition 4.1.3 A **square matrix** is a matrix with the same number of rows as columns. Here are two examples.

$$\begin{pmatrix} 4 & -2 & 8 \\ -3 & -3 & -3 \\ 0 & 0 & -1 \end{pmatrix} \qquad \begin{pmatrix} 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \end{pmatrix}$$

$\Diamond$

Definition 4.1.4 The **zero matrix** is the unique matrix (one for every size $m \times n$) where all the coefficients are all zero.

$$\begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix} \qquad \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$\Diamond$

Definition 4.1.5 The **diagonal entries** of a matrix are all entries a_{ii} where the row and column indices are the same. A **diagonal matrix** is a matrix where all the non-diagonal entries are zero.

$$\begin{pmatrix} 5 & 0 \\ 0 & 2 \\ 0 & 0 \end{pmatrix} \qquad \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -4 & 0 & 0 \\ 0 & 0 & 8 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$\Diamond$

Definition 4.1.6 The **identity matrix** is the unique $n \times n$ matrix (one for each n) where the diagonal entries are all 1, and all other entries are 0. There are several notations for the identity. When the size n is understood, it can be written as I or Id. When I want to clarify the size of the matrix, I will write I_n for the $n \times n$ identity matrix.

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \qquad \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \qquad \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$\Diamond$

When I defined vectors, I also defined addition term-by-term. I can do the same for matrices.

Definition 4.1.7 If A and B are both $n \times m$ matrices, then the sum $A + B$ is the $n \times m$ matrix formed by simply adding the entries of the two matrices in

their apporpriate position. ◊

The definition of vector addition is extremely important to the geometry of vectors. It turns out that matrix addition is less central to the geometry presented in this course. This operation will show up in a couple of places, but it is much less important that the operation of matrix multiplication, which will be presented later in Section 6.2,

Sometimes it is useful add separation to the organization of the matrix.

Definition 4.1.8 An **extended matrix** is a matrix with a vertical division separating the columns into two groups.

$$\left(\begin{array}{ccc|c} -3 & 0 & 6 & 1 \\ 4 & -2 & 2 & -1 \\ 0 & 0 & 3 & -7 \end{array}\right)$$

◊

Definition 4.1.9 The **transpose** of a matrix is the matrix formed by switching the rows and columns. Alternatively, it is the mirror of the matrix over the diagonal. If $A = a_{ij}$ is a matrix expressed in indices, then the transpose is written A^T and has entries a_{ji} with indices switched.

$$A = \begin{pmatrix} 3 & 0 & 6 & -1 \\ 4 & -2 & -7 & 0 \\ 6 & -6 & -1 & -2 \end{pmatrix} \qquad A^T = \begin{pmatrix} 3 & 4 & 6 \\ 0 & -2 & -6 \\ 6 & -7 & -1 \\ -1 & 0 & -2 \end{pmatrix}$$

$$A = \begin{pmatrix} 0 & 0 & 8 & -5 \\ 0 & 1 & 1 & -2 \\ 0 & 6 & 6 & -1 \\ 0 & 0 & -4 & -4 \end{pmatrix} \qquad A^T = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 6 & 0 \\ 8 & 1 & 6 & -4 \\ -5 & -2 & -1 & -4 \end{pmatrix}$$

◊

Definition 4.1.10 The set of all $n \times m$ matrices with real coefficients is written $M_{n,m}(\mathbb{R})$. For square matrices $(n \times n)$, I simply write $M_n(\mathbb{R})$. If I wanted to change the set of scalars to some other number set S (such as $\mathbb{Z}$), we would write $M_{n,m}(S)$ or $M_n(S)$. ◊

4.2 Solving Systems and Row Reduction

4.2.1 System of Linear Equations

So far, the focus of this course has been geometric. In this section, I change my perspective to look at the algebraic side of linear algebra. A major problem in algebra is solving systems of equations: given several equations in several variables, can I find values for each variable that satisfy all the equations? In general, algebra considers any kind of equation.

Example 4.2.1

$$\begin{aligned} x^2 + y^2 + z^2 &= 3 \\ xy + 2xz - 3yz &= 0 \\ x + y + z &= 3 \end{aligned}$$

This system is solved by $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ When substituted in the equations, these values satisfy all three. No other triples satisfy, so this is a unique solution. $\square$

Solving arbitrary systems of equations is a very difficult problem. There are two initial techniques, which are usually taught in high-school mathematics: isolating and replacing variables; or performing arithmetic with the equations themselves. Both are useful techniques. However, when the equations become quite complicated, both methods can fail. In that case, solutions to systems can only be approached with approximation methods. The methodology of these approximations is a whole branch of mathematics in itself.

I am going to restrict myself to a class of systems that are more approachable: systems of linear equations. Recall the definition of a linear equation from Definition 3.2.2.

Definition 4.2.2 Let a_i and c be real numbers. A **linear equation** in the variables $x_1, x_2 \ldots, x_n$ is an equation of the following form.

$$a_1x_1 + a_2x_2 + \ldots + a_nx_n = c$$

$\Diamond$

Given a fixed set of variables $x_1, x_2, \ldots, x_n$, I want to consider systems of linear equations in those variables. While much simpler than the general case, linear systems are a common and useful type of system to consider. Such systems also work well with arithmetic solution techniques since I can add two linear equations together and still have a linear equation.

This turns out to be the key idea: I can do arithmetic with linear equations. Specifically, there are three things I can do with a system of linear equations. These three techniques *preserve the solutions*; that is, they don't alter the values of the variables that solve the system.

- Multiply a single equation by a (non-zero) constant. (Multiplying both sides of the equation, of course).
- Change the order of the equations.
- Add one equation to another.

If I combine the first and third, I could restate the third as: add a multiple of one equation to another. In practice, I often think of the third operation this way.

4.2.2 Matrix Represetation of Systems of Linear Equations

If I worked with the equations directly, I would find, with careful application of the three techniques, I could always solve linear systems. However, the notation becomes cumbersome for large systems. Therefore, matrices are used to encode the information in a nicely organized way.

Consider a general system with m equations in the variables $x_1, x_2, \ldots, x_n$, where a_{ij} and c_i are real numbers.

$$\begin{aligned}
a_{11}x_1 + a_{12}x_2 + \ldots a_{1n}x_n &= c_1 \\
a_{21}x_1 + a_{22}x_2 + \ldots a_{2n}x_n &= c_2 \\
\vdots \qquad & \quad \vdots \\
a_{m1}x_1 + a_{m2}x_2 + \ldots a_{mn}x_n &= c_m
\end{aligned}$$

I encode this as an extended matrix by taking the a_{ij} and the c_i as the matrix coefficients. I drop the x_i, keeping track of them implicitly by their positions in the matrix. The result is a $m \times (n+1)$ extended matrix where the vertical line notes the position of the equals sign in the original equation.

$$\left(\begin{array}{cccc|c} a_{11} & a_{12} & \dots & a_{1n} & c_1 \\ a_{21} & a_{22} & \dots & a_{2n} & c_2 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} & c_1 \end{array}\right)$$

Example 4.2.3

$$\begin{aligned} -x + 3y + 6z &= 1 \\ 2x + 2y + 2z &= -6 \\ 5x - 5y + z &= 0 \end{aligned}$$

I transfer the coefficients into the matrix representation.

$$\left(\begin{array}{ccc|c} -1 & 3 & 6 & 1 \\ 2 & 2 & 2 & -6 \\ 5 & -5 & 1 & 0 \end{array}\right)$$

□

Example 4.2.4

$$\begin{aligned} -3x - 10y + 15z &= -34 \\ 20x - y + 19z &= 25 \\ 32x + 51y - 31z &= 16 \end{aligned}$$

I transfer the coefficients into the matrix representation.

$$\left(\begin{array}{ccc|c} -3 & -10 & 15 & -34 \\ 20 & -1 & 19 & 25 \\ 32 & 51 & -31 & 16 \end{array}\right)$$

□

Example 4.2.5 Sometimes not every equation explicitly mentions each variable.

$$\begin{aligned} x - 2z &= 0 \\ 2y + 3z &= -1 \\ -3x - 4y &= 9 \end{aligned}$$

This system is clarified by adding the extra variables with coefficient 0.

$$\begin{aligned} x + 0y - 2z &= 0 \\ 0x + 2y + 3z &= -1 \\ -3x - 4y + 0z &= 9 \end{aligned}$$

Then it can be clearly encoded as a matrix.

$$\left(\begin{array}{ccc|c} 1 & 0 & -2 & 0 \\ 0 & 2 & 3 & -1 \\ -3 & -4 & 0 & 9 \end{array}\right)$$

□

In this way, I can change any system of linear equations into an extended matrix (with one column after the vertical line) and any such extended matrix into a system of equations. The columns represent the hidden variables. In the examples above, the first column is the x column; the second is the y column; the third is the z column; and the column after the vertical line is the column of constants.

4.2.3 Row Operations and Gaussian Elimination

I defined three operations on systems of equations. Now that I have encoded systems as matrices, I need to understand the equivalent operations for matrices. Each equation in the system gives a row in the matrix; therefore, equation operations become row operations.

- Multiply an equation by a non-zero constant $\implies$ multiply a row by a non-zero constant.
- Change the order of the equations $\implies$ exchange two rows of a matrix.
- Add (a multiple of) an equation to another equation $\implies$ add (a multiple of) one row to another row.

Since the original operations didn't change the solution of a system of equations, the row operations on matrices also preserve the solution of the associated system.

In addition, I need to know what a solution looks like in matrix form.

Example 4.2.6 This example shows the encoding of a direction solution.

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & -3 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 8 \end{array}\right)$$

The system has three columns before its vertical line; therefore it corresponds to a system of equations in x, y, z. If I translate this back into equations, I get the following system.

$$\begin{aligned} x + 0y + 0z &= -3 \\ 0z + y + 0z &= -2 \\ 0x + 0y + z &= 8 \end{aligned}$$

Removing the zero terms gives $x = -3$, $y = -2$ and $z = 8$. This equation delivers its solution directly: no work is necessary. I just read the solution off the page. □

There is a name for the special form of a matrix where I can directly read the solutions of the associated linear system.

Definition 4.2.7 A matrix is a **reduced row-echelon** matrix or is said to be in **reduced row-echelon form** if the following things are true:

- The first non-zero entry in each row is one. (A row entirely of zeros is fine). These entries are called **leading ones**.
- Each leading one is in a column where all the other entries are zero.

◊

As long as an extended matrix is in reduced row-echelon form, I can always directly read off the solution to the system.

Knowing how to recognize solutions in matrix form and knowing the row operations, I can now describe the full matrix-based approach to solving linear systems. First, I translate a system into a matrix. Then, I use row operations (which don't change the solution at all) to turn the matrix into reduced row-echelon form, where I can just read off the solution. Row operations will always be able to accomplish this.

Definition 4.2.8 The process of using row operations to change a matrix into reduced row-echelon form is called **Gaussian elimination** or **row reduction**. This process proceeds in the following steps.

- Choose a row that has a non-zero entry in the first column. (Often, I exchange this row with the first so that I am working on the first row).
- Multiply by a constant to get a leading one in the chosen row.
- Now that there is a leading one, I want to clear all the other entries in the column containing that leading one. Therefore, I add multiples of the row with the leading one to the other rows, one by one, to make those column entries zero.
- The steps so far produce a column with a leading one in the first column and sets all other first column entries to zero.
- Then I proceed to the next column and repeat the process all over again. Any column, which is all zeros is skipped.
- The process stops when every row either starts with a leading one or is a row entirely of zeros.

$\Diamond$

The leading ones are a key feature of the reduced row-echelon form. It will be useful in the future to keep track of the number of these leading ones. Here is a definition that does this.

Definition 4.2.9 The **rank** of a matrix is the number of leading ones in its reduced row-echelon form. $\Diamond$

4.2.4 Examples of Row Reduction

Example 4.2.10 Solve this system with row reduction.

$$\begin{aligned} x + z &= 4 \\ 3x - y + 2z &= 0 \\ y - 5z &= -3 \end{aligned}$$

I first turn the system into a matrix.

$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & 4 \\ 3 & -1 & 2 & 0 \\ 0 & 1 & -5 & -3 \end{array}\right)$$

Then I will row reduce. For the examples here, I'm going to be very explicit and show each little step. (As I do more and more row reductions, this level of detail will be unnecessary.) There is already a leading one in the first row, so I try to clear its column. There is a three below it. To remove that three, I

subtract three times the first row from the second.

$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & 4 \\ 0 & -1 & -1 & -12 \\ 0 & 1 & -5 & -3 \end{array}\right)$$

Then the first column is finished. Now I want a leading one in the second row. To get this, I'll exchange the second and third rows.

$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & 4 \\ 0 & 1 & -5 & -3 \\ 0 & -1 & -1 & -12 \end{array}\right)$$

Then I have a leading one in the second row. I want to clear its column. There is already a zero above, but I need to clear the -1 below. To do that, I add the second row to the third.

$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & 4 \\ 0 & 1 & -5 & -3 \\ 0 & 0 & -6 & -15 \end{array}\right)$$

That finishes the second column. Now I want a leading one in the third row. To get that, I divide the third row by -6.

$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & 4 \\ 0 & 1 & -5 & -3 \\ 0 & 0 & 1 & \frac{5}{2} \end{array}\right)$$

Then I need to clear the column above this leading one. I first subtract the third row from the first row.

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & \frac{3}{2} \\ 0 & 1 & -5 & -3 \\ 0 & 0 & 1 & \frac{5}{2} \end{array}\right)$$

Then I add 5 times the third row to the second.

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & \frac{3}{2} \\ 0 & 1 & 0 & \frac{19}{2} \\ 0 & 0 & 1 & \frac{5}{2} \end{array}\right)$$

The matrix is now in reduced row-echelon form and I can read off the unique solution.

$$x = \frac{3}{2} \qquad y = \frac{19}{2} \qquad z = \frac{5}{2}$$

□

Example 4.2.11 Solve this system with row reduction.

$$\begin{aligned} 2x + y - 3z &= 1 \\ 4x + z &= 7 \\ -3x - 3y + 2z &= 0 \end{aligned}$$

I turn the system to a matrix.

$$\left(\begin{array}{ccc|c} 2 & 1 & -3 & 1 \\ 4 & 0 & 1 & 7 \\ -3 & -3 & 2 & 0 \end{array}\right)$$

I want to make a leading one in the first row, so I divide the first row by 2.

$$\left(\begin{array}{ccc|c} 1 & \frac{1}{2} & \frac{-3}{2} & \frac{1}{2} \\ 4 & 0 & 1 & 7 \\ -3 & -3 & 2 & 0 \end{array}\right)$$

Then I want to clear the column under this leading one. I start by subtracting 4 times the first row from the second row.

$$\left(\begin{array}{ccc|c} 1 & \frac{1}{2} & \frac{-3}{2} & \frac{1}{2} \\ 0 & -2 & 7 & 5 \\ -3 & -3 & 2 & 0 \end{array}\right)$$

Then I add 3 times the first row to the third row.

$$\left(\begin{array}{ccc|c} 1 & \frac{1}{2} & \frac{-3}{2} & \frac{1}{2} \\ 0 & -2 & 7 & 5 \\ 0 & \frac{-3}{2} & \frac{-5}{2} & \frac{3}{2} \end{array}\right)$$

That finishes the first column. Next, I want a leading one in the second row. To get that, I divide the second row by -2.

$$\left(\begin{array}{ccc|c} 1 & \frac{1}{2} & \frac{-3}{2} & \frac{1}{2} \\ 0 & 1 & \frac{-7}{2} & \frac{-5}{2} \\ 0 & \frac{-3}{2} & \frac{-5}{2} & \frac{3}{2} \end{array}\right)$$

Then I want to clear the column above and below this leading one. First, I'll subtract $\frac{1}{2}$ times the second row from the first row.

$$\left(\begin{array}{ccc|c} 1 & 0 & \frac{1}{4} & \frac{7}{4} \\ 0 & 1 & \frac{-7}{2} & \frac{-5}{2} \\ 0 & \frac{-3}{2} & \frac{-5}{2} & \frac{3}{2} \end{array}\right)$$

Then I'll add $\frac{3}{2}$ times the second row to the third row.

$$\left(\begin{array}{ccc|c} 1 & 0 & \frac{1}{4} & \frac{7}{4} \\ 0 & 1 & \frac{-7}{2} & \frac{-5}{2} \\ 0 & 0 & \frac{-31}{4} & \frac{-9}{4} \end{array}\right)$$

That finishes the second column. Now I need a leading one in the third row. To get that, I'll divide the third row by $\frac{-31}{4}$.

$$\left(\begin{array}{ccc|c} 1 & 0 & \frac{1}{4} & \frac{7}{4} \\ 0 & 1 & \frac{-7}{2} & \frac{-5}{2} \\ 0 & 0 & 1 & \frac{9}{31} \end{array}\right)$$

Now I need to clear the column above the leading one. (The arithmetic is starting to get pretty annoying here, but I will press on. I made good use of a computer algebra system to do the fraction arithmetic for me.) First, I subtract $\frac{1}{4}$ times the third row from the first row.

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & \frac{52}{31} \\ 0 & 1 & \frac{-7}{2} & \frac{-5}{2} \\ 0 & 0 & 1 & \frac{9}{31} \end{array}\right)$$

Then I add $\frac{7}{2}$ times the third row to the second row.

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & \frac{52}{31} \\ 0 & 1 & 0 & \frac{-46}{31} \\ 0 & 0 & 1 & \frac{9}{31} \end{array}\right)$$

This is the reduced row echelon form. Now I can read off the solution.

$$x = \frac{52}{31} \qquad y = \frac{-46}{31} \qquad z = \frac{9}{31}$$

□

Example 4.2.12 Solve this system with row reduction.

$$\begin{aligned} 2x - 6y + z &= 9 \\ x + 4y - 2z &= -2 \\ 5x + 6y - 5z &= 3 \end{aligned}$$

I turn the system into a matrix.

$$\left(\begin{array}{ccc|c} 2 & -6 & 1 & 9 \\ 1 & 4 & -2 & -2 \\ 5 & 6 & -5 & 3 \end{array}\right)$$

Then I want to row reduce this. I start by interchanging rows one and two, to get a leading one in the first row.

$$\left(\begin{array}{ccc|c} 1 & 4 & -2 & -2 \\ 2 & -6 & 1 & 9 \\ 5 & 6 & -5 & 3 \end{array}\right)$$

Then I want to clear the column below the leading one. I subtract 2 times the first row from the second.

$$\left(\begin{array}{ccc|c} 1 & 4 & -2 & -2 \\ 0 & -14 & 5 & 13 \\ 5 & 6 & -5 & 3 \end{array}\right)$$

Then I subtract 5 times the first row from the third row.

$$\left(\begin{array}{ccc|c} 1 & 4 & -2 & -2 \\ 0 & -14 & 5 & 13 \\ 0 & -14 & 5 & 13 \end{array}\right)$$

Immediately, I notice something odd. I can subtract the second row from the third to clear it entirely.

$$\left(\begin{array}{ccc|c} 1 & 4 & -2 & -2 \\ 0 & -14 & 5 & 13 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

Then I can keep going, by dividing the second row by -14.

$$\left(\begin{array}{ccc|c} 1 & 4 & -2 & -2 \\ 0 & 1 & \frac{-5}{14} & \frac{-13}{14} \\ 0 & 0 & 0 & 0 \end{array}\right)$$

Then I need to clear above this new leading one. I subtract 4 times the second row from the first row.

$$\left(\begin{array}{ccc|c} 1 & 0 & \frac{-24}{7} & \frac{-40}{7} \\ 0 & 1 & \frac{-5}{14} & \frac{-13}{14} \\ 0 & 0 & 0 & 0 \end{array}\right)$$

This matrix is now in reduced row-echelon form. There is nothing more to be done. However, reading off the solution isn't as obvious in this case. I will go over this kind of solution in the next section of the notes. □

4.3 Description by Parameters

4.3.1 Solution Spaces and Free Parameters

Definition 4.3.1 If I have a system of equations in the variables $x_1, x_2, \ldots, x_n$, the set of values for these variables which satisfy all the equations is called the **solution space** of the system. Since each set of values is a vector, I think of the solution space as a subset of $\mathbb{R}^n$. For linear equations, the solution space will always be an affine subspace. If I have encoded the system of equations in an extended matrix, we can refer to the solution space of the matrix instead of the system.

The term 'solution space' is a conventional term in algebra, since it refers to the solution of some equations. However, geometrically, I haven't actually done anything knew. The set of points that satisfy some number of equation is the locus of those equations. So solutions spaces are just another name for loci. The description of solutions spaces in this section can also be understood as descriptions of loci. ◊

Definition 4.3.2 Since solution spaces are affine, they can be written as $u+L$ where u is a fixed offset vector, and L is a linear space. Since L is linear, it has a basis $v_1, v_2, \ldots v_k$, and any vector in L is a linear combination of the v_i. I can write any and all solutions to the system in the following fashion (where $a_i \in \mathbb{R}$).

$$u + a_1v_1 + a_2v_2 + a_3v_3 + \ldots + a_kv_k$$

In such a description of the solution space, I call the a_i **free parameters**. One of the goals in solving systems is to determine the number of free parameters for each solution space. ◊

I want to show how to go from reduced row echelon matrices to descriptions of solutions spaces by parameters. This is best to do with a series of examples showing all the various posibilities.

Example 4.3.3 Let me start with a matrix that encodes a unique solution.

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & -3 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 8 \end{array}\right)$$

This example is the easiest case: there is a leading one in each row and column left of the horizontal line, and I just read off the solution, $x = -3$, $y = -2$ and $z = 8$. There are no parameters, since the solution is unique. The 'description by parameters' is just the statement of the unique solution. □

Example 4.3.4 I might have a row of zeros in the reduced row-echelon form.

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & -3 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 8 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

This row of zeros doesn't actually change the solution at all: it is still $x = -3$, $y = -2$ and $z = 8$. The last line just corresponds to an equation that says $0x + 0y + 0z = 0$ or just $0 = 0$, which is always true. Rows of zeros arise from redundant equations in the original system. This still have a unique solution and no free parameters. □

Example 4.3.5 There might be a column of zeros. Consider something similar to the previous examples, but now in four variables w, x, y, z. (In that order, so that the first column is the w column, the second is the x column, and so on).

$$\left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & -3 \\ 0 & 1 & 0 & 0 & -2 \\ 0 & 0 & 1 & 0 & 8 \end{array}\right)$$

The translation gives $w = -3$, $x = -2$ and $y = 8$, but the matrix says nothing about z. A column of zeros corresponds to a free variable: any value of z solves the system as long as the other variables are set. This system has one free paramter. A description of the solution by parameters here would show that the first three coefficients are constant, but z can be anything. I can express this as a sum of vectors.

$$\begin{pmatrix} w \\ x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -3 \\ -2 \\ 8 \\ 0 \end{pmatrix} + z \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

□

Example 4.3.6 The columns of free variables need not contain all zeros. The following matrix is still in reduced row-echelon form.

$$\left(\begin{array}{cccc|c} 1 & 0 & 0 & 2 & -3 \\ 0 & 1 & 0 & -1 & -2 \\ 0 & 0 & 1 & -1 & 8 \end{array}\right)$$

Reduced row-echelon form needs a leading one in each non-zero row, and all other entries in a column with a lead one must be zero. This matrix satisfies all the conditions. The fourth column corresponds to the z variable but lacks a leading one. Any column lacking a leading one corresponds to a free variable. Look at the translation of this matrix back into a linear system.

$$\begin{aligned} w + 2z &= -3 \\ x - z &= -2 \\ y - z &= 8 \end{aligned}$$

I move the z terms to the right side.

$$\begin{aligned} w &= -2z - 3 \\ x &= z - 2 \end{aligned}$$

$$\begin{aligned} y &= z + 8 \\ z &= z \end{aligned}$$

I have included the last equation, which is trivial to satisfy, to show how all the terms depend on z. This makes it clear that z is a free parameter. I can write the solution as follows in terms of that free parameter.

$$\begin{pmatrix} w \\ x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -3 \\ -2 \\ 8 \\ 0 \end{pmatrix} + z \begin{pmatrix} -2 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

Any choice of z will solve this system. If I want a specific solution, I take specific values of z. For example, if $z = 1$, these equations give $w = -5, x = -1$ and $y = 9$.

Moreover, this description by parameters describes the solutions as an offset span. The columns of constants is the offset and the free variable gives the span, in this case, just the span of one vector. □

Example 4.3.7 The following matrix is also in reduced row-echelon form (again, using w, x, y and z as variables in that order).

$$\left(\begin{array}{cccc|c} 1 & 0 & 3 & 2 & -3 \\ 0 & 1 & -1 & -1 & -2 \end{array}\right)$$

Here, both the third and the fourth columns have no leading one; therefore, both y and z are free variables. I translate the matrix back into a linear system.

$$\begin{aligned} w &= -3y - 2z - 3 \\ x &= y + z - 2 \\ y &= y \\ z &= z \end{aligned}$$

Then I can write the solution using the free parameters. I can express the four equations as one equation of vectors; each equation is the equation of one of the four components of the vectors, filling in zeros for anything not mentioned in the equations.

$$\begin{pmatrix} w \\ x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -3 \\ 2 \\ 0 \\ 0 \end{pmatrix} + y \begin{pmatrix} -3 \\ 1 \\ 1 \\ 0 \end{pmatrix} + z \begin{pmatrix} -2 \\ 1 \\ 0 \\ 1 \end{pmatrix}$$

Any choice of y and z gives a solution. For example, if $y = 0$ and $z = 1$ then $w = -5$ and $x = -1$ complete a solution. □

Example 4.3.8 I can also have a reduced row-echelon matrix of the following form.

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & -3 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 8 \\ 0 & 0 & 0 & 1 \end{array}\right)$$

The first three rows translate into equations without issue. However, the fourth row translates to $0x + 0y + 0z = 1$ or $0 = 1$. Obviously, this can never be satisfied. Any row which translates to $0 = 1$ leads to a contradiction. A

contradition means that the system has no solutions, since the $0 = 1$ can never be satisfied. □

I will summarize the three possible situations for reduced row-echelon matrices.

- If there is a row that translates to $0 = 1$, then there is a contradiction, and the system has no solutions. This overrides any other information: whether or not there are free variables elsewhere, there are still no solutions.
- If there are no $0 = 1$ rows and all columns left of the vertical line have leading ones; then there is a unique solution. There are no columns representing free variables, and each variable has a specific value that can be ead directly from the matrix.
- If there are no $0 = 1$ rows and there is at least one column left of the vertical line without a leading one; then each such column represents a free variable. The remaining variables can be expressed in terms of the free variables, and any choices of the free variables lead to a solution. There are infinitely many solutions. The solutions space can always be expressed as an offset span.

In particular, note that there are only three basic cases for the number of solutions: none, one, or infinitely many. No linear system has exactly two or exactly three solutions. In the case of infinitely many solutions, the interesting question is the dimension of the solution space. This question, however, is easily answered.

Proposition 4.3.9 *The dimension of a solution space of a linear system of equations is the number of free variables. Equivalently, the dimension of a locus is the number of free varialbes in its solution when expressed as a system of equations.*

4.4 Week 4 Activity

4.4.1 Row Reduction

Activity 4.4.1 Row reduce this matrix.

$$\begin{pmatrix} 1 & -3 & 0 \\ 1 & 0 & 7 \\ 0 & -4 & -2 \end{pmatrix}$$

Solution. I proceed step-by-step through the row reduction algorithm. I'll be very explicit about the row operation in each step. The first row already has a leading one in the first column, so I don't need to create one. When there is a leading one, I want to clear its column. I will start by subtracting row 1 from row 2.

$$\begin{pmatrix} 1 & -3 & 0 \\ 0 & 3 & 7 \\ 0 & -4 & -2 \end{pmatrix}$$

This has cleared the first column other than the leading 1. Now I want to create a leading one in the second column if I can. To get that, I'll divide the second row by 3.

$$\begin{pmatrix} 1 & -3 & 0 \\ 0 & 1 & \frac{7}{3} \\ 0 & -4 & -2 \end{pmatrix}$$

Now I have a leading 1, so I want to clear its column. I'll add 3 times row 2 to row 1 to clear the entry above the leading 1.

$$\begin{pmatrix} 1 & 0 & 7 \\ 0 & 1 & \frac{7}{3} \\ 0 & -4 & -2 \end{pmatrix}$$

Then I'll add 4 times row 2 to row 3 to clear the entry below the leading 1.

$$\begin{pmatrix} 1 & 0 & 7 \\ 0 & 1 & \frac{7}{3} \\ 0 & 0 & \frac{22}{3} \end{pmatrix}$$

That clears the second column. Now I want to create a leading 1 in the third column if I can. I can divide the last row by $\frac{22}{3}$ to make a leading 1.

$$\begin{pmatrix} 1 & 0 & 7 \\ 0 & 1 & \frac{7}{3} \\ 0 & 0 & 1 \end{pmatrix}$$

Then I need to clear the column above this leading 1. I subtract 7 times the third row from the first row.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & \frac{7}{3} \\ 0 & 0 & 1 \end{pmatrix}$$

Finally, I subtract $\frac{7}{3}$ times the third row from the second row.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

This is the row reduced form.

Activity 4.4.2 Row reduce this matrix.

$$\begin{pmatrix} 4 & 4 & 0 & 1 \\ -1 & -1 & 2 & 1 \\ 6 & 6 & -4 & -1 \end{pmatrix}$$

Solution. I proceed step-by-step through the row reduction algorithm. I'll be very explicit about the row operation in each step.

$$\begin{pmatrix} 4 & 4 & 0 & 1 \\ -1 & -1 & 2 & 1 \\ 6 & 6 & -4 & -1 \end{pmatrix}$$

First, I need a leading one in the first column. I could divide the first row by 4, but I'd like to avoid introducing fractions this early if I can. Instead, I'll multiply the second row by (-1) and then'll interchange the first and second row.

$$\begin{pmatrix} 1 & 1 & -2 & -1 \\ 4 & 4 & 0 & 1 \\ 6 & 6 & -4 & -1 \end{pmatrix}$$

Now I have a leading one in the first column. I need to clear the other entries in this column. I'll subtract 4 times the first row from the second row.

$$\begin{pmatrix} 1 & 1 & -2 & -1 \\ 0 & 0 & 8 & 5 \\ 6 & 6 & -4 & -1 \end{pmatrix}$$

Then I'll subtract 6 times the first row from the third row.

$$\begin{pmatrix} 1 & 1 & -2 & -1 \\ 0 & 0 & 8 & 5 \\ 0 & 0 & 8 & 5 \end{pmatrix}$$

Now I've cleared the first column. Next I'll try to make a leading one in the second column. Looking at the matrix, though, this is impossible. The only non-zero entry in the second column is already behind a leading one -- it can never be the first non-zero entry in its row. Since it is impossible to make a leading one in the second column, I move on to the third column. The first thing I notice is that the second row and the third row are the same. So here, I'll subtract the second row from the third row.

$$\begin{pmatrix} 1 & 1 & -2 & -1 \\ 0 & 0 & 8 & 5 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Now I want to make sure the second row has a leading one in the third column and so I divide the second row by 8.

$$\begin{pmatrix} 1 & 1 & -2 & -1 \\ 0 & 0 & 1 & \frac{5}{8} \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Now I want to clear the third column above the leading one. To do so I will add 2 times the second row to the first row.

$$\begin{pmatrix} 1 & 1 & 0 & \frac{1}{4} \\ 0 & 0 & 1 & \frac{5}{8} \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

This is the row reduced form.

Activity 4.4.3 Row reduce this matrix.

$$\begin{pmatrix} 2 & -1 & -3 & 0 \\ -6 & 3 & 5 & 2 \\ -4 & 2 & -2 & 2 \end{pmatrix}$$

Solution. I want a leading one in the first row, so I'll divide the first row by 2. (There is no avoiding fractions in this row reduction.)

$$\begin{pmatrix} 1 & \frac{-1}{2} & \frac{-3}{2} & 0 \\ -6 & 3 & 5 & 2 \\ -4 & 2 & -2 & 2 \end{pmatrix}$$

Now I want to clear the first column under the leading one. I add 6 times the

first row to the second.

$$\begin{pmatrix} 1 & \frac{-1}{2} & \frac{-3}{2} & 0 \\ 0 & 0 & -4 & 2 \\ -4 & 2 & -2 & 2 \end{pmatrix}$$

Then I add 4 times the first row to the third.

$$\begin{pmatrix} 1 & \frac{-1}{2} & \frac{-3}{2} & 0 \\ 0 & 0 & -4 & 2 \\ 0 & 0 & -8 & 2 \end{pmatrix}$$

Now the first column is finished. It is impossible to make a leading one in the second column, so I move to the third column. I'll divide the second row by (-4)

$$\begin{pmatrix} 1 & \frac{-1}{2} & \frac{-3}{2} & 0 \\ 0 & 0 & 1 & \frac{-1}{2} \\ 0 & 0 & -8 & 2 \end{pmatrix}$$

Then I want to clear the third column above and below the leading one. I add $\frac{3}{2}$ times the second row to the first row.

$$\begin{pmatrix} 1 & \frac{-1}{2} & 0 & \frac{-3}{4} \\ 0 & 0 & 1 & \frac{-1}{2} \\ 0 & 0 & -8 & 2 \end{pmatrix}$$

Then I'll add 8 times the second row to the third row.

$$\begin{pmatrix} 1 & \frac{-1}{2} & 0 & \frac{-3}{4} \\ 0 & 0 & 1 & \frac{-1}{2} \\ 0 & 0 & 0 & -2 \end{pmatrix}$$

That finishes the third column, so I move to the fourth. I can make a leading one in the third row. I divide the third row by (-2).

$$\begin{pmatrix} 1 & \frac{-1}{2} & 0 & \frac{-3}{4} \\ 0 & 0 & 1 & \frac{-1}{2} \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Then I want to clear the fourth column. I add $\frac{1}{2}$ times the third row to the second row.

$$\begin{pmatrix} 1 & \frac{-1}{2} & 0 & \frac{-3}{4} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Finally, I add $\frac{3}{4}$ times the third row to the first row.

$$\begin{pmatrix} 1 & \frac{-1}{2} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

This is the reduced form.

Activity 4.4.4 Row reduce this matrix.

$$\begin{pmatrix} 1 & 0 & -4 & 2 & 9 \\ 0 & 3 & -2 & 0 & 7 \\ 7 & 0 & 0 & 2 & -1 \end{pmatrix}$$

Solution. I proceed step-by-step through the row reduction algorithm. I'll be very explicit about the row operation in each step.

$$\begin{pmatrix} 1 & 0 & -4 & 2 & 9 \\ 0 & 3 & -2 & 0 & 7 \\ 7 & 0 & 0 & 2 & -1 \end{pmatrix}$$

There is already a leading one in the first row, and there is already a zero in the second row below it. I need to clear the first entry in the third row, so I subtract 7 times the first row from the third row.

$$\begin{pmatrix} 1 & 0 & -4 & 2 & 9 \\ 0 & 3 & -2 & 0 & 7 \\ 0 & 0 & 28 & -12 & -64 \end{pmatrix}$$

There is one non-zero entry in the second column, so I'll use that for the leading one. I divide the second row by 3.

$$\begin{pmatrix} 1 & 0 & -4 & 2 & 9 \\ 0 & 1 & \frac{-2}{3} & 0 & \frac{7}{3} \\ 0 & 0 & 28 & -12 & -64 \end{pmatrix}$$

That already finishes the second column. Now I work on the third column. To make a leading one, I divide the third row by 28. (The numbers get a bit unwieldy here, but that often happens with row reduction).

$$\begin{pmatrix} 1 & 0 & -4 & 2 & 9 \\ 0 & 1 & \frac{-2}{3} & 0 & \frac{7}{3} \\ 0 & 0 & 1 & \frac{-3}{7} & \frac{-16}{7} \end{pmatrix}$$

Then I need to clear the third column above the leading one. I add $\frac{2}{3}$ times the third row to the second row.

$$\begin{pmatrix} 1 & 0 & -4 & 2 & 9 \\ 0 & 1 & 0 & \frac{-2}{7} & \frac{17}{21} \\ 0 & 0 & 1 & \frac{-3}{7} & \frac{-16}{7} \end{pmatrix}$$

Then I add 4 times the third row to the first row.

$$\begin{pmatrix} 1 & 0 & 0 & \frac{2}{7} & \frac{-1}{7} \\ 0 & 1 & 0 & \frac{-2}{7} & \frac{17}{21} \\ 0 & 0 & 1 & \frac{-3}{7} & \frac{-16}{7} \end{pmatrix}$$

This is the row reduced form.

4.4.2 Solving Linear Systems

Activity 4.4.5 Solve this linear system using a matrix and row reduction. Describe the solution as a locus, as a span, and in a description by free parameters.

$$\begin{aligned} y + 3z &= 0 \\ -2x - 2y + z &= 0 \\ 4x + y - 3z &= 1 \end{aligned}$$

Solution. First, I write the matrix of the system.

$$\left(\begin{array}{ccc|c} 0 & 1 & 3 & 0 \\ -2 & -2 & 1 & 0 \\ 4 & 1 & -3 & 1 \end{array}\right)$$

Then I row reduce this matrix. After the previous activity for row reducing matrices, I'm not going to show all the steps; I'll just give the row reduced form calculated by a computer. You are welcome to do the same for all row reductions for the rest of the course. Here is the row reduced matrix.

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & \frac{7}{16} \\ 0 & 1 & 0 & \frac{-3}{8} \\ 0 & 0 & 1 & \frac{1}{8} \end{array}\right)$$

Now I need to interpret this in the three ways required by the question. The solution as a locus doesn't need any interpretation -- the original set of three equations defined a locus since a locus is nothing but the solutions of some number of equations in the coordinate variables.

A description by parameters looks for free parameters, which are found in columns left of the dividing line without leading ones. Then I can interpret the matrix as a new system, using these free parameters. Here there are no free parameters, and there is a unique solution

$$\begin{aligned} x &= \frac{7}{16} \\ y &= \frac{-3}{8} \\ z &= \frac{1}{8} \end{aligned}$$

Finally, a description as a span is a rewriting of the description by free parameters. For every parameter, we think of the coefficients of that parameter as a vector. The vector of constants becomes the offset vector. Grouping the variables into a vector lets me write the solution as an (offset) span. Since this is a unique solution, there is only the the offset vector and no span.

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \frac{7}{16} \\ \frac{-3}{8} \\ \frac{1}{8} \end{pmatrix}$$

Activity 4.4.6 Solve this linear system using a matrix and row reduction. Describe the solution as a locus, as a span, and in a description by free parameters.

$$\begin{aligned} 4x - y - z &= 5 \\ -3x + 2z &= 1 \\ -2x - y + 3z &= 9 \end{aligned}$$

Solution. First I write the matrix of the system.

$$\left(\begin{array}{ccc|c} 4 & -1 & -1 & 5 \\ -3 & 0 & 2 & 1 \\ -2 & -1 & 3 & 9 \end{array}\right)$$

Then I asked a computer for the row reduced form of the matrix.

$$\left(\begin{array}{ccc|c} 1 & 0 & \frac{-2}{3} & 0 \\ 0 & 1 & \frac{-5}{3} & 0 \\ 0 & 0 & 0 & 1 \end{array}\right)$$

I have a row that translates to $0 = 1$, which is a contradiction. Therefore, this system has no solutions. As there are no solutions, there is no description of those solutions, either as a locus, by parameters, or as a span.

Activity 4.4.7 Solve this linear system using a matrix and row reduction. Describe the solution as a locus, as a span, and in a description by free parameters.

$$\begin{aligned} -x - y + z &= 2 \\ 2x + 3z &= 1 \\ -4y + 7z &= 6 \end{aligned}$$

Solution. First, I write the matrix of the system.

$$\left(\begin{array}{ccc|c} -1 & -1 & 1 & 2 \\ 2 & 0 & 3 & 1 \\ 0 & -4 & 7 & 6 \end{array}\right)$$

Then I asked a computer for the row reduced form of the matrix.

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & \frac{-3}{2} \\ 0 & 1 & 0 & \frac{5}{6} \\ 0 & 0 & 1 & \frac{4}{3} \end{array}\right)$$

Now I need to interpret this in the three ways required by the question. The solution as a locus doesn't need any interpretation -- the original set of three equations defined a locus since a locus is nothing but the solutions of some number of equations in the coordinate variables.

A description by parameters looks for free parameters, which are found in columns left of the dividing line without leading ones. Then I can interpret the matrix as a new system, using these free parameters. Here, there are no free parameters and there is an exact solution.

$$\begin{aligned} x &= \frac{-3}{2} \\ y &= \frac{5}{6} \\ z &= \frac{4}{3} \end{aligned}$$

Finally, description as a span is a rewriting of the description by free parameters. For every parameter, I think of the coefficients of that parameter as a vector. The vector of constants becomes the offset vector. Grouping the variables into a vector lets us write the solution as an (offset) span. Since there is a unique solution, the description only includes the offset vector and no span.

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \frac{-3}{2} \\ \frac{5}{6} \\ \frac{4}{3} \end{pmatrix}$$

Activity 4.4.8 Solve this linear system using a matrix and row reduction. Describe the solution as a locus, as a span, and in a description by free parameters.

$$\begin{aligned} 2x - y + 6z &= 1 \\ 3x - 7z &= 1 \\ y + 4z &= 2 \end{aligned}$$

Solution. First, I write the matrix of the system.

$$\left(\begin{array}{ccc|c} 2 & -1 & 6 & 1 \\ 3 & 0 & -7 & 1 \\ 0 & 1 & 4 & 2 \end{array}\right)$$

Then I asked a computer for the row reduced form of the matrix.

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & \frac{31}{44} \\ 0 & 1 & 0 & \frac{15}{11} \\ 0 & 0 & 1 & \frac{7}{44} \end{array}\right)$$

Now I need to interpret this in the three ways required by the question. The solution as a locus doesn't need any interpretation -- the original set of three equations defined a locus, since a locus is nothing but the solutions of some number of equations in the coordinate variables.

A description by parameters looks for free parameters, which are found in columns left of the dividing line without leading ones. Then I can interpret the matrix as a new system, using these free parameters. Here there are no free parameters and a unique solution.

$$\begin{aligned} x &= \frac{31}{44} \\ y &= \frac{15}{11} \\ z &= \frac{7}{44} \end{aligned}$$

Finally, description as a span is a rewriting of the description by free parameters. For every parameter, we think of the coefficients of that parameter as a vector. The vector of constants becomes the offset vector. Grouping the variables into a vector lets us write the solution as an (offset) span. Since there is a unique solution, there is only the offset vector and no span at all.

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \frac{31}{44} \\ \frac{15}{11} \\ \frac{7}{44} \end{pmatrix}$$

Activity 4.4.9 Solve this linear system using a matrix and row reduction. Describe the solution as a locus, as a span, and in a description by free parameters.

$$\begin{aligned} x - y + 2z &= 0 \\ 2x - 3z &= 1 \\ 4x - 2y + z &= 1 \\ x + 3y - 12z &= 2 \end{aligned}$$

Solution. First, I write the matrix of the system.

$$\left(\begin{array}{ccc|c} 1 & -1 & 2 & 0 \\ 2 & 0 & -3 & 1 \\ 4 & -2 & 1 & 1 \\ 1 & 3 & -12 & 2 \end{array}\right)$$

Then I asked a computer for the row reduced form of the matrix.

$$\left(\begin{array}{ccc|c} 1 & 0 & \frac{-3}{2} & \frac{1}{2} \\ 0 & 1 & \frac{-7}{2} & \frac{1}{2} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

Now I need to interpret this in the three ways required by the question. The solution as a locus doesn't need any interpretation -- the original set of three equations defined a locus, since a locus is nothing but the solutions of some number of equations in the coordinate variables.

A description by parameters looks for free parameters, which are found in columns left of the dividing line without leading ones. Then I can interpret the matrix as a new system, using these free parameters. Here, the third column has no leading ones, so z is a free parameter.

$$\begin{aligned} x &= \frac{1}{2} + \frac{3}{2}z \\ y &= \frac{1}{2} + \frac{7}{2}z \\ z &= z \end{aligned}$$

Finally, a description as a span is a rewriting of the description by free parameters. For every parameter, we think of the coefficients of that parameter as a vector. The vector of constants becomes the offset vector. Grouping the variables into a vector lets me write the solution as an (offset) span.

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \\ 0 \end{pmatrix} + z \begin{pmatrix} \frac{3}{2} \\ \frac{7}{2} \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \\ 0 \end{pmatrix} + \text{Span}\left\{ \begin{pmatrix} \frac{3}{2} \\ \frac{7}{2} \\ 1 \end{pmatrix} \right\}$$

If I want, I can scale the span to clear the fraction (since a span is all multiple, applying any multiple of the vectors in the span has no effect on the span.)

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \\ 0 \end{pmatrix} + \text{Span}\left\{ \begin{pmatrix} 3 \\ 7 \\ 2 \end{pmatrix} \right\}$$

Activity 4.4.10 Solve this linear system using a matrix and row reduction. Describe the solution as a locus, as a span, and in a description by free parameters. In $\mathbb{R}^4$, the order of variables that I am using in this course is w, x, y, z.

$$\begin{aligned} w - y - z &= 4 \\ 5w - x + 2z &= 2 \\ 6x - 3y &= 1 \end{aligned}$$

Solution. First, I write the matrix of the system.

$$\left(\begin{array}{cccc|c} 1 & 0 & -1 & -1 & 4 \\ 5 & -1 & 0 & 2 & 2 \\ 0 & 6 & -3 & 0 & 1 \end{array}\right)$$

Then I asked a computer for the row reduced form of the matrix.

$$\left(\begin{array}{cccc|c} 1 & 0 & 0 & \frac{5}{9} & \frac{1}{27} \\ 0 & 1 & 0 & \frac{7}{9} & \frac{-49}{27} \\ 0 & 0 & 1 & \frac{14}{9} & \frac{-107}{27} \end{array}\right)$$

Now I need to interpret this in the three ways required by the question. The solution as a locus doesn't need any interpretation -- the original set of three equations defined a locus, since a locus is nothing but the solutions of some number of equations in the coordinate variables.

A description by parameters looks for free parameters, which are found in columns left of the dividing line without leading ones. Then I can interpret the matrix as a new system, using these free parameters. Here the fourth column has no leading one, so I can use z as the free parameter.

$$\begin{aligned} w &= \frac{1}{27} - \frac{5}{9}z \\ x &= \frac{-49}{27} - \frac{7}{9}z \\ y &= \frac{-107}{27} - \frac{14}{9}z \\ z &= z \end{aligned}$$

Finally, description as a span is a rewriting of the description by free parameters. For every parameter, I think of the coefficients of that parameter as a vector. The vector of constants becomes the offset vector. Grouping the variables into a vector lets me write the solution as an (offset) span.

$$\begin{pmatrix} w \\ x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \frac{1}{27} \\ \frac{-49}{27} \\ \frac{-107}{27} \\ 0 \end{pmatrix} + z \begin{pmatrix} \frac{-5}{9} \\ \frac{-7}{9} \\ \frac{-14}{9} \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{27} \\ \frac{-49}{27} \\ \frac{-107}{27} \\ 0 \end{pmatrix} + \text{Span}\left\{ \begin{pmatrix} \frac{-5}{9} \\ \frac{-7}{9} \\ \frac{-14}{9} \\ 1 \end{pmatrix} \right\}$$

I can scale the term in the span if I want to clear the denominators.

$$\begin{pmatrix} w \\ x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \frac{1}{27} \\ \frac{-49}{27} \\ \frac{-107}{27} \\ 0 \end{pmatrix} + \text{Span}\left\{ \begin{pmatrix} -5 \\ -7 \\ -14 \\ 9 \end{pmatrix} \right\}$$

Activity 4.4.11 Solve this linear system using a matrix and row reduction. Describe the solution as a locus, as a span, and in a description by free parameters. In $\mathbb{R}^4$, the order of variables that I am using in this course is

$$\begin{aligned} -3w + 2x - z &= 3 \\ x + y - 4z &= 2 \\ 2w + x &= 3 \\ -w + 4x + y - 5z &= 8 \end{aligned}$$

Solution. First, I write the matrix of the system.

$$\left(\begin{array}{cccc|c} -3 & 2 & 0 & -1 & 3 \\ 0 & 1 & 1 & -4 & 2 \\ 2 & 1 & 0 & 0 & 3 \\ -1 & 4 & 1 & -5 & 8 \end{array}\right)$$

Then I asked a computer for the row reduced form of the matrix.

$$\left(\begin{array}{cccc|c} 1 & 0 & 0 & \frac{1}{7} & \frac{3}{7} \\ 0 & 1 & 0 & \frac{-2}{7} & \frac{15}{7} \\ 0 & 0 & 1 & \frac{-26}{7} & \frac{-1}{7} \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

Now I need to interpret this in the three ways required by the question. The solution as a locus doesn't need any interpretation -- the original set of three equations defined a locus, since a locus is nothing but the solutions of some number of equations in the coordinate variables.

A description by parameters looks for free parameters, which are found in columns left of the dividing line without leading ones. Then I can interpret the matrix as a new system, using these free parameters. Since the fourth column has no leading one, I can use z as a free parameter.

$$\begin{aligned} w &= \frac{3}{7} - \frac{1}{7}z \\ x &= \frac{15}{7} + \frac{2}{7}z \\ y &= \frac{-1}{7} - \frac{26}{7}z \\ z &= z \end{aligned}$$

Finally, description as a span is a rewriting of the description by free parameters. For every parameter, I think of the coefficients of that parameter as a vector. The vector of constants becomes the offset vector. Grouping the variables into a vector lets me write the solution as an (offset) span.

$$\begin{pmatrix} w \\ x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \frac{3}{7} \\ \frac{15}{7} \\ \frac{-1}{7} \\ 0 \end{pmatrix} + z \begin{pmatrix} \frac{-1}{7} \\ \frac{2}{7} \\ \frac{-26}{7} \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{3}{7} \\ \frac{15}{7} \\ \frac{-1}{7} \\ 0 \end{pmatrix} + \text{Span}\left\{ \begin{pmatrix} \frac{-1}{7} \\ \frac{2}{7} \\ \frac{-26}{7} \\ 1 \end{pmatrix} \right\}$$

I can scale the vector in the span if I want to clear denominators.

$$\begin{pmatrix} w \\ x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \frac{3}{7} \\ \frac{15}{7} \\ \frac{-1}{7} \\ 0 \end{pmatrix} + \text{Span}\left\{ \begin{pmatrix} -1 \\ 2 \\ -26 \\ 7 \end{pmatrix} \right\}$$

4.4.3 Conceptual Review Questions

- What is a linear system?
- What does a matrix encode a linear system?
- What is row reduction?

- What is description by parameters?

Chapter 5

Week 5

5.1 Dimensions of Spans

5.1.1 Rank

Let me remind you of a definition from Section 4.2 since it will be useful in this section.

Definition 5.1.1 Let A be an $m \times n$ matrix. The **rank** of A is the number of leading ones in its reduced row-echelon form. $\Diamond$

5.1.2 Dimensions of Spans

I can now present techniques to solve the dimension problems presented in Section 3.1 and Section 3.2. For loci and spans, I didn't have a way of determining what information was redundant, hence what the dimensions of the objects should be. Row-reduction of matrices solves this problem.

In this section, I'll present the solution for spans; details for loci follow in Section 5.3. Given some vectors $\{v_1, v_2, \ldots, v_k\}$, what is the dimension of the span? By definition, it is the number of linearly independent vectors in the set. Now I can test for linearly independent vectors.

Proposition 5.1.2 *Let $\{v_1, v_2, \ldots, v_k\}$ be a set of vectors in $\mathbb{R}^n$. If I make these vectors the rows of a matrix A, then the rank of the matrix A is the number of linearly independent vectors in the set. Moreover, if I do the row reduction without exchanging rows (which is always possible), then the vectors which correspond to rows with leading ones in the reduced row-echelon will form a maximal linearly independent set, i.e., a basis. Any vector corresponding to a row without a leading one is a redundant vector in the span. The set is linearly independent if and only if the rank of A is k.*

5.1.3

Throughout this section, all row reductions were calculated by Wolfram Alpha.

Example 5.1.3 Find the dimension of this span.

$$\text{Span}\left\{\begin{pmatrix}5\\-1\\0\end{pmatrix}, \begin{pmatrix}-2\\6\\1\end{pmatrix}, \begin{pmatrix}3\\4\\1\end{pmatrix}\right\}$$

To find the dimensions of the span, I put the vectors of the span into a matrix (as rows, not columns).

$$\begin{pmatrix} 5 & -1 & 0 \\ -2 & 6 & 1 \\ 3 & 4 & 1 \end{pmatrix}$$

This row reduces to the following matrix.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Since there are three leading ones, all three vectors are linearly independent. The rank of the matrix is three, and the dimension of the span is three. □

Example 5.1.4 Find the dimension of this span.

$$\text{Span}\left\{ \begin{pmatrix} 7 \\ -2 \\ -3 \end{pmatrix}, \begin{pmatrix} 2 \\ -2 \\ 5 \end{pmatrix}, \begin{pmatrix} 1 \\ 4 \\ -18 \end{pmatrix} \right\}$$

To find the dimension of the span, I put the vectors of the span into a matrix (as rows, not columns).

$$\begin{pmatrix} 7 & -2 & 3 \\ 2 & -2 & 5 \\ 1 & 4 & -12 \end{pmatrix}$$

This row reduces to the following matrix.

$$\begin{pmatrix} 1 & 0 & \frac{-2}{5} \\ 0 & 1 & \frac{-29}{10} \\ 0 & 0 & 0 \end{pmatrix}$$

There are only two leading ones, so only two of the original vectors were linearly independent. The rank is two and the dimension of the span is two. □

Example 5.1.5 Find the dimension of this span.

$$\text{Span}\left\{ \begin{pmatrix} 4 \\ -1 \\ 0 \\ 2 \\ -8 \\ 3 \end{pmatrix}, \begin{pmatrix} 12 \\ 4 \\ -3 \\ 7 \\ 15 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 5 \\ -2 \\ -2 \\ -14 \\ 3 \end{pmatrix}, \begin{pmatrix} 16 \\ 8 \\ -5 \\ 7 \\ -9 \\ 7 \end{pmatrix}, \begin{pmatrix} 12 \\ -6 \\ 1 \\ 11 \\ 43 \\ -5 \end{pmatrix}, \begin{pmatrix} 8 \\ 8 \\ -4 \\ 0 \\ -44 \\ 12 \end{pmatrix} \right\}$$

To find the dimension of the span, I put the vectors of the span into a matrix (as rows, not columns).

$$\begin{pmatrix} 4 & -1 & 0 & 2 & -8 & 3 \\ 12 & 4 & -3 & 7 & 15 & 1 \\ 0 & 5 & -2 & -2 & -14 & 3 \\ 16 & 8 & -5 & 7 & -9 & 7 \\ 12 & -6 & 1 & 11 & 43 & -5 \\ 8 & 8 & -4 & 0 & -44 & 12 \end{pmatrix}$$

This row reduces to the following matrix.

$$\begin{pmatrix} 1 & 0 & 0 & \frac{-3}{2} & 0 & 7 \\ 0 & 1 & 0 & -8 & 0 & 25 \\ 0 & 0 & 1 & -19 & 0 & 61 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

There are three leading ones, so only three of the original six vectors are linearly independent. The rank of the matrix is three and the dimension of the span is three. □

5.2 Linear Combinations

5.2.1 Writing Linear Combinations

In Section 3.5, I asked you to write some vectors as linear combinations of other vectors. In those calculations, you had to directly solve a system to find the correct coefficients for the linear combinations. In the present section, I want to return to that problem now that I have defined matrices and row reduction as tools. However, I also want to re-interpret the problem.

5.2.2 Bases for Spans

In Subsection 3.1.3, I defined bases for linear subspaces; to recap, a basis is a linearly independent set of vectors which spans the subspace. Since the basis spans the subspace, any vector in the subspace can be written as a linear combination of the vectors in the basis. A basis gives a way of describing a subspace.

If $\{v_1, v_2, v_3\}$ are a basis for a three-dimensional subspace in $\mathbb{R}^n$, then the vectors in this subspace are precisely all the vectors of the form $av_1 + bv_2 + cv_3$ where a, b and c are any real constants. To talk about a specific vector in a subspace, I need to specify the three constants. If I do, I can say that the vector is presenting **in terms of the basis**.

Let me be more concrete and consider the vector $v = \begin{pmatrix} 4 \\ 5 \end{pmatrix}$ in $\mathbb{R}^2$. I can express this as a linear combination of the standard basis.

$$v = 4e_1 + 5e_2$$

However, I can express v in any of the infinitely-many other bases of $\mathbb{R}^2$ as well. Here is one example.

$$v = 1\begin{pmatrix} 4 \\ 0 \end{pmatrix} - 1\begin{pmatrix} 0 \\ -5 \end{pmatrix}$$

Here is yet another example.

$$v = \frac{9}{2}\begin{pmatrix} 1 \\ 1 \end{pmatrix} + \frac{1}{2}\begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

The problem of finding an expression of a vector as a linear combination of other vectors is the problem of expressing a vector in terms of a given basis. Depending on the situation, different bases for the same subspaces are often

useful, and I often want to change the presentation of vectors from one basis to another.

Example 5.2.1 The subspace Span $\left\{ \begin{pmatrix} -2 \\ 1 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ -2 \\ 3 \end{pmatrix} \right\}$ is a plane in $\mathbb{R}^3$. The vector $\begin{pmatrix} -4 \\ -4 \\ 7 \end{pmatrix}$ is a vector on this plane. Write this vector in terms of the given basis for the span.

To do this, I need to find the numbers a and b that satisfy the following:

$$\begin{pmatrix} -4 \\ -4 \\ 7 \end{pmatrix} = a \begin{pmatrix} -2 \\ 1 \\ -1 \end{pmatrix} + b \begin{pmatrix} 0 \\ -2 \\ 3 \end{pmatrix}.$$

I'll write each component of this vector equation separately.

$$\begin{aligned} -4 &= -2a + 0b \\ -4 &= a + (-2)b \\ 7 &= -1a + 3b \end{aligned}$$

This is just a new linear system, with three equations and two variables, a and b. I can solve it using matrices and row reduction. The matrix representing this system is

$$\left(\begin{array}{cc|c} -2 & 0 & -4 \\ 1 & -2 & -4 \\ -1 & 3 & 7 \end{array} \right).$$

This row reduces to

$$\left(\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{array} \right).$$

This is solvable and the solution is $a = 2$ and $b = 3$. This lets me write the vector in terms of the basis of the plane.

$$\begin{pmatrix} -4 \\ -4 \\ 7 \end{pmatrix} = 2 \begin{pmatrix} -2 \\ 1 \\ -1 \end{pmatrix} + 3 \begin{pmatrix} 0 \\ -2 \\ 3 \end{pmatrix}$$

If the system wasn't solvable (if the last row of the matrix turned out to be equivalent to $0 = 1$), then the vector I started with wouldn't have been on the plane. Only vectors on the plane can be expressed in terms of the basis of the plane. □

Example 5.2.2 Take the basis $\left\{ \begin{pmatrix} 4 \\ -1 \\ -1 \end{pmatrix}, \begin{pmatrix} -2 \\ 0 \\ -3 \end{pmatrix}, \begin{pmatrix} 0 \\ 2 \\ 7 \end{pmatrix} \right\}$ for $\mathbb{R}^3$. Express the vector $\begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$ in terms of this basis.

To do this, I need to find constants a, b and c that solve

$$\begin{pmatrix}1\\1\\2\end{pmatrix} = a\begin{pmatrix}4\\-1\\-1\end{pmatrix} + b\begin{pmatrix}-2\\0\\-3\end{pmatrix} + c\begin{pmatrix}0\\2\\7\end{pmatrix}.$$

Written in components, this gives the following linear system.

$$\begin{aligned}3a - 2b + 0c &= 1\\ -a + 0b + 2c &= 1\\ -a + 3b + 7c &= 2\end{aligned}$$

I translate this into a matrix.

$$\left(\begin{array}{ccc|c}3 & -2 & 0 & 1\\ -1 & 0 & 2 & 1\\ -1 & 3 & 7 & 2\end{array}\right)$$

I row reduce this matrix to get the reduced row-echelon form.

$$\left(\begin{array}{ccc|c}1 & 0 & 0 & 0\\ 0 & 1 & 0 & \frac{-1}{2}\\ 0 & 0 & 1 & \frac{1}{2}\end{array}\right)$$

I conclude that $a = 0$, $b = \frac{-1}{2}$ and $c = \frac{1}{2}$, which lets me write the vector in terms of the basis.

$$\begin{pmatrix}1\\1\\2\end{pmatrix} = 0\begin{pmatrix}3\\-1\\-1\end{pmatrix} - \frac{1}{2}\begin{pmatrix}-2\\0\\-3\end{pmatrix} + \frac{1}{2}\begin{pmatrix}0\\2\\7\end{pmatrix}$$

□

5.3 Dimensions of Loci

As with spans, I can use row-reduction of matrices to find the dimension of a locus. In particular, I will be able to determine if any of the equations are redundant. Recall that I described any affine or linear subspace of $\mathbb{R}^n$ as the locus of some finite number of linear equations: the set of points that satisfy a list of equations.

$$\begin{aligned}a_{11}x_1 + a_{12}x_2 + \ldots a_{1n}x_n &= c_1\\ a_{21}x_1 + a_{22}x_2 + \ldots a_{2n}x_n &= c_2\\ &\ldots\\ a_{m1}x_1 + a_{m2}x_2 + \ldots a_{mn}x_n &= c_m\end{aligned}$$

This is just a system of linear equations, which has a corresponding matrix.

$$\left(\begin{array}{cccc|c}a_{11} & a_{12} & \ldots & a_{1n} & c_1\\ a_{21} & a_{22} & \ldots & a_{2n} & c_2\\ \vdots & \vdots & \ldots & \vdots & \vdots\\ a_{11} & a_{12} & \ldots & a_{1n} & c_1\end{array}\right)$$

So, by definition, the locus of a set of linear equations is just the geometric version of the solution space of the system of linear equations. Loci and solution

spaces are exactly the same things; only loci are geometry, and solution spaces are algebra. The dimension of a locus is the same dimension as a solution space of a system. In Section 4.3, the dimension of a solution space was determined by the free variables in the reduced form of the matrix. Let me restate the result.

Proposition 5.3.1 *Consider a locus defined by a set of linear equations. Then the dimension of the locus is the number of free variables in the solution space of the matrix corresponding to the system of the equations. If the matrix contains a row that leads to a contradiction* $0 = 1$*, then the locus is empty (and the dimension is not defined).*

To understand a locus, I just solve the associated system. If it has solutions, I count the free variables: that's the dimension of the locus. I can get even more information from this process. When I row reduce the matrix A corresponding to the linear system, the equations corresponding to rows with leading ones (keeping track of exchanging rows, if necessary) are equations that are necessary to define the locus. Those that end up as rows without leading ones are redundant, and the locus is unchanged if those equations are removed.

If the ambient space is $\mathbb{R}^n$, then the equations have n variables. That is, there are n columns to the left of the vertical line in the extended matrix. If the rank of A is k, and there are no rows that reduce to $0 = 1$, then there will be $n - k$ columns without leading ones, so $n - k$ free variables. The dimension of the locus is the ambient dimension n minus the rank of the matrix k.

This fits the intuition for spans and loci. Spans are built up: their dimension is equal to the rank of an associated matrix since each leading one corresponds to a unique direction in the span, adding to the dimension by one. Loci are restrictions down from the total space: their dimension is the ambient dimension minus the rank of an associated matrix, since each leading one corresponds to a real restriction on the locus, dropping the dimensions by one. In either case, the question of calculating dimension boils down to the rank of an associated matrix.

Example 5.3.2 I'm going to repeat Example 4.3.7 to make the point that nothing new is happening here: I'm just treating loci as linear systems and looking at their solutions spaces. Consider a locus in $\mathbb{R}^4$ defined by two equations.

$$\begin{aligned} w + 3y + 2z &= -3 \\ x - y - z &= -2 \end{aligned}$$

The translation of this into a matrix is precisely the matrix in Example 4.3.7.

$$\left(\begin{array}{cccc|c} 1 & 0 & 3 & 2 & -3 \\ 0 & 1 & -1 & -1 & -2 \end{array}\right)$$

The equation is already in reduced row-echolon form. There are two columns (other than the constant column) without leading ones; therefore y and z are free variables. The solution, in a description with free parameters, is

$$\begin{pmatrix} w \\ x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -3 \\ -2 \\ 0 \\ 0 \end{pmatrix} + y \begin{pmatrix} -3 \\ 1 \\ 1 \\ 0 \end{pmatrix} + z \begin{pmatrix} -2 \\ 1 \\ 0 \\ 1 \end{pmatrix}.$$

This is an affine subspace with dimension 2. The number of variables (4) minus the rank of the matrix (2) also confirms this dimension. □

5.4 Week 5 Activity

5.4.1 Dimensions and Bases of Spans

Activity 5.4.1 Calculate the dimension of this span and give a basis.

$$\text{Span}\left\{\begin{pmatrix}-3\\-6\\1\end{pmatrix},\begin{pmatrix}2\\5\\-3\end{pmatrix},\begin{pmatrix}1\\4\\-5\end{pmatrix}\right\}$$

Solution. I need to determine if there are any redundancies in the span. To do this, I put the vectors in the span as *rows* of a matrix.

$$\begin{pmatrix}-3 & -6 & 1\\2 & 5 & -3\\1 & 4 & -5\end{pmatrix}$$

I row reduce this matrix. I will either row reduce without interchanging rows or keeping track of row exchanges. (If done by computer, as is the usual practice, I'll have to be careful choosing a basis. Because of this, I'll recheck the bases after I find them.)

$$\begin{pmatrix}1 & 0 & \frac{13}{3}\\0 & 1 & \frac{-7}{3}\\0 & 0 & 0\end{pmatrix}$$

Every row that has a leading one is a row that is necessary for the span; every row without a leading one is redundant. Here, there is one vector that is redundant, since there is a row of zeros. There are two leading ones, so the dimension of this span is 2. I can take the vectors that corresponded to rows with leading ones as a basis.

$$\left\{\begin{pmatrix}-3\\-6\\1\end{pmatrix},\begin{pmatrix}2\\5\\-3\end{pmatrix}\right\}$$

I can check with a matrix that these four are independent using the same steps as before (I've not shown that work). I need this check to make sure rows weren't exchanged during row reduction.

Activity 5.4.2 Calculate the dimension of this span and give a basis.

$$\text{Span}\left\{\begin{pmatrix}1\\1\\0\end{pmatrix},\begin{pmatrix}3\\0\\-4\end{pmatrix},\begin{pmatrix}0\\0\\6\end{pmatrix},\begin{pmatrix}-1\\-1\\-1\end{pmatrix}\right\}$$

Solution. I need to determine if there are any redundancies in the span. To do this, I put the vectors in the span as *rows* of a matrix.

$$\begin{pmatrix}1 & 1 & 0\\3 & 0 & -4\\0 & 0 & 6\\-1 & -1 & -1\end{pmatrix}$$

I row reduce this matrix. I will either row reduce without interchanging rows or keeping track of row exchanges. (If done by computer, as is the usual practice,

I'll have to be careful choosing a basis. Because of this, I'll recheck the bases after I find them.)

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

Every row that has a leading one is a row that is necessary for the span; every row without a leading one is redundant. Here, the fourth vector is all zeros, so there is a redundant vector. There are three leading ones, so the dimension of this span is 3. I can take the vectors that corresponded to rows with leading ones as a basis.

$$\left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 3 \\ 0 \\ -4 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 6 \end{pmatrix} \right\}$$

I can check with a matrix that these four are independent using the same steps as before (I've not shown that work). I need this check to make sure rows weren't exchanged during row reduction.

Activity 5.4.3 Calculate the dimension of this span and give a basis.

$$\text{Span} \left\{ \begin{pmatrix} 7 \\ 0 \\ -5 \\ -5 \end{pmatrix}, \begin{pmatrix} 0 \\ -3 \\ -3 \\ 4 \end{pmatrix}, \begin{pmatrix} 7 \\ -3 \\ -8 \\ -1 \end{pmatrix} \right\}$$

Solution. I need to determine if there are any redundancies in the span. To do this, I put the vectors in the span as *rows* of a matrix.

$$\begin{pmatrix} 7 & 0 & -5 & -5 \\ 0 & -3 & -3 & 4 \\ 7 & -3 & -8 & -1 \end{pmatrix}$$

I row reduce this matrix. I will either row reduce without interchanging rows or keeping track of row exchanges. (If done by computer, as is the usual practice, I'll have to be careful choosing a basis. Because of this, I'll recheck the bases after I find them.)

$$\begin{pmatrix} 1 & 0 & \frac{-5}{7} & \frac{-5}{7} \\ 0 & 1 & 1 & \frac{-4}{3} \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Every row that has a leading one is a row that is necessary for the span; every row without a leading one is redundant. Here I see that there is a row of zeros, so there is a redundant vector. There are two leading ones, to the dimension of this span is I can take the vectors that corresponded to rows with leading ones as a basis.

$$\left\{ \begin{pmatrix} 7 \\ 0 \\ -5 \\ -5 \end{pmatrix}, \begin{pmatrix} 0 \\ -3 \\ -3 \\ 4 \end{pmatrix} \right\}$$

I can check with a matrix that these four are independent using the same steps as before (I've not shown that work). I need this check to make sure rows weren't exchanged during row reduction.

Activity 5.4.4 Calculate the dimension of this span and give a basis.

$$\text{Span}\left\{\begin{pmatrix}6\\-1\\-1\\4\\7\end{pmatrix},\begin{pmatrix}0\\4\\3\\0\\-5\end{pmatrix},\begin{pmatrix}-4\\10\\2\\-5\\-8\end{pmatrix},\begin{pmatrix}-8\\-6\\3\\4\\0\end{pmatrix},\begin{pmatrix}-6\\7\\7\\3\\-6\end{pmatrix}\right\}$$

Solution. I need to determine if there are any redundancies in the span. To do this, I put the vectors in the span as *rows* of a matrix.

$$\begin{pmatrix}6&-1&-1&4&7\\0&4&3&0&-5\\-4&10&2&-5&-8\\-8&-6&3&4&0\\-6&7&7&3&-6\end{pmatrix}$$

I row reduce this matrix. I will either row reduce without interchanging rows or keeping track of row exchanges. (If done by computer, as is the usual practice, I'll have to be careful choosing a basis. Because of this, I'll recheck the bases after I find them.)

$$\begin{pmatrix}1&0&0&0&\frac{-1}{4}\\0&1&0&0&\frac{23}{62}\\0&0&1&0&\frac{-67}{31}\\0&0&0&1&\frac{52}{31}\\0&0&0&0&0\end{pmatrix}$$

Every row that has a leading one is a row that is necessary for the span; every row without a leading one is redundant. Here I see that the fifth row is a row of zeros, so one vector is redundant. There are four leading ones, so the dimension of this span is 4. I can take the vectors that corresponded to rows with leading ones as a basis.

$$\left\{\begin{pmatrix}6\\-1\\-1\\4\\7\end{pmatrix},\begin{pmatrix}0\\4\\3\\0\\-5\end{pmatrix},\begin{pmatrix}-4\\10\\2\\-5\\-8\end{pmatrix},\begin{pmatrix}-8\\-6\\3\\4\\0\end{pmatrix}\right\}$$

I can check with a matrix that these four are independent using the same steps as before (I've not shown that work). I need this check to make sure rows weren't exchanged during row reduction.

Activity 5.4.5 Calculate the dimension of this span and give a basis.

$$\text{Span}\left\{\begin{pmatrix}4\\0\\-3\\-3\end{pmatrix},\begin{pmatrix}1\\2\\2\\0\end{pmatrix},\begin{pmatrix}-3\\2\\5\\3\end{pmatrix},\begin{pmatrix}8\\0\\-7\\1\end{pmatrix}\right\}$$

Solution. I need to determine if there are any redundancies in the span. To

do this, I put the vectors in the span as *rows* of a matrix.

$$\begin{pmatrix} 4 & 0 & -3 & -3 \\ 1 & 2 & 2 & 0 \\ -3 & 2 & 5 & 3 \\ 8 & 0 & -7 & 1 \end{pmatrix}$$

I row reduce this matrix. I will either row reduce without interchanging rows or keeping track of row exchanges. (If done by computer, as is the usual practice, I'll have to be careful choosing a basis. Because of this, I'll recheck the bases after I find them.)

$$\begin{pmatrix} 1 & 0 & 0 & -6 \\ 0 & 1 & 0 & 10 \\ 0 & 0 & 1 & -7 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Every row that has a leading one is a row that is necessary for the span; every row without a leading one is redundant. I see that there is a row of zeros, so one vector is redundant. There are three leading ones, so the dimension of this span is 3. I can take the vectors that corresponded to rows with leading ones as a basis.

$$\left\{ \begin{pmatrix} 4 \\ 0 \\ -3 \\ -3 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 2 \\ 0 \end{pmatrix}, \begin{pmatrix} -3 \\ 2 \\ 5 \\ 3 \end{pmatrix} \right\}$$

If I check, however, I find that this is still redundant. This is not a basis, so one of these rows must have been redundant and the row reduction must have involved some exchange of rows. If I take the fourth vector instead of the third, I get this basis.

$$\left\{ \begin{pmatrix} 4 \\ 0 \\ -3 \\ -3 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 8 \\ 0 \\ -7 \\ 1 \end{pmatrix} \right\}$$

If I check, I will find that there are no redundancies now, so this is a basis.

5.4.2 Dimensions and Bases of Loci

Activity 5.4.6 Calculate the dimension of this locus.

$$\begin{aligned} 4x - y + 7z &= 3 \\ 2x + z &= 0 \\ 2x - y + 6z &= 3 \end{aligned}$$

Solution. I change the locus into a matrix.

$$\left(\begin{array}{ccc|c} 4 & -1 & 7 & 3 \\ 2 & 0 & 1 & 0 \\ 2 & -1 & 6 & 3 \end{array} \right)$$

I row reduce this matrix.

$$\left(\begin{array}{ccc|c} 1 & 0 & \frac{1}{2} & 0 \\ 0 & 1 & -5 & -3 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

Then the dimension is determined by the number of free variables, as long as there isn't a row equivalent to $0 = 1$. Here, there is one free variable, so the locus has dimension 1.

Activity 5.4.7 Calculate the dimension of this locus.

$$\begin{aligned} x - y - z &= 7 \\ 2x + 5y - 3z &= 2 \\ 3x + 4y - 4z &= 4 \end{aligned}$$

Solution. I change the locus into a matrix.

$$\left(\begin{array}{ccc|c} 1 & -1 & -1 & 7 \\ 2 & 5 & -3 & 2 \\ 3 & 4 & -4 & 4 \end{array}\right)$$

I row reduce this matrix.

$$\left(\begin{array}{ccc|c} 1 & 0 & \frac{-8}{7} & 0 \\ 0 & 1 & \frac{-1}{7} & 0 \\ 0 & 0 & 0 & 1 \end{array}\right)$$

Then the dimension is determined by the number of free variables, as long as there isn't a row equivalent to $0 = 1$. Here, the last row is such a row, so this locus is empty and has no dimension. (Note this is not dimension 0, which is the dimension of a point. The empty set simply doesn't have any dimension.)

Activity 5.4.8 Calculate the dimension of this locus. (In this course, the order of the variables in $\mathbb{R}^4$ is w, x, y, z.)

$$\begin{aligned} 9w - 3x + 5y - z &= 0 \\ 3w - 12x + 4y &= 0 \\ 3w + 21x - 3y - z &= 0 \end{aligned}$$

Solution. I change the locus into a matrix.

$$\left(\begin{array}{cccc|c} 9 & -3 & 5 & -1 & 0 \\ 3 & -12 & 4 & 0 & 0 \\ 3 & 21 & -3 & -1 & 0 \end{array}\right)$$

I row reduce this matrix.

$$\left(\begin{array}{cccc|c} 1 & 0 & \frac{16}{33} & \frac{-4}{33} & 0 \\ 0 & 1 & \frac{-7}{33} & \frac{-1}{33} & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

Then the dimension is determined by the number of free variables, as long as there isn't a row equivalent to $0 = 1$. Here, there are two free variables, so the locus has dimension 2.

Activity 5.4.9 Calculate the dimension of this locus. (In this course, the order of the variables in $\mathbb{R}^4$ is w, x, y, z.)

$$\begin{aligned} w - 2x + y - 3z &= 0 \\ 2w - 4x + 2y - 6z &= 0 \\ 3w - 6x + 3y - 9z &= 0 \end{aligned}$$

Solution. I change the locus into a matrix.

$$\left(\begin{array}{cccc|c} 1 & -2 & 1 & -3 & 0 \\ 2 & -4 & 2 & -6 & 0 \\ 3 & -6 & 3 & -9 & 0 \end{array}\right)$$

I row reduce this matrix.

$$\left(\begin{array}{cccc|c} 1 & -2 & 1 & -3 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

Then the dimension is determined by the number of free variables, as long as there isn't a row equivalent to $0 = 1$. Here, there are three free variables, so the dimension of the locus is three.

5.4.3 Calculating Linear Combinations

Activity 5.4.10 Write the first vector as a linear combination of the vectors in the span.

$$\begin{pmatrix} 2 \\ -2 \end{pmatrix} \qquad \text{Span}\left\{\begin{pmatrix} 4 \\ 9 \end{pmatrix}, \begin{pmatrix} -3 \\ 12 \end{pmatrix}\right\}$$

Solution. I can convert into an equation with unknowns by writing out the definition of a linear combination.

$$\begin{pmatrix} 2 \\ -2 \end{pmatrix} = a\begin{pmatrix} 4 \\ 9 \end{pmatrix} + b\begin{pmatrix} -3 \\ 12 \end{pmatrix}$$

Then I can translate this into a system of equation in the unknowns a, b.

$$\begin{aligned} 2 &= 4a - 3b \\ -2 &= 9a + 12b \end{aligned}$$

Then I can translate this into a matrix.

$$\left(\begin{array}{cc|c} 4 & -3 & 2 \\ 9 & 12 & -2 \end{array}\right)$$

Then I row reduce this matrix.

$$\left(\begin{array}{cc|c} 1 & 0 & \frac{6}{25} \\ 0 & 1 & \frac{-26}{75} \end{array}\right)$$

This has a unique solution, which gives me the coefficient of the linear combination. Finally, I can write the original vector as a linear combination of the spanning vectors.

$$\begin{pmatrix} 2 \\ -2 \end{pmatrix} = \frac{6}{25}\begin{pmatrix} 4 \\ 9 \end{pmatrix} + \frac{-26}{75}\begin{pmatrix} -3 \\ 12 \end{pmatrix}$$

Activity 5.4.11 Write the first vector as a linear combination of the vectors in the span.

$$\begin{pmatrix} 1 \\ 0 \\ 4 \end{pmatrix} \qquad \text{Span}\left\{\begin{pmatrix} 4 \\ 0 \\ 13 \end{pmatrix}, \begin{pmatrix} -1 \\ -1 \\ 3 \end{pmatrix}, \begin{pmatrix} 0 \\ -5 \\ 4 \end{pmatrix}\right\}$$

Solution. I can convert into an equation with unknown by writing out the definition of a linear combination.

$$\begin{pmatrix}1\\0\\4\end{pmatrix} = a\begin{pmatrix}4\\0\\13\end{pmatrix} + b\begin{pmatrix}-1\\-1\\3\end{pmatrix} + c\begin{pmatrix}0\\-5\\4\end{pmatrix}$$

Then I can translate this into a system of equation in the unknowns a, b.

$$\begin{aligned}1 &= 4a - b\\0 &= -b - 5c\\4 &= 13a + 3b + 4c\end{aligned}$$

Then I can translate this into a matrix.

$$\left(\begin{array}{ccc|c}4 & -1 & 0 & 1\\0 & -1 & -5 & 0\\13 & 3 & 4 & 4\end{array}\right)$$

Then I row reduce this matrix.

$$\left(\begin{array}{ccc|c}1 & 0 & 0 & \frac{31}{109}\\0 & 1 & 0 & \frac{15}{109}\\0 & 0 & 1 & \frac{-3}{109}\end{array}\right)$$

This has a unique solution, which gives me the coefficient of the linear combination. Finally, I can write the original vector as a linear combination of the spanning vectors.

$$\begin{pmatrix}1\\0\\4\end{pmatrix} = \frac{31}{109}\begin{pmatrix}4\\0\\13\end{pmatrix} + \frac{15}{109}\begin{pmatrix}-1\\-1\\3\end{pmatrix} + \frac{-3}{109}\begin{pmatrix}0\\-5\\4\end{pmatrix}$$

Activity 5.4.12 Write the first vector as a linear combination of the vectors in the span.

$$\begin{pmatrix}-6\\9\\-25\\16\end{pmatrix} \qquad \text{Span}\left\{\begin{pmatrix}0\\5\\-3\\8\end{pmatrix}, \begin{pmatrix}-1\\-1\\4\\3\end{pmatrix}, \begin{pmatrix}-5\\-5\\0\\2\end{pmatrix}\right\}$$

Solution. I can convert into an equation with unknown by writing out the definition of a linear combination.

$$\begin{pmatrix}-6\\9\\-25\\16\end{pmatrix} = a\begin{pmatrix}0\\5\\-3\\8\end{pmatrix} + b\begin{pmatrix}-1\\-1\\4\\3\end{pmatrix} c\begin{pmatrix}-5\\-5\\0\\2\end{pmatrix}$$

Then I can translate this into a system of equation in the unknowns a, b.

$$\begin{aligned}-6 &= -b - 5c\\9 &= 5a - b - 5c\\-25 &= -3a + 4b\\16 &= 8a + 3b + 2c\end{aligned}$$

Then I can translate this into a matrix.

$$\left(\begin{array}{ccc|c} 0 & -1 & -5 & -6 \\ 5 & -1 & -5 & 9 \\ -3 & 4 & 0 & -25 \\ 8 & 3 & 2 & 16 \end{array}\right)$$

Then I row reduce this matrix.

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & -4 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

This has a unique solution, which gives me the coefficient of the linear combination. Finally, I can write the original vector as a linear combination of the spanning vectors.

$$\begin{pmatrix} -6 \\ 9 \\ -25 \\ 16 \end{pmatrix} = 3 \begin{pmatrix} 0 \\ 5 \\ -3 \\ 8 \end{pmatrix} - 4 \begin{pmatrix} -1 \\ -1 \\ 4 \\ 3 \end{pmatrix} + 2 \begin{pmatrix} -5 \\ -5 \\ 0 \\ 2 \end{pmatrix}$$

5.4.4 Conceptual Review Questions

- What are dimensions?
- How are spans “built up” but loci are “restricted down”?

Chapter 6

Week 6

6.1 Linear Transformations

6.1.1 Definitions

After a good definition of the environment ($\mathbb{R}^n$) and its objects (lines, planes, hyperplanes, etc.), the next mathematical step is to understand the functions that live in the environment and affect its objects. First, I need to generalize the simple notion of a function to linear spaces. Algebra and calculus are concerned with functions of real numbers. These functions are rules $f : A \to B$ which go between subsets of real numbers. The function f assigns to each *number* in A a unique *number* in B. These functions include the very familiar $f(x) = x^2$, $f(x) = \sin(x)$, $f(x) = e^x$ and many others.

Definition 6.1.1 Let A and B be subsets of $\mathbb{R}^n$ and $\mathbb{R}^m$, respectively. A **function** between linear spaces is a rule $f : A \to B$, which assigns to each *vector* in A a unique *vector* in B. ◊

Example 6.1.2 I can define a function $f : \mathbb{R}^3 \to \mathbb{R}^3$ by $f\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x^2 \\ y^2 \\ z^2 \end{pmatrix}$. □

Example 6.1.3 Another function $f : \mathbb{R}^3 \to \mathbb{R}^2$ could be $f\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x - y \\ z - y \end{pmatrix}$. □

Definition 6.1.4 A **linear function** or **linear transformation** from $\mathbb{R}^n$ to $\mathbb{R}^m$ is a function $f : \mathbb{R}^n \to \mathbb{R}^m$ such that for two vectors, $u, v \in \mathbb{R}^n$ and any scalar $a \in \mathbb{R}$, the function must obey two rules.

$$\begin{aligned} f(u + v) &= f(u) + f(v) \\ f(au) &= af(u) \end{aligned}$$

Often I will say that the function **respects** the two main operations on linear spaces: addition of vectors and multiplication by scalars. If I perform addition before or after the function, I get the same result. Likewise for scalar multiplication. ◊

By inspection and testing, I could determine that Example 6.1.2 example above fails these two rules, but Example 6.1.3 satisfies them. Therefore, the later is a linear transformation but the former is not.

This definition creates a restrictive but important class of linear functions. I could easily define linear algebra as a study of these transformations. The defintiion gives so far is an algebraic definition; however, there is also an elegant geometric description of these functions.

Proposition 6.1.5 *A function $f : \mathbb{R}^n \to \mathbb{R}^m$ is linear if and only if it sends linear objects to linear objects.*

Under a linear transformation, points, lines, planes and etc. are changed to other points, lines, planes and etc. A line can't be bent into an arc or broken into two different lines. Hopefully, some of the major ideas of the course are starting to fit together: the two basic operations of addition and scalar multiplication give rise to spans, which are flat objects. Linear transformations preserve those operations, so they preserve flat objects. Exactly *how* they change these objects can be tricky to determine.

Lastly, because of scalar multiplication, if I take $a = 0$, I find that $f(0) = 0$. Under a linear transformation, the origin is always sent to the origin. So, in addition to preserving flat objects, linear transformations can't move the origin. I could drop this condition of preserving the origin to get another class of functions.

Definition 6.1.6 An **affine transformation** from $\mathbb{R}^n$ to $\mathbb{R}^m$ is a transformation that preserves affine subspaces. These transformations preserve flat objects but may move the origin. $\Diamond$

Though they are interesting, I don't spend much time with affine transformations. They can always be realized as a linear transformation combined with a shift or displacement of the entire space by a fixed vector. Since shifts are relatively simple, I can usually reduce problems of affine transformations to problems of linear transformations.

6.1.2 Composition

Composition is a very important idea for the study of real-valued functions in calculus. It likewise is very important for linear transformations.

Definition 6.1.7 Let $f : \mathbb{R}^n \to \mathbb{R}^m$ and $g : \mathbb{R}^m \to \mathbb{R}^l$ be linear transformations. Then $g \circ f : \mathbb{R}^n \to \mathbb{R}^l$ is the linear transformation formed by first applying f and then g. Note that the $\mathbb{R}^m$ has to match: f outputs to $\mathbb{R}^m$, which is the input for g. Also note that the notation is written right-to-left: In $g \circ f$, the transformation f happens first, followed by g. This new transformation is called the **composition** of f and g. $\Diamond$

6.1.3 Symmetry

Notice that I've defined linear functions by the objects and/or properties they preserve. This is a very general technique in mathematics. Very frequently, functions are classified by what they preserve. As discussed in the very first chapter, I use the word 'symmetry' to describe this perspective: the symmetries of a function are the objects or algebraic properties preserved by the function. A function exhibits more symmetry if it preserves more objects or more properties. The conventional use of symmetry in English relates more to a shape than a function: what are the symmetries of a hexagon? I can connect the two ideas: asking for the symmetries of the hexagon can be thought of as asking for the transformations of $\mathbb{R}^2$ that preserve a hexagon. This is a bit of a reverse: the standard usage of the word talks about transformation as the symmetries of a shape. Here we start with a transformation and talk about

the shape as a symmetry of the transformation: the hexagon is a symmetry of rotation by one-sixth of a full turn.

6.2 Matrices and Transformations

6.2.1 Matrix Representation of Linear Transformations

Now I come to the second major application of matrices. In addition to succinctly encoding linear systems, matrices can also be used very efficiently to encode linear transformations. This is done by defining how a matrix can act on a vector.

Definition 6.2.1 Let $A = a_{ij}$ be a $m \times n$ matrix and let v be a vector in $\mathbb{R}^n$. There is an action of A on v, written Av, which defines a new vector in $\mathbb{R}^m$. That action is given in the following formula.

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix} = \begin{pmatrix} a_{11}v_1 + a_{12}v_2 + \ldots + a_{1n}v_n \\ a_{21}v_1 + a_{22}v_2 + \ldots + a_{2n}v_n \\ \vdots \\ a_{n1}v_1 + a_{n2}v_2 + \ldots + a_{nn}v_n \end{pmatrix}$$

This is a bit troubling to work out in general. Let me show what it looks like slightly more concretely in $\mathbb{R}^2$ and $\mathbb{R}^3$.

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} ax + by \\ cx + dy \end{pmatrix}$$

$$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} ax + by + cz \\ dx + ey + fz \\ gx + hy + iz \end{pmatrix}$$

In this way, all $m \times n$ matrices determine a method of sending vectors in $\mathbb{R}^n$ to $\mathbb{R}^m$: a function $\mathbb{R}^n \to \mathbb{R}^m$. It is not at all obvious from the definition, but matrices completely describe all linear transformations. $\Diamond$

Proposition 6.2.2 *If A is an $m \times n$ matrix, then the associated function defined by the matrix action is a linear function $\mathbb{R}^n \to \mathbb{R}^m$. Moreover, all linear functions $\mathbb{R}^n \to \mathbb{R}^m$ can be encoded this way. Finally, each linear function is encoded uniquely, i.e., each $m \times n$ matrix corresponds to a different transformation.*

In this way, the set of linear transformations $\mathbb{R}^n \to \mathbb{R}^m$ is *exactly* the same as the set of $m \times n$ matrices. This is a very powerful result: in order to understand linear transformations of Euclidean space, I only have to understand matrices and their properties.

I'll start with some important examples, using 3×3 matrices for transformations from $\mathbb{R}^3$ to itself.

Example 6.2.3 Let me interpret the matrix action of the zero matrix (all coefficient are zero).

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0x + 0y + 0z \\ 0x + 0y + 0z \\ 0x + 0y + 0z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

The zero matrix corresponds to the transformation that sends all vectors to the origin. $\square$

Example 6.2.4 Let me also interpret the matrix action of the identity matrix.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1x + 0y + 0z \\ 0x + 1y + 0z \\ 0x + 0y + 1z \end{pmatrix} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

The identity matrix corresponds to the transformation which doesn't change anything. Appropriately, we called this the identity transformation. □

Example 6.2.5 I can also interpret a more general diagonal matrix. Let $a, b, c \in \mathbb{R}$ be scalars.

$$\begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} ax + 0y + 0z \\ 0x + by + 0z \\ 0x + 0y + cz \end{pmatrix} = \begin{pmatrix} ax \\ by \\ cz \end{pmatrix}$$

This is a dilation: the x direction is stretched by the factor a, the y direction by the factor b and the z direction by the factor c. (If any of the three constants are zero, that entire axis direction is collapsed). Diagonal matrices are dilations in the axis directions, with the possibility of completely collapsing an axis as well. □

6.2.2 Composition and Matrix Multiplication

In Subsection 6.1.2, I defined the composition of linear transformations. Composition allows me to combine transformations. However, since matrices represent transformations, this composition should somehow be accounted for in the matrix representation. If A is the matrix of the transformation S and B is the matrix of the transformation T, what is the matrix of $S \circ T$? The answer is given by matrix multiplication.

Definition 6.2.6 Let A be a $k \times m$ matrix and B an $m \times n$ matrix. I can think of the **rows** of A as vectors in $\mathbb{R}^m$, and the **columns** of B as vectors in $\mathbb{R}^m$ as well. To emphasize this perspective, I write the following, using u_i for the rows of A and v_i for the columns of B.

$$A = \begin{pmatrix} \rightarrow & u_1 & \rightarrow \\ \rightarrow & u_2 & \rightarrow \\ \rightarrow & u_3 & \rightarrow \\ \vdots & \vdots & \vdots \\ \rightarrow & u_k & \rightarrow \end{pmatrix} \qquad B = \begin{pmatrix} \downarrow & \downarrow & \downarrow & \dots & \downarrow \\ v_1 & v_2 & v_3 & \dots & v_n \\ \downarrow & \downarrow & \downarrow & \dots & \downarrow \end{pmatrix}$$

With this notation, the **matrix multiplication** of A and B is the $k \times n$ matrix where the entries are the dot products of rows and columns.

$$AB = \begin{pmatrix} u_1 \cdot v_1 & u_1 \cdot v_2 & u_1 \cdot v_3 & \dots & u_1 \cdot v_n \\ u_2 \cdot v_1 & u_2 \cdot v_2 & u_2 \cdot v_3 & \dots & u_2 \cdot v_n \\ u_3 \cdot v_1 & u_3 \cdot v_2 & u_3 \cdot v_3 & \dots & u_3 \cdot v_n \\ \vdots & \vdots & \vdots & \dots & \vdots \\ u_k \cdot v_1 & u_k \cdot v_2 & u_k \cdot v_3 & \dots & u_k \cdot v_n \end{pmatrix}$$

◊

This operation has the desired property: the product of the matrices repesents the composition of the transformations. (This remarkable fact is pre-

sented here without proof; I'll leave it to you to wonder why this weird combination of dot products has the desired geometric interpretation.) Remember that the composition still works from right to left so that the matrix multiplication AB represents the transformation associated with B first, followed by the transformation associated with A. When I defined matrices acting on vectors, I wrote the action on the right: Av. Now when a composition acts, as in ABv, the closest matrix gets to act first.

I have defined a new algebraic operation. As with the new products for vectors (dot and cross), I want to know the properties of this new operation.

Proposition 6.2.7 *Let A be a $k \times l$ matrix, B an $l \times m$ matrices, and C an $m \times n$ matrix. Also, let I_i be the identity matrix in $\mathbb{R}^i$ for any $i \in \mathbb{N}$. Then, there are two important properties of matrix multiplication I want to state at the start.*

$$A(BC) = (AB)C \qquad \textit{Associative}$$
$$AI_l = I_k A = A \qquad \textit{Identity}$$

Note that commutativity is not on this list. In general, $AB \neq BA$. In fact, if A and B are not square matrices, if AB is defined then BA will not be, since the indices will not match. Not only am I unable to exchange the order of matrix multiplication, sometimes that multiplication doesn't even make sense as an operation. Matrix multiplication is a very important example of a non-commutative product.

6.3 Examples in $\mathbb{R}^2$

6.3.1 Five Types

It is useful to look at transformations $\mathbb{R}^2 \to \mathbb{R}^2$ in order to build experience and intuition. As I described earlier, linear transformations preserve flat objects. What can be done to $\mathbb{R}^2$ that preserves lines and preserves the origin?

Proposition 6.3.1 *There are five basic types of linear transformation of $\mathbb{R}^2$.*

- *Rotations* about the origin (either clockwise or counter-clockwise) preserve lines. I can't rotate around any other point since that would move the origin. Since I can choose any angle, there are infinitely many such rotations. Also, since rotating by θ radians clockwise is the same as $2\pi - \theta$ radians counter-clockwise, I typically choose to only deal with counter-clockwise rotations. Counter-clockwise rotations are considered positive and clockwise rotations are considered negative.

- *Reflections* over lines through the origin preserve lines. The line of reflection must pass through the origin or else the reflection will move the origin. Since there are infinitely many different lines through the origin, there are infinitely many such reflections.

- *Skews* are a little trickier to visualize. A skew is a transformation that takes either vertical or horizontal lines (but not both) and tilts them diagonally. It changes squares into parallelograms. The tilted lines are still lines, so it is a linear transformation.

- *Dilations* are transformations which stretch or shrink in various directions.

- *Projections* are transformations that collapse $\mathbb{R}^2$ down to a line through the origin. Two important examples are the projections onto either axis. Projection onto the x axis sends a point (a, b) to $(a, 0)$, removing the y component entirely. Likewise, projection onto the y axis sends a point (a, b) to $(0, b)$, removing the x component. In a similar manner, I can project onto any line through the origin by sending each point to the closest point on the line. Finally, there is the projection to the origin which sends all points to the origin.

I present the following interesting theorem without proof.

Theorem 6.3.2 *All linear transformations of $\mathbb{R}^2$ are generated by the composition of transformations of the previous five types.*

6.3.2 Matrices of the Five Types

The rotation in $\mathbb{R}^2$ by the angle θ (counter-clockwise) around the origin given by the matrix

$$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

Following this formula, a quarter turn is

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Likewise, a half turn is

$$\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Finally, a three-quarter turn is

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

If $\begin{pmatrix} a \\ b \end{pmatrix}$ is a *unit* vector in $\mathbb{R}^2$, then the reflection over the line in the direction of $\begin{pmatrix} a \\ b \end{pmatrix}$ is given by the matrix

$$\begin{pmatrix} a^2 - b^2 & 2ab \\ 2ab & b^2 - a^2 \end{pmatrix}$$

Reflection over the x axis is

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Reflection over the y axis is

$$\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Reflection over the the line $y = x$ is

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Reflection over the line $y = -x$ is

$$\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}.$$

There isn't necessarily an easily describable form for all skews. However, I can still give some examples. A horizontal skew that moves the positive y axis in the positive x direction is

$$\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

Similarly, a skew in the positive y direction is

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Dialations have two stretch factors: a in the x direction and b in the y direction. The matrix for this is

$$\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}.$$

Again, let $\begin{pmatrix} a \\ b \end{pmatrix}$ be a *unit* vector. This unit vector defines a line through the origin by taking all multiples of this vector. The projection onto this line is given by the matrix

$$\begin{pmatrix} a^2 & ab \\ ab & b^2 \end{pmatrix}.$$

In addition, there is one other projection: the projection to the origin. This sends everything to zero and its matrix is the zero matrix.

6.4 Examples in $\mathbb{R}^3$

6.4.1 Examples in $\mathbb{R}^3$

I would also like to talk about transformations of $\mathbb{R}^3$, though it is not as easy to give a complete account of the basic types. However, all of the types listed for $\mathbb{R}^2$ generalize.

- Rotations in $\mathbb{R}^3$ are no longer about the origin. Instead, I have to choose an axis. Any line through the origin will do for an axis of rotation. Any rotation in $\mathbb{R}^3$ is determined by an axis of rotation and an angle of rotation about that axis.
- Reflections are also altered: instead of reflecting over a line, I have to reflect over a plane through the origin. Any plane through the origin determines a reflection.
- Skews are similarly defined: one or two directions are fixed, and the remaining directions are tilted.
- Dilations are also similar, though there are three axis directions in which to stretch or compress.
- Like $\mathbb{R}^2$, I can project onto the origin, sending everything to zero, or onto a line, sending every point to the closest point on a line. Examples include projection onto the axes. However, I can also project onto planes. Sending $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ to $\begin{pmatrix} a \\ b \\ 0 \end{pmatrix}$, for example, removes the z coordinate; this is projection onto the xy plane.

In addition, other types of transformations exist. One example is the 'reflection through the origin': this transformation multiplies all vectors by (-1), sending them to the opposite point on the other side of the origin. In $\mathbb{R}^2$, this turned out to be the same as the rotation by half a turn. However, in $\mathbb{R}^3$, this is neither a rotation nor a reflection.

6.5 Proofs - Matrix Actions

6.5.1 General Properties of the Matrix Action

I have defined linear transformations. At this point, I could state and try to prove a variety of properties about linear transformations and their matrix representations. Most obviously, perhaps, I could prove that the matrix action is actually a linear transformation, as I stated in Section 6.2, and the converse that all linear transformations can be represented as matrices. I could also prove the two properties in Proposition 6.2.7. I am opting not to present these general proofs. They are not conceptually difficult, but to do them, in general, is notationally difficult. It involves a lot of matrix coefficients, vector coefficients and many subscripts. To include those proofs wasn't useful for building up experience with proofs (other than perhaps to show that notation is a major challenge in writing mathematical proof).

6.5.2 Properties of Transformations in $\mathbb{R}^2$

Instead of proving the most general statements about matrix transformations, I'll prove some more specific statements about the types of transformations I defined in Section 6.3.

Proposition 6.5.1 *Any two rotations in $\mathbb{R}^2$ commute.*

Proof. To commute means that if the order of the composition (or matrix multiplication) is reversed, the outcome is the same. That is, if A and B are the matrix representations of two rotations, then $AB = BA$ is a true statement about matrix multiplication.
I want my proof to be completely general, so I want to work with arbitrary rotations. Let's say that A is a rotation by the angle θ, so it has the matrix

$$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

Let's also say that B is a rotation by the angle ϕ, so it has the matrix

$$\begin{pmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{pmatrix}.$$

Now I need to calculate the two matrix multiplications and show that the result is the same regardless of the order. First I will calculate AB.

$$\begin{aligned} AB &= \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{pmatrix} \\ &= \begin{pmatrix} \cos\theta\cos\phi - \sin\theta\sin\phi & -\cos\theta\sin\phi - \sin\theta\cos\phi \\ \sin\theta\cos\phi + \cos\theta\sin\phi & -\sin\theta\sin\phi + \cos\theta\cos\phi \end{pmatrix}. \end{aligned}$$

Now I'll do the other order of the multiplication.

$$BA = \begin{pmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{pmatrix} \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

$$= \begin{pmatrix} \cos\phi\cos\theta - \sin\phi\sin\theta & -\cos\phi\sin\theta - \sin\phi\cos\theta \\ \sin\phi\cos\theta + \cos\phi\sin\theta & -\sin\phi\sin\theta + \cos\phi\cos\theta \end{pmatrix}.$$

I can take this second result and switch the order of all the multiplications (which is valid since these are just multiplications of ordinary numbers). I can also reorder a couple of the additions/subtractions.

$$\begin{pmatrix} \cos\theta\cos\phi - \sin\theta\sin\phi & -\cos\theta\sin\phi - \sin\theta\cos\phi \\ \sin\theta\cos\phi + \cos\theta\sin\phi & -\sin\theta\sin\phi + \cos\theta\cos\phi \end{pmatrix}.$$

This is the same as the result of AB. This concludes the proof, showing that the outcome of either order of matrix multiplication is the same. ■

You might wonder, at this point, about the structure of this proof. I've tried to say several times that to prove an equality, I need to start at one side and end at the other. However, here I've tried to prove $AB = BA$ and done so by reducing to a common form. Am I disobeying my own rules?

I am not contradicting myself, though it is a very good question. To understand why, look at the statement of the proposition. The composition of the transformation is a process where two transformations combine into one new one. There are two ways to do this: AB or BA. The proposition says that the outcome of both orders is the same. Therefore, I can calculate the outcome of both orders and compare them. This is different from a proposition that simply states an equation; for these propositions, reducing each side of the equation to a common expression can result in a false proof (for example, by multiplying both sides by zero). Here, I'm not reducing an equation to a common form but doing two processes and comparing the result.

Proposition 6.5.2 *Prove that a projection in $\mathbb{R}^2$ composed with itself is just the same projection. (That is, for any projection A, $AA = A$.)*

Proof. All I need to do for this proof is take the general matrix for a projection and multiply it by itself. The general matrix is

$$A = \begin{pmatrix} a^2 & ab \\ ab & b^2 \end{pmatrix},$$

remembering that the vector defining the line is a unit vector direction. Since it is a unit vector direction, I know that $a^2 + b^2 = 1$, which I will need for this proof. Now I'll take this matrix and multiply it by itself.

$$A^2 = \begin{pmatrix} a^2 & ab \\ ab & b^2 \end{pmatrix}\begin{pmatrix} a^2 & ab \\ ab & b^2 \end{pmatrix} = \begin{pmatrix} (a^2)(a^2) + (ab)(ab) & (a^2)(ab) + (ab)(b^2) \\ (ab)(a^2) + (b^2)(ab) & (ab)(ab) + (b^2)(b^2) \end{pmatrix}$$

In each component of this matrix, I can factor out a^2, b^2, or ab.

$$= \begin{pmatrix} a^2(a^2 + b^2) & ab(a^2 + b^2) \\ ab(a^2 + b^2) & b^2(a^2 + b^2) \end{pmatrix}$$

Then I can use the fact that $a^2 + b^2 = 1$.

$$= \begin{pmatrix} a^2 & ab \\ ab & b^2 \end{pmatrix} = A$$

This is the same as the starting matrix. I've proved that $AA = A$, which was the goal of the proposition. ■

6.6 Week 6 Activity

6.6.1 Matrix Action

Activity 6.6.1 In Subsection 6.3.2, I gave the matrices for rotations by $\frac{\pi}{2}$ radians, π radians, and $\frac{3\pi}{2}$ radians. I also defined reflections over the x-axis, y-axs, the line $y = x$, and the line $y = -x$. For all seven of these transformations, check that the matrix action acts as expected on the vectors $\begin{pmatrix}1\\0\end{pmatrix}$, $\begin{pmatrix}0\\1\end{pmatrix}$, $\begin{pmatrix}1\\1\end{pmatrix}$ and $\begin{pmatrix}1\\-1\end{pmatrix}$.

Solution. I have seven matrices, each of which acts on the four given vectors. I'll go through the actions one by one, starting with rotation by $\frac{\pi}{2}$.

$$\begin{pmatrix}0 & -1\\1 & 0\end{pmatrix}\begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}0+0\\1+0\end{pmatrix} = \begin{pmatrix}0\\1\end{pmatrix}$$
$$\begin{pmatrix}0 & -1\\1 & 0\end{pmatrix}\begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}0-1\\0+0\end{pmatrix} = \begin{pmatrix}-1\\0\end{pmatrix}$$
$$\begin{pmatrix}0 & -1\\1 & 0\end{pmatrix}\begin{pmatrix}1\\1\end{pmatrix} = \begin{pmatrix}0-1\\1+0\end{pmatrix} = \begin{pmatrix}-1\\1\end{pmatrix}$$
$$\begin{pmatrix}0 & -1\\1 & 0\end{pmatrix}\begin{pmatrix}1\\-1\end{pmatrix} = \begin{pmatrix}0+1\\1+0\end{pmatrix} = \begin{pmatrix}1\\1\end{pmatrix}$$

All four vectors are indeed sent to their rotations by a quarter turn. Now look at the rotation by π.

$$\begin{pmatrix}-1 & 0\\0 & -1\end{pmatrix}\begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}-1+0\\0+0\end{pmatrix} = \begin{pmatrix}-1\\0\end{pmatrix}$$
$$\begin{pmatrix}-1 & 0\\0 & -1\end{pmatrix}\begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}0+0\\0-1\end{pmatrix} = \begin{pmatrix}0\\-1\end{pmatrix}$$
$$\begin{pmatrix}-1 & 0\\0 & -1\end{pmatrix}\begin{pmatrix}1\\1\end{pmatrix} = \begin{pmatrix}-1+0\\0+-1\end{pmatrix} = \begin{pmatrix}-1\\-1\end{pmatrix}$$
$$\begin{pmatrix}-1 & 0\\0 & -1\end{pmatrix}\begin{pmatrix}1\\-1\end{pmatrix} = \begin{pmatrix}-1+0\\0+1\end{pmatrix} = \begin{pmatrix}-1\\1\end{pmatrix}$$

All of these vectors are indeed rotated by half a turn. (Notice that this is the same as "reflection" about the origin: each vector is sent to its negative. Now look at the rotation by $\frac{3\pi}{2}$.

$$\begin{pmatrix}0 & 1\\-1 & 0\end{pmatrix}\begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}0+0\\-1+0\end{pmatrix} = \begin{pmatrix}0\\-1\end{pmatrix}$$
$$\begin{pmatrix}0 & 1\\-1 & 0\end{pmatrix}\begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}0+1\\0+0\end{pmatrix} = \begin{pmatrix}1\\0\end{pmatrix}$$
$$\begin{pmatrix}0 & 1\\-1 & 0\end{pmatrix}\begin{pmatrix}1\\1\end{pmatrix} = \begin{pmatrix}0+1\\-1+0\end{pmatrix} = \begin{pmatrix}1\\-1\end{pmatrix}$$
$$\begin{pmatrix}0 & 1\\-1 & 0\end{pmatrix}\begin{pmatrix}1\\-1\end{pmatrix} = \begin{pmatrix}0-1\\-1+0\end{pmatrix} = \begin{pmatrix}-1\\-1\end{pmatrix}$$

All of these vectors are sent to their rotation by three quarters of a turn. Notice that I could also interpret this a rotation by a quarter turn clockwise. Now

look at reflection over the x-axis.

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1+0 \\ 0+0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$
$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0+0 \\ 0-1 \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \end{pmatrix}$$
$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1+0 \\ 0+-1 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$
$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1+0 \\ 0+1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

The sign of the y coordinate of each vector is changed, which is exactly what I expect for reflection over the x-axis. Now look at reflection over the y-axis.

$$\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -1+0 \\ 0+0 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$$
$$\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0+0 \\ 0+1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$
$$\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1+0 \\ 0+1 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$
$$\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} -1+0 \\ 0-1 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \end{pmatrix}$$

The x coordinate has now changed in sign, which is as expected. Now look at the reflection over the line $y = x$.

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0+0 \\ 1+0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$
$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0+1 \\ 0+0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$
$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0+1 \\ 1+0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$
$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 0-1 \\ 1+0 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

The two coordinates are exchanged, which is exactly what I would expect for this reflection. Finally, look at the reflection over the line $y = -x$.

$$\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0+0 \\ -1+0 \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \end{pmatrix}$$
$$\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0-1 \\ 0+0 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$$
$$\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0-1 \\ -1+0 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \end{pmatrix}$$
$$\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 0+1 \\ -1+0 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

The coordinates are switched and the signs are changed, which is what I would expect with this reflection.

Activity 6.6.2 Calculate this matrix action.

$$\begin{pmatrix} -5 & 0 & 3 & 2 \\ 4 & -5 & -7 & 1 \\ 1 & 2 & 0 & -8 \end{pmatrix} \begin{pmatrix} 4 \\ -1 \\ -1 \\ -1 \end{pmatrix}$$

Solution. I apply the matrix action algorithm: going along the rows of the matrix and down the column vector, multiplying paired terms and adding the products together.

$$\begin{pmatrix} -5 & 0 & 3 & 2 \\ 4 & -5 & -7 & 1 \\ 1 & 2 & 0 & -8 \end{pmatrix} \begin{pmatrix} 4 \\ -1 \\ -1 \\ -1 \end{pmatrix}$$
$$= \begin{pmatrix} (-5)(4) + (0)(-1) + (3)(-1) + (2)(-1) \\ (4)(4) + (-5)(-1) + (-7)(-1) + (1)(-1) \\ (1)(4) + (2)(-1) + (0)(-1) + (-8)(-1) \end{pmatrix} = \begin{pmatrix} -25 \\ 27 \\ 10 \end{pmatrix}$$

Activity 6.6.3 Calculate this matrix action.

$$\begin{pmatrix} -1 & 6 & 0 & -4 \\ 0 & 3 & 3 & -12 \\ -5 & -6 & -2 & 0 \\ 0 & -2 & -2 & 5 \end{pmatrix} \begin{pmatrix} 0 \\ 10 \\ -5 \\ -5 \end{pmatrix}$$

Solution. I apply the matrix action algorithm: going along the rows of the matrix and down the column vector, multiplying paired terms and adding the products together.

$$\begin{pmatrix} -1 & 6 & 0 & -4 \\ 0 & 3 & 3 & -12 \\ -5 & -6 & -2 & 0 \\ 0 & -2 & -2 & 5 \end{pmatrix} \begin{pmatrix} 0 \\ 10 \\ -5 \\ -5 \end{pmatrix}$$
$$= \begin{pmatrix} (-1)(0) + (6)(10) + (0)(-5) + (-4)(-5) \\ (0)(0) + (3)(10) + (3)(-5) + (-12)(-5) \\ (-5)(0) + (-6)(10) + (-2)(-5) + (0)(-5) \\ (0)(0) + (-2)(10) + (-2)(-5) + (5)(-5) \end{pmatrix} = \begin{pmatrix} 80 \\ 75 \\ -50 \\ -35 \end{pmatrix}$$

Activity 6.6.4 Calculate this matrix action.

$$\begin{pmatrix} 5 & -3 & 9 & 0 & 2 & -7 \\ 2 & -2 & 0 & -5 & -6 & -9 \end{pmatrix} \begin{pmatrix} 0 \\ -3 \\ -3 \\ 7 \\ 2 \\ 4 \end{pmatrix}$$

Solution. I apply the matrix action algorithm: going along the rows of the matrix and down the column vector, multiplying paired terms and adding the

products together.

$$\begin{pmatrix} 5 & -3 & 9 & 0 & 2 & -7 \\ 2 & -2 & 0 & -5 & -6 & -9 \end{pmatrix} \begin{pmatrix} 0 \\ -3 \\ -3 \\ 7 \\ 2 \\ 4 \end{pmatrix}$$
$$= \begin{pmatrix} (5)(0) + (-3)(-3) + (9)(-3) + (0)(7) + (2)(2) + (-7)(4) \\ (2)(0) + (-2)(-3) + (0)(-3) + (-5)(7) + (-6)(2) + (-9)(4) \end{pmatrix}$$
$$= \begin{pmatrix} 0 + 9 - 27 + 0 + 4 - 28 \\ 0 + 6 + 0 - 35 - 12 - 36 \end{pmatrix} = \begin{pmatrix} -42 \\ -77 \end{pmatrix}$$

Activity 6.6.5 Calculate this matrix action.

$$\begin{pmatrix} -5 & 3 & 0 \\ 2 & 0 & -4 \\ -10 & 1 & 1 \\ -13 & -2 & 5 \\ 5 & 4 & -9 \end{pmatrix} \begin{pmatrix} 1 \\ -2 \\ 0 \end{pmatrix}$$

Solution. I apply the matrix action algorithm: going along the rows of the matrix and down the column vector, multiplying paired terms and adding the products together.

$$\begin{pmatrix} -5 & 3 & 0 \\ 2 & 0 & -4 \\ -10 & 1 & 1 \\ -13 & -2 & 5 \\ 5 & 4 & -9 \end{pmatrix} \begin{pmatrix} 1 \\ -2 \\ 0 \end{pmatrix}$$
$$= \begin{pmatrix} (-5)(1) + (3)(-2) + (0)(0) \\ (2)(1) + (0)(-2) + (-4)(0) \\ (-10)(1) + (1)(-2) + (1)(0) \\ (-13)(1) + (-2)(-2) + (5)(0) \\ (5)(1) + (4)(-2) + (-9)(0) \end{pmatrix} = \begin{pmatrix} -11 \\ 2 \\ -12 \\ -9 \\ -3 \end{pmatrix}$$

6.6.2 Matrix Multiplication

Activity 6.6.6 Multiply these two matrices.

$$\begin{pmatrix} 9 & -1 & -1 \\ 0 & 5 & -5 \\ -3 & -3 & -3 \end{pmatrix} \begin{pmatrix} 0 & -2 & -8 \\ 1 & 1 & -8 \\ 9 & 0 & 3 \end{pmatrix}$$

Solution. I use the algorithm, going across the rows of the first matrix and down the rows of the second.

$$\begin{pmatrix} 9 & -1 & -1 \\ 0 & 5 & -5 \\ -3 & -3 & -3 \end{pmatrix} \begin{pmatrix} 0 & -2 & -8 \\ 1 & 1 & -8 \\ 9 & 0 & 3 \end{pmatrix} = \begin{pmatrix} -10 & -19 & -67 \\ -40 & 5 & -55 \\ -30 & 3 & 39 \end{pmatrix}$$

Activity 6.6.7 Multiply these two matrices.

$$\begin{pmatrix} 7 & -1 & -1 & 3 \\ 5 & -9 & 0 & 1 \end{pmatrix} \begin{pmatrix} 4 & -4 \\ 1 & 0 \\ -3 & -3 \\ 1 & 5 \end{pmatrix}$$

Solution. I use the algorithm, going across the rows of the first matrix and down the rows of the second.

$$\begin{pmatrix} 7 & -1 & -1 & 3 \\ 5 & -9 & 0 & 1 \end{pmatrix} \begin{pmatrix} 4 & -4 \\ 1 & 0 \\ -3 & -3 \\ 1 & 5 \end{pmatrix} = \begin{pmatrix} 33 & -10 \\ 12 & -15 \end{pmatrix}$$

Activity 6.6.8 Take the matrix for rotation by $\frac{\pi}{2}$ radians in $\mathbb{R}^2$ and muliply it by itself to check that the result is the matrix for rotation by π radians. Multiply by the original matrix again and check that the result is now the matrix for rotation by $\frac{3\pi}{2}$.

Solution. The matrix for a quarter turn rotation is

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

To compose transformations we multiply matrices. To compose this transformation with itself, I mutiply the matrix with itself.

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$$

The result of the matrix multiplication is indeed the rotation matrix for rotation by π radians. To see what happens after a third rotation by a quarter turn, I multiply this matrix by the original quarter turn matrix.

$$\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

The result of the matrix multiplication is indeed the matrix of the rotation by $\frac{3\pi}{2}$ radians.

6.6.3 Proof Questions

Activity 6.6.9 Consider two rotations in $\mathbb{R}^2$, by angles θ and ϕ. Using the general form of a matrix for a rotation, prove that the composition of these two rotations is the rotation by the angle $(\theta + \phi)$. (You will need to use some trigonometric identities to prove this.)

Solution. The general form of a rotation by θ is

$$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

I can make the same matrix for the angle ϕ.

$$\begin{pmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{pmatrix}.$$

I compose these two matrices.

$$\begin{aligned}
&\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}\begin{pmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{pmatrix} \\
&= \begin{pmatrix} (\cos\theta)(\cos\phi) - (\sin\theta)(\sin\phi) & -(\cos\theta)(\sin\phi) - (\sin\theta)(\cos\phi) \\ (\sin\theta)(\cos\phi) + (\cos\theta)(\sin\phi) & -(\sin\theta)(\sin\phi) + (\cos\theta)(\cos\phi) \end{pmatrix}
\end{aligned}$$

Now I need trig identities. Each of these four matrix entries is a sum identity. Replacing those according to identities drawn from the reference material, I get this matrix.

$$\begin{pmatrix} \cos(\theta+\phi) & -\sin(\theta+\phi) \\ \sin(\theta+\phi) & \cos(\theta+\phi) \end{pmatrix}$$

This is precisely the matrix form for rotation by the angle $(\theta+\phi)$. Therefore, I have proven that the matrix multiplication, and hence the composition, of two rotations produces the rotation by the sum of the two angles.

Activity 6.6.10 Prove that any reflection in $\mathbb{R}^2$ composes with itself to produce the identity. (Use the general matrix for these reflections).

Solution. The general form for a reflection over the line spanned by the unit vector $\begin{pmatrix} a \\ b \end{pmatrix}$ is

$$\begin{pmatrix} a^2 - b^2 & 2ab \\ 2ab & b^2 - a^2 \end{pmatrix}.$$

Since the vector defining the line is a unit vector, I know $a^2 + b^2 = 1$. This fact will be important to the proof. Now I take this matrix and multiply it by itself to do the composition of the reflection with itself. I'll have to do some algebra with the resulting terms in the new matrix.

$$\begin{aligned}
&\begin{pmatrix} a^2 - b^2 & 2ab \\ 2ab & b^2 - a^2 \end{pmatrix}\begin{pmatrix} a^2 - b^2 & 2ab \\ 2ab & b^2 - a^2 \end{pmatrix} \\
&= \begin{pmatrix} (a^2-b^2)^2 + (2ab)(2ab) & (a^2-b^2)(2ab) + (2ab)(b^2-a^2) \\ (2ab)(a^2-b^2) + (b^2-a^2)(2ab) & (2ab)(2ab) + (b^2-a^2)^2 \end{pmatrix} \\
&= \begin{pmatrix} a^4 - 2a^2b^2 + b^4 + 4a^2b^2 & (a^2-b^2)(2ab) - (a^2-b^2)(2ab) \\ (2ab)(a^2-b^2) - (2ab)(a^2-b^2) & 4a^2b^2 + b^4 - 2a^2b^2 + a^4 \end{pmatrix} \\
&= \begin{pmatrix} a^4 + 2a^2b^2 + b^4 & 0 \\ 0 & b^2 + 2a^2b^2 + a^2 \end{pmatrix} \\
&= \begin{pmatrix} (a^2+b^2)^2 & 0 \\ 0 & (a^2+b^2)^2 \end{pmatrix} \\
&= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}
\end{aligned}$$

In the last step, I use the important fact about the unit vector. I end up with the identity matrix, which is what I wanted, so the proof is finished.

6.6.4 Conceptual Review Questions

- Why can matrices represent linear transformation?
- What is the matrix action?
- What is an associative binary operations? What is a commutative binary operation?

- Why isn't matrix multiplication commutative?

Chapter 7

Week 7

7.1 Inverse Linear Transforms and Matrix Inversion

7.1.1 Inverse Transforms

I've now defined linear transformations. It is possible that a linear transformation is invertible, much like a function of numbers. Let me give the definition.

Definition 7.1.1 Let $T : \mathbb{R}^n \to \mathbb{R}^n$. The inverse transformation of T is a transformation $S : \mathbb{R}^n \to \mathbb{R}^n$ which undoes what T did. Equivalently, it is a transformation S such that both the compositions $T \circ S$ and $S \circ T$ are the identity. For an arbitrary transformation, there is no guarantee that an inverse exists; if it does, it is written T^{-1}. $\Diamond$

Since I've encoded transformations as matrices, I can use matrices to understand the inverses of linear transformations.

7.1.2 Inverse Matrices

Before returning to inverse transformations, I'm going to talk about the algebraic idea of inversion and reciprocals. In number systems, when there is an identity, there usually is a way of getting back to that identity. Zero is the identity for addition. For any number a, there exists a number $(-a)$ so that $a + (-a) = 0$ gets back to the identity (this obviously works in numbers systems that allow negatives, such as $\mathbb{Z}$). One is the identity in multiplication. For any *non-zero* number a, there is the number $\frac{1}{a}$ so that $a\frac{1}{a} = 1$ gets back to the identity (this obviously works in numbers systems that allow reciprocals). For matrices, I have already defined the identity matrix, which is the identity for matrix multiplication. If A is a $k \times l$ matrix, then $AI_l = I_kA = A$. For matrices, multiplying by the identity doesn't change the matrix.

For matrices, I have the same question as I had for numbers systems. For any matrix M, is there another matrix N such that $MN = I$? Is there a way to get back to the identity? Multiplication of numbers already showed that no such number need exist, since I cannot divide by zero. For matrices, I have to be even more cautious. For now, here is the formal definition.

Definition 7.1.2 Let M be a $n \times n$ (square) matrix. Then the **inverse** of M is the **unique** matrix M^{-1} (if it exists) such that $MM^{-1} = M^{-1}M = I_n$. $\Diamond$

I should note a couple of thing about this definition. First, it only applies to square matrices. I don't even try to invert non-square matrices. Second, both

multiplications are required: MM^{-1} and $M^{-1}M$. Both are required because matrix multiplication is non-commutative. In general, these two orders could result in different products. In this definition, I insist that both orders produce the identity matrix.

7.1.3 Calculating Inverse Matrices

The definition is good, but I am left with the problem of determining which matrices have inverses and calculating those inverses. It turns out there is a convenient algorithm using techniques I have already presented. I will write an extended matrix $(M|I_n)$ with the original matrix and the identity matrix next to each other, separated by a vertical line. Then I row reduce, treating the conglomeration as one matrix with long rows. If the row reduction of M results in the identity on the left, then the matrix has the form $(I|M^{-1})$; the right-hand side will be the inverse.

This algorithm leads to two observations. First, obviously, it is a way to calculate inverses. But it also is a condition: an $n \times n$ matrix is invertible, in this algorithm, only if it row reduces to the identity. This is part of a general result, which is the first of several conditions for invertible matrices.

Proposition 7.1.3 *A $n \times n$ matrix is invertible if and only if it row reduces to the identity matrix I_n. Since the identity has n leading ones, this is equivalent to the matrix having rank n.*

Definition 7.1.4 The set of all intervible $n \times n$ matrices with real coefficient forms a group. It is called the **General Linear Group** and written $GL_n(\mathbb{R})$. If I wanted to change the coefficients to another set of scalars S, I would write $GL_n(S)$. When the coefficients are understood, the notation is sometimes simplified to GL_n or $GL(n)$. ◊

7.2 Transformations of Spans and Loci

7.2.1 Transformations of Spans

Now that I understand how matrices describe transformations, I want to know how transformations affect linear subspaces. I know that they preserve them (and each linear subspace is sent to some other linear subspace), but I want to be more specific: given a particular linear subspace, can I determine exactly where it goes under a transformation?

Definition 7.2.1 If $M : \mathbb{R}^n \to \mathbb{R}^m$ is a transformation (represented by a $m \times n$ matrix M) and L is a linear or affine subspace of $\mathbb{R}^n$, then the result of the matrix M acting on all vectors in L will be a linear or affine subspace of $\mathbb{R}^m$ called the **image of L under M**. The image of all of $\mathbb{R}^n$ under M is simply called the **image of M**. ◊

The most important property of linear transformations is that they preserve addition and scalar multiplication. That means they preserve the linear combinations. Let a_i be scalars and v_i be vectors in $\mathbb{R}^n$, with M a $m \times n$ matrix.

$$\begin{aligned} M(a_1v_1 + a_2v_2 + \ldots + a_kv_k) &= M(a_1v_1) + M(a_2v_2) + \ldots + M(a_kv_k) \\ &= a_1(Mv_1) + a_2(Mv_2) + \ldots + a_k(Mv_k) \end{aligned}$$

This calculation shows that the image of a linear combination is still a linear combination, just in the vectors Mv_i instead of v_i. Since all elements of a span

are linear combinations, everything in the span of the v_i is sent to the span of the vectors Mv_i. I can summarize these observations in a proposition.

Proposition 7.2.2 *A linear transformation represented by a $m \times n$ matrix M sends spans to spans. In particular, the span* $\text{Span}\{v_1, v_2, \ldots, v_k\}$ *is sent to* $\text{Span}\{Mv_1, Mv_2, \ldots, Mv_k\}$.

Matrices acting on spans are easy: I just figure out where the individual vectors go and the span of those vectors will be transformed into the span of their images. Offset spans are almost as easy: consider an affine subspace $u+\text{Span}\{v_1, v_2, \ldots, v_k\}$. Any element of this looks like $u+a_1v_1+a_2v_2+\ldots a_kv_k$. I can calculate where this linear combination goes under the action of M.

$$M(u + a_1v_1 + a_2v_2 + \ldots + a_kv_k) = Mu + a_1Mv_1 + a_2Mv_2 + \ldots + a_kMv_k$$

The offset span is still sent to an offset span, with offset Mu and spanning vectors Mv_i.

Be careful that the matrix need not preserve the dimension of the span. Even if the v_i are linearly independent and form a basis for the span, the vectors Mv_i may not be linearly independent. The dimension of the new span might be smaller. Linear combinations are sent to linear combinations, but linear independence may not be preserved.

7.2.2 Transformation of Loci

For loci, the picture is much more complicated. Equations do not transform nearly as pleasantly as spans. Planes in $\mathbb{R}^3$ are defined by a normal; I might hope that the new plane is defined by the image of the normal. Unfortunately, since the matrix may not preserve orthogonality, this will usually not happen. To determine the image of a locus, the best approach is to describe the locus as a span or offset span to find the image of that span.

7.3 Kernels and Images

7.3.1 Row and Column Spaces

Definition 7.3.1 Let M be an $m \times n$ matrix. The **rowspace** of the matrix is the span of the rows of the matrix (considered as vectors in $\mathbb{R}^n$). It is a subspace of $\mathbb{R}^n$. The **columnspace** of the matrix is the span of the columns of the matrix, considering those columns as vectors in $\mathbb{R}^m$. It is a subspace of $\mathbb{R}^m$. ◊

Proposition 7.3.2 *Let M be an $m \times n$ matrix. The columnspace and the row spaces of M are linear subspaces (of $\mathbb{R}^m$ and $\mathbb{R}^n$, respectively) with dimensions equal to the rank of M.*

7.3.2 Images

In Section 7.2, I defined the image of a matrix or transformation to be the whole collection of outputs. (This is what I call the range in single variable functions.) There is a nice proposition which tells me how to calculate images for linear transformations.

Proposition 7.3.3 *The image of a matrix M is the same as its columnspace.*

Proof. The image is the set of all outputs of M, acting on the whole space $\mathbb{R}^n$. I can think of $\mathbb{R}^n$ as the span of its standard basis $e_1, e_2, \ldots e_n$. The image, then, is the span of $Me_1, Me_2, \ldots Me_n$. These vectors are precisely the columns of

the matrix, so their span is the columnspace. ■

Example 7.3.4 Here is a 3×3 matrix acting on the standard basis of $\mathbb{R}^3$. I can demonstrate how the images of the basis vectors are exactly the columns of the matrix.

$$\begin{pmatrix} -2 & -2 & 0 \\ 7 & -3 & 2 \\ -4 & -1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} -2 \\ 7 \\ -4 \end{pmatrix}$$
$$\begin{pmatrix} -2 & -2 & 0 \\ 7 & -3 & 2 \\ -4 & -1 & 2 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -2 \\ -3 \\ -1 \end{pmatrix}$$
$$\begin{pmatrix} -2 & -2 & 0 \\ 7 & -3 & 2 \\ -4 & -1 & 2 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \\ 2 \end{pmatrix}$$

□

7.3.3 Kernels

Definition 7.3.5 Let M be an $m \times n$ matrix representing a transformation $\mathbb{R}^n \to \mathbb{R}^m$. For any $u \in \mathbb{R}^m$, the **preimage** of u is the set of all vectors in $\mathbb{R}^m$ which are mapped to u by M. It is writtem $M^{-1}\{u\}$. The **kernel** or **nullspace** of M is all vectors in $\mathbb{R}^n$ which are sent to the zero vector under the transformation associated to M. Combining the two terms, the kernel is the preimage of the zero vector. ◊

Proposition 7.3.6 *The preimage of u is the solution space of the system of equations associated with the extended matrix $(M|u)$ where the vector u is added as a column to the matrix M.*

Proof. The preimage is all vectors v in $\mathbb{R}^n$ that are sent to u. If I write v in coordinates $x_1, \ldots, x_n$, and apply the matrix action, I get a system of m equations with the entries of u as the constants. The matrix encoding of this system is precisely the extended matrix $(M|u)$. ■

I can look at $\mathbb{R}^2$ to build intuition. Conider a general matrix action in two variables.

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x_0 \\ y_0 \end{pmatrix}$$

If I write this out in terms of coordinates, we get a system.

$$ax + by = x_0$$
$$cx + dy = y_0$$

This is the system associated to the extended matrix with $\begin{pmatrix} x_0 \\ y_0 \end{pmatrix}$ as constants.

7.3.4 A Central Connection Idea

At this point in the course, I have talked about loci, solved linear systems, and now defined the preimage of a vector under a transformation. Loci are very geometric, systems are very algebraic, and preimages are again geometric but from a different perspective (transformations instead of static objects). Despite their different starting points, these three are essentially all talking about the

same thing. The connection between all three is a major conceptual goal in this course; if I can understand how these three novel and abstract definitions fit together, that points me to a deep and thorough understanding of the three concepts. Let me try to describe this connection.

Let me start with a linear system. This is a set of linear equations, implicitly asking what values of the variables (if any) solve all the equations simultaneously. To move to loci, all I have to do is think geometrically. A locus is the set of points that satisfy some equations (linear equations in this course). So a locus is just the solution of a system, seen geometrically. A locus is a solution space. I made this connection before when I calculated the dimension of loci: the dimension was the number of free parameters in the solution of the associated system.

So what about pre-images? Conisider a $m \times n$ matrix M, representing a transformation $\mathbb{R}^m \to \mathbb{R}^n$. For a vector u in $\mathbb{R}^n$, I can ask for the pre-image $M^{-1}\{u\}$. This is all the vectors in $\mathbb{R}^m$ which the matrix sends to u (this could possibly be empty if none of the vectors are sent to u). I can express this as a matrix equation: v is in the preimage only if

$$Mv = u.$$

But then I can look at the coordinates of this equation. Each coordinate gives me a linear equation, so I get a system of m linear equations (because $v \in \mathbb{R}^m$, so it has m coordinates). The constants of this linear system are the components of u. The matrix of this linear system is the extended matrix $(M|u)$. The solution to this linear system can be interpreted as a locus. This locus (line, plane, etc) is precisely all the points that M sends to u. In this way, solutions spaces to systems, loci, and preimages are all the same things and are all found by the matrix $(M|u)$.

Specifically, the kernel is the linear space of all vectors v with $Mv = 0$. This is a solution space for the linear system with matrix $(M|0)$, where the column after the dividing line is all zeroes. To calculate a kernel, I have to solve a system where the constants are all zero. The matrix of that system is just M, the matrix of the transformation.

7.4 Proofs - Inversion, Kernels, Images

7.4.1 Interactions between Definitions

In Section 7.3, I defined a number of terms: rowspace, columnspace, kernel, image and preimage. In this section, I want to prove some properties of these new definitions. All these terms, along with the matrix rank, fit togther very tightly.

Proposition 7.4.1 *Let M be an $m \times n$ matrix, representing a transformation $\mathbb{R}^n \to \mathbb{R}^m$. The dimension of the kernel of M and the dimension of the image of M add up to n, the dimension of the domain.*

Proof. I know the dimension of the image is the rank, which is the number of leading ones in the reduced row echelon form. I also know that the dimension of the kernel is the dimension of the system of equations represented by the extended matrix $(M|0)$. The dimension of a solution space to a system of equations is equal to the number of free parameters. Each column without a leading one in the extended matrix (left of the dividing line) is a free variable. Therefore, the dimension of the kernel is the number of columns without leading ones.

Each column in the row reduced form of M either has or does not have a leading one. The columns with leading ones contribute to the rank. The columns without leading ones are free parameters for the system. Therefore, the dimensions of the image (rank) plus the dimension of the kernel (free parameters) add up to the number of columns, which is n. ■

Proposition 7.4.2 *Let M be an $n \times n$ matrix. M is invertible if and only if its kernel is zero.*

Proof. For this if and only if statement, I need to prove in both directions. I'll start with the forward implication. For this direction of the proof, I must assume the matrix is invertible.
Since the matrix is invertible, it row reduces to the identity matrix. (I know this from the inversion algorithm, where I must row reduce the left side of the extended matrix to the identity.) That means it has rank n, since each row has a leading one. This means that the image has dimension n since the dimension of the image is equal to the rank. Using the previous proposition, this fact implies that the kernel has dimension 0. The only thing with dimension zero is a point, so the kernel is a point. The kernel always includes zero, since zero is also always sent to zero, so the kernel must just be the zero vector.
I have to prove the reverse implication as well. Luckily in this case, I can essentially reverse the same logic. A zero kernel has dimension zero. Therefore the image has to have dimension n. Therefore, there must be n leading ones. Therefore, the matrix must reduce to the identity matrix (the only $n \times n$ matrix with n leading ones). Thus, the matrix is invertible. ■

It turns out that there will be many ways to describe an invertible matrix. I'll start a list of equivalent properties of invertible matrices, which will be extended throughout the course. Assume A is an $n \times n$ matrix. The following properties are equivalent (each implies all the rest).

- A is invertible.
- A row reduces to the identity matrix.
- A has rank n.
- The image of A is $\mathbb{R}^n$.
- The columnspace of A is $\mathbb{R}^n$.
- The columns of A are linearly independent.
- The rowspace of A is $\mathbb{R}^n$.
- The rows of A are linearly independent.
- The kernel of A is zero.
- A preserves the dimension of any (linear or affine) object it acts upon
- $Au = v$ has a unique solution u for any choice of vector $v \in \mathbb{R}^n$.

7.5 Week 7 Activity

7.5.1 Inversion

All row reductions in these activity were done with Wolfram Alpha.

Activity 7.5.1 Calculate the inverse of this matrix. Check by matrix multiplication that the product of the matrix and its inverse (in both orders!) is the

identity.

$$\begin{pmatrix} 7 & 0 \\ -5 & -4 \end{pmatrix}$$

Solution. I write the extended matrix with the original matrix and the identity.

$$\left(\begin{array}{cc|cc} 7 & 0 & 1 & 0 \\ -5 & -4 & 0 & 1 \end{array}\right)$$

I row reduce this extended matrix.

$$\left(\begin{array}{cc|cc} 1 & 0 & \frac{4}{28} & 0 \\ 0 & 1 & \frac{-5}{28} & \frac{-1}{4} \end{array}\right)$$

The left side of the extended matrix reduces to the identity, so the right side is the inverse matrix. Then I check the two products.

$$\frac{1}{28}\begin{pmatrix} 4 & 0 \\ -5 & -7 \end{pmatrix}\begin{pmatrix} 7 & 0 \\ -5 & -4 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$
$$\begin{pmatrix} 7 & 0 \\ -5 & -4 \end{pmatrix}\frac{1}{28}\begin{pmatrix} 4 & 0 \\ -5 & -7 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Both products are the identity matrix, so this is the correct inverse.

Activity 7.5.2 Calculate the inverse of this matrix. Check by matrix multiplication that the product of the matrix and its inverse (in both orders!) is the identity.

$$\begin{pmatrix} 3 & 0 & 1 \\ -3 & -1 & 0 \\ 2 & 1 & -4 \end{pmatrix}$$

Solution. I write the extended matrix with the original matrix and the identity.

$$\left(\begin{array}{ccc|ccc} 3 & 0 & 1 & 1 & 0 & 0 \\ -3 & -1 & 0 & 0 & 1 & 0 \\ 2 & 1 & -4 & 0 & 0 & 1 \end{array}\right)$$

I row reduce this extended matrix.

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{4}{11} & \frac{1}{11} & \frac{1}{11} \\ 0 & 1 & 0 & \frac{-12}{11} & \frac{-14}{11} & \frac{-3}{11} \\ 0 & 0 & 1 & \frac{-1}{11} & \frac{-3}{11} & \frac{-3}{11} \end{array}\right)$$

The left side reduces to the identity matrix, so the right side is the inverse matrix. Then I check the two products.

$$\frac{1}{11}\begin{pmatrix} 4 & 1 & 1 \\ -12 & -14 & -3 \\ -1 & -3 & -3 \end{pmatrix}\begin{pmatrix} 3 & 0 & 1 \\ -3 & -1 & 0 \\ 2 & 1 & -4 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$
$$\begin{pmatrix} 3 & 0 & 1 \\ -3 & -1 & 0 \\ 2 & 1 & -4 \end{pmatrix}\frac{1}{11}\begin{pmatrix} 4 & 1 & 1 \\ -14 & -14 & -3 \\ -1 & -3 & -3 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Both products are the identity matrix, so this is the correct inverse.

Activity 7.5.3 Calculate the inverse of this matrix. Check by matrix multiplication that the product of the matrix and its inverse (in both orders!) is the identity.

$$\begin{pmatrix} 0 & -5 & 2 \\ 3 & 0 & 1 \\ 0 & -4 & 1 \end{pmatrix}$$

Solution. I write the extended matrix with the original matrix and the identity.

$$\left(\begin{array}{ccc|ccc} 0 & -5 & 2 & 1 & 0 & 0 \\ 3 & 0 & 1 & 0 & 1 & 0 \\ 0 & -4 & 1 & 0 & 0 & 1 \end{array}\right)$$

I row reduce this extended matrix.

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{-4}{9} & \frac{3}{9} & \frac{5}{9} \\ 0 & 1 & 0 & \frac{3}{9} & 0 & \frac{-6}{9} \\ 0 & 0 & 1 & \frac{12}{9} & 0 & \frac{-15}{9} \end{array}\right)$$

The left side reduces to the identity matrix, so the right side is the inverse matrix. Then I check the two products.

$$\frac{1}{9}\begin{pmatrix} -4 & 3 & 5 \\ 3 & 0 & -6 \\ 12 & 0 & -15 \end{pmatrix}\begin{pmatrix} 0 & -5 & 2 \\ 3 & 0 & 1 \\ 0 & -4 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 0 & -5 & 2 \\ 3 & 0 & 1 \\ 0 & -4 & 1 \end{pmatrix}\frac{1}{9}\begin{pmatrix} -4 & 3 & 5 \\ 3 & 0 & -6 \\ 12 & 0 & -15 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Both products are the identity matrix, so this is the correct inverse.

Activity 7.5.4 Calculate the inverse of this matrix. Check by matrix multiplication that the product of the matrix and its inverse (in both orders!) is the identity.

$$\begin{pmatrix} 1 & -6 & 2 \\ -3 & 4 & 1 \\ -5 & 2 & 4 \end{pmatrix}$$

Solution. I write the extended matrix with the original matrix and the identity.

$$\left(\begin{array}{ccc|ccc} 1 & -6 & 2 & 1 & 0 & 0 \\ -3 & 4 & 1 & 0 & 1 & 0 \\ -5 & 2 & 4 & 0 & 0 & 1 \end{array}\right)$$

I row reduce this extended matrix.

$$\left(\begin{array}{ccc|ccc} 1 & 0 & -1 & 0 & \frac{2}{14} & \frac{-4}{14} \\ 0 & 1 & \frac{-1}{2} & 0 & \frac{5}{14} & \frac{-3}{14} \\ 0 & 0 & 0 & \frac{14}{14} & \frac{28}{14} & \frac{-14}{14} \end{array}\right)$$

The right side does not reduce to the identity. Therefore, this matrix is not invertible.

7.5.2 Transformations of Spans

Activity 7.5.5 Calculate the image of the given span under the given matrix.

$$\text{Span}\left\{\begin{pmatrix}3\\-1\end{pmatrix}\right\} \qquad \begin{pmatrix}1 & -4\\-2 & 4\end{pmatrix}$$

Solution. To calculate the image of the span, I apply the matrix to the vector.

$$\begin{pmatrix}1 & -4\\-2 & 4\end{pmatrix}\begin{pmatrix}3\\-1\end{pmatrix} = \begin{pmatrix}7\\-10\end{pmatrix}$$

Then the image of the span is the span of the new vector.

$$\begin{pmatrix}1 & -4\\-2 & 4\end{pmatrix}\left[\text{Span}\left\{\begin{pmatrix}3\\-1\end{pmatrix}\right\}\right] = \text{Span}\left\{\begin{pmatrix}7\\-10\end{pmatrix}\right\}$$

Activity 7.5.6 Calculate the image of the given span under the given matrix.

$$\text{Span}\left\{\begin{pmatrix}1\\1\\1\end{pmatrix}\right\} \qquad \begin{pmatrix}-1 & -1 & 4\\0 & 2 & -4\\-3 & -5 & 1\end{pmatrix}$$

Solution. To calculate the image of the span, I apply the matrix to the vector.

$$\begin{pmatrix}-1 & -1 & 4\\0 & 2 & -4\\-3 & -5 & 1\end{pmatrix}\begin{pmatrix}1\\1\\1\end{pmatrix} = \begin{pmatrix}2\\-2\\-7\end{pmatrix}$$

Then the image of the span is the span of the new vector.

$$\begin{pmatrix}-1 & -1 & 4\\0 & 2 & -4\\-3 & -5 & 1\end{pmatrix}\left[\text{Span}\left\{\begin{pmatrix}1\\1\\1\end{pmatrix}\right\}\right] = \text{Span}\left\{\begin{pmatrix}2\\-2\\-7\end{pmatrix}\right\}$$

Activity 7.5.7 Calculate the image of the given span under the given matrix.

$$\text{Span}\left\{\begin{pmatrix}0\\-4\\0\end{pmatrix}, \begin{pmatrix}-1\\-1\\2\end{pmatrix}\right\} \qquad \begin{pmatrix}-2 & 4 & 7\\1 & 0 & 1\\1 & 6 & -3\end{pmatrix}$$

Solution. To calculate the image of the span, I apply the matrix to both vectors.

$$\begin{pmatrix}-2 & 4 & 7\\1 & 0 & 1\\1 & 6 & -3\end{pmatrix}\begin{pmatrix}0\\-4\\0\end{pmatrix} = \begin{pmatrix}-16\\0\\-24\end{pmatrix}$$

$$\begin{pmatrix}-2 & 4 & 7\\1 & 0 & 1\\1 & 6 & -3\end{pmatrix}\begin{pmatrix}-1\\-1\\2\end{pmatrix} = \begin{pmatrix}12\\1\\-13\end{pmatrix}$$

Then the image of the span is the span of the new vectors.

$$\begin{pmatrix}-2 & 4 & 7\\1 & 0 & 1\\1 & 6 & -3\end{pmatrix}\left[\text{Span}\left\{\begin{pmatrix}0\\-4\\0\end{pmatrix}, \begin{pmatrix}-1\\-1\\2\end{pmatrix}\right\}\right] = \text{Span}\left\{\begin{pmatrix}-16\\0\\-24\end{pmatrix}, \begin{pmatrix}12\\1\\-13\end{pmatrix}\right\}$$

Activity 7.5.8 Calculate the image of the given offset span under the given matrix.

$$\begin{pmatrix}4\\0\\-1\end{pmatrix}+\text{Span}\left\{\begin{pmatrix}2\\1\\-7\end{pmatrix},\begin{pmatrix}1\\-4\\1\end{pmatrix}\right\}\qquad\begin{pmatrix}0&-4&-4\\-2&6&1\\-5&3&1\end{pmatrix}$$

Solution. To calculate the image of the offset span, I apply the matrix to all three vectors.

$$\begin{pmatrix}0&-4&-4\\-2&6&1\\-5&3&1\end{pmatrix}\begin{pmatrix}4\\0\\-1\end{pmatrix}=\begin{pmatrix}4\\-9\\-21\end{pmatrix}$$
$$\begin{pmatrix}0&-4&-4\\-2&6&1\\-5&3&1\end{pmatrix}\begin{pmatrix}2\\1\\-7\end{pmatrix}=\begin{pmatrix}24\\-5\\-14\end{pmatrix}$$
$$\begin{pmatrix}0&-4&-4\\-2&6&1\\-5&3&1\end{pmatrix}\begin{pmatrix}1\\-4\\1\end{pmatrix}=\begin{pmatrix}12\\-25\\-16\end{pmatrix}$$

Then the image of the offset span is the offset span of the new vectors.

$$\begin{pmatrix}0&-4&-4\\-2&6&1\\-5&3&1\end{pmatrix}\left[\begin{pmatrix}4\\0\\-1\end{pmatrix}+\text{Span}\left\{\begin{pmatrix}2\\1\\-7\end{pmatrix},\begin{pmatrix}1\\-4\\1\end{pmatrix}\right\}\right]$$
$$=\begin{pmatrix}4\\-9\\-21\end{pmatrix}+\text{Span}\left\{\begin{pmatrix}24\\-5\\-14\end{pmatrix},\begin{pmatrix}12\\-25\\-16\end{pmatrix}\right\}$$

Activity 7.5.9 Calculate the image of the given offset span under the given matrix.

$$\begin{pmatrix}1\\0\\1\\1\end{pmatrix}+\text{Span}\left\{\begin{pmatrix}-1\\5\\6\\-1\end{pmatrix},\begin{pmatrix}0\\-3\\0\\-3\end{pmatrix},\begin{pmatrix}-1\\0\\5\\5\end{pmatrix}\right\}\qquad\begin{pmatrix}0&-5&1&1\\-1&-2&6&-3\\0&1&0&-4\end{pmatrix}$$

Solution. To calculate the image of the offset span, I apply the matrix to all four vectors.

$$\begin{pmatrix}0&-5&1&1\\-1&-2&6&-3\\0&1&0&-4\end{pmatrix}\begin{pmatrix}1\\0\\1\\1\end{pmatrix}=\begin{pmatrix}2\\2\\-4\end{pmatrix}$$
$$\begin{pmatrix}0&-5&1&1\\-1&-2&6&-3\\0&1&0&-4\end{pmatrix}\begin{pmatrix}-1\\5\\6\\-1\end{pmatrix}=\begin{pmatrix}-20\\30\\9\end{pmatrix}$$
$$\begin{pmatrix}0&-5&1&1\\-1&-2&6&-3\\0&1&0&-4\end{pmatrix}\begin{pmatrix}0\\-3\\0\\-3\end{pmatrix}=\begin{pmatrix}12\\15\\9\end{pmatrix}$$

$$\begin{pmatrix} 0 & -5 & 1 & 1 \\ -1 & -2 & 6 & -3 \\ 0 & 1 & 0 & -4 \end{pmatrix} \begin{pmatrix} -1 \\ 0 \\ 5 \\ 5 \end{pmatrix} = \begin{pmatrix} 10 \\ 16 \\ -20 \end{pmatrix}$$

Then the image of the offset span is the offset span of the new vector.

$$\begin{pmatrix} 0 & -5 & 1 & 1 \\ -1 & -2 & 6 & -3 \\ 0 & 1 & 0 & -4 \end{pmatrix} \left[\begin{pmatrix} 1 \\ 0 \\ 1 \\ 1 \end{pmatrix} + \text{Span} \left\{ \begin{pmatrix} -1 \\ 5 \\ 6 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ -3 \\ 0 \\ -3 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 5 \\ 5 \end{pmatrix} \right\} \right]$$

$$= \begin{pmatrix} 2 \\ 2 \\ -4 \end{pmatrix} + \text{Span} \left\{ \begin{pmatrix} -20 \\ 30 \\ 9 \end{pmatrix}, \begin{pmatrix} 12 \\ 15 \\ 9 \end{pmatrix}, \begin{pmatrix} 10 \\ 16 \\ -20 \end{pmatrix} \right\}$$

7.5.3 Kernels and Images

Activity 7.5.10 Calculate the kernel and image of this matrix.

$$\begin{pmatrix} 2 & 3 \\ -3 & 2 \end{pmatrix}$$

Solution. The kernel is the solution to the system with all constants as zero. Here is the matrix of that system.

$$\left(\begin{array}{cc|c} 2 & 3 & 0 \\ -3 & 2 & 0 \end{array} \right)$$

I row reduce this system.

$$\left(\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \end{array} \right)$$

The only solution is the zero vector.

$$\text{Ker} \begin{pmatrix} 2 & 3 \\ -3 & 2 \end{pmatrix} = \left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right\}$$

The image is the span of the columns. I put the columns as rows in a matrix and row reduce (without exchanging rows).

$$\begin{pmatrix} 2 & -3 \\ 3 & 2 \end{pmatrix}$$

I row reduce this matrix.

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

I see that all of these are linearly independent. I write the image as the span of the linearly independent columns.

$$\text{Im} \begin{pmatrix} 2 & 3 \\ -3 & 2 \end{pmatrix} = \text{Span} \left\{ \begin{pmatrix} 2 \\ -3 \end{pmatrix}, \begin{pmatrix} 3 \\ 2 \end{pmatrix} \right\}$$

Alternatively, I could simply say that the image is all of $\mathbb{R}^2$, since these two vectors span the whole space.

Activity 7.5.11 Calculate the kernel and image of this matrix.

$$\begin{pmatrix} 2 & 5 \\ -4 & -10 \end{pmatrix}$$

Solution. The kernel is the solution to the system with all constants as zero. Here is matrix of that system.

$$\left(\begin{array}{cc|c} 2 & 5 & 0 \\ -4 & -10 & 0 \end{array}\right)$$

I row reduce this system.

$$\left(\begin{array}{cc|c} 1 & \frac{5}{2} & 0 \\ 0 & 0 & 0 \end{array}\right)$$

I interpret the solution as a span. In the solution, $x + \frac{5}{2}y = 0$. If I take $y = 2$ then $x = -5$ gives us a vector to describe the span.

$$\operatorname{Ker}\begin{pmatrix} 2 & 5 \\ -4 & -10 \end{pmatrix} = \operatorname{Span}\left\{\begin{pmatrix} -5 \\ 2 \end{pmatrix}\right\}$$

The image is the span of the columns. I put the columns as rows in a matrix and row reduce (without exchanging rows).

$$\begin{pmatrix} 2 & -4 \\ 5 & -10 \end{pmatrix}$$

I row reduce this matrix.

$$\begin{pmatrix} 1 & -2 \\ 0 & 0 \end{pmatrix}$$

I see that only one of these is linearly independent. I write the image as the span of the linearly independent columns.

$$\operatorname{Im}\begin{pmatrix} 2 & 5 \\ -4 & -10 \end{pmatrix} = \operatorname{Span}\left\{\begin{pmatrix} 2 \\ -4 \end{pmatrix}\right\}$$

Activity 7.5.12 Calculate the kernel and image of this matrix.

$$\begin{pmatrix} 4 & 0 & -1 \\ -2 & -2 & -2 \\ -1 & 0 & 3 \end{pmatrix}$$

Solution. The kernel is the solution to the system with all constants as zero. Here is matrix of that system.

$$\left(\begin{array}{ccc|c} 4 & 0 & -1 & 0 \\ -2 & -2 & -2 & 0 \\ -1 & 0 & 3 & 0 \end{array}\right)$$

I row reduce this system.

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right)$$

All components must be set to zero, so the kernel is only the zero vector.

$$\operatorname{Ker}\begin{pmatrix} 4 & 0 & -1 \\ -2 & -2 & -2 \\ -1 & 0 & 3 \end{pmatrix} = \left\{ \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \right\}$$

The image is the span of the columns. I put the columns as rows in a matrix and row reduce (without exchanging rows).

$$\begin{pmatrix} 4 & -2 & -1 \\ 0 & -2 & 0 \\ -1 & -2 & 3 \end{pmatrix}$$

I row reduce this matrix.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

I see that all of these are linearly independent. I write the image as the span of the linearly independent columns.

$$\operatorname{Im}\begin{pmatrix} 4 & 0 & -1 \\ -2 & -2 & -2 \\ -1 & 0 & 3 \end{pmatrix} = \operatorname{Span}\left\{ \begin{pmatrix} 4 \\ -2 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ -2 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ -2 \\ 3 \end{pmatrix} \right\}$$

Alternatively, I can just saw that the image is $\mathbb{R}^3$, since these three vectors span the entire space.

Activity 7.5.13 Calculate the kernel and image of this matrix.

$$\begin{pmatrix} 0 & -5 & 2 \\ 3 & -4 & 1 \\ -3 & -1 & 1 \end{pmatrix}$$

Solution. The kernel is the solution to the system with all constants as zero. Here is matrix of that system.

$$\left(\begin{array}{ccc|c} 0 & -5 & 2 & 0 \\ 3 & -4 & 1 & 0 \\ -3 & -1 & 1 & 0 \end{array}\right)$$

I row reduce this system.

$$\left(\begin{array}{ccc|c} 1 & 0 & \frac{-1}{5} & 0 \\ 0 & 1 & \frac{-2}{5} & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

I interpret the solution as a span. There is one free variable. If I set $z = 5$ then $x = 1$ and $y = 2$ gives a vector to describe the span.

$$\operatorname{Ker}\begin{pmatrix} 0 & -5 & 2 \\ 3 & -4 & 1 \\ -3 & -1 & 1 \end{pmatrix} = \operatorname{Span}\left\{ \begin{pmatrix} 1 \\ 2 \\ 5 \end{pmatrix} \right\}$$

The image is the span of the columns. I put the columns as rows in a matrix and row reduce (without exchanging rows).

$$\begin{pmatrix} 0 & 3 & -3 \\ -5 & -4 & -1 \\ 2 & 1 & 1 \end{pmatrix}$$

I row reduce this matrix.

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}$$

I see that two of these are linearly independent. I write the image as the span of the linearly independent columns.

$$\operatorname{Im}\begin{pmatrix} 0 & -5 & 2 \\ 3 & -4 & 1 \\ -3 & -1 & 1 \end{pmatrix} = \operatorname{Span}\left\{\begin{pmatrix} 0 \\ 3 \\ -3 \end{pmatrix}\begin{pmatrix} -5 \\ -4 \\ -1 \end{pmatrix}\right\}$$

Activity 7.5.14 Calculate the kernel and image of this matrix.

$$\begin{pmatrix} 1 & 3 & 0 & 1 \\ -1 & -2 & 0 & 1 \end{pmatrix}$$

Solution. The kernel is the solution to the system with all constants as zero. Here is matrix of that system.

$$\left(\begin{array}{cccc|c} 1 & 3 & 0 & 1 & 0 \\ -1 & -2 & 0 & 1 & 0 \end{array}\right)$$

I row reduce this system.

$$\left(\begin{array}{cccc|c} 1 & 0 & 0 & -5 & 0 \\ 0 & 1 & 0 & 2 & 0 \end{array}\right)$$

I interpret the solution as a span. There are two free variables (I take the variables for $\mathbb{R}^4$ as w, x, y, z in that order). The other variables don't depend at all on y, which gives one spanning vector. Otherwise, if I set $y = 0$ and $z = 1$ then $w = 5$ and $x = -2$ produces another vector.

$$\operatorname{Ker}\begin{pmatrix} 1 & 3 & 0 & 1 \\ -1 & -2 & 0 & 1 \end{pmatrix} = \operatorname{Span}\left\{\begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}\begin{pmatrix} 5 \\ -2 \\ 0 \\ 1 \end{pmatrix}\right\}$$

The image is the span of the columns. I put the columns as rows in a matrix and row reduce (without exchanging rows).

$$\begin{pmatrix} 1 & -1 \\ 3 & -2 \\ 0 & 0 \\ 1 & 1 \end{pmatrix}$$

I row reduce this matrix.

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}$$

I see that two of these are linearly independent. I write the image as the span of the linearly independent columns.

$$\operatorname{Im}\begin{pmatrix} 1 & 3 & 0 & 1 \\ -1 & -2 & 0 & 1 \end{pmatrix} = \operatorname{Span}\left\{\begin{pmatrix} 1 \\ -1 \end{pmatrix}\begin{pmatrix} 3 \\ -2 \end{pmatrix}\right\}$$

7.5.4 Proof Questions

Activity 7.5.15 Let M and N be $n \times n$ invertible matrices. Prove that

$$(MN)^{-1} = N^{-1}M^{-1}$$

Solution. There are a number of ways to argue this. I could write out general matrices and argue algebraically, but that seems very difficult and cumbersome. Instead, I'd like to give a geometric argument. (Such an argument is not as formal but, depending on the context, could still be considered a proof.)

If I interpret MN as a transformation $\mathbb{R}^n \to \mathbb{R}^n$, I apply N first, and then I apply M (composition works from right to left). Since both are invertible matrices, I can reverse both processes. If I wanted to reverse the composition, I would have to start with the last thing I did, which is M. Therefore, the first step in the inverse would be M^{-1}. Then I would get back to the point after I applied N, so the next thing to do would be N^{-1}. The way to write the composition of M^{-1} and N^{-1} would be $N^{-1}M^{-1}$. Therefore, I have proven that the inverse of MN should be the composition $N^{-1}M^{-1}$.

Activity 7.5.16 Let M be a 3×3 matrix. Assume that the kernel of M is 2 dimensional. Prove that M can have at most one linearly independent column.

Solution. I can use the proposition that the kernel and the image have dimensions that add to the dimension of the domain. I know the dimension of the kernel is 2, and the dimension of the domain is 3. Therefore, the dimension of the image must be 1. But I also know that the image is the span of the columns. Since the image has dimension 1, the span of the columns has dimension 1. This means that there is only one linearly independent column.

7.5.5 Conceptual Review Questions

- What is an inverse matrix? What does it mean?
- What does it mean to transform a linear or affine subspace under a linear transformation?
- What is a kernel? What is an image? How do these two things give valuable information about a transformation.

Chapter 8

Week 8

8.1 Determinants

8.1.1 Concept of the Determinant

Linear transformations are defined by their symmetries: they preserve linear subspaces and linear operations. The associated matrices are used to understand these transformations. Matrices are an algebraic description; via matrices questions of transformations can be turned into algebraic problems. To figure out which points went to zero under a transformation, I showed an algebraic method of calculating the kernel as a solution space. Now that I know the algebraic description of transformations as matrices, I want to investigate what else the matrix can tell me about the transformation. In this section, I am asking two questions in particular.

First, what does the transformation do to the size of objects? I choose the vague word 'size' intentionally because I will work in various dimensions. In one dimension, size is the length. In two dimensions, size is area. In three dimensions, size is volume. And in higher dimensions, there is some extended notion of volume that fits that dimension; I can call this hyper-volume. It is important to note that size depends on the ambient dimension. A square in $\mathbb{R}^2$ has some area, some non-zero size. But if there is a flat square somewhere in $\mathbb{R}^3$, it has no volume, therefore no substantial size in that space.

Second, what does the transformation do to the orientation of an object? In Section 1.2, I defined orientation as a choice of axis directions relative to each other, but that isn't the most enlightening definition. To help, I will describe orientation for each dimension. In $\mathbb{R}$, orientation is a direction: moving in the positive or negative direction along the number line. There are only two directions of movement, so two orientations. If I have a transformation of $\mathbb{R}$, I can ask if it changes or preserves these directions. In $\mathbb{R}^2$, instead of moving in a line, I think of moving in closed loops or paths. These paths can be clockwise or counter-clockwise. Then I can ask if a transformation changes clockwise loops into other clockwise loops or into counter-clockwise loops. The axis system in $\mathbb{R}^3$ is, conventionally, given by a right-hand-rule. If I know the x and y directions, the right-hand-rule indicates the positive z direction. Then I can ask where these three directions go under a transformation and if a right-hand-rule still applies. If it does, the orientation is preserved. If it doesn't, and a left-hand-rule would work instead, the transformation reverses orientation. In higher dimensions, there are other extensions of the notion of orientation. In each case, the question is binary: a transformation either preserves or reverses orientation.

8.1.2 Definition of the Determinant

Definition 8.1.1 Let M be a square $n \times n$ matrix. The **determinant** of M is a real number, written $\det M$ or just $|M|$, with two properties. Its absolute value, $|\det M|$, measures the effect that M has on size (length, area, volume, hyper-volume). Its sign (positive or negative) measures the effect on orientation; if the sign is positive, orientation is preserved, and if the sign is negative, orientation is reversed. The notation $|M|$ is a little misleading, since the determinant can be a negative number; this is not an absolute value. However, this notation is convenient for actual matrices, where the conventional parantheses are replaced with straight lines to indicate the determinant. ◊

That definition is all well and good, but I need to show that such a thing can be constructed. Section 8.2 shows an algorithm for building determinants.

The following definition will be used in the activities in Section 8.5.

Definition 8.1.2 An **upper triangular** matrix is a matrix where all entries below the diagonal are zero. A **lower triangular** matrix is a matrix where all entries above the diagonal are zero. Below are two examples of upper triangular matrices.

$$\begin{pmatrix} -1 & 3 & 5 \\ 0 & 2 & -9 \\ 0 & 0 & 1 \end{pmatrix} \qquad \begin{pmatrix} 0 & 0 & 2 & 2 \\ 0 & -7 & 4 & 0 \\ 0 & 0 & 9 & -4 \\ 0 & 0 & 0 & -4 \end{pmatrix}$$

◊

8.2 Cofactor Expansion

8.2.1 2×2 determinants

The determinant of a (square) matrix is constructed recursively, starting with 2×2 matrices. Therefore, I need to first show how to calculate the determinant of a 2×2 matrix.

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

Example 8.2.1 For the matrix $\begin{pmatrix} 1 & -3 \\ 2 & -1 \end{pmatrix}$, the determinant is $(1)(-1) - (2)(-3) = -1 + 6 = 5$. Therefore, I know that this matrix multiplies all areas by a factor of 5 and preserves orientation. □

8.2.2 The Cofactor Algorithm

Definition 8.2.2 The algorithm for calculating determinants is called **Co-Factor Expansion**. It is a recursive algorithm that reduces determinant calculations to determinants of smaller square matrices, eventually down to 2×2 matrices whose determinants have already been defined. ◊

Co-factor expansion proceeds in this way: I choose *any* column or row of the matrix. I take the coefficients from that column or row. For each of the coefficients, I multiply that coefficient by the determinant of the matrix formed by removing both the column and row containing that coefficient. Then I add up the determinants of these small matrices multiplied by matching coefficients, with a pattern of $\pm$ signs. That pattern of $\pm$ signs is a checkerboard pattern,

as in the following 5×5 matrix.

$$\begin{pmatrix} + & - & + & - & + \\ - & + & - & + & - \\ + & - & + & - & + \\ - & + & - & + & - \\ + & - & + & - & + \end{pmatrix}$$

That's a hard algorithm to intuit from the formal description. I'll do an example.

Example 8.2.3 Here is a 3×3 example where I choose the first row for co-factor expansion. For each element in the first row, I multiply the number by the determinant of the matrix with the first row and the matching column removed. I multiply by (-1) in the second coefficient to match the checkerboard pattern of signs. After the first step, I can just calculate the three smaller determinants and simplify the arithmetic.

$$\begin{aligned} &\begin{vmatrix} 5 & -2 & 0 \\ -3 & 3 & -2 \\ 1 & -5 & 3 \end{vmatrix} = +5 \begin{vmatrix} 3 & -2 \\ -5 & 3 \end{vmatrix} - (-2) \begin{vmatrix} -3 & -2 \\ 1 & 3 \end{vmatrix} + 0 \begin{vmatrix} -3 & 3 \\ 1 & -5 \end{vmatrix} \\ &= 5((3)(3) - (-2)(-5)) + 2((-3)(3) - (-2)(1)) + (0)((-3)(-5) - (3)(1)) \\ &= 5(-1) + 2(-7) + 0 = -19 \end{aligned}$$

□

If I did co-factor expansion of an arbitrary 3×3 matrix, I would get a direct expression for the determinant.

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = aei - ahf - bdi + bfg + cdh - ceg$$

I could use this formula for 3×3 matrices if I wished. I could do the same for larger matrices, but it starts to get quite complicated. The computational complexity of the recursive determinant algorithm grows very quickly. For a 3×3 matrix, I had 3 recursions to 2×2 matrices, each with 2 multiplication terms in the determinant, giving 6 terms. Each term was the multiple of three coefficients, giving 12 total multiplications. For a 4×4, the recursion gives 24 terms, each with 4 coefficients, so $24 \cdot 3 = 72$ multiplications. For a 5×5 matrix, I need to recurse five times to a 4×4 matrix, for 120 terms each with 5 coefficients, which is $120 \cdot 4 = 600$ multiplications. The pattern continues: there are $n!$ terms in the determinant of a $n \times n$ matrix, each with n coefficients, for $n!(n-1)$ multiplications. This is computationally terrifying, making determinants of large matrices computationally very difficult.

8.3 Properties of Determinants

8.3.1 Determinants and Composition

Before talking about composition, there is a property of determinant that follows from the cofactor expansion. I'll just state it here.

Proposition 8.3.1 *If A is an $n \times n$ matrix, then* $\det A^T = \det A$.

Now I can talk about composition. Algebra considers sets with structure. On the set $M_n(\mathbb{R})$ of $n \times n$ matrices, the determinant is a new algebraic structure. I would like to investigate how it interacts with existing structures, starting with matrix multiplication.

To compose two transformations, I multiply the matrices. In this composition, the effects on size should compound. If the first transformation doubles size and the second triples it, the composition should multiply the size by a factor of 6. Similarly, with orientation: if both preserve orientation, the composition should as well. If both reverse orientation, reversing twice should return to the normal orientation. So, two positive or two negatives should result in a positive, and one positive and one negative should result in a negative. These two observations (compounding effect on sze, reversing orientation) are captured in this proposition.

Proposition 8.3.2 *Let A and B be two $n \times n$ matrices. Then* $\det(AB) = \det(A)\det(B)$.

Here is a similar proposition about the properties of determinants.

Proposition 8.3.3 *If A is an invertible $n \times n$ matrix, then* $\det A^{-1} = \frac{1}{\det A}$.

In particular, for A to be invertible, it must have a non-zero determinant. It turns out this property is sufficient as well as necessary. A non-zero determinant preserves the dimension: it may increase or decrease size, but it doesn't entirely destroy it. Therefore, there is always a way to go back and to reverse the process.

Proposition 8.3.4 *A $n \times n$ matrix A is invertible if and only if* $\det A \neq 0$.

I can expand the list of criteria for invertibility by including this new determinant condition. Here is the list that I previously presented in Section 7.4 with the new condition added at the end.

Proposition 8.3.5 *Let A be an $n \times n$ matrix. All of the following properties are equivalent.*

- *A is invertible.*
- *A has rank n.*
- *A row reduces to the identity matrix.*
- $\mathrm{Ker}(A) = \{0\}$.
- $\mathrm{Im}(A) = \mathbb{R}^n$.
- *The columnspace of A is $\mathbb{R}^n$.*
- *The columns of A are linearly independent.*
- *The rowspace of A is $\mathbb{R}^n$.*
- *The rows of A are linearly independent.*
- *$Au = v$ has a unique solution u for any choice of $v \in \mathbb{R}^n$.*
- $\det A \neq 0$.

8.3.2 Other Algebraic Properties

Here are two more properties of determinants.

Proposition 8.3.6 *Let A and B be $n \times n$ matrices and let $\alpha \in \mathbb{R}$ be a scalar.*

$$\det(AB) = \det(BA)$$
$$\det \alpha A = \alpha^n \det A$$

Row reduction is an important operation on matrices. It is convenient to know how the determinant changes under row operations.

Proposition 8.3.7 *Let A be an $n \times n$ matrix. If two rows are exchanged, the determinant changes sign (positive to negative, negative to positive). If a row is multiplied by a constant α, the determinant is multiplied by the same constant α. If a multiple of one row is added to another, the determinant is (surprisingly) unchanged.*

All of the properties of determinant listed so far have been multiplicative. The situation for matrix addition and determinants is less elegant: $\det(A+B)$ has no pleasant identity. This is an interesting contrast from many of the other things in this course: determinants are not linear functions $M_n(\mathbb{R}) \to \mathbb{R}$ since they do not act nicely with addition. Instead, they act nicely with multiplication. On sets with various algebraic structures, it is common for an operation to interact well with one structure but poorly with the others. It is not surprisingly that the determinant doesn't work well with matrix multiplication based on the original observation that matrix multiplication doesn't have a clear geometric meaning. The determinant is a very geometric definition, so an operation that is simply algebraic without a geometric meaning is unlikely to easily cooperate with a strongly geometric construction.

8.4 Orthogonality and Symmetry

8.4.1 Preserving Lengths and Angles

Definition 8.4.1 An $n \times n$ matrix A is called **orthogonal** if it preserves lengths. That is, for all $v \in \mathbb{R}^n$, $|v| = |Av|$. $\Diamond$

Preservation of lengths is not the only way to define orthogonal matrices. The following proposition shows the rich structure of this group of matrices. Before stating the proposition, here is a useful definition.

Definition 8.4.2 Let i, j be two indices which range from 1 to n. The **Kronecker delta** is a twice-indexed set of numbers δ_{ij}, which evaluates to 0 whenever $i \neq j$ and 1 whenever $i = j$.

$$\delta_{ij} = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}$$

$\Diamond$

The Kronecker delta is a convenient short-hand to indicate that matching indices are important. The expressions $\delta_{5,6}$, $\delta_{4,10}$ and $\delta_{9,3}$ all evaluate to 0 since the indices do not match. The expressions $\delta_{4,4}$, $\delta_{17,17}$ and $\delta_{1,1}$ all evaluate to 1 since the indices do match.

Proposition 8.4.3 *Let A be an $n \times n$ matrix. Write $A = (a_1, a_2, \ldots, a_n)$ where the a_i are column vectors (for each a_i is a vector in $\mathbb{R}^n$). Then the following seven statements are all equivalent, and all can be taken as the definition of an orthogonal matrix.*

1. *A preserves lengths: $|v| = |Av|$ for all $v \in \mathbb{R}^n$.*

2. *A preserves dot products:* $u \cdot v = (Au) \cdot (Av)$ *for all* $u, v \in \mathbb{R}^n$.
3. $A^T = A^{-1}$.
4. $A^T A = \mathrm{Id}_n$.
5. $a_i \perp a_j$ *for all* $i \neq j$ *and* $|a_i| = 1$.
6. $a_i \cdot a_j = \delta_{ij}$
7. *A preserves angles (the angle between* u *and* v *is the same as the angle between* Au *and* Av *for all* $u, v \in \mathbb{R}^n$*)* and $\det A = \pm 1$.

There are several very surprising statements here. First, preserving angles and preserving lengths are nearly equivalent (preserving angles needs the additional determinant condition). This is a little odd; there isn't a strong intuition that exactly the same transformations should preserve both properties. Second, there is a convenient algebraic method of identifying orthogonal matrices in property d): I can multiply by the transpose: if I get the identity, I know the matrix is orthogonal. The equivalence of property d) and property f) is seen in simply calculating the matrix multiplication; it involves the dot products of all the columns, which must evaluate to 0 or 1 exactly matching the Kronecker delta.

Orthogonal matrices can be thought of as rigid-body transformations. Since they preserve both lengths and angles, they preserve the shape of anything formed of vertices and lines: any polygon, polyhedron, or higher-dimensional analogue. They may not preserve the position (they may be moved around, rotated, reflected, etc.), but they will preserve the shape. This makes orthogonal matrices extremely useful since many applications want to use transformations that don't destroy polygons or polyhedra. Here is a proposition that gathers some other properties of orthogonal matrices.

Proposition 8.4.4

- *In* $\mathbb{R}^2$*, the only orthogonal transformations are the identity, the rotations and the reflections.*
- *If* A *is an orthogonal matrix, so is* A^{-1}.
- *All orthogonal matrices have determinant* ± 1.
- *If* A *and* B *are orthogonal matrices, then so are* AB *and* BA.

Proof. The proofs of these statements will be part of the activity for this week. ■

Example 8.4.5 Even in $\mathbb{R}^3$, orthogonality already gets a bit trickier than just the rotations/reflection of $\mathbb{R}^2$. Consider the following matrix.

$$\begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

This matrix is orthogonal, but it isn't a rotation or reflection in $\mathbb{R}^3$. It is some kind of 'reflection through the origin' where every point is sent to the opposite point with respect to the origin. (The equivalent in $\mathbb{R}^2$ is a rotation by π radians). It isn't even a physical transformation: it's not something I can do with a physical object without destroying it. (If the object were, say, an inflatable beach-ball, this transformation would be equivalent to collasping the ball, turning it inside-out, and reinflating it that way). However, the

transformations still satisfies the condition of orthogonality. □

Example 8.4.6 Here is another example of an orthogonal matrix in $\mathbb{R}^3$.

$$\begin{pmatrix} \frac{5}{\sqrt{38}} & \frac{-3}{\sqrt{67}} & \frac{-23}{\sqrt{2546}} \\ \frac{-3}{\sqrt{38}} & \frac{-7}{\sqrt{67}} & \frac{-9}{\sqrt{2546}} \\ \frac{-2}{\sqrt{38}} & \frac{3}{\sqrt{67}} & \frac{-44}{\sqrt{2546}} \end{pmatrix}$$

To check the orthogonality, I would just take the dot products of the columns with each other. I should get 0 for each of these dot products to show that the columns are perpendicular to each other. Then I would check that the length of each column is 1. □

8.5 Week 8 Activity

8.5.1 Calculating Determinants

Activity 8.5.1 Calculate the determinant of this matrix. Show the steps in the cofactor expansion.

$$\begin{pmatrix} 4 & 0 & -8 \\ -1 & -2 & 3 \\ 5 & 0 & 0 \end{pmatrix}$$

Solution. I do cofactor expansion along the second column to take advantage of the zeros in that column.

$$\begin{aligned} &\begin{vmatrix} 4 & 0 & -8 \\ -1 & -2 & 3 \\ 5 & 0 & 0 \end{vmatrix} \\ &= (-1)(0)\begin{vmatrix} -1 & 3 \\ 5 & 0 \end{vmatrix} + (1)(-2)\begin{vmatrix} 4 & -8 \\ 5 & 0 \end{vmatrix} + (-1)(0)\begin{vmatrix} 4 & -8 \\ -1 & 3 \end{vmatrix} + \end{aligned}$$

The terms multiplied by zero are obviously zero. For the middle term, I use the formula for a 2×2 determinant.

$$\begin{aligned} &= 0 - 2\left[(4)(0) - (5)(-8)\right] + 0 \\ &= -2(40) = -80 \end{aligned}$$

Activity 8.5.2 Calculate the determinant of this matrix. Show the steps in the cofactor expansion.

$$\begin{pmatrix} 0 & -2 & -2 \\ 1 & -6 & 1 \\ -3 & 0 & 1 \end{pmatrix}$$

Solution. I do cofactor expansion along the first row.

$$\begin{aligned} &\begin{vmatrix} 0 & -2 & -2 \\ 1 & -6 & 1 \\ -3 & 0 & 1 \end{vmatrix} \\ &= (1)(0)\begin{vmatrix} -6 & 1 \\ 0 & 1 \end{vmatrix} + (-1)(-2)\begin{vmatrix} 1 & 1 \\ -3 & 1 \end{vmatrix} + (1)(-2)\begin{vmatrix} 1 & -6 \\ -3 & 0 \end{vmatrix} \end{aligned}$$

The term multiplied by zero goes away. For the other two terms, I use the formula for 2×2 determinants.

$$\begin{aligned} &= 0 + 2\left[(1)(1) - (1)(-3)\right] + (-2)\left[(1)(0) - (-3)(-6)\right] \\ &= 0 + 8 + 36 = 44 \end{aligned}$$

Activity 8.5.3 Calculate the determinant of this matrix. Show the steps in the cofactor expansion.

$$\begin{pmatrix} 6 & -1 & 3 \\ -4 & -2 & 0 \\ -2 & -5 & 3 \end{pmatrix}$$

Solution. I do cofactor expansion along the third column, to take advantage of the zero term.

$$\begin{aligned} &\begin{vmatrix} 6 & -1 & 3 \\ -4 & -2 & 0 \\ -2 & -5 & 3 \end{vmatrix} \\ &= (1)(3)\begin{vmatrix} -4 & -2 \\ -2 & -5 \end{vmatrix} + (-1)(0)\begin{vmatrix} 6 & -1 \\ -2 & -5 \end{vmatrix} + (1)(3)\begin{vmatrix} 6 & -1 \\ -4 & -2 \end{vmatrix} \end{aligned}$$

The term multiplied by zero goes away. For the other two terms, I use the formula for the determinant of a 2×2 matrix.

$$\begin{aligned} &= (3)\left[(-4)(-5) - (-2)(-2)\right] + 0 + (3)\left[(6)(-2) - (-1)(-4)\right] \\ &= 48 - 48 = 0 \end{aligned}$$

Activity 8.5.4 Calculate the determinant of this matrix. Show the steps in the cofactor expansion.

$$\begin{pmatrix} 6 & 2 & -1 & 3 \\ 0 & 4 & 8 & 2 \\ 0 & 0 & -3 & 9 \\ 0 & 0 & 0 & -7 \end{pmatrix}$$

Solution. This is a upper triangular matrix. I know for any diagonal or triangular matrix, the determinant is just the product of the diagonal entries. I don't need to do cofactor expansion.

$$\begin{vmatrix} 6 & 2 & -1 & 3 \\ 0 & 4 & 8 & 2 \\ 0 & 0 & -3 & 9 \\ 0 & 0 & 0 & -7 \end{vmatrix} = (6)(4)(-3)(-7) = 504$$

Activity 8.5.5 Calculate the determinant of this matrix. Show the steps in the cofactor expansion.

$$\begin{pmatrix} 4 & -1 & -1 & 6 \\ 0 & -3 & 3 & -2 \\ 0 & 1 & 1 & -2 \\ -2 & 0 & 0 & 3 \end{pmatrix}$$

Solution. I do cofactor expand along the first column to take advantage of

the zero terms.

$$\begin{vmatrix} 4 & -1 & -1 & 6 \\ 0 & -3 & 3 & -2 \\ 0 & 1 & 1 & -2 \\ -2 & 0 & 0 & 3 \end{vmatrix}$$
$$= (1)(4)\begin{vmatrix} -3 & 3 & -2 \\ 1 & 1 & -2 \\ 0 & 0 & 3 \end{vmatrix} + (-1)(0)\begin{vmatrix} -1 & -1 & 6 \\ 1 & 1 & -2 \\ 0 & 0 & 3 \end{vmatrix}$$
$$+ (1)(0)\begin{vmatrix} -1 & -1 & 6 \\ -3 & 3 & -2 \\ 0 & 0 & 3 \end{vmatrix} + (-1)(-2)\begin{vmatrix} -1 & -1 & 6 \\ -3 & 3 & -2 \\ 1 & 1 & -2 \end{vmatrix}$$

The two terms multiplied by zero go away. There are two remaining 3×3 determinants. I'll do these separately and then put their determinant values back into the original equation. Here is the first one; I do cofactor expansion along the third row.

$$\begin{vmatrix} -3 & 3 & -2 \\ 1 & 1 & -2 \\ 0 & 0 & 3 \end{vmatrix}$$
$$= (1)(0)\begin{vmatrix} 3 & -2 \\ 1 & -2 \end{vmatrix} + (-1)(0)\begin{vmatrix} -3 & -2 \\ 1 & -2 \end{vmatrix} + (1)(3)\begin{vmatrix} -3 & 3 \\ 1 & 1 \end{vmatrix}$$
$$= 0 + 0 + 3\left[(-3)(1) - (3)(1)\right]$$
$$= -18$$

Here is the second one; I do cofactor expansion along the first row.

$$\begin{vmatrix} -1 & -1 & 6 \\ -3 & 3 & -2 \\ 1 & 1 & -2 \end{vmatrix}$$
$$= (1)(-1)\begin{vmatrix} 3 & -2 \\ 1 & -2 \end{vmatrix} + (-1)(-1)\begin{vmatrix} -3 & -2 \\ 1 & -2 \end{vmatrix} + (1)(6)\begin{vmatrix} -3 & 3 \\ 1 & 1 \end{vmatrix}$$
$$= (-1)\left[(3)(-2) - (1)(-2)\right] + (1)\left[(-3)(-2) - (1)(-2)\right]$$
$$+ (6)\left[(-3)(1) - (1)(3)\right]$$
$$= 4 + 8 - 36 = -24$$

Now I put these back into the full expression.

$$\begin{vmatrix} 4 & -1 & -1 & 6 \\ 0 & -3 & 3 & -2 \\ 0 & 1 & 1 & -2 \\ -2 & 0 & 0 & 3 \end{vmatrix}$$
$$= (1)(4)\begin{vmatrix} -3 & 3 & -2 \\ 1 & 1 & -2 \\ 0 & 0 & 3 \end{vmatrix} + (-1)(0)\begin{vmatrix} -1 & -1 & 6 \\ 1 & 1 & -2 \\ 0 & 0 & 3 \end{vmatrix}$$
$$+ (1)(0)\begin{vmatrix} -1 & -1 & 6 \\ -3 & 3 & -2 \\ 0 & 0 & 3 \end{vmatrix} + (-1)(-2)\begin{vmatrix} -1 & -1 & 6 \\ -3 & 3 & -2 \\ 1 & 1 & -2 \end{vmatrix}$$
$$= 4(-18) + 0 + 0 + 2(-24) = -120$$

8.5.2 Orthogonal Matrices

Activity 8.5.6 Determine which of these matrices are orthogonal. For the matrices which are not orthogonal, clearly state how they fail the orthogonality criteria.

$$a)\begin{pmatrix} 1 & 1 & -1 \\ -1 & 1 & -1 \\ 0 & 1 & 2 \end{pmatrix}$$

$$b)\begin{pmatrix} \frac{1}{\sqrt{38}} & \frac{3}{\sqrt{19}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{38}} & \frac{3}{\sqrt{19}} & \frac{-1}{\sqrt{2}} \\ \frac{-6}{\sqrt{38}} & \frac{1}{\sqrt{19}} & 0 \end{pmatrix}$$

$$c)\begin{pmatrix} \frac{2}{\sqrt{6}} & \frac{3}{\sqrt{29}} & \frac{-2}{\sqrt{174}} \\ \frac{-1}{\sqrt{6}} & \frac{4}{\sqrt{29}} & \frac{7}{\sqrt{174}} \\ \frac{1}{\sqrt{6}} & \frac{-2}{\sqrt{29}} & \frac{11}{\sqrt{174}} \end{pmatrix}$$

$$d)\begin{pmatrix} \frac{3}{\sqrt{17}} & \frac{-2}{\sqrt{13}} & \frac{-6}{\sqrt{173}} \\ \frac{-2}{\sqrt{17}} & 0 & \frac{-11}{\sqrt{173}} \\ \frac{-2}{\sqrt{17}} & \frac{3}{\sqrt{13}} & \frac{4}{\sqrt{173}} \end{pmatrix}$$

Solution. In a), the columns are pairwise orthogonal; I can check that all the dot products are zero.

$$\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = 0$$

$$\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ -1 \\ 2 \end{pmatrix} = 0$$

$$\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ -1 \\ 2 \end{pmatrix} = 0$$

However, the columns are not unit vectors. The length of the first column is $\sqrt{2}$, the second is $\sqrt{3}$ and the third if $\sqrt{6}$. An orthogonal matrix has to have columns of length one, so this is not an orthogonal matrix.

In b), the columns are pairwise orthogonal; I can check that all the dot products are zero. (Not that for these checks, I can scale the vectors as I wish, since whether or not the dot product is zero doesn't depend on the scaling. I've scaled the columns to remove the square roots to make these calculations easier.)

$$\begin{pmatrix} 1 \\ 1 \\ -6 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 3 \\ 1 \end{pmatrix} = 0$$

$$\begin{pmatrix} 1 \\ 1 \\ -6 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} = 0$$

$$\begin{pmatrix} 3 \\ 3 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} = 0$$

In addition, the column vectors all have length one.

$$\left| \begin{matrix} \frac{1}{\sqrt{38}} \\ \frac{1}{\sqrt{38}} \\ \frac{-6}{\sqrt{38}} \end{matrix} \right| = \frac{1}{\sqrt{38}} \left| \begin{matrix} 1 \\ 1 \\ -6 \end{matrix} \right| = \frac{1}{\sqrt{38}} \sqrt{1^2 + 1^2 + (-6)^2} = \frac{\sqrt{38}}{\sqrt{38}} = 1$$

$$\left| \begin{matrix} \frac{3}{\sqrt{19}} \\ \frac{3}{\sqrt{19}} \\ \frac{1}{\sqrt{19}} \end{matrix} \right| = \frac{1}{\sqrt{19}} \left| \begin{matrix} 3 \\ 3 \\ 1 \end{matrix} \right| = \frac{1}{\sqrt{19}} \sqrt{3^2 + 3^2 + 1^2} = \frac{\sqrt{19}}{\sqrt{19}} = 1$$

$$\left| \begin{matrix} \frac{1}{\sqrt{2}} \\ \frac{-1}{\sqrt{2}} \\ 0 \end{matrix} \right| = \frac{1}{\sqrt{2}} \left| \begin{matrix} 1 \\ -1 \\ 0 \end{matrix} \right| = \frac{1}{\sqrt{2}} \sqrt{1^2 + (-1)^2 + 0^2} = \frac{\sqrt{2}}{\sqrt{2}} = 1$$

In c), the columns are pairwise orthogonal; I can check that all the dot products are zero. (Note that for these checks, I can scale the vectors as I wish, since whether or not the dot product is zero doesn't depend on the scaling. I've scaled the columns to remove the square roots to make these calculations easier.)

$$\begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 4 \\ -2 \end{pmatrix} = 0$$

$$\begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 7 \\ 11 \end{pmatrix} = 0$$

$$\begin{pmatrix} 3 \\ 4 \\ -2 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 7 \\ 11 \end{pmatrix} = 0$$

In addition, the column vectors all have length one.

$$\left| \begin{matrix} \frac{\sqrt{2}}{\sqrt{6}} \\ \frac{-1}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \end{matrix} \right| = \frac{1}{\sqrt{6}} \left| \begin{matrix} 2 \\ -1 \\ 1 \end{matrix} \right| = \frac{1}{\sqrt{6}} \sqrt{2^2 + (-1)^2 + 1^2} = \frac{\sqrt{6}}{\sqrt{6}} = 1$$

$$\left|\begin{matrix}\frac{3}{\sqrt{29}}\\ \frac{4}{\sqrt{29}}\\ \frac{-2}{\sqrt{29}}\end{matrix}\right| = \frac{1}{\sqrt{29}}\left|\begin{matrix}3\\4\\-2\end{matrix}\right| = \frac{1}{\sqrt{29}}\sqrt{3^2+4^2+(-2)^2} = \frac{\sqrt{29}}{\sqrt{29}} = 1$$

$$\left|\begin{matrix}\frac{-2}{\sqrt{174}}\\ \frac{7}{\sqrt{174}}\\ \frac{11}{\sqrt{174}}\end{matrix}\right| = \frac{1}{\sqrt{174}}\left|\begin{matrix}-2\\7\\11\end{matrix}\right| = \frac{1}{\sqrt{174}}\sqrt{(-2)^2+7^2+(11)^2} = \frac{\sqrt{174}}{\sqrt{174}} = 1$$

In d), the columns are not pairwise orthogonal; I can check to see that this fails with the first and third columns.

$$\begin{pmatrix}3\\-2\\-2\end{pmatrix}\cdot\begin{pmatrix}-6\\-11\\4\end{pmatrix}$$
$$= (3)(-6)+(-2)(-11)+(-2)(-4) = -18+22+8 = 12 \neq 0$$

Since the columns are not orthogonal, this cannot be an orthogonal matrix. (I could check that the columns are unit vectors, which indeed they are, but I don't need to since I've already shown that this cannot match the criteria for othogonality).

Activity 8.5.7 For this incomplete matrix, add a final column which produces an orthogonal matrix.

$$\begin{pmatrix}\frac{5}{\sqrt{41}} & \cdot\\ \frac{4}{\sqrt{41}} & \cdot\end{pmatrix}$$

Solution. I simply need to add a vector of length one which is perpendicular to the given column. If I switch the terms in the vector and make one of them negative, I can accomplish this. (This isn't an algorithmic solution; this is essentially by inspection. I'm looking at the matrix and making a guess based on my familiarity with vectors in $\mathbb{R}^2$.)

$$\begin{pmatrix}\frac{5}{\sqrt{41}} & \frac{-4}{\sqrt{41}}\\ \frac{4}{\sqrt{41}} & \frac{5}{\sqrt{41}}\end{pmatrix}$$

Note there are two possible answers; if I choose to make the other component negative, the following is another option.

$$\begin{pmatrix}\frac{5}{\sqrt{41}} & \frac{4}{\sqrt{41}}\\ \frac{4}{\sqrt{41}} & \frac{-5}{\sqrt{41}}\end{pmatrix}$$

These are, however, the only two possible solutions.

Activity 8.5.8 For this incomplete matrix, add a final column which produces

an orthogonal matrix.

$$\begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{-3}{\sqrt{22}} & \cdot \\ \frac{1}{\sqrt{2}} & \frac{3}{\sqrt{22}} & \cdot \\ 0 & \frac{-2}{\sqrt{22}} & \cdot \end{pmatrix}$$

Solution. Unlike the previous matrix, this is much harder to do by inspection. Fortunately, I have an algorithmic approach. I want a new vector in $\mathbb{R}^3$ which is orthogonal to two given vector (the two columns). In $\mathbb{R}^3$, the cross product gives exactly this desired vector. I'll take the cross product of the two columns. For ease of calculation, I'll scale the vectors to remove the denominators to find this third vector, since all I care about at this point is orthogonality.

$$\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \times \begin{pmatrix} -3 \\ 3 \\ -2 \end{pmatrix} = \begin{pmatrix} -2 \\ 2 \\ 6 \end{pmatrix}$$

This produces a vector othogonal to both columns. Now I want to scale it so that it is unit vector. The length of this vector is $\sqrt{(-2)^2 + 2^2 + 6^2} = \sqrt{44}$, so I divide by this to make a unit vector. That produces this orthogonal matrix.

$$\begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{-3}{\sqrt{22}} & \frac{-2}{\sqrt{44}} \\ \frac{1}{\sqrt{2}} & \frac{3}{\sqrt{22}} & \frac{2}{\sqrt{44}} \\ 0 & \frac{-2}{\sqrt{2}} & \frac{6}{\sqrt{44}} \end{pmatrix}$$

Note, again, that there are two possible answers. If I did the cross-product in the other order, I would get the same vector multiplied by (-1). That produces this other answer.

$$\begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{-3}{\sqrt{22}} & \frac{2}{\sqrt{44}} \\ \frac{1}{\sqrt{2}} & \frac{3}{\sqrt{22}} & \frac{-2}{\sqrt{44}} \\ 0 & \frac{-2}{\sqrt{2}} & \frac{-6}{\sqrt{44}} \end{pmatrix}$$

These are the only two solutions.

Activity 8.5.9 For this incomplete matrix, add a final column which produces an orthogonal matrix.

$$\begin{pmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} & \cdot \\ 0 & \frac{1}{\sqrt{2}} & 0 & \cdot \\ \frac{-1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} & \cdot \\ 0 & \frac{-1}{\sqrt{2}} & 0 & \cdot \end{pmatrix}$$

Solution. Now I again lack an algorithmic approach, since I don't have a cross product in $\mathbb{R}^4$. So I have to try to work by inspection again, which is tricky. Looking at the first and third columns, these are only non-zero in the first and third coordinates. The second column is non-zero only in the second and fourth coordinate. I might try to match the second column and take zeros in the first and third components; that guarantees orthogonality with the first

and third vector. Then I need to choose components to be orthogonal to the second column; that can be done by copying the second colum and setting one of the components to be negative. That gives the following answer.

$$\begin{pmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} & 0 \\ 0 & \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ \frac{-1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} & 0 \\ 0 & \frac{-1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{pmatrix}$$

8.5.3 Proof Questions

Activity 8.5.10 Prove that all rotations in $\mathbb{R}^2$ have determinant 1.

Solution. The general rotation by an angle θ in $\mathbb{R}^2$ has matrix

$$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

I can simply calculate the determinant of this matrix.

$$\begin{vmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{vmatrix} = \cos^2\theta + \sin^2\theta = 1$$

That completes the proof.

Activity 8.5.11 Prove that all reflections in $\mathbb{R}^2$ have determinant 1.

Solution. The general reflection in the (unit vector) direction $\begin{pmatrix} a \\ b \end{pmatrix}$ in $\mathbb{R}^2$ has matrix

$$\begin{pmatrix} a^2 - b^2 & 2ab \\ 2ab & b^2 - a^2 \end{pmatrix}.$$

I can simply calculate the determinant of this matrix.

$$\begin{vmatrix} a^2 - b^2 & 2ab \\ 2ab & b^2 - a^2 \end{vmatrix} = (a^2 - b^2)(b^2 - a^2) - (2ab)(2ab)$$
$$= -a^4 + 2a^2b^2 - b^4 - 4a^2b^2 = -(a^4 + 2a^2b^2 + b^2) = -(a^2 + b^2)^2$$

Since the description uses a unit vector for the direction, $a^2 + b^2 = 1$. With that fact, I can see that the determinant is -1, which completes the proof.

Activity 8.5.12 Let M be a square matrix. Prove that $\det M = \det M^T$.

Solution. I can argue this fact using cofactor expansion. Since cofactor expansions works along rows or columns, if I do cofactor expansion along a row of M, that is the same as cofactor expansion along the matching column of M (since the transpose turns rows into columns). Since both the row and column are removed in the cofactor expansion, the effect on the matrix M and its transpose is the same.

Using this cofactor expansion argument, I can reduce the question down to only 2×2 matrices. Here, I can simply check by calculating the determinant

of a general 2×2 matrix and its transpose.

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

$$\begin{vmatrix} a & c \\ b & d \end{vmatrix} = ad - cb = ad - bc$$

This completes the proof.

Activity 8.5.13 Prove that the inverse of an orthogonal transformation is also orthogonal.

Solution. This is relatively easy to argue based on the concept and main definition. Orthogonal matrices preserve lengths. If lengths do not change when I do a transformation forward, they also can't change when I unto the transformation and go backwards. Undoing no change is still no change.

Activity 8.5.14 Prove that the composition of orthogonal transformations is still orthogonal.

Solution. Like the previous proof, I can argue this conceptually using the basic definition. If two transformations preserve lengths, then doing them one after another will also preserve lengths. Lengths can't change in the first step and can't change in the second step, so they can't change at all. Therefore, the composition (doing one transformation after the other) must also be orthogonal.

Activity 8.5.15 Prove that all orthogonal matrices have determinant ± 1.

Solution. For this proof, I'm going to use one of the algebraic properties of orthogonal matrices.

$$A^T A = \mathrm{Id}$$

Then I'll take the determinant of both sides of this matrix equation.

$$\det(A^T A) = \det(\mathrm{Id})$$

The determinant of the identity is 1. On the left side, the determinant is multiplicative.

$$\det(A^T)\det(A) = 1$$

One of the properties of the determinant is that the determinant of a transpose is the same at the determinant of the original.

$$\det(A)\det(A) = 1 \implies \det(A)^2 = 1$$

So the number $\det(A)$ is a number which squares to 1. The only numbers with this property are 1 and -1. That completes the proof.

Activity 8.5.16 Prove that the rotations and the reflections are the only orthogonal transformations in $\mathbb{R}^2$. (This one is a lot trickier than the previous proofs. Give it a shot, but don't distress too much if the approach is difficult to find.)

Solution. This proof is substantially more difficult to approach than the three previous one. There are many transformations in $\mathbb{R}^2$. I can't even just prove this for the five basic types, since there are complicated transformations that are combinations of the types and the combination might be orthogonal even when the pieces aren't. I need to find a trick or idea that lets me approach this.

The idea I thought to use was acting on the standard basis: $\{e_1, e_2\}$. Since

these two vectors span all of $\mathbb{R}^2$, if I know what happens to these two vectors, I know the entire transformation. First, the transformation does something to e_1, sending it to Me_1. Since length is preserved, this is some other unit vector.

Now where can e_2 go? There are, in fact, only two choices. Me_2 must also be a unit vector, since lengths are preserved. Also, angles are preserved, so the angle between the Me_1 and Me_2 must still be $\frac{\pi}{2}$ (they are still perpendicular). That gives only two choices. If I face in the direction of Me_2, then I can have the unit vector pointing in the perpendicular direction to the right or to the left.

If Me_2 is the perpendicular direction to the left, then M is a rotation. The relative positions of the two vectors are preserved and they are still unit vectors, so the are moved around the circle by whatever the angle is between e_1 and Me_1.

If Me_2 is the perpendicular direction to the right, then M is a reflection. This is a bit trickier to see, but the line of reflection will be the line halfway between e_1 and Me_1. That line will flip e_1 into Me_1 by construction. But the flip will take anything perpendicular on the right and flip it to something perpendicular on the left. So e_2, which starts as the perpendicular vector on the right, ends up as the perpendicular direction on the left.

M must have one of these two behaviours, so M must be a rotation or a reflection.

8.5.4 Conceptual Review Questions

- What is a determinant and what does it mean?
- Why is the determinant multiplicative but not linear?
- What is an orthogonal matrix?
- Why are there so many equivalent descriptions of an orthogonal matrix?

Chapter 9

Week 9

9.1 Dihedral Groups

9.1.1 Dihedral Groups

The case of regular polygons, such as the hexagon, is a very useful place to talk about symmetry. I want to know which linear transformations preserve the regular polygons. By 'preserve', in this section I mean that the new shape is precisely the same as the old: same size and same positions. Some of the vertices and edges may have been moved, but the result ends up tracing over exaclty the same original shape.

Definition 9.1.1 A **Dihedral Group** is the group of linear symmetries of a regular polygon. If the polygon has n edges, then I write D_n for the associated dihedral group. Thus, D_3 is the group of symmetries of the triangle, D_4 is the group of symmetries of the square, D_5 is the group of symmetries of the pentagon, and so on. (Some texts use D_{2n} instead of D_n.) $\Diamond$

You may have noticed that I used the word 'group' instead of 'set' to describe this collection. 'Group' is a technical word in mathematics, referring to a set with some extra properties. Let me give the definition in somewhat formal language. (In this definition, a binary operation is a process that takes two elements and combines them into one. It is a general term that includes addition of numbers, multiplication of numbers, or composition of transformations: in each operation, two things are combined together).

Definition 9.1.2 Let G be a set with an (associative) binary operation. I represent the operation by writing two elements next to each other (as is common with multiplication of variables). G is a group if it satisfies these properties.

- There is an element e, called the identity element. If g is any other element of the group, then $ge = eg = g$. (Multiplication by the identity doesn't do anything to any group element).
- For each element of the group g, there is a unique element g^{-1} such that $gg^{-1} = g^{-1}g = e$. (Everything in a group has an inverse).

$\Diamond$

You already know a number of groups.

- $\mathbb{Z}$ is a group under the operation of addition. 0 is the identity element, and the inverse of any a is $-a$.

- $\mathbb{Q} \setminus \{0\}$ is a group under multiplication. 1 is the identity, and the inverse of any fraction $\frac{a}{b}$ is $\frac{b}{a}$. Obviously, I needed to exclude 0 since I cannot divide by 0, and therefore it wouldn't have an inverse in the group.
- $\mathbb{R} \setminus \{0\}$ is also a group under multiplication.
- In Definition 7.1.4, I defined the general linear group $GL_n(\mathbb{R})$. This was all $n \times n$ invertible matrices. The identity matrix (of size n) is the identity element; the composition is the 'multiplication,' and the matrix inverse is the inverse operation. By restriction to only invertible matrices of a fixed size, all elements can be composed together, and all elements have inverses.
- In Definition 9.2.1, I defined another matrix group: the special linear group $SL_n(\mathbb{R})$. This was like the general linear group, but with the added restriction of determinant 1.
- In Definition 9.2.2, I defined two more matrix groups: the orthogonal group $O_n(\mathbb{R})$ and the special orthogonal group $SO_n(\mathbb{R})$. These are smaller sets of invertible matrices with particular properties. Each of them contains the identity matrix and has inverses, which are matrices with the same restricted properties.

In this course, I will primarily discuss dihedral groups to understand what groups are all about. In other institutions with larger mathematics programs, courses are devoted exclusively to group theory, which is an active and important area of mathematical research.

9.1.2 The Group D_4

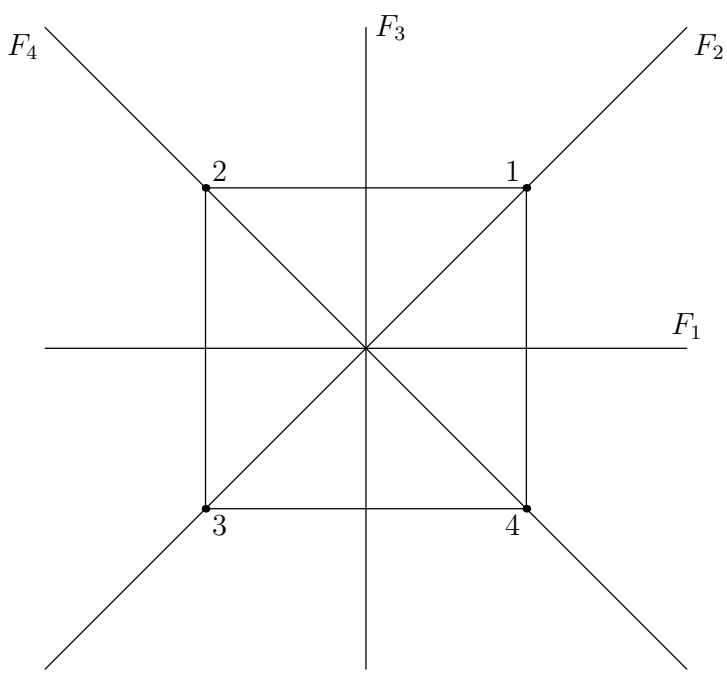

Figure 9.1.3 Symmetries of the Square

Consider the symmetries of the square. I assume the origin is at the centre of the square, and I can put the vertices of the square at $(1, 1)$, $(1, -1)$, $(-1, -1)$ and $(-1, 1)$. Conveniently, the vertices are all I need to think about: since

linear transformations preserve lines, if I know where the vertices go, then I know that the lines connecting them are preserved, and hence the shape is preserved. Which transformations preserve the vertices?

There are two relatively obvious classes that might occur. First, there is rotation about the origin. If I rotate by $\pi/2$ radians (a quarter turn), the vertices are preserved. Likewise for rotations by π and $3\pi/2$ radians. If I am really paying attention, I might also think of rotation by 2π radians, a full turn. This might seem silly, but the identity transformation (the transformation that leaves everything fixed) is a very important transformation and should be included in the list. So, for now, the symmetries of the square include the identity and three rotations. I'll call these three rotations R_1, R_2 and R_3, in the order I defined them.

I might also think about reflections. Looking at the square, I can find four lines of reflection: the x-axis, the y-axis, the line $x = y$ and the line $x = -y$. I'd like to label the reflection by taking the lines of reflection in counter-clockwise order starting from the x-axis: the reflection over the x-axis is F_1; the reflection over $y = x$ is F_2; the reflection over the y-axis is F_3; and the reflection over $y = -x$ is F_4. The lines in Figure 9.1.3 are labelled with these reflection names.

That brings the total number of transformations to eight: the identity, three rotations, and four reflections. These eight are, in fact, the complete group.

So I've constructed a group of symmetries of the square. I have eight elements, one of which is the identity and the operation of composition. The abstract question that mathematicians ask now is: what is the structure of this group? By 'structure,' I mean the elements but, more importantly, how the elements relate to each other when composed. To show this structure, I can build a multiplication table for the group. Such a table would explicitly show all the compositions. It is structured very much like the multiplication tables for numbers that you learned in elementary school — you are now just studying a new kind of multiplication.

To make the multiplication table, I need to know how to calculate the composition of elements. I could do this by writing their matrices and doing the matrix multiplication. That would work, but I want to show you an easier method — one much more connected to the geometry of the square. I'm going to represent each element of the group simply by how it permutes the vertices. In Figure 9.1.3, I labelled the vertices. By showing where each vertex goes, I can describe each transformation. For example, the rotation R_1 sends each vertex to the next vertex counterclockwise. I can represent this by the following vertex movements.

$$\begin{aligned} 1 &\mapsto 2 \\ 2 &\mapsto 3 \\ 3 &\mapsto 4 \\ 4 &\mapsto 1 \end{aligned}$$

The reflection F_2 is over the line $y = x$. It fixes the two vertices which lie on that line and switches the other two. I can represent this by the following vertex movements.

$$\begin{aligned} 1 &\mapsto 1 \\ 2 &\mapsto 4 \\ 3 &\mapsto 3 \\ 4 &\mapsto 2 \end{aligned}$$

Now I can use these vertex movements to understand composition. Consider $R_1 \circ F_2$. Since composition works from right to left, this is doing the reflection

first, then the rotation. I can put the two sets of vertex movements together.

$$1 \mapsto 1 \mapsto 2$$
$$2 \mapsto 4 \mapsto 1$$
$$3 \mapsto 3 \mapsto 4$$
$$4 \mapsto 2 \mapsto 3$$

Then I can ask: what is this transformation? This switches vertices 1 and 2, and also switches the other two vertices. This must be reflected over the y-axis, which is F_3. Therefore, $R_1 \circ F_2 = F_3$. I've calculated a multiplication. What happens if I reverse the order: $F_2 \circ R_1$? Now the rotation happens first, but I can still calculate the vertex movements by doing the rotation then the reflection.

$$1 \mapsto 2 \mapsto 4$$
$$2 \mapsto 3 \mapsto 3$$
$$3 \mapsto 4 \mapsto 2$$
$$4 \mapsto 1 \mapsto 1$$

This transformation switches vertices 1 and 4, and also switches vertices 2 and 3. It must be the reflection over the x-axis, which is called F_1. Therefore, $F_2 \circ R_1 = F_1$.

In this way, I can calculate all the compositions and put them into a multiplication table. In this table, each entry is the composition of the row and then the column; therefore, the column entry happens first (again, right to left) and the row entry second.

Table 9.1.4

$\circ$	Id	R_1	R_2	R_3	F_1	F_2	F_3	F_4
Id	Id	R_1	R_2	R_3	F_1	F_2	F_3	F_4
R_1	R_1	R_2	R_3	Id	F_2	F_3	F_4	F_1
R_2	R_2	R_3	Id	R_1	F_3	F_4	F_1	F_2
R_3	R_3	Id	R_1	R_2	F_4	F_1	F_2	F_3
F_1	F_1	F_4	F_3	F_2	Id	R_3	R_2	R_1
F_2	F_2	F_1	F_4	F_3	R_1	Id	R_3	R_2
F_3	F_3	F_2	F_1	F_4	R_2	R_1	Id	R_3
F_4	F_4	F_3	F_2	F_1	R_3	R_2	R_1	Id

From the table, I can see a number of patterns, all of which are part of the structure of this group. Even with eight elements, there is already a fair bit to talk about. The composition of any two rotations is another rotation. Moreover, rotations commute with each other (they can be composed in either order without changing the result). The composition of any two reflections is also a rotation. The composition of any rotation and any reflection is some other reflection. The rotations and reflections do not commute, nor do reflections commute with each other. In each row and each column, each group element shows up exactly once.

9.2 Matrix Groups

9.2.1 Symmetry, Again

A major theme of this course is symmetry. When I defined linear transformations, I defined them to be transformations that preserved something. Geomet-

rically, they preserved linear subspaces: each linear subspace is sent to another linear subspace. Algebraically, they preserved the linear operations of addition and scalar multiplication: I can do vector addition or scalar multiplication before or after the transformation and the result is the same. This the general notion of symmetry that I want to develop in this course: transformations that preserve something (either algebraic or geometric).

This general idea of symmetry continues to develop as the course moves deeper into the discipline of linear algebra. The classification of matrices, at least geometrically, is almost always a question of symmetry. In this section, I'm going to define some new classes of matrices in terms of their symmetries: in terms of what they preserve.

Recall the notation I've already established for matrix sets and groups (with real coefficients).

- The set of $n \times m$ matrices is written $M_{n,m}(\mathbb{R})$.
- The set of square $(n \times n)$ matrices is written $M_n(\mathbb{R})$.
- The group of invertible matrices is called the general linear group and written $GL_n(\mathbb{R})$ or $GL(n)$.

9.2.2 Transformations that Preserve Size and Orientation

Determinants measure two geometric properties: size and orientation. These are both potential symmetries. If a matrix has determinant ± 1, I know that it preserves size. If a matrix has positive determinant, I know that it preserves orientation. If a matrix has determinant one, it preserves both.

Definition 9.2.1 The group of matrices of determinant one with real coefficients is called the **Special Linear Group** and is written $SL_n(\mathbb{R})$ or $SL(n)$. $\Diamond$

Definition 9.2.2 The group of orthogonal $n \times n$ matrices is called the **Orthogonal Group** and is written $O_n(\mathbb{R})$ or $O(n)$. The group of orthogonal matrices, which also preserve orientation (have determinant one) is called the **Special Orthogonal Group** and is written $SO_n(\mathbb{R})$ or $SO(n)$. $\Diamond$

9.3 Abstract Vector Spaces

9.3.1 Abstract and Euclidean Space

In Subsection 9.1.1, I defined the abstract structure of a group: an idea that ties together patterns of transformations. Here, I want to do much the same. Instead of talking about groups, I want to talk much more holistically about what we've accomplished so far in this course. This course began with $\mathbb{R}^n$, real euclidean space, and I presented it all as a a geometry course: I wanted to understand euclidean space and its structures. What structures have I found? Several, but notable these: addition of vectors; scalar multiplication; linear combinations; linear independence; spans; subspaces; and transformations.

Now I want to abstract these structures. The same way a group was an abstraction to talk about collections of invertible tranformations, I want an abstraction to talk about these linear algebra definition. What is the pattern that I need to use these terms, these defnition? How can I use these ideas as broadly as possible>

This is the main strength of an abstraction: an abstraction is difficult to undersatnd, being removed from its source and turned into just a pattern, but

the abstraction applies more broadly than its source. If I abstract a structure, I can use it in many places, even if it is harder to understand than just a concrete instance.

I'm looking to abstract euclindean space, the space of vectors. What has the same kind of structure? It all leads to this definition.

Definition 9.3.1 A set V is an **abstract vector space** if it satisfies three properties.

1. There is an addition operation on V. For $u, v \in V$, I can perform $u + v$ and get some other element of $V/$. This addition is associative and commutative.

2. There is a scalar multiplication operation on V. For $u \in V$ and for any real number $a \in \mathbb{R}$, there is way to calculate av and get some new element of V.

3. The scalar multiplication is distributive over the addition. That is, for any real number $a \in \mathbb{R}$ and $u, v \in V$, it is true that $a(u + v) = au + av$.

$\Diamond$

This is how I extend the idea vector space. The things in V are usually no longer vectors. But since they have the linear opeartions of addition and scalar multiplication, they act enough like vectors to use the language of linear algebra. Just these properties is enough to talk about linear combinations, linear independence, span, bases, transformations, image, kernels, and many other terms from the course. This adds great value to everything we've done so far in this course, because suddenly there are all these new environment for the use of these terms and ideas. In many way, this definition justified the important place that linear algebra has in university mathematics. Vectors and their structures are important, but linear algebra is so central, in great part, because of ubiquitous its terms are used in other settings, in other abstract vector spaces.

Before, I get to examples, I must also note that something is lost. In any abstraction process, by abstracting a structure, some particulars are left behind. What is lost here?

- Mostly obvioulsy, the things in an abstract vector space are no longer vectors: list of numbers, positions in space.

- Loosing explicit vectors means loosing geometric interpretation, for the most part. The elements of an abstract vector space may no longer have the geometric meaning of vectors. In particular, the whole structure of loci may be gone, or at least radically changed.

- The matrix interpretation of transformations came from the matrix action on vectors. For other objects, that action might not work any more, or at least not as easily. Linear transformation may no longer be matrices.

- Since both matrices and geometry are lost, ideas like impact on volume and orientation probably don't make sense anymore, so determinants are also lost.

- Finally and most interestingly, the finite dimensional nature of $\mathbb{R}^n$ may also be lost. Many asbtract vectors spaces are infinite dimension, as I'll show shortly in the examples.

9.3.2 Examples of Abstract Vector Spaces

Example 9.3.2 The set $P(\mathbb{R})$ consists of all polynomial in the variable x with real coefficient.

- I can add polynomials: if $p(x)$ and $q(x)$ are polynomials, then $p(x)+q(x)$ is still a polynomial.
- I can multiply polynomials by scalars: if $a \in \mathbb{R}$ is a real number and $p(x)$ is a polynomial, the $ap(x)$ is still a polyonomial.
- Since all the opeartions here are inherited from ordinary arithmetic of numbers and variables, the distributive law of multiplication over addition applies to scalar multiplication and polynomial addition.

Having satisfied the three criteria, this set of polynomials is an abstract vector space. Therefore, I can make claims like the following.

- The polynomial $4x^2 - 4$ is in Span$\{x + 4, x^2 + x + 3\}$. The span is all lienar combination, so I can justify this claim by a demonstration.

  $$4x^2 + 4 = 4(x^2 + x + 3) - 4(x - 4)$$

  This equation shows that $4x^2 + 4$ is a linar combination of the spanning vectors, therefor a member of the span.
- The set of polynomials $\{x^3 - 4, x^5 - x^2, 3x^4\}$ is linearly independent. (To prove this, I would write the equation $a(x^3 - 4) + b(x^5 - x^2) + c(3x^4) = 0$ and prove that the only solution is $a = b = c = 0$.)
- The set Span$\{x^3, x^7, x^10, x^{12}\}$ has dimension four. (To prove this, I would have to prove that the four polynomials are a basis, which I would do by proving that they are linearly independent.)

As I have already mentioned, one of the stranger things about abstract vectors spaces is that they can be infinite dimensions. This space of polynomials is the first example of an infinite dimensional vector space. What does that mean? Well, the definition of dimension came from a basis: dimensions is the number of elements in a basis. So, what is a basis for the polynomials?

A basis is a spanning set, so I need some core elements such that any other polymomial can be built out of them. One clear choice are the monomials: $\{1, x, x^2, x^3, x^4, x^5, \ldots\}$. Is this a linearly independent set? Yes, it is. There is not way to get a monomial as a sum of other monomials with different degreres. (For example, there is not way to write x^4 as $a + bx + cx^2 + dx^3 + ex^4$ and so on. The only way to get the power x^4 from some other monomial is to multiply or divide by the variable, and linear combinations are just multiplicatons by scalars and addition. This is a linearly independent set.

So I have this linear independent set $\{1, x, x^2, x^3, \ldots\}$. By writing it this way, I imply that the set never stops, that is include x^n for all $n \in \mathbb{N}$. I implies that it is an infinite set. Does it need to be? Well, if it were finite, then there would be a highest monomial, some x^k. Even if k is a very, very large number, I can still ask about the polynomial $p(x) = x^{k+1}$. There is no limit on the degree of a polynomial. Therefore, no finite subset work. All of these infinitely many monomials are necessary to build all the polynomials. Therefore, they are a basis. Since the basis is infinite, the abstract vector space is infinite dimensional. $\square$

Example 9.3.3 The set $C^\infty(\mathbb{R})$ is the set of all convergent power series in the variable x with domain $\mathbb{R}$.

- I can add power series: if $\sum_{n=0}^{\infty} f_x x^n$ and $\sum_{n=0}^{\infty} g_x x^n$ are power series, then their addition is $\sum_{n=0}^{\infty}(f_n + g_n)x^n$ is still a power series which converges for all or $\mathbb{R}$.

- I can multiply power series by scalars: if $a \in \mathbb{R}$ is a real number and $\sum_{n=0}^{\infty} f_n x^n$ is an infinite series, the $a\sum_{n=0}^{\infty} f_n x^n = \sum_{n=0}^{\infty} a f_n x^n$ is still a convergent power series.

- Since all the opeartions here are inherited from ordinary arithmetic of numbers and variables, the distributive law of multiplication over addition applies to scalar multiplication and power series addition.

Having satisfied the three criteria, this set of polynomials is an abstract vector space. As with polynomials, I could make linear combinations and spans, ask about linear independence and bases, and otherwise use the tool of linear algebra to talk about series.

Like the polynomials, this is also an infinite dimensional abstract vector space. Justifying this is not too difficult, since any polynomials is also a zero by letting all the highter coefficient be zero, so $\text{Span}\{1, x, x^2, x^3 \ldots, \}$ is already contained within the infinite series. If it contains infinite dimensions already, then it itself must be infinite dimensional. However, actually talking about the basis for series is tricky. First and foremost, why doesn't $\text{Span}\{1, x, x^2, x^3 \ldots\}$ produce all the infinite series? A series is, after all, a sum $f(x) = f_0 + f_1 x + f_2 x^2 + f_3 x^3 + f_4 x^4 + \ldots$. This looks like something I can get from the span of monomials. Can't I> I can't, but the reason is a a strange subtlety, one that I haven't had to deal with yet. In all of the sums of linear algebra, such as the sums that make liner combinations, the sums must be finite. I never had to worry about this with actual vectors, since I never had to consider infinite sums of vector. But here, an infinite defined by an infinite sum. Therefore, the monomial are not a basis: series are not (finite) linear combinations of monomials.

So what is a basis for the infinite series? It would have to be some infinite set of linearly independent infinite series such that any series could be written as a (finite!) some of some of the basis elements. Such a basis would be very difficult to find. □

Example 9.3.4 The set $C^0(\mathbb{R})$ is the set of all continuous function with domain $\mathbb{R}$

- I can add continuous function: if $f(x)$ and $g(x)$ are two continuous function defines for all real number, then $f(x) + g(x)$ is still a function, still has domain $\mathbb{R}$, and is still continuous.

- I can multiply continuous function by scalars: if $a \in \mathbb{R}$ is a real number and $f(x)$ is a continuous functionn, then $af(x)$ is also a continuous function define for all real number.

- Since all the opeartions here are inherited from ordinary arithmetic of numbers and variables, the distributive law of multiplication over addition applies to scalar multiplication and the addition of continuous functions.

Having satisfied the three criteria, this set of continuous functions is an abstract vector space. As with polynomials and series above, I could make

linear combinations and spans, ask about linear independence and bases, and otherwise use the tool of linear algebra to talk about continuous function.

Like the polynomials and infinite series, this is also an infinite dimensional abstract vector space. Again, I can justify this claim simply by the fact that polynomials themselves are continuous function defined on all real numbers. Since there is a subset which is already infinite dimensional, the whole set must also be infinite dimensional.

The basis question here is extremely fraught, even more so than for infinite series. There are many, many, many continuous functions: any path that I can draw in R^2 that extends to infinite in both directions and satisfies a vertical line test is the graph of a continuous function. The variety here is, literally, endless. It's difficult to even consider the idea of a basis. If I were to investigate this question, I would find that this abstract vector space is, in fact, uncountably infinite (for those who know what that means). The question as to whether such an abstract vector space even has a basis is not at all clear: this relates to some very deep and important questions at the very core of mathematics. For more information, ask me sometime about the Axiom of Choice and its implications.

Even though this abstract vector space of continuous functions is unbelievably enormous, considering it as an abstract vector space is still very useful. In particular, the solutions to differential equations are often continuous function. The space of solution (since there are often infnintely many solutions) is a set of function. In many instances, this set can be described as a span of finitely many functions: it has a dimension. To describe these solutions, I need a basis: I need to find linearly independent functions to span this solutions space. In this way, linear algebra language gets used to describe solutions to differential equations, an solution to differential equations benefit from this description. □

9.4 Week 9 Activity

9.4.1 Dihedral Groups

Activity 9.4.1 Make a multiplication table for the dihedral group D_5. (Label the rotations R_i; the index i indicates how many fifth of a full turn for the rotation. Label the reflections F_i; the index indicates reflection over a line through the i vertex, counting counterclockwise. Then start calculating the compositions using vertex opertaions. One you have done a number of these, you can start looking for patterns in the multiplication table and potentially use those patterns to finish the table.)

Solution. There are many multiplications to calculate. I recommend calculating matrices by vertex operations. If I number the vertices of the pentagon counterclockwise from 1 to 5, then R_2 (rotation by $\frac{2}{5}$ of a turn) can be represented by this movement of vertices.

$$\begin{aligned} 1 &\mapsto 3 \\ 2 &\mapsto 4 \\ 3 &\mapsto 5 \\ 4 &\mapsto 1 \\ 5 &\mapsto 2 \end{aligned}$$

Reflection F_3 (reflection over the line through vertex 3 and the middle of the

opposite edge)) can be represented by this action on the vertices.

$$1 \mapsto 5$$
$$2 \mapsto 4$$
$$3 \mapsto 3$$
$$4 \mapsto 2$$
$$5 \mapsto 1$$

To calculate $R_2 \circ F_3$, I can use these vertex changes. In composition, I do the right transformation first. Therefore, starting at vertex 1, F_3 send it to vertex 5, and then R_2 sends vertex 5 to vertex 2. Therefore, the whole composition sends vertex 1 to vertex 2. I calculate the movement of the other four vertices similarly.

$$1 \mapsto 2$$
$$2 \mapsto 1$$
$$3 \mapsto 5$$
$$4 \mapsto 4$$
$$5 \mapsto 3$$

Then I have to identify this transformation, since is still should be something in the group. It cannot be a rotation, since vertex 4 is fixed. The only rotation that fixes vertex 4 is F_4. Therefore, I have $R_2 \circ F_3 = F_4$. Note that the multiplication is not commutative. If I did $F_3 \circ R_2$, the result would be these vertex transformations.

$$1 \mapsto 3$$
$$2 \mapsto 2$$
$$3 \mapsto 1$$
$$4 \mapsto 5$$
$$5 \mapsto 4$$

This is the reflection F_2, so $F_3 \circ R_2 = F_2$. This is just a demonstration calculation: you'll have to many of these to produce the whole table. Between calculating compositions and using the patterns of the table to fill in remaining entries, you should end up with this multiplication for D_5.

Table 9.4.1

$\circ$	Id	R_1	R_2	R_3	R_4	F_1	F_2	F_3	F_4	F_5
Id	Id	R_1	R_2	R_3	R_4	F_1	F_2	F_3	F_4	F_5
R_1	R_1	R_2	R_3	R_4	Id	F_4	F_5	F_1	F_2	F_3
R_2	R_2	R_3	R_4	Id	R_1	F_2	F_3	F_4	F_1	F_2
R_3	R_3	R_4	Id	R_1	R_2	F_5	F_1	F_2	F_3	F_4
R_4	R_4	Id	R_1	R_2	R_3	F_3	F_4	F_5	F_1	F_2
F_1	F_1	F_3	F_5	F_2	F_4	Id	R_3	R_1	R_4	R_2
F_2	F_2	F_4	F_1	F_3	F_5	R_2	Id	R_3	R_1	R_4
F_3	F_3	F_5	F_2	F_4	F_1	R_4	R_2	Id	R_3	R_1
F_4	F_4	F_1	F_3	F_5	F_2	R_1	R_4	R_2	Id	R_3
F_5	F_5	F_2	F_4	F_1	F_3	R_3	R_1	R_4	R_2	Id

9.4.2 Linear Algebra of Polynomials

Activity 9.4.2 Write $5x^4 + 2x^3$ as a linear combination of $x^4 - x^3 + 7x$ and $x^3 - 5x$. (Write the linear combination with unknown coefficients and work with the polynomials to try to figure out what the two unknown coefficients should be.)

Solution. I need to find two constants for this equation.

$$5x^4 + 2x^3 = a(x^4 - x^3 + 7x) + b(x^3 - 5x)$$

The only power of x^4 in either of the two polynomials is in the first. Therefore, the only way to get 5x^4 in the result is for $a = 5$. Let me put that value in and see what the situation is.

$$\begin{aligned} 5x^4 + 2x^3 &= 5(x^4 - x^3 + 7x) + b(x^3 - 5x) = \\ &= 5x^4 + (-5 + b)x^3 + (35 - 5b)x \end{aligned}$$

The coefficient of x^3 needs to be 2. The only way to do this is to make $b = 7$. Let me check if this works.

$$\begin{aligned} 5x^4 + 2x^3 &= 5(x^4 - x^3 + 7x) + (-5 + 7)x^3 + (35 - 5(7))x \\ &= 5x^4 + 2x^3 + 0x = 5x^4 + 2x^3 \end{aligned}$$

This works.

Activity 9.4.3 Write $7x^5 - 7x^2 + 14$ as a linear combination of $4x^6 - x^3 + 1$, $x^5 + x^3$ and $x^3 + x^2 - 2$. (Write the linear combination with unknown coefficients and work with the polynomials to try to figure out what the two unknown coefficients should be.)

Solution. I need to find three constants for this equation.

$$7x^5 - 7x^2 + 14 = a(4x^6 - x^3 + 1) + b(x^5 + x^3) + c(x^3 + x^2 - 2)$$

There is no x^6 in the polynomial I want to build, not in any of the other terms. Therefore, $a = 0$ is the only possibility to get ride of the x^6.

$$7x^5 - 7x^2 + 14 = b(x^5 + x^3) + c(x^3 + x^2 - 2)$$

Then to match the power of x^5, I need $b = 7$. Let me put that in and see what it looks like.

$$7x^5 - 7x^2 + 14 = 7x^5 + (7 + c)x^3 + cx^2 - 2c$$

Then to get 0 as the coefficient of x^3, I need c = -7

$$7x^5 - 7x^2 + 14 = 7x^5 + (7 - 7)x^3 - 7x^2 + 14 = 7x^5 - 7x^2 + 14$$

The remaining coefficients all work out, so $a = 0$, $b = 7$ and $c = -7$ works.

Activity 9.4.4 What is the dimension of $\text{Span}\{x^2 - 4, x, 4, x - 1, x^2 + 4x + 3\}$. (You can do this algebraically, but it's better to try to argue from the standard basis of polynomials that I talked about in the notes and the video.)

Solution. The dimension is the number of linear independent polynomials that I need to make this span. I can use the fact, from the notes and the videos, that the monomials are enough to span the polynomials. I have 4, x and $x^2 - 4$, which gets me the first three monoials (I can get x^2 by subtracting 4 from $x^2 - 4$.) Between the first three monomials, nothing else is needed to

make any degree 2 or less polynomials. Therefore, this span has dimension 3.

Activity 9.4.5 What is the dimension of the set of all polynomials with degree 6 or less?

Solution. Again, I'm going to use the monomials from the video. To get degree 6 polynomials, I need these monomials: $\{1, x, x^2, x^3, x^4, x^5, x^6\}$. These are all linearly independent and their are seven of them. Therefore, the dimension of all polynomials with degree 6 or less is 7.

Activity 9.4.6 What's the dimension of the set of all odd-degree polynomials?

Solution. Since the odd degree are unbounded, this degree can be as high as I want. It includes all the monomials $\{x, x^3, x^5, x^7 \ldots\}$. Since there are infinitely many of these monomials and they are all linearly independent, the dimension of the odd-degree polynomials is infinite.

Activity 9.4.7 If the set $\{x^6, x^4, x^3, x^2, 4x^6 + 3x^4 - 9x^3 - 4x^2 - 15\}$ linearly independent?

Solution. The first four terms are different monomials, so they are independent. The real question here is the last polynomial. It has all the same monomials as the first four. The only different is the constant, -15. Does that make it independent?

It does make the last polynomials independent. Using the first first, I can build $4x^6 + 3x^4 - 9x^3 - 4x^2$, but there is no way that I can make a linear combination of the monomials and get the -15. I only get multiples of the powers -- linear combinations cannot add another constant. Therefore, the fifth polynomials is independent, and this whole set is a linearly independent set.

9.4.3 Conceptual Review Questions

- What is a group? What is a matrix group? What is a dihedral group?
- What does it mean to make a multiplication table?
- What is an abstract vector spaces?
- How to the definitions and ideas of linear algebra extend from vectors to other environments?
- Why do polynomials, series and functions form abstract vector spaces?

Chapter 10

Week 10

10.1 Eigenvectors, Eigenvalues and Spectra

10.1.1 Definitions

Definition 10.1.1 Let A be an $n \times n$ matrix. A non-zero vector $v \in \mathbb{R}^n$ is an **eigenvector** for A with **eigenvalue** λ if $Av = \lambda v$. The set of all eigenvalues for the matrix A is called the **spectrum** of A. $\Diamond$

Eigenvectors for a matrix are vectors which do not change direction. They may be dilated by a factor λ (including a flip if λ is negative), but they still point in the same direction. Projections are also included since $\lambda = 0$ is allowed; such an eigenvector would represent a direction sent entirely to zero under the transformation. Note that if v is an eigenvector, then so is αv for any $\alpha \neq 0$, $\alpha \in \mathbb{R}$.

At first glance, this definition may seem insignificant; interesting, perhaps, but not necessarily central. It turns out that eigenvectors and eigenvalues are one of the most useful and important definitions in linear algebra. Many problems in applied mathematics depend upon finding the eigenvalues of a particular matrix.

Before I show how to calculate eigenvectors and eigenvalues, let me go through some obvious examples.

Example 10.1.2 The identity is the easiest: for any vector v, $\mathrm{Id}v = v$. Therefore, all vectors are eigenvectors of the identity with eigenvalue $\lambda = 1$. $\square$

Example 10.1.3 The zero matrix is similarly easy: it sends all vectors to the origin. All vectors are eigenvectors with eigenvalue $\lambda = 0$. $\square$

Example 10.1.4 Consider a dialation matrix in $\mathbb{R}^3$:

$$\begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} a \\ 0 \\ 0 \end{pmatrix} = a \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ b \\ 0 \end{pmatrix} = b \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ c \end{pmatrix} = c \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Here, all three of the standard basis vectors are eigenvectors. e_1 has eigen-

value a, e_2 has eigenvalue b and e_3 has eigenvalue c. If a, b and c are all distinct numbers, there are no other eigenvectors (up to scale). □

Example 10.1.5 Consider a rotation in $\mathbb{R}^2$. Assuming the rotation is not trivial ($\theta \neq 0$), then there are no preserved directions. This rotation has no eigenvectors. However, for a rotation in $\mathbb{R}^3$, I have to choose an axis. The axis direction is fixed, so a vector in that direction would be an eigenvector with eigenvalue one. □

Example 10.1.6 Consider a reflection in $\mathbb{R}^2$. A vector along the line of reflection is unchanged, so it would be an eigenvector with eigenvalue one. A vector perpendicular to the line of reflection is flipped exactly, so it would be an eigenvector with eigenvalue -1. In $\mathbb{R}^3$, reflection happens over a plane through the origin. Any vector in the plane of reflection is unchanged so that vector is an eigenvector with eigenvalue one. Any vector perpendicular to the plane of reflection is precisely flipped, so that vector is an eigenvector with eigenvalue -1. □

Example 10.1.7 Finally, consider projections in $\mathbb{R}^2$. Projection onto the x-axis preserves the x axis, so any vector $\begin{pmatrix} a \\ 0 \end{pmatrix}$ is an eigenvector with eigenvalue one. However, it sends the y-axis direction to the origin, so any vector $\begin{pmatrix} 0 \\ b \end{pmatrix}$ is an eigenvector with eigenvalue zero. □

10.1.2 Calculation of Eigenvalues and Eigenvectors

I would like to demonstrate an algorithm for finding eigenvectors and eigenvalues. Remember the definition: v is an eigenvector of A if $Av = \lambda v$ for some real number λ. I can write the scalar multiplication λv as $\lambda \mathrm{Id} v$, since the matrix $\lambda \mathrm{Id}$ multiplies each entry by λ. Therefore, the equation becomes $Av = \lambda \mathrm{Id} v$ or $Av - \lambda \mathrm{Id} v = 0$, where the right-hand side is the zero vector. Since matrix action is linear, this is the same as $(A - \lambda \mathrm{Id})v = 0$.

The definition of an eigenvector states that v is non-zero. Therefore, the matrix $(A - \lambda \mathrm{Id})$ sends a non-zero vector v to the zero vector. This implies $(A - \lambda \mathrm{Id})$ has a non-trivial kernel, hence it must have a zero determinant. Therefore, a necessary condition for the existence of an eigenvector with eigenvalue λ is that $\det(A - \lambda \mathrm{Id}) = 0$. I start the algorithm by computing this determinant.

This determinant has terms that involve the entries of A and λ. Since A is an $n \times n$ matrix, each term can be the product of n entries. That means that the determinant will be a degree n polynomial in the variable λ.

Definition 10.1.8 Let A be an $n \times n$ matrix. The polynomial $\det(A - \lambda \mathrm{Id})$ in the variable λ is called the **characteristic polynomial** of the matrix. ◊

I need an important definition concerning the roots of polynomials. Recall that α is a root of a polynomial $p(\lambda)$ if and only if $(\lambda - \alpha)$ is a factor of the polynomial. However, it may be true that $(\lambda - \alpha)^m$ is a factor for some integer $m > 1$.

Definition 10.1.9 Let $p(\lambda)$ be a polynomial in the variable λ and α a root. Let m be the largest positive integer such that $(\lambda - \alpha)^m$ is a factor of $p(\lambda)$. Then m is called the **multiplicity** of the root α. ◊

A degree n polynomial has at most n real roots, so I can find at most n values for λ where the determinant vanishes. These will be the possible eigenvalues. However, the polynomial also has at most n factors, so the sum of all the multiplicities of the roots is also at most n. That implies that if

there are roots with higher multiplicities, the polynomial will have fewer than n roots.

Now I return to the eigenvalue/eigenvector algorithm. When there is an eigenvalue λ, I want to find the matching eigenvectors. These vectors will be in the kernel of $(A - \lambda \mathrm{Id})$, so I just have to calculate this kernel.

The second step will always find eigenvectors: the determinant of the matrix is zero, so it must have a non-trivial kernel. However, I don't know how many I will find (I don't know the dimension of the kernel). In the first step, I am looking for the roots of a polynomial of degree n. It may have as many as n distinct real roots, but it may have far fewer or even none. Recall the identity and the zero matrix, where all vectors in $\mathbb{R}^n$ are eigenvectors for the appropriate eigenvalues, and also the rotations in $\mathbb{R}^2$ where there were no eigenvectors for any eigenvalue. Also, note that solving for roots of polynomials, particularly in high degrees, is a very hard problem.

Here is a definition that helps keep track of this information.

Definition 10.1.10 Let A be an $n \times n$ matrix and let λ be an eigenvalue of A. Then the **eigenspace** of λ, written $E(\lambda)$, is the set of all eigenvectors for the eigenvalue λ. Equivalently, it is the kernel of the matrix $A - \lambda \mathrm{Id}$. The dimension of the eigenspace associated with λ is bounded by the multiplicity of λ as a root of the characteristic polynomial. $\Diamond$

Not all matrices have the maximum number of eigenvalues or eigenvectors. However, there is a special class of matrices that always achieves this maximum number. I defined the transpose of a matrix in Section 4.1. The transpose leads to a new definition in this chapter.

Definition 10.1.11 An $n \times n$ matrix A is **symmetric** if $A = A^T$. $\Diamond$

Example 10.1.12 Here are two symmetric matrices.

$$\begin{pmatrix} 5 & 1 & -3 \\ 1 & 2 & 0 \\ -3 & 0 & 4 \end{pmatrix} \qquad \begin{pmatrix} 1 & 4 & -2 & -9 \\ 4 & -4 & 3 & 3 \\ -2 & 3 & 8 & -1 \\ -9 & 3 & -1 & 7 \end{pmatrix}$$

$\square$

I defined symmetric matrices in this section because they behave very well with respect to eigenvalues and eigenvectors.

Proposition 10.1.13 *A symmetric $n \times n$ matrix always has n real eigenvalues, counted with multiplicity. Moreover, the eigenspaces all have the maximum dimension, which is the multiplicity of the eigenvalue.*

I'll end this section with some examples of this algorithm for calculating eigenvalues and eigenvectors.

Example 10.1.14

$$A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix} \implies A - \lambda \mathrm{Id} = \begin{pmatrix} 3-\lambda & 1 \\ 1 & 3-\lambda \end{pmatrix}$$

The determinant of this is $9 - 6\lambda + \lambda^2 - 1 = 8 - 6\lambda + \lambda^2 = (\lambda - 4)(\lambda - 2)$. So the eigenvalues are $\lambda = 4$ and $\lambda = 2$. I can calculate the kernel of $A - \lambda \mathrm{Id}$ first for $\lambda = 2$.

$$A - 2\mathrm{Id} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

The kernel of this matrix is $\mathrm{Span}\left\{\begin{pmatrix} 1 \\ -1 \end{pmatrix}\right\}$. All multiples of $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$ are

eigenvectors for $\lambda = 2$. Next, I move on to $\lambda = 4$.

$$A - 4\text{Id} = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}$$

The kernel of this matrix is $\text{Span}\left\{\begin{pmatrix} 1 \\ 1 \end{pmatrix}\right\}$. All multiples of $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ are eigenvectors for $\lambda = 4$. □

Example 10.1.15

$$A = \begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix} \implies A - \lambda\text{Id} = \begin{pmatrix} -\lambda & -2 \\ 2 & -\lambda \end{pmatrix}$$

The characteristic polynomial is $\lambda^2 + 4 = 0$, which has no roots. Therefore, there are no eigenvalues. □

Example 10.1.16 Consider a diagonal matrix.

$$A = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -3 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \implies A - \lambda\text{Id} = \begin{pmatrix} 1-\lambda & 0 & 0 & \\ 0 & -3-\lambda & 0 & 0 \\ 0 & 0 & 4-\lambda & 0 \\ 0 & 0 & 0 & -\lambda \end{pmatrix}$$

The characteristic polynomial is $(1-\lambda)(-3-\lambda)(4-\lambda)(-\lambda)$, which clearly has solutions $\lambda = 1$, -3, 4 and 0. The eigenvectors are just the axis vectors e_1, e_2, e_3 and e_4, respectively. In general, all diagonal matrices have axis vectors as eigenvectors and diagonal entries as eigenvalues. □

10.2 Eigenfunctions

10.2.1 Differential Operators

To Be Written

10.3 Week 10 Activity

10.3.1 Eigenvalues and Eigenvectors

Activity 10.3.1 Calculate the eigenvalue and eigenvectors of this matrix. (Give the solutions in exact values).

$$\begin{pmatrix} -2 & 0 & 1 \\ 0 & -2 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

Solution. I follow the algorithm for eigenvalue and eigenvectors. The first step is writing the matrix $A - \lambda\text{Id}$.

$$\begin{pmatrix} -2-\lambda & 0 & 1 \\ 0 & -2-\lambda & 1 \\ 1 & 1 & -\lambda \end{pmatrix}$$

Then I calculate the determinant of this matrix. I use cofactor expansion on

the first row.

$$\begin{aligned}
&\begin{vmatrix} -2-\lambda & 0 & 1 \\ 0 & -2-\lambda & 1 \\ 1 & 1 & -\lambda \end{vmatrix} \\
&= (1)(-2-\lambda)\begin{vmatrix} -2-\lambda & 1 \\ 1 & -\lambda \end{vmatrix} + (-1)(0)\begin{vmatrix} 0 & 1 \\ 1 & -\lambda \end{vmatrix} + (1)(1)\begin{vmatrix} 0 & -2-\lambda \\ 1 & 1 \end{vmatrix} \\
&= (-2-\lambda)\left[(-2-\lambda)(-\lambda)-1\right] + 0 + \left[(0)(1)-(-2-\lambda)(1)\right] \\
&= (-2-\lambda)(\lambda^2+2\lambda-1)+2+\lambda \\
&= -\lambda^3-2\lambda^2-2\lambda^2-4\lambda+\lambda+2+2+\lambda \\
&= -\lambda^3-4\lambda^2-2\lambda+4
\end{aligned}$$

By inspection or by computer, I can see that $\lambda = -2$ is a root, so this polynomial factors as follows.

$$-\lambda^3-4\lambda^2-2\lambda+4 = -(\lambda+2)(\lambda^2+2\lambda-2)$$

The quadratic piece does not factor into integer roots. The quadratic formula gives the other roots as $\lambda = -1 \pm \sqrt{3}$. Now I calculate the kernel of each $A - \lambda Id$ to find the matching eigenvectors. I'll start with $\lambda = -2$.

$$\left(\begin{array}{ccc|c} 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 1 & 2 & 0 \end{array}\right)$$

I row reduce to find the kernel.

$$\left(\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

To show the kernel, I need to express the solution indicated by this matrix in terms of parameters. y is free and I can interpret the row of the matrix as expressions for x and z in terms of y. (Though, in this matrix, the second line simply indicatres that $z = 0$.) By choosing a specific value of y, I can get specific values of x, and thus a specific eigenvector. I choose $y = -1$ for this particular system.

$$\text{Ker}\begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 2 \end{pmatrix} = \text{Span}\left\{\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}\right\}$$

That gives me an eigenvector for $\lambda = -2$. Next, I'll look at $\lambda = -1+\sqrt{3}$.

$$\left(\begin{array}{ccc|c} -1-\sqrt{3} & 0 & 1 & 0 \\ 0 & -1-\sqrt{3} & 1 & 0 \\ 1 & 1 & 1-\sqrt{3} & 0 \end{array}\right)$$

I row reduce to find the kernel.

$$\left(\begin{array}{ccc|c} 1 & 0 & \frac{-1}{1+\sqrt{3}} & 0 \\ 0 & 1 & \frac{-1}{1+\sqrt{3}} & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

To show the kernel, I need to express the solution indicated by this matrix in terms of parameters. z is free and I can interpret the row of the matrix

as expressions for x and y in terms of z. By choosing a specific value of z, I can get specific values of x and y, and thus a specific eigenvector. I choose $z = 1 + \sqrt{3}$ for this particular system, to clear the denominator.

$$\operatorname{Ker}\begin{pmatrix} -1-\sqrt{3} & 0 & 1 \\ 0 & -1-\sqrt{3} & 1 \\ 1 & 1 & 1-\sqrt{3} \end{pmatrix} = \operatorname{Span}\left\{\begin{pmatrix} 1 \\ 1 \\ 1+\sqrt{3} \end{pmatrix}\right\}$$

That gives me an eigenvector for $\lambda = -1 + \sqrt{3}$. Next, I'll look at $\lambda = -1 - \sqrt{3}$.

$$\left(\begin{array}{ccc|c} -1+\sqrt{3} & 0 & 1 & 0 \\ 0 & -1+\sqrt{3} & 1 & 0 \\ 1 & 1 & 1+\sqrt{3} & 0 \end{array}\right)$$

I row reduce to find the kernel.

$$\left(\begin{array}{ccc|c} 1 & 0 & \frac{-1}{1-\sqrt{3}} & 0 \\ 0 & 1 & \frac{-1}{1-\sqrt{3}} & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

To show the kernel, I need to express the solution indicated by this matrix in terms of parameters. z is free and I can interpret the row of the matrix as expressions for x and y in terms of z. By choosing a specific value of z, I can get specific values of x and y, and thus a specific eigenvector. I choose $z = -1 - \sqrt{3}$ for this particular system, to clear the denominator.

$$\operatorname{Ker}\begin{pmatrix} -1+\sqrt{3} & 0 & 1 \\ 0 & -1+\sqrt{3} & 1 \\ 1 & 1 & 1+\sqrt{3} \end{pmatrix} = \operatorname{Span}\left\{\begin{pmatrix} 1 \\ 1 \\ 1-\sqrt{3} \end{pmatrix}\right\}$$

The spanning vector is an eigenvector for $\lambda = -1 - \sqrt{3}$.

Activity 10.3.2 Calculate the eigenvalue and eigenvectors of this matrix. (Give the solutions in exact values).

$$\begin{pmatrix} 0 & -6 & 0 & 6 \\ 0 & -2 & 0 & 2 \\ 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 2 \end{pmatrix}$$

Solution. I follow the algorithm for eigenvalue and eigenvectors. The first step is writing the matrix $A - \lambda \mathrm{Id}$.

$$\begin{pmatrix} -\lambda & -6 & 0 & 6 \\ 0 & -2-\lambda & 0 & 2 \\ 0 & 0 & 2-\lambda & 2 \\ 0 & 0 & 0 & 2-\lambda \end{pmatrix}$$

Then I calculate the determinant of this matrix. This is an upper triangular matrix, so the determinant is the product of the diagonal entries.

$$\begin{vmatrix} -\lambda & -6 & 0 & 6 \\ 0 & -2-\lambda & 0 & 2 \\ 0 & 0 & 2-\lambda & 2 \\ 0 & 0 & 0 & 2-\lambda \end{vmatrix} = -\lambda(-2-\lambda)(2-\lambda)^2$$

This already in factored form, so I see that the eigenvalue are $\lambda = 0$, $\lambda = 2$ and $\lambda = -2$. For each eigenvalue, I calculate the kernel of $A - \lambda \mathrm{Id}$ to find the eigenvalues. I'll start with $\lambda = 0$.

$$\left(\begin{array}{cccc|c} 0 & -6 & 0 & 6 & 0 \\ 0 & -2 & 0 & 2 & 0 \\ 0 & 0 & 2 & 2 & 0 \\ 0 & 0 & 0 & 2 & 0 \end{array}\right)$$

I row reduce to find the kernel.

$$\left(\begin{array}{cccc|c} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

To show the kernel, I need to express the solution indicated by this matrix in terms of parameters. In this matrix, w is free, but all other coordinates are zero. I choose $w = 1$ for this particular system to get a unique eigenvector.

$$\mathrm{Ker}\begin{pmatrix} 0 & -6 & 0 & 6 \\ 0 & -2 & 0 & 2 \\ 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 2 \end{pmatrix} = \mathrm{Span}\left\{\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}\right\}$$

That gives me an eigenvector for $\lambda = 0$. Next, I'll look at $\lambda = 2$.

$$\left(\begin{array}{cccc|c} -2 & -6 & 0 & 6 & 0 \\ 0 & -4 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

I row reduce to find the kernel.

$$\left|\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

To show the kernel, I need to express the solution indicated by this matrix in terms of parameters. In this matrix, y is free and all the other variables are zero. I choose $y = 1$ for this particular system to get a unique eigenvector.

$$\mathrm{Ker}\begin{pmatrix} -2 & -6 & 0 & 6 \\ 0 & -4 & 0 & 2 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix} = \mathrm{Span}\left\{\begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}\right\}$$

That gives me an eigenvector for $\lambda = 2$. Next, I'll look at $\lambda = -2$.

$$\left(\begin{array}{cccc|c} 2 & -6 & 0 & 6 & 0 \\ 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 4 & 2 & 0 \\ 0 & 0 & 0 & 4 & 0 \end{array}\right)$$

I row reduce to find the kernel.

$$\left|\begin{array}{cccc|c} 1 & -3 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

To show the kernel, I need to express the solution indicated by this matrix in terms of parameters. x is free and I can interpret the row of the matrix as expressions for the other variables in terms of x. However, two of the lines simply indicate that y and z are zero, so only w depends on x I choose $x = 1$ for this particular system.

$$\text{Ker}\begin{pmatrix} 2 & -6 & 0 & 6 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 4 & 2 \\ 0 & 0 & 0 & 4 \end{pmatrix} = \text{Span}\left\{\begin{pmatrix} 3 \\ 1 \\ 0 \\ 0 \end{pmatrix}\right\}$$

The spanning vector is an eigenvector for $\lambda = -2$.

Activity 10.3.3 Calculate the eigenvalues and eigenvectors of this matrix. Use approximate values (to three significant figures), but show all the steps in the algorithm.

$$\begin{pmatrix} 0.294 & 0.343 & 0.945 & 0 \\ 0.812 & 0 & 0.310 & 0.050 \\ 0.110 & 0.745 & 0.901 & 0.874 \\ 0.314 & 0 & 0 & 0.041 \end{pmatrix}$$

Solution. I ask a computer for the determinant of $A - \lambda Id$.

$$\begin{vmatrix} 0.294-\lambda & 0.343 & 0.945 & 0 \\ 0.812 & -\lambda & 0.310 & 0.050 \\ 0.110 & 0.745 & 0.901-\lambda & 0.874 \\ 0.314 & 0 & 0 & 0.041-\lambda \end{vmatrix}$$
$$= x^4 - 1.235x^3 - 0.300x^2 - 0.515x - 0.025$$

I ask a computer for the roots of this polynomial. There are two real roots: $\lambda = 1.622$ and $\lambda = -0.049$. I calculate the kernel of $A - \lambda\text{Id}$ for both eigenvalues. I start with $\lambda = 1.622$

$$\text{Ker}\begin{pmatrix} -1.328 & 0.343 & 0.945 & 0 \\ 0.812 & -1.622 & 0.310 & 0.050 \\ 0.110 & 0.745 & -0.721 & 0.874 \\ 0.314 & 0 & 0 & -1.581 \end{pmatrix}$$
$$= \text{Span}\left\{\begin{pmatrix} 5.035 \\ 3.651 \\ 5.752 \\ 1 \end{pmatrix}\right\}$$

The spanning vector is an eigenvector for $\lambda = 1.622$. Then I do the same for $\lambda = -0.049$.

$$\text{Ker}\begin{pmatrix} 0.343 & 0.343 & 0.945 & 0 \\ 0.812 & 0.049 & 0.310 & 0.050 \\ 0.110 & 0.745 & 0.950 & 0.874 \\ 0.314 & 0 & 0 & 0.090 \end{pmatrix}$$

$$= \text{Span}\left\{\begin{pmatrix}-0.286\\-2.351\\0.957\\1\end{pmatrix}\right\}$$

This is an eigenvalue for $\lambda = -0.049$.

Activity 10.3.4 Calculate the eigenvalues and eigenvectors of this matrix. Use approximate values (to three significant figures), but show all the steps in the algorithm.

$$\begin{pmatrix}2.30 & -0.29 & -0.92 & -3.17\\-1.01 & 1.31 & 3.51 & 4.52\\-3.68 & -5.42 & 8.54 & -4.62\\-9.54 & -3.33 & -7.54 & 2.50\end{pmatrix}$$

Solution. I ask a computer for the determinant of $A - \lambda Id$.

$$\begin{vmatrix}2.30-\lambda & -0.29 & -0.92 & -3.17\\-1.01 & 1.31-\lambda & 3.51 & 4.52\\-3.68 & -5.42 & 8.54-\lambda & -4.62\\-9.54 & -3.33 & -7.54 & 2.50-\lambda\end{vmatrix}$$
$$= x^4 - 14.65x^3 - 29.54x^2 - 35.81x + 73.34$$

I ask a computer for the roots of this polynomial. There are two real roots: $\lambda = 12.47$ and $\lambda = 2.26$. I calculate the kernel of $A - \lambda\text{Id}$ for both eigenvalues. I start with $\lambda = 12.47$

$$\text{Ker}\begin{pmatrix}-10.17 & -0.29 & -0.92 & -3.17\\-1.01 & -11.16 & 3.51 & 4.52\\-3.68 & -5.42 & -4.02 & -4.62\\-9.54 & -3.33 & -7.54 & -9.97\end{pmatrix}$$
$$= \text{Span}\left\{\begin{pmatrix}-0.22\\0.08\\-1.09\\1\end{pmatrix}\right\}$$

The spanning vector is an eigenvector for $\lambda = 12.47$. Then I do the same for $\lambda = 2.26$.

$$\text{Ker}\begin{pmatrix}0.04 & -0.29 & -0.92 & -3.17\\-1.01 & -0.95 & 3.51 & 4.52\\-3.68 & -5.42 & 6.28 & -4.62\\-9.54 & -3.33 & -7.54 & 0.24\end{pmatrix}$$
$$= \text{Span}\left\{\begin{pmatrix}3.15\\-5.01\\-1.74\\1\end{pmatrix}\right\}$$

This is an eigenvalue for $\lambda = 2.26$.

10.3.2 Eigenfunctions

Activity 10.3.5 Let $n \in \mathbb{N}$. Prove that multiplication by x^n is a linear transformation on the space $C^\infty(\mathbb{R})$ of differentiable function.

Solution. I need to check the two properties of linearity. First I look at addition of function.

$$x^n(f(x) + g(x))$$

I can just use the ordinary distribution law. This applies to function since it is a pointwise operation.

$$x^n(f(x) + g(x)) = x^n f(x) + x^n g(x)$$

That's already one of the two properties. Now I look at scalar multiplication.

$$x^n(af(x))$$

This is just a series of multiplication of functions and constants. I know from the study of function that I do these in which ever order I wish. Let me change the order and the brackets.

$$x^n(af(x)) = a(x^n f(x))$$

This is the second property finish. Multiplication x^n is a linear transformation.

Activity 10.3.6 Calculate the eigenfunction of the linear operator $\frac{x^2}{30}\frac{d^2}{dx^2}$ with eigenvalue $\lambda = 1$.. (Hint: the solution is some kind of polynomial. Do this by inspection or by trial and error: there isn't a general algorithm here to rely on.)

Solution. I know I'm looking for a polynomial here. First, let me think about monomials. The second derivative of a monomial x^n is $n(n-1)x^{n-2}$. Then multiplying by $\frac{x^2}{30}$ produces $\frac{n(n-1)x^n}{30}$. This is only going to equal the original (eigenvalue of 1) if $\frac{n(n-1)}{30} = 1$. The only natural number that satisfies this is $n = 6$. Therefore, $p(x) = ax^6$ is an eigenfunction for this operator, for any constant a.

Activity 10.3.7 Consider the linear transformation $\frac{d^4}{dx^4}$ on $C^\infty(\mathbb{R})$. Verify that the function $f(x) = a\sin(\sqrt{3}x) + b\cos(\sqrt{3}x)$ is an eigenfunction for this transformation with eigenvalue $\lambda = 9$. Explain why the square root terms are necessary inside the trig functions. Are their eigenfunctions that don't involve trig for this transformation?

Solution. To verify, I just need to do four derivative of the function. Here are those four derivatives.

$$\begin{aligned}
\frac{d}{dx}a\sin(\sqrt{3}x) + b\cos(\sqrt{3}x) &= \sqrt{3}a\cos(\sqrt{3}x) - \sqrt{3}b\sin(\sqrt{3}x)\\
\frac{d^2}{dx^2}a\sin(\sqrt{3}x) + b\cos(\sqrt{3}x) &= -3a\sin(\sqrt{3}x) - 3b\cos(\sqrt{3}x)\\
\frac{d^3}{dx^3}a\sin(\sqrt{3}x) + b\cos(\sqrt{3}x) &= -3\sqrt{3}a\cos(\sqrt{3}x) + 3\sqrt{3}b\sin(\sqrt{3}x)\\
\frac{d^4}{dx^4}a\sin(\sqrt{3}x) + b\cos(\sqrt{3}x) &= 9a\sin(\sqrt{3}x) + 9b\cos(\sqrt{3}x)
\end{aligned}$$

This is, indeed, exactly 9 times the original function, so the eigenfunction is verified.

As the previous calculation shows, the $\sqrt{3}$ is necessary. In each derivative,

the derivative of the inside is multiplied by the function. So, after four derivative, I have multiplied by $\sqrt{3}$ 4 times. That's precisely the same as multiplying by 9, which was the desired eigenvalue. The square root inside is the only way to produce this eigenvalue out of successive derivatives.

There are other eigenfunctions for this transformation and eigenvalue. Among them are $\cosh(\sqrt{3}x)$, $\sinh(\sqrt{3}x)$ and $e^{\sqrt{3}x}$.

Activity 10.3.8 Prove that any polynomial is an eigenfunction for some derivative transformation with eigenvalue $\lambda = 0$.

Solution. The key idea here is that if I take more and more higher derivatives, eventuall the result is zero. So, let p be some polynomial. The polynomial has a degree n, which is the highest polynomial power. Each derivative drops the degree of a polynomial by one. That means after n derivative, the degree of the polynomial is zero, a constant. One more derivative gives zero, since the derivative of a constant is zero. This means that the operator $\frac{d^{n+1}}{dx^{n+1}}$ will send the polynomial to zero. That's the definition of an eigenfunction with eigenvalue 0: it gets sent to zero. Therefore, p is an eigenfunction of $\frac{d^{n+1}}{dx^{n+1}}$ with eigenvalue 0.

10.3.3 Proof Questions

Activity 10.3.9 Prove that a square matrix A is invertible if and only if $\lambda = 0$ is not an eigenvalue of A.

Solution. There are a variety of ways to argue this. First, I could say that invertible matrices have to preserve the dimension of the space. They can't have any directions which are collapsed to zero. But an eigenvalue of $\lambda = 0$ is precisely a direction which is collapsed to zero. That's what sending v to $\lambda v = 0v = 0$ does to the eigenvector. Therefore, invertible matrices can't have 0 as an eigenvalue. To prove the other part of the implication, a non-invertible matrix must collapse at least one vector to zero. By linearity, then, the whole line is collapsed to zero, and any vector on that line is an eigenvector with eigenvalue 0, showing that an invertible matrix must have 0 as an eigenvalue.

Alternatively, I could use a property like the fact that the determinant of a matrix is the product of the eigenvalues. With this fact, the determinant is zero if and only if one of the eigenvalues is zero. But a matrix is invertible if and only if its determinant is not zero. Putting these together, a matrix is invertible if and only if 0 is not an eigenvalue.

Activity 10.3.10 Let A be an invertible matrix. Prove that λ is an eigenvalue of A if and only if $\frac{1}{\lambda}$ is an eigenvalue of A^{-1}.

Solution. I have to prove both implications. Let me start with the forward implication. I assume that λ is an eigenvalue, so there exists a vector v with

$$Av = \lambda v$$

Now I apply A^{-1} to the equation.

$$A^{-1}Av = A^{-1}\lambda v$$

On the left, the matrices cancel, and I just have v. on the first I can use linearity.

$$v = \lambda A^{-1}v$$

Then I can divide by λ.

$$\frac{1}{\lambda}v = A^{-1}v$$

This is the definition of an eigenvalue, so $\frac{1}{\lambda}$ is an eigenvalue of A^{-1}

For the reverse implication. I can just reverse the steps. All of the steps of the argument are entirely reversible.

Activity 10.3.11 Let A be a 3×3 matrix. Prove that A has at least one real eigenvalue.

Solution. The eigenvalues are the zeros of the characteristic polynomial $\det(A - \lambda \mathrm{Id})$. Since this is a 3×3 matrix, the polynomial is a cubic. Any cubic has a real root (since cubics are continuous and the limits going to $\pm\infty$ are positive in one direction and negative in the other -- meaning the graph must cross the axis). Therefore, there is at least one real eigenvalue.

Activity 10.3.12 Prove that rotations in $\mathbb{R}^2$ have no real eigenvalues. (Exclude the trivial rotation, which is just the identity matrix.)

Solution. I could do this algebraically by applying the algorithm to the generic rotation matrix. However, this is easier to argue conceptually. A rotation takes any direction and moves it by some angle θ in a counter-clockwise direction. Therefore, the direction cannot be preserved. Since this is true for any direction, and thus any vector, an eigenvector is impossible. Since an eigenvector is impossible, an eigenvalue is likewise impossible.

Activity 10.3.13 Prove that reflections in $\mathbb{R}^2$ have 1 and (-1) as their two eigenvalues and that their eigenvectors are perpendicular. Also, argue that these two conditions completely describe reflections (i.e., any matrix with these two properties must be a reflection).

Solution. As with the rotations, this is easier to argue conceptually. A reflection in $\mathbb{R}^2$ is a reflection over a line. Anything on the line of reflection doesn't move. Therefore, these vectors are eigenvectors with eigenvalue $\lambda = 1$. The perpendicular direction to the line is directly flipped. Flipping a vector is the same as multiplication by -1, so these vectors are eigenvectors with eigenvalue $\lambda = -1$. Since there can be at most two eigenvalues, this finishes the argument. By construction, the eigenvectors are perpendicular.

If I started with two perpendicular eigenvectors, one with eigenvalue $\lambda = 1$ and one with eigenvalue $\lambda = -1$, then I can basically repeat the argument to give a reflection. One line is fixed, and the line perpendicular is flipped. Any other vector can be written as a combination of the two eigenvectors: so a piece along the line is preserved, and the piece perpendicular to the line is flipped. This is exactly a reflection over the fixed-line.

10.3.4 Conceptual Review Questions

- What are eigenvalues and eigenvectors?
- How do determinants help to calculate eigenvalues?
- What is an eigenspace? What is a spectrum?

Chapter 11

Week 11

11.1 Dynamical Systems

11.1.1 Theory of Dynamical Systems

Consider an applied mathematics problem where a vector $v \in \mathbb{R}^n$ somehow represents the state of the situation under consideration. This state could be a position, keeping with the geometry of $\mathbb{R}^n$, but typically it is just a list of numbers with some other interpretation. They could be the statistics of a countries economy, the concentration of a number of different chemicals in the same solutions, or any other list of quantities that are part of the same system. The state of the system isn't static: it changes over time. Unlike the continuous systems in calculus, sometimes it is convenient to measure the system after some fixed time intervals. This interval is called a timestep. Finally, since the state is a vector in $\mathbb{R}^n$, the change over one timestep is a sequence of vectors. If the system is in state v_k, then after the next second/minute/day/year/timestep, in the next state v_{k+1}. The model is essentially a sequence $\{v_k\}_{k=1}^{\infty}$ of states in $\mathbb{R}^n$.

How does the model move from one state to the next? If the process is linear, its progression can be described by a matrix action. That is, there exists a matrix A such that $v_{k+1} = Av_k$ for all $k \in \mathbb{N}$.

Definition 11.1.1 A sequence of vectors $\{v_k\}_{k=1}^{\infty}$ which satisfy the matrix equation $v_{k+1} = Av_k$ is called a **(discrete) linear dynamical system**. $\Diamond$

Dynamical systems are powerful and important mathematical models. However, they have some limitations, which are good to state up front. First, obviously, the relationship is linear. Linearity is a good starting point for models, but not all processes in the world are linear. It may be that the relationship between one state and the next state is non-linear, in which case, a matrix would not be able to calculate the state transition.

Another limitation is that each state of the dynamical system comes directly from the state before it by the matrix action. The matrix is the same for each transition, so the only thing that affects any particular state is the state before it. A dynamical system has no memory: states other than the directly previous state can have no effect on the next state. This again is good for some models, but many systems also build up some kind of memory that needs to be taken into account.

11.1.2 Long Term Behaviour of Dynamical Systems

When I build a dynamical system to model some situation in the world, it's natural to ask about the long-term behaviour of the system. From a certain starting value, where will the dynamical system go? This question is mostly answered by eigenvectors and eigenvalues.

Consider a dynamical system described by a matrix A. If v is an eigenvector with eigenvalue λ, then $Av = \lambda v$. The matrix action is a change of state over a timestep. If the state is an eigenvector, then the only effect of the timestep is to multiply all the quantities in the state vector by the same constant. What does this mean? First, it means that the *ratio* between the state quantities is fixed. The values are changed, but if the state is an eigenvector, the relative values of the quantities are fixed. This is a kind of stabilization. Second, the long term values are determined by λ in six cases.

- If $\lambda = 0$, then this is a collapsing state and all future states are simply the zero vector.
- If $0 < |\lambda| < 1$, then the sequence of states displays exponential decay. The long term behaviour of $A^k v$ is an exponential decay of the original vector.
- If $\lambda = -1$, then there is a 2-period oscillating state. The sequence begins with the state v and jumps back and forth between v and $-v$.
- If $\lambda = 1$, then v is a steady state: the sequence never changes. These steady states are often very important in modelling.
- If $\lambda > 1$, then the states display exponential growth. The long term behaviour of $A^k v$ is exponential growth of the original vector.
- If $\lambda < -1$, then the states display exponential growth with an oscillation. The sign of the vector with flip back and forth while the absolute value of the state quantities will grow exponentially.

As an aside, I could ask what happens for complex eigenvalues, since those will naturally also occur. Since the characteristic polynomial will have real coefficients, these complex eigenvalues will come in conjugate pairs. As you might learn in other courses, such pairs and the exponential behaviour (λ^n) give rise to sinusoidal behaviour after clever linear combinations. The reasons for this will have to wait for another course.

This is all well for eigenvectors, but what if the starting state isn't an eigenvector? Well, the eigenvectors still end up controlling the long term behaviour, but it can get a bit more complicated. Ideally, there a full set of eigenvectors and I can write any starting state v as a linear combination of eigenvectors $v = a_1v_1 + a_2v_2 + \ldots + a_nv_n$, where each eigenvector v_i has eigenvalue λ_i. Then, the long term behaviour of the system with the starting state v is indeed given by the eigenvalues.

$$A^k v = A^k(a_1v_1 + a_2v_2 + \ldots + a_nv_n) = \lambda_1^k a_1 v_1 + \lambda_2^k a_2 v_2 + \ldots + \lambda_n^k a_n v_n$$

If any of the λ_i have an absolute value less than one, those terms will decay away. The eigenvalues which have an absolute value greater than one will grow. The largest eigenvalue (in absolute value) will lead to the largest growth, so it will dominate.

This can still be quite complicated, particularly if there are many eigenvalues with absolute values greater than one. For a completely arbitrary matrix A,

it's pretty difficult to use this to understand the system. However, the types of matrices which show up in dynamical systems usually have special properties, which make for more predictable and understandable systems.

In many models, the coefficients of the matrix will be probabilities, transition terms, growth rates or other *positive* real numbers. The matrix A will very often be a matrix with all non-negative entries. There is a powerful theorem that tells us what to expect for the eigenvalues/eigenvectors of such a matrix: the Perron-Frobenius theorem. There are some technical details in the assumptions for the theorem; I'll state a weak version first.

Theorem 11.1.2 *Let A be an $n \times n$ matrix with non-negative coefficients. Then there is a largest non-negative eigenvalue λ_1 with an eigenvector that has all non-negative entries. All other eigenvalues λ satisfy $|\lambda| \leq |\lambda_1|$.*

The stronger version of the theorem needs an new definition.

Definition 11.1.3 Let A be an $n \times n$ matrix. Then A is called **irreducible** if for all i and j, there exists a positive integer m such that the ijth entry of A^m is non-zero. $\Diamond$

This definition roughly captures the idea that all the coefficients in the states are somehow related. Using the definition, here is a stronger version of the Perron-Frobenius theorem.

Theorem 11.1.4 *Let A be an $n \times n$ irreducible matrix with non-negative coefficients. Then there is a* unique *largest* positive *eigenvalue λ_1 with a 1-dimensional eigenspace and an eigenvector that has all* positive *entries. All other eigenvalues λ satisfy $|\lambda| < |\lambda_1|$. Moreover, if r_i is the sum of the entries of A in the ith row, then λ is bounded above and below by the largest and the smallest of the $|r_i|$.*

For dynamical systems with irreducible non-negative matrices, the long term behaviour is really controlled and understood by this unique largest positive eigenvalue and the corresponding eigenvector. You will see this clearly in the next section, where I develop the theory of Leslie matrices, a nice example of a dynamical system.

11.2 Leslie Matrices

11.2.1 Age-Structured Populations

One of the starting points of mathematical biology is the study of population growth. Though many of the most interesting systems involve multiple species with various interactions (symbiotic, competition, predator-prey, etc.), the theory starts with single-species dynamics. How does a species grow or decline when considered by itself? For those who've taken calculus with me, I used population growth as a major motivator for that course, working with the differential equations that describe exponential and logistic growth.

Sometimes I want to know more details about the population than simply the count of its members. One way of describing a population is by age categories; this is called an age-structured population. This is pretty common in human geography, where the population of a country (or other human groups) is given in ten-year increments. (Usually, these graphs are segregated by sex as well.) This can also be very useful in animal populations since there are often important age-defined subsets that control the growth or decline of a population. In conservation biology, resources can be much more carefully used if a model can focus on a specific age category in a population.

In the simplest single-species models, there is a birth and death rate. For an age-structured population, the situation is more complicated. Instead of just asking for the death rate, I now need to know the survival rate from each age category to the next (assuming that no one survives from the oldest age category). If there are n age categories, that's $(n-1)$ survival rates (again, excluding the oldest age category). Likewise, instead of a generic birth rate, I need to know the birth rate due to each age category. These are called fecundity rates. There may be fewer of these (or, rather, some might be set to zero) since only certain age categories may be fertile, But again, in theory, I have up to n new rates.

How do I track all this information? I can do it with the dynamical system described in the next section.

11.2.2 Leslie Matrices

A Leslie matrices is a dynamical system that models an age-structured population. To demonstrate, I'll assume there are four age categories. The state vector v_k is the current population in each age category: so that v_1 is the population of the youngest category, v_2 of the next, v_3 of the next, and finally v_4 of the oldest category. The matrix C, called a Leslie matrix, is a 4×4 matrix, and the action of a matrix (like for all dynamical systems) takes a state vector and moves it ahead by one time step. If I use coordinates w, x, y, z in $\mathbb{R}^4$, here is the dynamical system.

$$\begin{pmatrix} w_{k+1} \\ x_{k+1} \\ y_{k+1} \\ z_{k+1} \end{pmatrix} = \begin{pmatrix} f_1 & f_2 & f_3 & f_4 \\ s_1 & 0 & 0 & 0 \\ 0 & s_2 & 0 & 0 \\ 0 & 0 & s_3 & 0 \end{pmatrix} \begin{pmatrix} w_k \\ x_k \\ y_k \\ z_k \end{pmatrix}$$

Nine of the entries must be set to zero, since they are age categories that cannot transfer to other age categories in one time step (i.e., something in age category 1 cannot jump directly to age category 4, nor can something in the 3rd age category suddenly get younger and move into the 2nd age category). I want to interpret the remaining 7 non-zero entries. The first row outputs to the first category, representing the creation of new members of the population. Each f_i is the *fecundity* of each population state: its rate of producing offspring. Of course, some of these might be zero if only certain age categories produce offspring. The s_i, on the other hand, are transitions from one age category to the next. These are *survival rates*: s_1 is the rate of survival from category 1 to category 2, s_2 from category 2 to 3 and s_3 from category 3 to 4.

These Leslie matrices are irreducible (recall that irreducible meant that each state connects to the other; by survival and fertility, it is possible to pass from any age category to another over several years). This means that there is a unique largest eigenvalue λ with a positive eigenvector. This is the value I care about: in the long run, the largest eigenvalue dominates, and its eigenvector gives the stable age distribution. If $\lambda = 1$, I expect a stable population. If $\lambda > 1$, I expect exponential growth, and if $\lambda < 1$, I expect exponential decay. In addition, I can look for the limit of the ratios between the age categories. As the population either grows or decays, those ratios will approach fixed values, which are represented by the eigenvector matching the dominant eigenvalue.

11.2.3 Examples of Leslie Matrices

All these examples will be four-stage population models, using the description as above. For each, I will determine the long term behaviour by analyzing the

dominant eigenvalue and its matching eigenvector.

Example 11.2.1

$$\begin{pmatrix} 0 & 0 & 0.7 & 0.6 \\ 0.8 & 0 & 0 & 0 \\ 0 & 0.6 & 0 & 0 \\ 0 & 0 & 0.7 & 0 \end{pmatrix}$$

Solution. Here is the dominant eigenvalue and the matching eigenvector (calculated by computer).

$$\lambda = 0.83 \qquad\qquad v = \begin{pmatrix} 1.72 \\ 1.65 \\ 1.19 \\ 1 \end{pmatrix}$$

The popluation is not viable, since the dominant eigenvalue is less than 1; the population will decay. The age ratios will approach $1.72 : 1.65 : 1.19 : 1$ from youngest to oldest. □

Example 11.2.2 This example has only changed one fecundity coefficient and one survival coefficient from the previous example. What is the effect?

$$\begin{pmatrix} 0 & 0 & 1.1 & 0.6 \\ 0.8 & 0 & 0 & 0 \\ 0 & 0.9 & 0 & 0 \\ 0 & 0 & 0.7 & 0 \end{pmatrix}$$

Solution. Here is the dominant eigenvalue and the matching eigenvector (calculated by computer).

$$\lambda = 1.03 \qquad\qquad v = \begin{pmatrix} 2.15 \\ 1.68 \\ 1.49 \\ 1 \end{pmatrix}$$

The population is viable; since the dominant eigenvalue is larger than one, the population will grow. The age ratios will approach $2.15 : 1.68 : 1.49 : 1$. Compared with the previous example, two coefficients have been changed: the fecundity coefficient of the 3rd age category and the survival coefficient from the 2nd to the 3rd age category. These were both increased sufficiently to change the non-viable population into a viable population. □

Example 11.2.3

$$\begin{pmatrix} 0 & 0 & 2 & 1 \\ 0.4 & 0 & 0 & 0 \\ 0 & 0.9 & 0 & 0 \\ 0 & 0 & 0.9 & 0 \end{pmatrix}$$

Solution. Here is the dominant eigenvalue and the matching eigenvector (calculated by computer).

$$\lambda = 1.01 \qquad\qquad v = \begin{pmatrix} 3.21 \\ 1.27 \\ 1.16 \\ 1 \end{pmatrix}$$

This is a viable population; since the dominant eigenvalue is positive, this population will grow. That said, since the dominant eigenvalue is only slightly above 1, the population growth will be relatively slow. As the population grows, the age ratios will approach 3.21 : 1.27 : 1.16 : 1. □

Example 11.2.4

$$\begin{pmatrix} 0 & 0 & 150 & 50 \\ 0.03 & 0 & 0 & 0 \\ 0 & 0.9 & 0 & 0 \\ 0 & 0 & 0.9 & 0 \end{pmatrix}$$

Solution. Here is the dominant eigenvalue and the matching eigenvector (calculated by computer).

$$\lambda = 1.63 \qquad\qquad v = \begin{pmatrix} 196.37 \\ 3.50 \\ 1.87 \\ 1 \end{pmatrix}$$

The population is viable, since the dominant eigenvalue is larger than 1. In addition, this is quite a fast growth rate, since the eigenvalue is significantly above one. The age ratios will approach 196.37 : 3.50 : 1.87 : 1. Note that these ratios are quite different from previous examples: this is a population with many many young members, very few of which reach maturity. □

Example 11.2.5 This question has reduced the fecundity compared with the previous example.

$$\begin{pmatrix} 0 & 0 & 50 & 30 \\ 0.03 & 0 & 0 & 0 \\ 0 & 0.9 & 0 & 0 \\ 0 & 0 & 0.9 & 0 \end{pmatrix}$$

Solution. Here is the dominant eigenvalue and the matching eigenvector (calculated by computer).

$$\lambda = 1.25 \qquad\qquad v = \begin{pmatrix} 79.63 \\ 1.92 \\ 1.38 \\ 1 \end{pmatrix}$$

The population is viable, since the dominant eigenvalue is greater than 1. The age ratios will approach 79.63 : 1.92 : 1.38 : 1. Even with the reduced fecundity coefficient (significantly reduced!), the population is still viable. □

Example 11.2.6 This example has reduced the first survival coefficient compared to the previous example. What is the effect?

$$\begin{pmatrix} 0 & 0 & 50 & 30 \\ 0.01 & 0 & 0 & 0 \\ 0 & 0.9 & 0 & 0 \\ 0 & 0 & 0.9 & 0 \end{pmatrix}$$

Solution. Here is the dominant eigenvalue and the matching eigenvector (calculated by computer).

$$\lambda = 0.90 \qquad\qquad v = \begin{pmatrix} 89.012 \\ 0.993 \\ 0.996 \\ 1 \end{pmatrix}$$

The population is not viable, since the dominant eigenvalue is less than 1. The age ratios will approach 89.012 : 0.993 : 0.993 : 1. Compared to the previous example, the viable population has now become non-variable. The only coefficient that has changed is the survival coefficient from the 1st to the 2nd age category. Previously, 3% of the youngest survived for a time interval. Now, only 1% survive. That change is significant enough to drop the population from viable to non-viable. □

11.3 Stochastic Matrices and Markov Chains

11.3.1 Graphs and Probability

Dynamical systems are good at modelling certainly probability situations. Like all dynamical systems, there is a vector $v \in \mathbb{R}^n$ of the states of the system. In a probability model, the states represent possibilities of the system, and in each time step, the system can transition from one state to another. (This is different from previous examples, where the states represented different but interacting quantities). The action of the matrix A represents the probability of shifting from one state to another in a time step. This is best illustrated by a graph with labelled edges. Figure 11.3.1 is an example with three vertices.

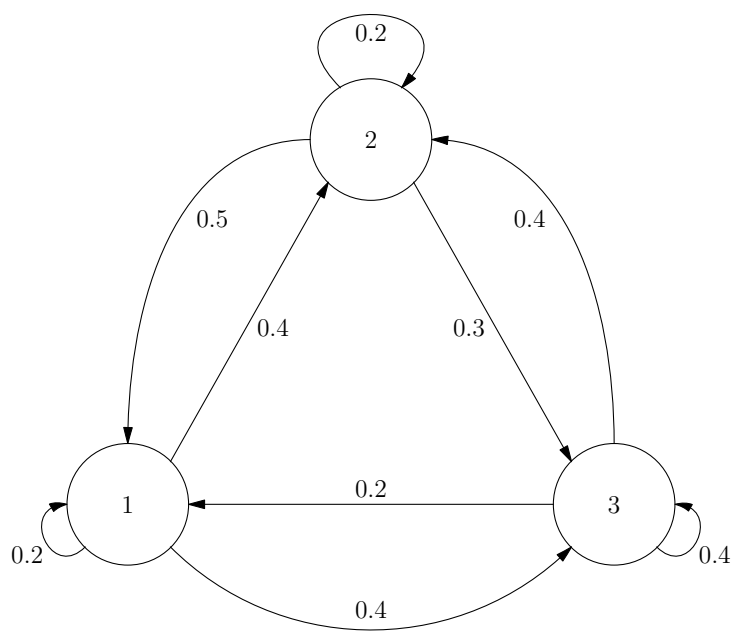

Figure 11.3.1 Probability Graph 1

In Figure 11.3.1, the arrows are labelled with the transition probabilities. This means that if the system is in vertex 1, there is a 20% probability that it stays in vertex 1 (the arrow pointing from the vertex back to itself); there is a 40% probability that it transitions to vertex 2 (the arrow pointing from

vertex 1 to vertex 2; and a 40% probability that it transition to vertex 3 (the arrow pointing from vertex 1 to vertex 3). The total probability of going out of vertex 1 is 100%. You can check the same is true for the other vertices: the outgoing probabilities always add to 1. The incoming probability does not need to satisfy this.

Now let me write this as a matrix action. The vector $v_k = \begin{pmatrix} x_k \\ y_k \\ z_k \end{pmatrix}$ is a probability vector: x_k is the probability that we are currently in vertex 1, y_k for vertex 2 and z_k for vertex 3. Since the total probability must be 1, the vector entries must satisfy $x+y+z=3$. Then the matrix A acts on the vector, taking v_k to v_{k+1} in the next timestep. The entry A_{ij} in the matrix is the probability of transition from vertex j to vertex i, so the number on the arrow from j to i.

$$\begin{pmatrix} 0.2 & 0.5 & 0.2 \\ 0.4 & 0.2 & 0.4 \\ 0.4 & 0.3 & 0.4 \end{pmatrix} \begin{pmatrix} x_k \\ y_k \\ z_k \end{pmatrix} = \begin{pmatrix} x_{k+1} \\ y_{k+1} \\ z_{k+1} \end{pmatrix}$$

The columns of this matrix add to 1. That's the same fact that I mentioned in the previous paragraph, that the outgoing probabilities from any vertex must add to one. The columns are all the outgoing probabilities from a vertex. The rows are all the incoming probabilities for a vertex.

Let me be even more explicit with this example. Say that we start on vertex 2 with probability 1. This means the starting vector is $v_0 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$. What happens to this as the system changes over a series of timesteps. Calculating the matrix action, here is the sequence of states. (I've not written the matrix actions, which I calculated by computer.)

$$v_0 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \qquad v_1 = Av_0 = \begin{pmatrix} 0.5 \\ 0.2 \\ 0.3 \end{pmatrix}$$

$$v_2 = A^2 v_0 = \begin{pmatrix} 0.26 \\ 0.36 \\ 0.38 \end{pmatrix} \qquad v_3 = A^3 v_0 = \begin{pmatrix} 0.308 \\ 0.328 \\ 0.364 \end{pmatrix}$$

$$v_4 = A^4 v_0 = \begin{pmatrix} 0.2984 \\ 0.3344 \\ 0.3672 \end{pmatrix} \qquad v_5 = A^5 v_0 = \begin{pmatrix} 0.30032 \\ 0.33312 \\ 0.36656 \end{pmatrix}, \ldots$$

It looks like this might be stabilizing into a set list of probabilities, with a roughly 30% chance that we end up at vertex 1; a roughly 33% chance that we end up back at vertex 2; and a roughly 37% chance that we end up at vertex 3

Hopefully this example helped to illustrate the idea of modelling probability on graph transitions. I'm going to take this example and turn it into a series of formal definitions.

11.3.2 Definitions

I am going to start with formalizing the type of vector and the type of matrix that I used in the example.

Definition 11.3.2 A vector $v \in \mathbb{R}^n$ is a **probability vector** or a **stochastic vector** if all its entries v_i are non-negative and all these entires sum to 1. $\Diamond$

Definition 11.3.3 A non-negative $n \times n$ matrix where all the columns sum to one is called a **(left) stochastic matrix**. (There is a similar definition of a right stochastic matrix where all the rows sum to one; it is, in fact, the more common definition. However, it doesn't mesh that well with our conventions of column vectors and matrix actions, so we will work with left stochastic matrices and suppress the word 'left' in what follows.) $\Diamond$

These vectors and matrices are the appropriate matrices for probability models for the following reason (which I present without proof for now).

Proposition 11.3.4 *Let A be a stochastic $n \times n$ matrix and let $v \in \mathbb{R}^n$ be a stochastic vector. Then Av is also a stochastic vector.*

Stochastic vectors are those which make sense for probabilities. This proposition tells us that the matrix action of a stochastic matrix preserves this, so the vectors can still be interpreted as probabilities after the matrix action. With this proposition, I can define the particular type of dynamical system I am presenting in this section.

Definition 11.3.5 A **Markov chain** is a linear dynamical system where the the matrix A is a stochastic matrix, and the starting state v is a stochastic vector. $\Diamond$

Now that they are defined, I want to understand the dynamics and the long term behaviour of Markov chains.

11.3.3 Markov Longterm Behaviour

In the example that I did above, it looked like the probabilities were approaching a stable configuration. Can I know that they are doing so? Can I expect that they are always doing so? The answers to this depend on the long-term behaviour of the Markov chain, which I analyze using eigenvalues and eigenvectors.

A stochastic matrix has all non-negative entries, so it satisfies the weak form of Perron-Frobenius. It is not guaranteed to be irreducible. However, for the purposes of this course, I can mostly restrict to only irreducible stochastic matrices. Thinking in terms of probability graphs, like the starting example, the stochastic matrix will be irreducible as long as the graph is connected and has no dead ends: that is, there is a path from any vertex to any other vertex. If the graph is disconnected, there the various disconnected pieces are essentially independent probability problems, and I could treat them individually. I will have an example later with dead ends, which I'll deal with when I get there. Otherwise, I will assume irreducible matrices.

The Perron-Frobenius theorem tells us that an irreducible stochastic matrix will have a unique largest positive eigenvalue with a matching vector with all positive entries. However, since stochastic matrices are a very special matrix form, there is an even stronger result, which I present here without proof.

Proposition 11.3.6 *Let A be an irreducible stochastic matrix. Then the dominant eigenvalue is $\lambda = 1$, and all other eigenvalues (real or complex) have $|\lambda| < 1$.*

Since I can scale the eigenvector v matching $\lambda = 1$ as I wish, I can choose v such that it is a stochastic vector. Since all other eigenvalues have $|\lambda| < 1$, their contributions to the system will decay over time. The long term probability behaviour is given by the coefficients of this vector v. This is *independent* of the starting situation. Markov chains (at least with connected probability graphs) will stabilize to a fixed probability distribution regardless of the starting state (though it might take longer to converge depending on where we start).

This is a pretty good description of the long term behaviour of a Markov chain. I just need to calculate and interpret the dominant eigenvector.

11.3.4 Markov Chain Examples

Example 11.3.7

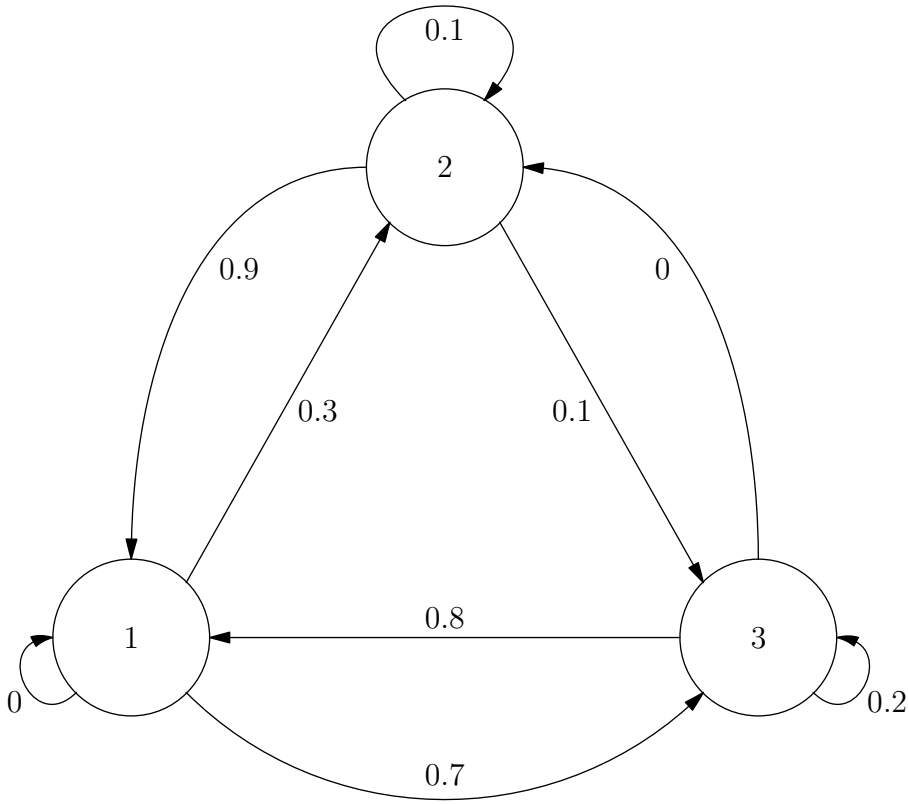

Figure 11.3.8 Probability Graph Example 2

Figure 11.3.8 shows a probability graph. Calculate the long term probability of ending up in each vertex.

Solution. First, I need to write down the stochastic matrix.

$$\begin{pmatrix} 0 & 0.9 & 0.8 \\ 0.3 & 0 & 0 \\ 0.7 & 0.1 & 0.2 \end{pmatrix}$$

Then I calculate (by computer) the dominant eigenvector. I scale the result the computer gave me to give a stochastic vector.

$$\begin{pmatrix} 1.09589 \\ 0.328786 \\ 1 \end{pmatrix} \mapsto \begin{pmatrix} 0.452 \\ 0.136 \\ 0.412 \end{pmatrix}$$

This gives me the long-term probability distribution. There is an approximately 45.2% chance of ending up on vertex 1, an approximately 13.6% chance of ending up on vertex 2, and an approximately 41.2% chance of ending up on vertex 3. □

Example 11.3.9

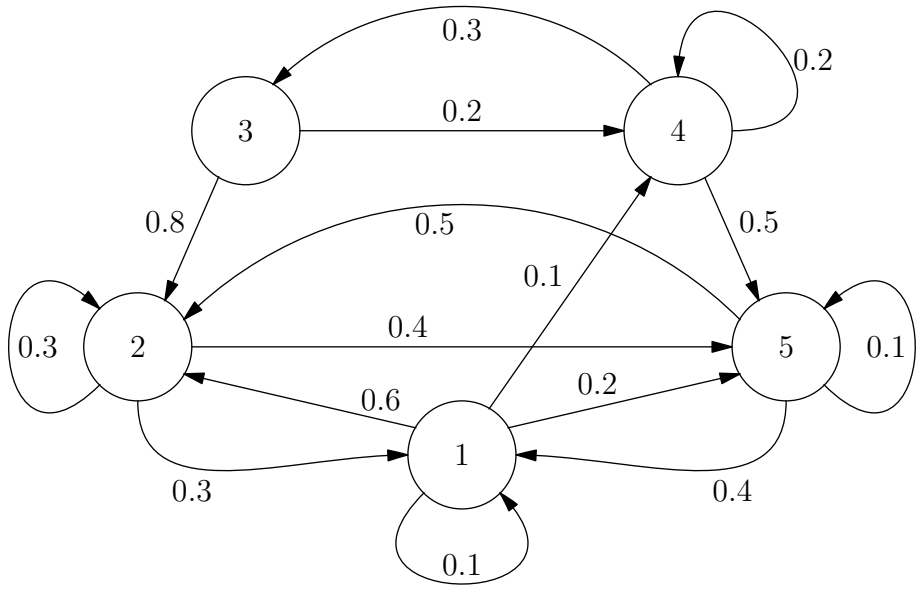

Figure 11.3.10 Probability Graph Example 3

Figure 11.3.10 shows a probability graph. (Any connections without lines between vertices have probability zero). Calculate the long-term probability of ending up in each vertex.

Solution. First, I need to write down the stochastic matrix.

$$\begin{pmatrix} 0.1 & 0.3 & 0 & 0 & 0.4 \\ 0.6 & 0.3 & 0.8 & 0 & 0.5 \\ 0 & 0 & 0 & 0.3 & 0 \\ 0.1 & 0 & 0.2 & 0.2 & 0 \\ 0.2 & 0.4 & 0 & 0.5 & 0.1 \end{pmatrix}$$

Then I calculate (by computer) the dominant eigenvector. (I scale the result given to me from the computer to get a stochastic vector.

$$\begin{pmatrix} 0.976673 \\ 1.59669 \\ 0.0395948 \\ 0.131983 \\ 1 \end{pmatrix} \mapsto \begin{pmatrix} 0.261 \\ 0.426 \\ 0.011 \\ 0.035 \\ 0.267 \end{pmatrix}$$

This gives me the long-term probability distribution. The approximate percentage chance of ending at each vertex is, in order, 26.1%, 42.6%, 1.1%, 3.5% and 26.7%. □

11.3.5 Games of Chance

Probablistic models like Markov chains are very common in game theory. In this section, I want to look at very simple games of chance (though the theory extends well to more complicated games). I want to build to a useful (and possibly morally interesting) example called the Gambler's Ruin.

I am going to model a pretty simple game of chance. In this game, there is a set stake that can be won or lost in each round (each time step, in the model). In each round, there is a winning and losing probability, which doesn't depend

on anything else. The player starts with some multiple of the stake, which will give the vertices on the graph. Then there is a stopping condition: the player stops either when they get to zero stakes, or they get to some winning number of stakes.

I'll start with a winning condition of 4 stakes. That gives me a five-stake model: the player can be at 0. 1, 2, 3 or 4 stakes. At 0 and 4 stakes, the game is finished; in the Markov chain, this is represented by a probability of 1 for staying at that vertex. For the middle three vertices, there is a probability of winning (going up a stake) and losing (going down a stake). Figure 11.3.11 shows the probability graph for this game, with W being the win probability and L being the loss probability.

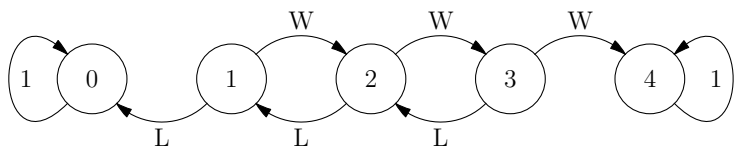

Figure 11.3.11 Probability Graph for a Game of Chance

Here is the accompanying stochastic matrix.

$$\begin{pmatrix} 1 & L & 0 & 0 & 0 \\ 0 & 0 & L & 0 & 0 \\ 0 & W & 0 & L & 0 \\ 0 & 0 & W & 0 & 0 \\ 0 & 0 & 0 & W & 1 \end{pmatrix}$$

Now we need to work out the model with actual winning and losing percentages. First, I'll start with a completly fair game so that $W = 0.5$ and $L = 0.5$. Here is that stochastic matrix.

$$\begin{pmatrix} 1 & 0.5 & 0 & 0 & 0 \\ 0 & 0 & 0.5 & 0 & 0 \\ 0 & 0.5 & 0 & 0.5 & 0 \\ 0 & 0 & 0.5 & 0 & 0 \\ 0 & 0 & 0 & 0.5 & 1 \end{pmatrix}$$

I asked a computer for the dominant eigenvalues. This is a little tricky, since there are actually two linearly independent eigenvalueS. (The problem is that this matrix is not actually irreducible, since there is no path from vertex 0 to vertex 4 this is a dead-end). Here are the two eigenvectors.

$$\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \qquad \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

These make sense. If the player is at the final losing or winning state, they stay there. However, what if they start in the middle? The final state will be some linear combination of these eigenvectors: some probability of the final winning and the final losing state. What linear combination? For this Markov model, I'll have to rely on actually calculating the matrix action. What I'll do is calculate the state after twenty iterations. That won't be a perfect final result but will give a pretty good idea of what is going on.

For this fair game, the player can start with 1, 2 or 3 stakes. I asked a computer to calculate the probabilities after twenty iterations for each of these starting points.

$$\begin{pmatrix} 1 & 0.5 & 0 & 0 & 0 \\ 0 & 0 & 0.5 & 0 & 0 \\ 0 & 0.5 & 0 & 0.5 & 0 \\ 0 & 0 & 0.5 & 0 & 0 \\ 0 & 0 & 0 & 0.5 & 1 \end{pmatrix}^{20} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0.7495 \\ 0.0005 \\ 0 \\ 0.0005 \\ 0.2495 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0.5 & 0 & 0 & 0 \\ 0 & 0 & 0.5 & 0 & 0 \\ 0 & 0.5 & 0 & 0.5 & 0 \\ 0 & 0 & 0.5 & 0 & 0 \\ 0 & 0 & 0 & 0.5 & 1 \end{pmatrix}^{20} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0.4995 \\ 0 \\ 0.0010 \\ 0 \\ 0.4995 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0.5 & 0 & 0 & 0 \\ 0 & 0 & 0.5 & 0 & 0 \\ 0 & 0.5 & 0 & 0.5 & 0 \\ 0 & 0 & 0.5 & 0 & 0 \\ 0 & 0 & 0 & 0.5 & 1 \end{pmatrix}^{20} \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0.2495 \\ 0.0005 \\ 0 \\ 0.0005 \\ 0.7495 \end{pmatrix}$$

This is something I can interpret. If I round to whole percentages (or even tenths), the situation is pretty clear. Starting with one stake, there is a 25% chance of winning and a 75% chance of losing. Starting with two stakes, there is an equal chance of winning and losing. Starting with three stakes, there is a 75% chance of winning and a 25% chance of losing. This seems pretty reasonable.

Now I can ask the interesting question: what is the effect of changing the individual round winning/losing probabilities on the final winning/losing percentages. If I make the game slightly unfair, say $W = 0.48$ and $L - 0.52$, how much will the other percentages change? I can run the model again.

$$\begin{pmatrix} 1 & 0.52 & 0 & 0 & 0 \\ 0 & 0 & 0.52 & 0 & 0 \\ 0 & 0.48 & 0 & 0.52 & 0 \\ 0 & 0 & 0.48 & 0 & 0 \\ 0 & 0 & 0 & 0.48 & 1 \end{pmatrix}^{20} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0.7787 \\ 0.0005 \\ 0 \\ 0.0004 \\ 0.2204 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0.52 & 0 & 0 & 0 \\ 0 & 0 & 0.52 & 0 & 0 \\ 0 & 0.48 & 0 & 0.52 & 0 \\ 0 & 0 & 0.48 & 0 & 0 \\ 0 & 0 & 0 & 0.48 & 1 \end{pmatrix}^{20} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0.5394 \\ 0 \\ 0.0010 \\ 0 \\ 0.4596 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0.52 & 0 & 0 & 0 \\ 0 & 0 & 0.52 & 0 & 0 \\ 0 & 0.48 & 0 & 0.52 & 0 \\ 0 & 0 & 0.48 & 0 & 0 \\ 0 & 0 & 0 & 0.48 & 1 \end{pmatrix}^{20} \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0.2802 \\ 0.0005 \\ 0 \\ 0.0005 \\ 0.7188 \end{pmatrix}$$

I changed the fairnes of the game by 2%. The effect on the final winning percentages is about 3% starting with one or three stakes, and 4% for starting with two stakes. The effect is slightly exaggerated.

What about if I make the game very unfair. Let me set $W = 0.35$ and $L = 0.65$ and run the system again.

$$\begin{pmatrix} 1 & 0.65 & 0 & 0 & 0 \\ 0 & 0 & 0.65 & 0 & 0 \\ 0 & 0.35 & 0 & 0.65 & 0 \\ 0 & 0 & 0.35 & 0 & 0 \\ 0 & 0 & 0 & 0.35 & 1 \end{pmatrix}^{20} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0.9211 \\ 0.0002 \\ 0 \\ 0.0001 \\ 0.0786 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0.65 & 0 & 0 & 0 \\ 0 & 0 & 0.65 & 0 & 0 \\ 0 & 0.48 & 0 & 0.65 & 0 \\ 0 & 0 & 0.35 & 0 & 0 \\ 0 & 0 & 0 & 0.35 & 1 \end{pmatrix}^{20} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0.7750 \\ 0 \\ 0.0004 \\ 0 \\ 0.2247 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0.65 & 0 & 0 & 0 \\ 0 & 0 & 0.65 & 0 & 0 \\ 0 & 0.35 & 0 & 0.65 & 0 \\ 0 & 0 & 0.35 & 0 & 0 \\ 0 & 0 & 0 & 0.35 & 1 \end{pmatrix}^{20} \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0.5035 \\ 0.0003 \\ 0 \\ 0.0002 \\ 0.4960 \end{pmatrix}$$

Again, the result are exagerated. A game that is 65% to lose in each round leads to a 78% chance of losing overall when starting in the middle.

11.3.6 The Gambler's Ruin

Now I am going to return to the fair game, with $W = L = 0.5$. Instead of changing the fairness of the game, I'm going to change the stopping condition. Let's say the players always start swith 2 stakes, but I'll change the number of stakes at which the player stops playing. In the original game, having 2 led to a winning percentage of 50%. Let me increase the stopping condition to five stakes. Here is the new matrix, with the action (with twenty iterations) on the starting vector with w stakes.

$$\begin{pmatrix} 1 & 0.5 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.5 & 0 & 0 & 0 \\ 0 & 0.5 & 0 & 0.5 & 0 & 0 \\ 0 & 0 & 0.5 & 0 & 0.5 & 0 \\ 0 & 0 & 0 & 0.5 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0.5 & 1 \end{pmatrix}^{20} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0.5924 \\ 0 \\ 0.0104 \\ 0 \\ 0.0065 \\ 0.3907 \end{pmatrix}$$

This hasn't entirely stabilized after twenty iterations, but I can see that the winning percentage has dropped from 50% to somewhere around 39%. Now I will increase the stopping condition to six stakes and rerun the system (I'll need to go to thirty iterations to get it to stabilize to a tenth of a percent).

$$\begin{pmatrix} 1 & 0.5 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.5 & 0 & 0 & 0 & 0 \\ 0 & 0.5 & 0 & 0.5 & 0 & 0 & 0 \\ 0 & 0 & 0.5 & 0 & 0.5 & 0 & 0 \\ 0 & 0 & 0 & 0.5 & 0 & 0.5 & 0 \\ 0 & 0 & 0 & 0 & 0.5 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0.5 & 1 \end{pmatrix}^{30} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0.6600 \\ 0 \\ 0.007 \\ 0 \\ 0.007 \\ 0 \\ 0.3267 \end{pmatrix}$$

Now the winning percentage has dropped again to about 33%.

I can continue this process (though doing the calculations, even by computer, becomes pretty tedious; however, I hope this is enough to make the point). If I play a fair game with fixed starting stakes, as the stopping point increases, the winning chance from the fixed starting point drops. In the limit (as the stopping condition goes to infinity) the winning chance will drop to zero. If I take the limit and let the stopping condition go to infinity, it actually doesn't matter what the starting number of stakes is -- the winning percentage will still eventually drop to zero.

This is called the Gambler's Ruin. It says that even if you play a perfectly fair game indefinitely, you will eventually lose with 100% probability.

11.4 Week 11 Activity

11.4.1 Leslie Matrices

Activity 11.4.1 This is a Leslie matrix for a four-stage population model. Determine the long term behaviour by analyzing the eigenvalues and eigenvectors. Describe the eventual age ratios and determine whether or not the population is viable.

$$\begin{pmatrix} 0 & 0 & 0.4 & 0.9 \\ 0.8 & 0 & 0 & 0 \\ 0 & 0.4 & 0 & 0 \\ 0 & 0 & 0.9 & 0 \end{pmatrix}$$

Solution. Here is the dominant eigenvalue and the matching eigenvector (calculated by computer).

$$\lambda = 0.77 \qquad\qquad v = \begin{pmatrix} 1.61 \\ 1.66 \\ 0.86 \\ 1 \end{pmatrix}$$

The population is not viable, since the eigenvalue is less than one. The age ratios will approach 1.61 : 1.66 : 0.86 : 1 from youngest to oldest as the population decays.

Activity 11.4.2 In the previous question, how much would you need to change the first fecundity coefficient to make the population viable.

Solution. I mostly need to guess and test. Let me change the coefficient into 0.7

$$\begin{pmatrix} 0 & 0 & 0.7 & 0.9 \\ 0.8 & 0 & 0 & 0 \\ 0 & 0.4 & 0 & 0 \\ 0 & 0 & 0.9 & 0 \end{pmatrix}$$

The new dominant eigenvalue is 0.82. This is not sufficient. Let me change the coefficient into 1.0.

$$\begin{pmatrix} 0 & 0 & 1.0 & 0.9 \\ 0.8 & 0 & 0 & 0 \\ 0 & 0.4 & 0 & 0 \\ 0 & 0 & 0.9 & 0 \end{pmatrix}$$

The new dominant eigenvalue is 0.85, which is still not sufficient. Let me

change the coefficient into 2.0.

$$\begin{pmatrix} 0 & 0 & 2.0 & 0.9 \\ 0.8 & 0 & 0 & 0 \\ 0 & 0.4 & 0 & 0 \\ 0 & 0 & 0.9 & 0 \end{pmatrix}$$

The new dominant eigenvalue is 0.97. This is still not sufficient, but getting better. Let me change the coefficient into 3.0.

$$\begin{pmatrix} 0 & 0 & 3.0 & 0.9 \\ 0.8 & 0 & 0 & 0 \\ 0 & 0.4 & 0 & 0 \\ 0 & 0 & 0.9 & 0 \end{pmatrix}$$

The new dominant eigenvalue is 1.06, so the population is now viable. I've overshot here by a bit, but this is good enough.

Activity 11.4.3 This is a Leslie matrix for a four-stage population model. Determine the long term behaviour by analyzing the eigenvalues and eigenvectors. Describe the eventual age ratios and determine whether or not the population is viable.

$$\begin{pmatrix} 0 & 15 & 200 & 100 \\ 0.07 & 0 & 0 & 0 \\ 0 & 0.21 & 0 & 0 \\ 0 & 0 & 0.12 & 0 \end{pmatrix}$$

Solution. Here is the dominant eigenvalue and the matching eigenvector (calculated by computer).

$$\lambda = 1.69 \qquad\qquad v = \begin{pmatrix} 2731.2 \\ 113.2 \\ 14.1 \\ 1 \end{pmatrix}$$

The population is viable. The age ratios will approach $2731.2 : 113.2 : 14.1 : 1$ from youngest to oldest.

Activity 11.4.4 In the previous question, how much would you have to reduce the first survival coefficient to make the population no longer viable.

Solution. I mostly need to guess and test. I'll first set the survival to 0.05.

$$\begin{pmatrix} 0 & 15 & 200 & 100 \\ 0.05 & 0 & 0 & 0 \\ 0 & 0.21 & 0 & 0 \\ 0 & 0 & 0.12 & 0 \end{pmatrix}$$

The new dominant eigenvector is $\lambda = 1.49$ This is still not enough. Now I'll set the survival to 0.01.

$$\begin{pmatrix} 0 & 15 & 200 & 100 \\ 0.01 & 0 & 0 & 0 \\ 0 & 0.21 & 0 & 0 \\ 0 & 0 & 0.12 & 0 \end{pmatrix}$$

The new dominant eigenvector is $\lambda = 0.83$. The popuation is no longer viable.

I may have overshot a bit, so I'll also try 0.02.

$$\begin{pmatrix} 0 & 15 & 200 & 100 \\ 0.02 & 0 & 0 & 0 \\ 0 & 0.21 & 0 & 0 \\ 0 & 0 & 0.12 & 0 \end{pmatrix}$$

The new dominant eigenvector is $\lambda = 1.06$, so perhaps I didn't overshoot that much. I could go to coefficients between 0.01 and 0.02, but this is sufficient for now.

Activity 11.4.5 This is a Leslie matrix for a four-stage population model. Determine the long term behaviour by analyzing the eigenvalues and eigenvectors. Describe the eventual age ratios and determine whether or not the population is viable.

$$\begin{pmatrix} 0 & 0.32 & 0.36 & 0.29 \\ 0.95 & 0 & 0 & 0 \\ 0 & 0.8 & 0 & 0 \\ 0 & 0 & 0.9 & 0 \end{pmatrix}$$

Solution. Here is the dominant eigenvalue and the matching eigenvector (calculated by computer).

$$\lambda = 0.92 \qquad\qquad v = \begin{pmatrix} 1.12 \\ 1.16 \\ 1.02 \\ 1 \end{pmatrix}$$

The population is not viable, since the dominant eigenvalue is less than 1. The age ratios will approach $1.12 : 1.16 : 1.02 : 1$ from youngest to oldest.

Activity 11.4.6 In the previous example, how can you change the survival to make the population viable. (You may change any of the three survival coefficients).

Solution. The survival coefficients here are already quite high. The maximum they can be is 1, since no more than 100% of an age category can survive. I'm first going to check what happens when I set all the survival coefficients equal to 1.

$$\begin{pmatrix} 0 & 0.32 & 0.36 & 0.29 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

The new dominant eigenvalue is 0.989. Even with the maximum possible survival, the population isn't viable. The fecundity rates are simply too low for any hope for this population.

11.4.2 Markov Chains for Probability Graphs

Activity 11.4.7

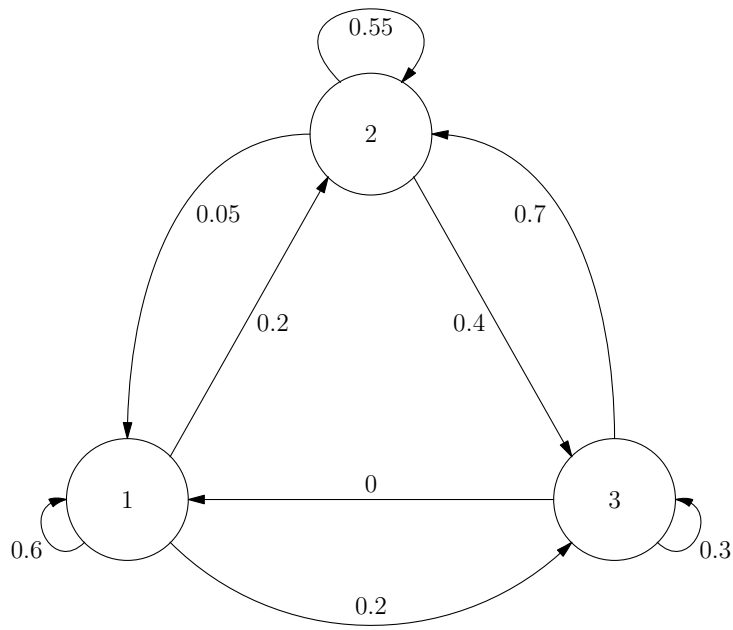

Figure 11.4.1 Probability Graph 1

Figure 11.4.1 shows a probability graph. Calculate the long term probability of ending up in each vertex.

Solution. First, I need to write down the stochastic matrix.

$$\begin{pmatrix} 0.6 & 0.05 & 0 \\ 0.2 & 0.55 & 0.7 \\ 0.2 & 0.4 & 0.3 \end{pmatrix}$$

Then I calculate (by computer) the dominant eigenvector.

$$\begin{pmatrix} 0.0722 \\ 0.5773 \\ 0.3505 \end{pmatrix}$$

This gives me the long-term probability distribution. There is roughly a 7.2% chance of ending on the first vertex; a 57.7% change of ending on the second vertex; and a 35.1% chance of ending on the third vertex.

Activity 11.4.8

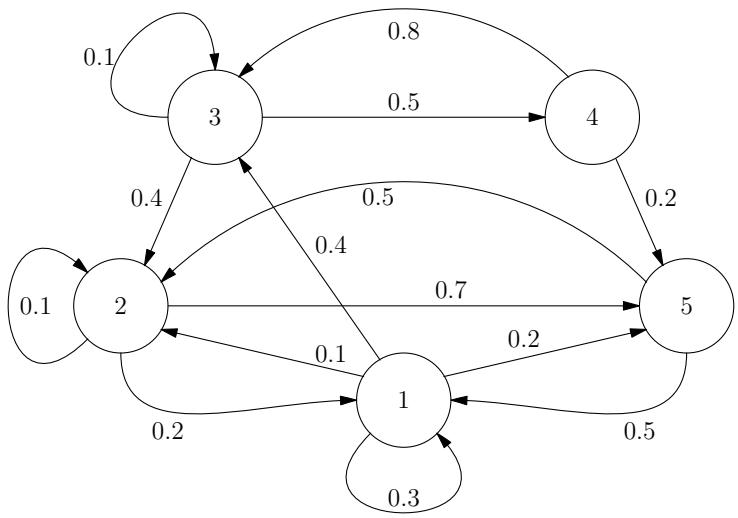

Figure 11.4.2 Probability Graph 1

Figure 11.4.2 shows a probability graph. Calculate the long term probability of ending up in each vertex. (Assume any possibile path between vertices that doesn't have an edge in the graph has a probability of zero).

Solution. First, I need to write down the stochastic matrix.

$$\begin{pmatrix} 0.3 & 0.2 & 0 & 0 & 0.5 \\ 0.1 & 0.1 & 0.4 & 0 & 0.5 \\ 0.4 & 0 & 0.1 & 0.8 & 0 \\ 0 & 0 & 0.5 & 0 & 0 \\ 0.2 & 0.7 & 0 & 0.2 & 0 \end{pmatrix}$$

Then I calculate (by computer) the dominant eigenvector.

$$\begin{pmatrix} 0.2375 \\ 0.2418 \\ 0.1900 \\ 0.0950 \\ 0.2358 \end{pmatrix}$$

This gives me the long-term probability distribution. There is roughly a 23.7% change of ending on vertex 1; a 24.2% change of ending on vertex 2; a 19.0% change of ending on vertex 3; a 9.5% change of ending on vertex 4; and a 23.6% change of ending on vertex 5.

Activity 11.4.9

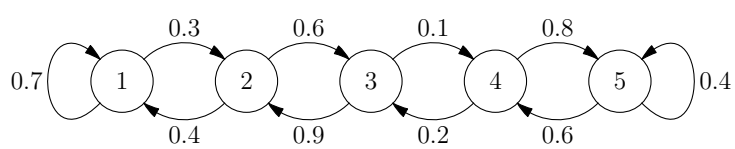

Figure 11.4.3 Probability Graph 3

Figure 11.4.3 shows a probability graph. Calculate the long term probability of ending up in each vertex.

Solution. First, I need to write down the stochastic matrix.

$$\begin{pmatrix} 0.7 & 0.4 & 0 & 0 & 0 \\ 0.3 & 0 & 0.9 & 0 & 0 \\ 0 & 0.6 & 0 & 0.2 & 0 \\ 0 & 0 & 0.1 & 0 & 0.6 \\ 0 & 0 & 0 & 0.8 & 0.4 \end{pmatrix}$$

Then I calculate (by computer) the dominant eigenvector.

$$\begin{pmatrix} 0.3529 \\ 0.2647 \\ 0.1764 \\ 0.0882 \\ 0.1176 \end{pmatrix}$$

This gives me the long-term probability distribution. There is roughly a 35.3% change of ending on vertex 1; a 26.5% change of ending on vertex 2; a 17.6% change of ending on vertex 3; a 8.8% change of ending on vertex 4; and a 11.8% change of ending on vertex 5.

Activity 11.4.10 Here is a model of a game of chance, in the same setup we used at the end of Section 11.3. Instead of looking at the long run probability of winning, what is the probability of still playing after 5 rounds when starting from 1, 2 or 3 stakes. What about after 10 rounds?

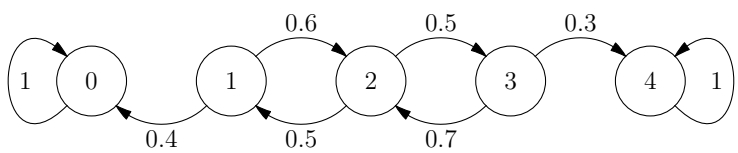

Figure 11.4.4 Probability Graph 4

Solution. First, I need to write down the stochastic matrix.

$$\begin{pmatrix} 1 & 0.4 & 0 & 0 & 0 \\ 0 & 0 & 0.5 & 0 & 0 \\ 0 & 0.6 & 0 & 0.7 & 0 \\ 0 & 0 & 0.5 & 0 & 0 \\ 0 & 0 & 0 & 0.3 & 1 \end{pmatrix}$$

To get the probabilities when starting from 1 stake after 5 rounds, I calculate the matrix action five times on the vector that represents 1 stake.

$$\begin{pmatrix} 1 & 0.4 & 0 & 0 & 0 \\ 0 & 0 & 0.5 & 0 & 0 \\ 0 & 0.6 & 0 & 0.7 & 0 \\ 0 & 0 & 0.5 & 0 & 0 \\ 0 & 0 & 0 & 0.3 & 1 \end{pmatrix}^5 \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0.598 \\ 0 \\ 0.2535 \\ 0 \\ 0.1485 \end{pmatrix}$$

The probability of still playing is the sum of the three middle probabilities, since the game stops when we are at 0 or 4 stakes. Here that sum is 25.4%.

To get the probabilities when starting from 2 stakes after 5 rounds, I calcu-

late the matrix action five times on the vector that represents 2 stakes.

$$\begin{pmatrix} 1 & 0.4 & 0 & 0 & 0 \\ 0 & 0 & 0.5 & 0 & 0 \\ 0 & 0.6 & 0 & 0.7 & 0 \\ 0 & 0 & 0.5 & 0 & 0 \\ 0 & 0 & 0 & 0.3 & 1 \end{pmatrix}^5 \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0.33 \\ 0.21125 \\ 0 \\ 0.21125 \\ 0.2475 \end{pmatrix}$$

The probability of still playing is the sum of the three middle probabilities, since the game stops when we are at 0 or 4 stakes. Here that sum is 42.3%.

To get the probabilities when starting from 3 stakes after 5 rounds, I calculate the matrix action five times on the vector that represents 3 stakes.

$$\begin{pmatrix} 1 & 0.4 & 0 & 0 & 0 \\ 0 & 0 & 0.5 & 0 & 0 \\ 0 & 0.6 & 0 & 0.7 & 0 \\ 0 & 0 & 0.5 & 0 & 0 \\ 0 & 0 & 0 & 0.3 & 1 \end{pmatrix}^5 \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0.231 \\ 0 \\ 0.29575 \\ 0 \\ 0.47325 \end{pmatrix}$$

The probability of still playing is the sum of the three middle probabilities, since the game stops when we are at 0 or 4 stakes. Here that sum is 29.6%.

To get the probabilities when starting from 1 stake after 10 rounds, I calculate the matrix action five times on the vector that represents 1 stake.

$$\begin{pmatrix} 1 & 0.4 & 0 & 0 & 0 \\ 0 & 0 & 0.5 & 0 & 0 \\ 0 & 0.6 & 0 & 0.7 & 0 \\ 0 & 0 & 0.5 & 0 & 0 \\ 0 & 0 & 0 & 0.3 & 1 \end{pmatrix}^{10} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0.68165 \\ 0.0535519 \\ 0 \\ 0.0535519 \\ 0.211241 \end{pmatrix}$$

The probability of still playing is the sum of the three middle probabilities, since the game stops when we are at 0 or 4 stakes. Here that sum is 10.7%.

To get the probabilities when starting from 2 stakes after 10 rounds, I calculate the matrix action five times on the vector that represents 2 stakes.

$$\begin{pmatrix} 1 & 0.4 & 0 & 0 & 0 \\ 0 & 0 & 0.5 & 0 & 0 \\ 0 & 0.6 & 0 & 0.7 & 0 \\ 0 & 0 & 0.5 & 0 & 0 \\ 0 & 0 & 0 & 0.3 & 1 \end{pmatrix}^{10} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0.505125 \\ 0 \\ 0.116029 \\ 0 \; 0.378845 \end{pmatrix}$$

The probability of still playing is the sum of the three middle probabilities, since the game stops when we are at 0 or 4 stakes. Here that sum is 11.6%.

To get the probabilities when starting from 3 stakes after 10 rounds, I calculate the matrix action five times on the vector that represents 3 stakes.

$$\begin{pmatrix} 1 & 0.4 & 0 & 0 & 0 \\ 0 & 0 & 0.5 & 0 & 0 \\ 0 & 0.6 & 0 & 0.7 & 0 \\ 0 & 0 & 0.5 & 0 & 0 \\ 0 & 0 & 0 & 0.3 & 1 \end{pmatrix}^{10} \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0.328598 \\ 0.0624772 \\ 0 \\ 0.0624772 \\ 0.546448 \end{pmatrix}$$

The probability of still playing is the sum of the three middle probabilities, since the game stops when we are at 0 or 4 stakes. Here that sum is 12.5%.

11.4.3 Conceptual Review Questions

- What is a recurrence relation and how it is translated into a matrix?
- What is a dynamical system? What does it mean for such a system to be discrete and linear?
- How can matrices be used as models?
- Why are eigenvalues and eigenvectors important in matrix models?
- What is a Leslie matrix and what does it measure?
- What is a Markov chain and how does it model probability?

Colophon

This book was authored in PreTeXt.